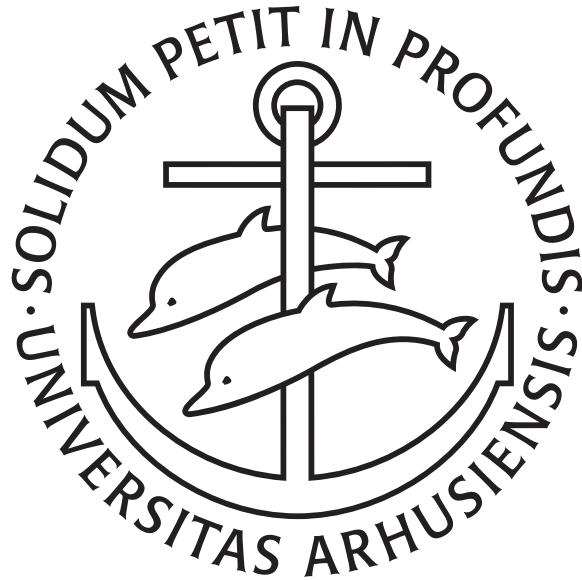


Aarhus University

Department of Economics and Business Economics

Master's Thesis



Survey Expectations and Finite-Sample Bias in a Macro-Finance Gaussian Affine Term Structure Model

Sune Grøn Pedersen[†]

Frederik Bendix Skammelsen[‡]

Supervisor

Martin Møller Andreasen

264,271 characters (incl. spaces) $\approx$ 110.11 pages

1st of June, 2026

[†]Student-ID: 202104950

[‡]Student-ID: 202107418

Abstract

We find that the finite-sample bias which is prevalent in a four-factor Gaussian Affine Term Structure Model, with two latent factors and inflation and unemployment as observable factors, can be partially mitigated by including survey expectations of the macro factors. The bias in the persistence parameter of inflation is greatly reduced, while in contrast the surveys overcorrect the bias reduction for unemployment. Furthermore, we find that the magnitude of the finite-sample bias is larger when using a shorter sample size both in the case with and without survey expectations incorporated. Lastly, the bias mitigation from surveys are most useful when using one-quarter ahead forecasts compared to using the noisier four-quarters ahead counterparts.

Contents

1	Introduction	1
2	Data	3
2.1	U.S. Yield Curve Data	3
2.2	Realised Macroeconomic Variables	4
2.2.1	Inflation	4
2.2.2	Unemployment	5
2.3	Survey Expectations	6
2.3.1	Timing and Horizon Structure of the SPF	6
2.3.2	Survey Inflation	6
2.3.3	Survey Unemployment	7
2.4	Data Preparation and Sample Alignment	7
2.5	Descriptive Analysis	8
2.5.1	Yield Curve Descriptive Statistics	8
2.5.2	Yield Surface Visualisation	9
2.5.3	Survey Versus Realised Macro Comparisons	10
2.6	Sample Choices and Alternatives	15
3	Theoretical framework	16
3.1	Yields	16
3.2	GATSM Framework	17
3.3	Physical Measure $\mathbb{P}$	18
3.4	$\mathbb{P}$ and $\mathbb{Q}$ Dynamics	19
3.4.1	Real-world Measure to Risk-Free Measure	19
3.5	Solving the Model	20
3.5.1	Yield Decomposition	22
4	Two-Factor Model	24
4.1	Identification	24
4.2	State-Space Representation	27
4.3	Kalman Filter	28
4.4	Two-Factor Starting Parameters	30
4.5	Choice of Optimiser	33
4.6	Two-Factor GATSM Result	35

4.6.1	Campbell-Shiller	41
4.7	Monte Carlo Study Two-Factor	43
5	Macro Factors	46
5.1	Macro-Finance Model Specification	46
5.2	Four-Factor Estimation Methodology	47
5.2.1	Step 1: First-Stage Reduced VAR OLS	47
5.2.2	Step 2: Structural Parameters	51
5.2.3	Step 3: Recovering $\mu^{\mathbb{Q}}$ and α Numerically	53
5.2.4	Step 4: Recovering $\mu^{\mathbb{P}}$ Analytically	54
5.2.5	Bond Pricing Loadings and Kalman Filter	54
5.2.6	Identification Test	55
5.3	Macro-Finance Results	57
5.3.1	Yield Curve Decomposition	61
6	Survey Expectations	63
6.1	Motivation for Surveys	63
6.2	Survey Incorporation	64
6.2.1	Surveys and Data	64
6.2.2	OLS Stacking	65
6.2.3	Downstream Estimation Effects	67
6.3	Survey Results	69
6.3.1	Yield Decomposition Surveys	72
6.3.2	Four-Factor Campbell-Shiller	74
7	Monte Carlo Finite Sample Test	75
7.1	Survey-Augmented Monte Carlo Study	75
7.1.1	Data-Generating Process	75
7.1.2	Survey Measurement and Calibration	75
7.1.3	Simulation Sampling	79
7.1.4	Three Estimators and Bias Metrics	81
7.2	1Q Monte Carlo Result	82
7.3	Monte Carlo Results Small Sample	85
7.4	Noisy Surveys	88

8	Short Versus Long Horizon Surveys	91
8.1	Monte Carlo Results of Differing Forecast Horizons	91
8.2	Short Versus Long Small Sample	94
9	Impulse Response Analysis	96
9.1	How IRFs Are Computed	96
9.2	Yield Response of Structural Shocks	99
9.3	State Vector Responses	104
9.4	Variance Decomposition	107
10	Discussion	109
10.1	Overview of Findings	109
10.2	Relation to the Literature	110
10.3	Alternative Data	115
10.4	Alternative Horizon or Timing Choice	117
10.5	Robustness and Sensitivity	118
10.6	Policy Implications	119
10.7	Limitations and Future Research	120
11	Conclusion	123
12	Appendices	1
12.1	Appendix A1: MCSE Yield Partition	1
12.2	Appendix A2: MCSE Conditions	3
12.3	Appendix A3: $\mathbb{P}$ to $\mathbb{Q}$	4
12.4	Appendix A4: FEVD	5
12.5	Appendix A5: $\mathbb{P}$ Recursion	7
12.6	Appendix A6: Alternative Term Premium	8
12.7	Appendix A7: Solving the Model	10
12.8	Appendix B1: Extra Figures	14
12.9	Appendix C1: Main Survey Script	20

1 Introduction

The term structure of interest rates is a central topic in both academic research and monetary policy practice. Central banks, financial institutions, and investors rely on yield curve models to decompose long-term interest rates into expected future short-rates and term premia, with each component carrying distinct implications for portfolio decisions, monetary policy decisions and the broader macroeconomic outlook. The reliability of this decomposition depends critically on how well the underlying model captures the true dynamics of interest rates, making the accuracy of term structure estimation a central question of both academic and practical importance.

A persistent challenge in the estimation of Gaussian affine term structure models (GATSM), a standard framework for yield curve decomposition, is the finite-sample bias in the physical measure dynamics which primarily stems from the fact that interest rates are highly persistent and empirical samples contain only a limited number of business cycles. Consequently, OLS estimates of the underlying VAR systematically understate the degree of mean reversion in the true process. This has implications for the economy as the model expects interest rates to revert to their long-run level too quickly, compressing the expected short-rate component and pushing a large share of yield variation into the term premium. Central banks assessing the market conditions and investors pricing long-term bonds risk drawing systematically misleading conclusions if the yield curve decomposition is distorted.

This paper investigates whether survey expectations of macroeconomic variables can help reduce this finite-sample bias in a macro-finance term structure model, where we proceed through a stepwise model extension approach. Starting from a two-factor latent GATSM estimated via Kalman filtering maximum likelihood, we document the existence of the finite-sample bias in the physical dynamics through a Monte Carlo study. We then extend the model to a four-factor macro-finance GATSM, following the framework of [Ang and Piazzesi \(2003\)](#), by incorporating inflation and unemployment as directly observable state variables alongside the two latent yield factors from the baseline model. The four-factor model and its further modifications are estimated using the minimum chi-square approach proposed in [Hamilton and Wu \(2012\)](#). The finite-sample bias persists in this extended model, so we augment the model with one-quarter-ahead SPF forecasts of inflation and unemployment, drawing on the survey augmentation insight of [Kim and Orphanides \(2012\)](#). The key motivation for this is that professional forecasters form their expectations about future macroeconomic outcomes in the real world, and their forecasts are therefore expected to carry information about the physical dynamics that the yield cross-section alone cannot provide. Stacking

survey observations alongside realised macro data in the estimation of the physical macro block, allows the model to incorporate forward-looking information beyond what the yield cross-section alone can provide.

This paper yields the following main findings. The Monte Carlo study confirms a systematic negative bias in the persistence parameters of the physical dynamics, which persists after the extension to the four-factor macro-finance specification. Incorporating one-quarter-ahead SPF forecasts substantially reduces the bias in inflation persistence, with the correction becoming more pronounced at shorter sample sizes where the conventional estimator is most unreliable. The bias reduction comes with a trade-off; the unemployment persistence estimate over-corrects and turns positive, suggesting that SPF forecasters systematically expect unemployment to mean-revert faster than it actually does. The survey augmentation also elevates the implied term premium considerably relative to the no-survey model, as the surveys pull down the expected short-rate component. Finally, a comparison between the one-quarter-ahead and four-quarter-ahead survey forecasts show that forecast horizon matter considerably for survey augmentation effectiveness, with shorter horizons strongly outperforming longer ones due to their substantially higher signal-to-noise ratio.

The paper is structured as follows. Section 2 describes the data. Section 3 presents the main theoretical framework of the GATSM. Section 4 introduces the two-factor latent GATSM and its estimation results. Section 5 extends the model to the four-factor macro-finance specification. Section 6 presents the survey augmentation and its results. Section 7 conducts the extensive Monte Carlo study. Section 8 examines the short versus long horizon survey comparison. Section 9 presents the impulse response analysis. Section 10 discusses the findings and their implications. Finally, Section 11 concludes the paper.

2 Data

This section describes and analyses the data used in this paper. The analysis focuses on three major U.S. data components: zero-coupon government bond yields, realised macroeconomic variables, and survey expectations of the included macroeconomic variables.

The U.S. dataset is constructed so that the three different data components are mutually consistent in timing, frequency, and variable definitions. In particular, the realised inflation measure is chosen as the GDP deflator, to match the PGDP from survey expectations, which has the largest observation period of the inflation survey measures. Furthermore, the frequency of data is matched. For example, the unemployment macroeconomic data is available in monthly frequency, while the corresponding survey expectation is only available quarterly. These are aggregated to quarterly frequency to match the frequency of the survey data.

The analysis focuses on two macroeconomic variables: inflation and unemployment. Inflation is measured using the GDP deflator, including quarterly survey data from 1958:Q4 to 2026:Q1. Unemployment is included because it captures an important aspect of business-cycle conditions. The final sample period for the U.S. analysis is 1972:Q1 to 2025:Q4 when merging the yield, macro variable and survey expectation datasets. The yield data are available daily from 1961 onward, and the Survey of Professional Forecasters (SPF) begins in 1968. However, the sample is restricted to 1972 onward to ensure sufficiently long maturity coverage.

2.1 U.S. Yield Curve Data

Zero-coupon government bond yields are used to represent the term structure of interest rates.

This dataset is obtained from [Fed \(2026\)](#)'s nominal yield curve dataset. This dataset provides daily yield curves from 1961 to the present. The yield curves are estimated using the [Svensson \(1994\)](#) methodology from 1980 onward, while the [Nelson and Siegel \(1987\)](#) methodology is used prior to 1980 because there were not enough Treasury securities to fit the Svensson model [Fed \(2019\)](#).

For the analysis, the daily data are converted to a monthly frequency using the last available end-of-month observation for each month. The code first sorts the daily observations by date, groups them by calendar month, removes rows containing missing values and selects the final remaining observation within each month. This ensures that the monthly yield

curves used in estimation are observed on a single end-of-month date. This is done to avoid excessive noise in the data and to better match the quarterly macroeconomic values used later in the analysis.

The initial yield curve dataset includes daily maturities ranging from 1 to 30 years, with one year intervals in yields. However, the earlier part of the sample does not contain the full maturity. The analysis keeps maturities from 1 to 15 years and restricts the start date to 1972:Q1 rather than 1961:Q3 where the dataset begins. The first 10 years of the dataset only include maturities of 1 to 7 years. This truncation is done to ensure a proper trade-off between using the longest possible time-series and preserving enough maturities to conduct the later analysis.

The dataset includes data from over five decades, including several major events such as the dot-com bubble, the financial crisis and post-COVID inflation.

The monthly yield data is summarised using standard descriptive statistics for each maturity: mean, median, standard deviation, variance, minimum, maximum, and number of observations.

In addition to the descriptive tables, the data section includes a three-dimensional surface plot of the yield curve over time. The plot uses date and maturity on the first and second axis, and the yield level on the vertical axis. This figure is included to visualise the evolution of yield curves over time.

2.2 Realised Macroeconomic Variables

The macroeconomic data consists of realised inflation and unemployment. These variables are selected because they both capture core fundamentals of the business cycle, are available over a long period and are comparable to the survey expectations from SPF. Furthermore, they are both often introduced in macro-finance term structure models.

2.2.1 Inflation

Realised inflation is measured using the Gross Domestic Product: Implicit Price Deflator (GDPDEF) obtained from [Fed-St.Louis \(2026a\)](#). This captures price changes across all domestically produced goods and services, not including imports. Since yield curves are largely shaped by expectations about domestic economic conditions, the GDP deflator provides a relevant measure of inflation. The series is quarterly, seasonally adjusted, and expressed as an index with base year $2017 = 100$. This is chosen to ensure consistency with the survey

expectations, since the SPF reports forecasts for the GDP price index (PGDP). The GDP deflator is calculated using nominal GDP and real GDP using the following formula

$$GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} * 100,$$

To transform the GDP deflator to be compatible with the quarterly survey expectations, the index is converted using the same procedure as the SPF to transform levels to growth. It is converted into an annualised quarter-over-quarter growth rate as

$$\hat{g}_{t|t-1} = 100 \left[\left(\frac{Y_t}{Y_{t-1}} \right)^4 - 1 \right],$$

where Y_t denotes the level of GDP deflator.

This transformation is used to calculate forecast errors in the descriptive analysis later.

There is a potential issue regarding the base years of the realised data and the survey data. The base years are not the same across the two datasets. This issue is prevented by exclusively analysing growth rates, since growth rates are on the same scale.

2.2.2 Unemployment

Realised unemployment is measured using the civilian unemployment rate from [Fed-St.Louis \(2026b\)](#). This series is monthly and seasonally adjusted. Unemployment in this series is defined as the number of unemployed people as a percentage of the labour force, where the labour force includes people aged 16 and above living in the U.S. excluding those in institutions and those on active military duty.

To match the data frequency of realised unemployment to the frequency of the survey data of unemployment from SPF, the unemployment rate has been aggregated to quarterly frequency by taking the simple average of the three monthly observations within each quarter. This means that the quarter stated as 1972:Q1, is computed as the simple average of January, February and March. In cases of missing data, such as October 2025, the quarterly values are constructed using available months, meaning that 2025:Q4 is computed using only monthly data from November and December 2025.

2.3 Survey Expectations

Survey expectations are taken from the SPF published by the Federal Reserve Bank of Philadelphia [Fed-Philadelphia \(2026\)](#). The SPF is the longest-running quarterly survey of U.S. macroeconomic forecasts, beginning in 1968. It provides forecasts for a large number of macroeconomic variables at quarterly and annual horizons. For the variables used in this paper, SPF is used because of the long history of expectations included and because of the consistency with official macroeconomic variable definitions.

This paper uses the median forecast series rather than the mean forecast series. This is done due the median being less sensitive to outliers and to extreme responses from individual forecasters, which is useful in the earlier sample where the number of respondents may be smaller and individual responses weight higher.

2.3.1 Timing and Horizon Structure of the SPF

For the survey data, the interpretation of the forecast horizons are important. For the quarterly variables, the suffix "2" in the SPF data denotes the forecast for the current quarter (nowcast), while suffixes "3" to "6" denote forecasts for one to four quarters ahead. In the dataset included in the analysis, these horizons are coded as $h = 0, \dots, 4$ for better interpretation and to match the realised values of the target quarter implied by the horizon structure. In particular, horizon $h = 0$ corresponds to the nowcast horizon of suffix "2" in the SPF dataset and horizons $h = 1, \dots, 4$ correspond to one to four quarters ahead of suffix "3" to "6" in the SPF dataset.

For clarification, looking at the forecasts conducted in 1980:Q1, the nowcast for the current quarter ($h = 0$) is a forecast of 1980:Q1, whereas the four quarter ahead ($h = 1, \dots, 4$) are forecast of 1980:Q2-1981:Q1.

2.3.2 Survey Inflation

Survey inflation is measured using the SPF forecasts for the GDP price index (PGDP) [Fed-Philadelphia \(2026\)](#). This is defined as the level of the chain-weighted GDP price index. PGDP is described as the quarterly and annual level of the chain-weighted GDP price index with varying base years. From 1992 to 1995 it is listed as the GDP implicit deflator, and prior to 1992 as the GNP implicit deflator. PGDP is available from 1968:Q4 onward. This long history is one of the main reasons why PGDP is chosen over CPI inflation in the paper.

For the analysis, the quarterly inflation forecasts are the transformed survey variables

labeled "dpgdp", which is the quarter-over-quarter inflation at annualised rates. These are directly comparable to the realised GDP deflator measure constructed from FRED. Both are calculated using,

$$\hat{g}_{t|t-1} = 100 \left[\left(\frac{Y_t}{Y_{t-1}} \right)^4 - 1 \right],$$

where Y_t denotes the GDP deflator.

The transition from GNP to GDP is important to note. While the survey data transitions from GNP to GDP in 1992, the realised GDP deflator uses GDP throughout the entire sample. Because the analysis uses growth rates rather than levels, and because GNP and GDP are very similar for the US, this conceptual break is not expected to significantly affect results.

2.3.3 Survey Unemployment

Survey unemployment is measured using the SPF unemployment variable UNEMP, available from 1968:Q4 onward [Fed-Philadelphia \(2026\)](#). The UNEMP contains forecasts for the quarterly average and annual unemployment rate, measured in percentage points. The quarterly forecasts are the average of the underlying monthly values with the quarter. This matches the aggregation used for the realised unemployment series.

For the quarterly variables, the labels UNEMP2 - UNEMP6 represent the forecasts for the current quarter and the next four quarters. The annual variables labeled UNEMPA - UENMPD represent the current year forecast and the next three years forecast. The paper uses the quarterly forecasts because these are directly comparable with the realised unemployment and they are available for a longer horizon.

2.4 Data Preparation and Sample Alignment

The survey comparison dataset is constructed by reshaping the SPF forecasts into a long form, assigning each forecast a horizon h , and matching it to its target quarter. The realised macroeconomic variable for that target quarter is then merged onto the survey observation. Forecast errors are computed as the realized value minus the survey forecast. The long-horizon forecasts near the end of the sample do not yet have observed realised data, so these forecasts are excluded when describing the forecasting errors.

2.5 Descriptive Analysis

The descriptive analysis serves two purposes. First, it provides an overview of the properties of the yield curve data used to estimate the term structure models. Second, it evaluates the quality of the SPF survey expectations by comparing forecasts and realised macroeconomic outcomes. This is done to assess whether survey data can contribute to reducing finite-sample bias in the estimated models and acts as a preliminary test for any bias in the survey data.

The yield curve descriptive statistics summarises the distributional properties of zero-coupon yields across maturities and visualises the evolution of the term structure over the full sample period. The survey comparison analysis constructs a matched dataset of forecasts and realised values for inflation and unemployment, computes forecast errors defined as the realised outcome minus the forecast, and evaluates accuracy both in aggregate and broken down by forecast horizon. The horizon decomposition is of particular interest, as it reveals how forecast precision decays with the length of the prediction horizon — a pattern with direct implications for how much information survey expectations at different horizons can be expected to contribute to the model.

2.5.1 Yield Curve Descriptive Statistics

Table 1: Descriptive Statistics - U.S. Yield Curve

Maturity	Mean	Median	Std.	Var.	Min	Max	Obs.
1	4.843	4.920	3.499	12.240	0.0678	16.110	653
2	5.034	4.920	3.451	11.913	0.1132	15.782	653
3	5.201	5.045	3.385	11.457	0.1365	15.575	653
4	5.350	5.220	3.315	10.992	0.1841	15.350	653
5	5.484	5.363	3.250	10.561	0.2431	15.178	653
6	5.604	5.486	3.191	10.180	0.3063	15.061	653
7	5.712	5.618	3.138	9.849	0.3701	14.987	653
8	5.807	5.720	3.093	9.565	0.4327	14.940	653
9	5.891	5.749	3.054	9.324	0.4933	14.911	653
10	5.965	5.825	3.020	9.121	0.5517	14.892	653
11	6.030	5.895	2.992	8.950	0.6079	14.880	653
12	6.087	5.923	2.968	8.806	0.6621	14.871	653
13	6.137	6.008	2.947	8.687	0.7144	14.865	653
14	6.181	6.054	2.931	8.589	0.7649	14.860	653
15	6.219	6.055	2.917	8.508	0.8138	14.857	653

Table 1 reports descriptive statistics for U.S. zero-coupon government bond yields across maturities from 1 to 15 years over the full sample of 653 monthly observations. Several

features of the yield curve are immediately clear. First, the term structure is on average upward sloping: mean yields rise monotonically from 4.84% at the one-year maturity to 6.22% at the fifteen-year maturity, consistent with the typical positive term premium embedded in longer-horizon bonds. Second, yield volatility declines with maturity. The standard deviation is 3.50 for the shortest maturity and falls monotonically with 2.92 for the longest maturity, consistent with short-term rates being more sensitive to cyclical monetary policy movements than long-term rates. The minimum yields near zero at short maturities capture the post-financial-crisis and post-COVID low-rate environments, while maxima exceeding 14–16% correspond to the high-inflation period of the early 1980s. The observation count of 653 across all maturities confirms that the truncation to 1972:Q1 and maturities 1–15 years successfully eliminates the earlier incomplete portion of the dataset.

2.5.2 Yield Surface Visualisation

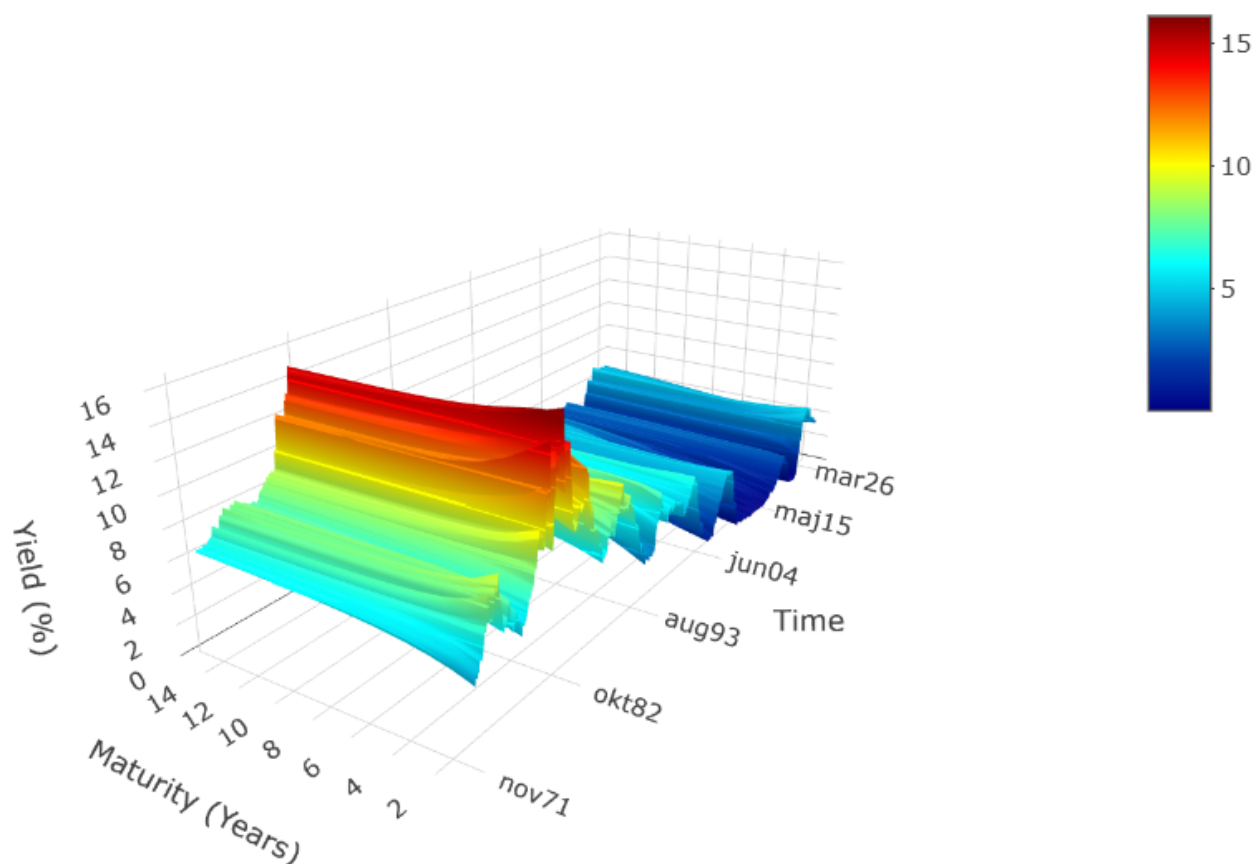


Figure 1: Surface Visualisation

Figure 1 plots the U.S. yield curve surface over the full sample period, with time on the horizontal axis, maturity on the depth axis, and yield level on the vertical axis. The surface reveals several well-known episodes in U.S. interest rate history. The most prominent feature is the sharp rise in yields across all maturities during the late 1970s and early 1980s, peaking around October 1982, which corresponds to the Federal Reserve’s aggressive monetary tightening to combat the double-digit inflation of the period. Following this peak, yields declined broadly and persistently over the subsequent four decades, with yields across all maturities falling to historically low levels during the post-financial-crisis, where the interest rate neared the zero-lower-bound, and again following the COVID-19 pandemic. The surface also illustrates the time-varying shape of the yield curve, with periods of steep upward slope alternating with episodes of flatness, reflecting how the balance between current short-rates and longer-run risk compensation shifts over time. The colour gradient, ranging from deep blue at low yield levels to red at high levels, provides a visual representation of the pronounced shift in the overall level of interest rates across the sample.

2.5.3 Survey Versus Realised Macro Comparisons

Table 2: Overall Accuracy - Inflation

Bias	MAE	RMSE	n
0.138	1.18	1.66	1130

Table 2 summarises overall forecast accuracy for inflation across all horizons and quarters. The sample of 1,130 observations exceeds the 653 monthly yield observations because the forecast evaluation dataset stacks observations across multiple forecast horizons, with each quarter contributing up to five observations corresponding to horizons $h = 0$ (nowcast) through $h = 4$ (four-quarters-ahead). The small positive bias of 0.138 percentage points indicates that SPF forecasters slightly underestimated realised GDP deflator inflation on average over the sample, though the magnitude is modest relative to the mean absolute error of 1.18 percentage points. The root mean squared error of 1.66 reflects the difficulty of predicting price dynamics over multi-quarter horizons, particularly during the high-inflation episodes of the 1970s and the post-COVID period.

Table 3: Horizon Accuracy - Inflation

Horizon	Bias	MAE	RMSE	n
0	0.106	0.861	1.15	229
1	0.160	1.08	1.48	228
2	0.161	1.22	1.69	227
3	0.158	1.35	1.88	226
4	0.105	1.39	1.97	220

Table 3 decomposes forecast accuracy by horizon and reveals a clear pattern of declining precision at longer horizons. The nowcast ($h = 0$) achieves the lowest RMSE of 1.15, while errors grow steadily to 1.97 at the four-quarter-ahead horizon. The increased errors by horizon are consistent with the well-known result that macroeconomic predictability decays as the forecast horizon extends, as uncertainty compounds over time. The bias remains small and positive across all horizons, ranging from 0.105 to 0.161, indicating that forecasters consistently underestimate realised inflation by a broadly similar margin regardless of the forecast horizon. Note that the number of observations is slightly declining in horizon. This is because the end of the sample does not have realised variables yet (forecasts for 2026:Q2 onward) so the surveys matching these are not included.

Table 4: Inflation Forecasts and Realisations

Horizon	mean	median	sd	var	min	max	n
Total	3.355	2.591	1.939	3.760	0.000	10.632	1130
0	3.411	2.658	2.155	4.643	0.569	10.632	229
1	3.347	2.548	2.008	4.032	0.420	10.195	228
2	3.343	2.613	1.855	3.441	1.236	9.461	227
3	3.339	2.572	1.840	3.385	0.000	9.000	226
4	3.332	2.475	1.826	3.336	1.331	9.622	220
Realised	3.493	2.672	2.499	6.244	-1.582	12.295	1130

Table 4 presents descriptive statistics for both SPF inflation forecasts and realised inflation, broken down by horizon. The "total" row reports pooled statistics across all forecast horizons. Across all horizons, survey forecasts exhibit mean values close to 3.3-3.4% with medians around 2.5-2.5%, pointing to a right-skewed distribution driven by the high-inflation episodes of the 1970s and early 1980s. The standard deviation of forecasts is relatively stable across horizons at approximately 1.8-2.2 percentage points, and is somewhat lower than that of realised inflation at 2.50 percentage points, consistent with forecasters smoothing their predictions and avoiding extreme values. The minimum realised inflation of -1.58% reflects to deflation during the financial crisis, while the maximum of 12.30% reflects the peak of the

high-inflation period of the early 1980s. Notably, the minimum forecast across all horizons is 0.000, suggesting that forecasters did not predict deflation even during periods where it was subsequently realised, consistent with the tendency to smooth predictions and avoid extreme values. The similarity in means between forecasts and realisations is consistent the modest bias reported in Tables 2 and 3.

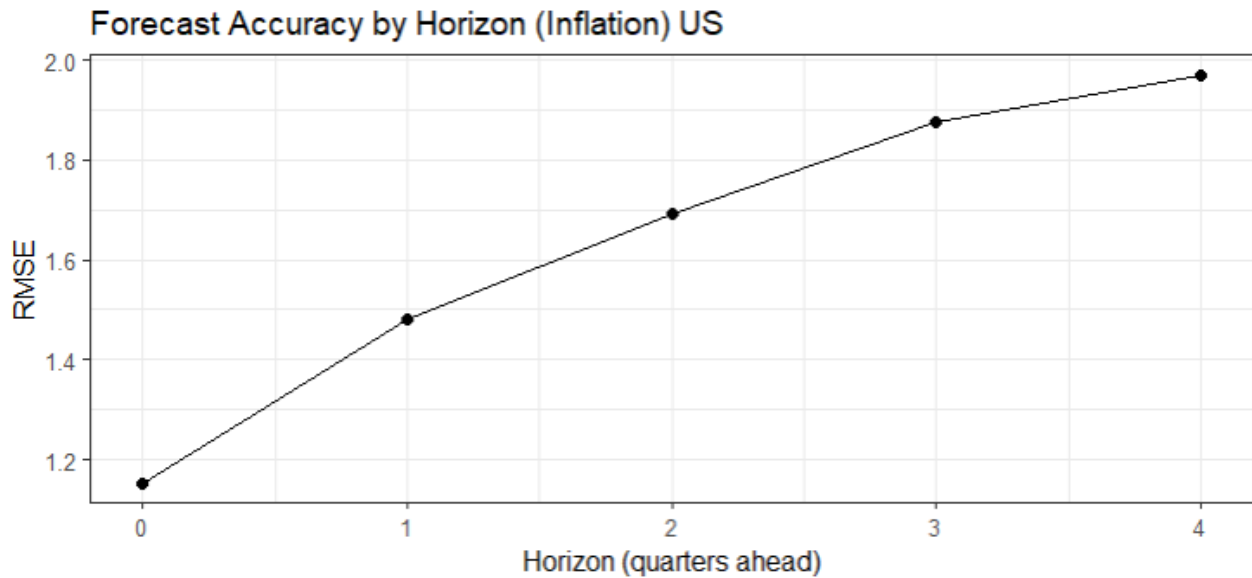


Figure 2: Forecast Accuracy by Horizon - Inflation

Figure 2 plots the RMSE of SPF inflation forecasts as a function of forecast horizon. The curve rises monotonically from 1.15 at the nowcast horizon to 1.97 at four-quarters-ahead, confirming the pattern of declining precision at longer horizons already documented in Table 3. The rate of increase gradually declines as the horizon extends, giving the curve a slightly concave shape with a modest flattening visible at the longer horizons.

Table 5: Overall Accuracy - Unemployment

Bias	MAE	RMSE	n
0.00491	0.439	0.904	1135

Table 5 reports overall SPF unemployment forecast accuracy across all horizons and quarters. The near-zero bias of 0.005 percentage points indicates that forecasters exhibit essentially no systematic directional error when evaluated over the full sample, suggesting that SPF unemployment forecasts are approximately unbiased on average. The mean absolute error of 0.44 percentage points and RMSE of 0.90 percentage points indicate moderate forecast

errors, which is not surprising given the considerable variation in realised unemployment over the sample, which includes both the low-unemployment expansions of the 1990s and mid-to-late 2010s and the sharp spikes during the financial crisis and COVID-19 pandemic.

Table 6: Horizon Accuracy - Unemployment

Horizon	Bias	MAE	RMSE	n
0	-0.0486	0.129	0.263	230
1	-0.0412	0.316	0.784	229
2	-0.0112	0.467	0.945	228
3	0.0464	0.594	1.06	227
4	0.0825	0.0703	1.19	221

Table 6 decomposes unemployment forecast accuracy by horizon. As with inflation, precision declines monotonically at longer horizons, with the nowcast RMSE of 0.26 rises to 1.19 at $h = 4$. A noteworthy feature is the change in sign of the bias: at short horizons ($h = 0$ and $h = 1$), the bias is slightly negative, implying that forecasters marginally overestimate unemployment in the near term. At longer horizons ($h = 3$ and $h = 4$), the bias turns positive, consistent with forecasters underestimating the extent to which unemployment can remain elevated following downturns.

Table 7: Unemployment Forecasts and Realisations

Horizon	mean	median	sd	var	min	max	n
Total	6.012	5.700	1.649	2.720	3.41	16.12	1135
0	6.042	5.725	1.810	3.276	3.50	16.12	230
1	6.046	5.700	1.713	2.936	3.50	12.94	229
2	6.027	5.700	1.639	2.686	3.50	11.00	228
3	5.981	5.650	1.569	2.463	3.50	10.00	227
4	5.963	5.650	1.503	2.260	3.40	9.80	221
Realised	6.017	5.700	1.719	2.955	3.40	13.00	1135

Table 7 presents descriptive statistics for both SPF unemployment forecasts and realised unemployment by horizon. Survey forecasts are centred around a mean of approximately 6.0% across horizons, with a median of 5.65–5.73%, reflecting the right-skewed nature of the unemployment distribution driven by recession periods. The standard deviation of forecasts at approximately 1.5–1.8 percentage points is marginally lower than that of realised unemployment at 1.72 percentage points, consistent with the general tendency of forecasters to smooth predictions and avoid extreme values, as noted earlier for inflation. The forecast range of 3.41% closely matches the minimum realised value of 3.40%, while the maximum

forecast of 16.12% exceeds the maximum realised of 13.00%. The similarity in means between forecasts and realisations is consistent with the modest bias reported in Tables 5 and 6.

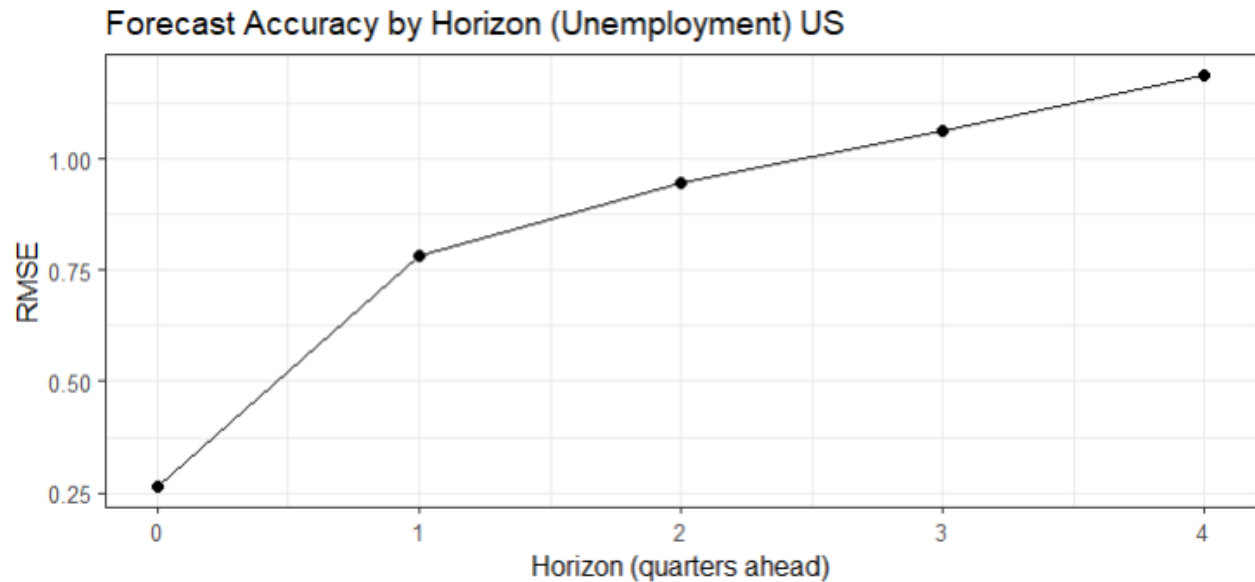


Figure 3: Forecast Accuracy by Horizon - Unemployment

Figure 3 plots the RMSE of SPF unemployment forecasts as a function of forecast horizon. The pattern is monotone: RMSE rises from approximately 0.26 percentage points at the nowcast horizon to 1.18 at four-quarters-ahead. The steepest increase occurs between $h = 0$ and $h = 1$, after which the increase in forecast errors continues at a more gradual pace, reflecting the informational advantage of the nowcast which benefits from partially within-quarter data releases. Compared to the inflation RMSE profile in Figure 2, two differences stand out. First, the level of errors is substantially lower throughout - unemployment RMSE is below inflation RMSE at every horizon - indicating better predictability of unemployment relative to inflation. Second, the biggest spike in RMSE occurs between $h = 0$ and $h = 1$ for unemployment, whereas for inflation the increase is more gradual and spread across all horizons.

2.6 Sample Choices and Alternatives

Several sample choices made in this paper involve trade-offs that are worth making explicit. The primary sample period is 1972:Q1 to 2025:Q4, with yield maturities from 1 to 15 years.

Inflation is measured using the GDP deflator rather than CPI or PCE, and unemployment is aggregated to quarterly frequency using simple averaging of monthly observations. The analysis uses the median rather than the mean SPF forecast series. The four-factor models are estimated at quarterly frequency to match the availability of SPF survey data, while the two-factor model is estimated at monthly frequency, resulting in the two models not being directly comparable in terms of estimation frequency. The reasoning behind these choices and the alternatives considered are discussed at the end of the paper.

With our data described, the next section will lay out the theoretical framework of the 'workhorse' of this paper, namely the GATSM.

3 Theoretical framework

3.1 Yields

When mentioning bonds and yields in this paper we are specifically referring to yields from zero-coupon government bonds. A zero-coupon bond is a bond which pays no periodically coupons for the bondholder, where instead the returns from bonds only stem from changes in the price of the bond when the bond issuer repays the debt (price of the bond) to the bondholder at the end of the maturity of the bond.

We denote the price of a bond as $P_t^{(k)}$ where t denotes the time period and k indicates the remaining periods until maturity of the bond. The price of a bond paying one unit at maturity is given by

$$P_t^{(k)} = \frac{1}{\left(1 + Y_t^{(k)}\right)^k}, \quad (1)$$

where the price is simply the discounted face value of the bond and $Y_t^{(k)}$ is the spot rate¹. Rewriting this we can derive an expression for the yield as

$$y_t^{(k)} = -\frac{1}{k} p_t^{(k)}, \quad (2)$$

where $y_t^{(k)} = \log\left(1 + Y_t^{(k)}\right)$ and $p_t^{(k)} = \log\left(P_t^{(k)}\right)$. We can then theoretically find the yield curve of bond yields by plugging in yields for different maturities and plotting the result. In reality, there is not issued zero-coupon bonds for all possible or relevant maturities, hence, researchers have turned to different methods to estimate the zero-coupon yield curve. A popular approach in many central banks for deriving the yield curve is to use the Svensson specification of [Svensson \(1994\)](#) which is an extension of the Nelson-Siegel parametric functional form in [Nelson and Siegel \(1987\)](#). All of the yield curve data used in this paper is derived using the Svensson model. The Nelson-Siegel yield curve is derived from the assumption that the instantaneous forward rate is the solution to a second-order differential equation with equal roots, i.e., a determinant of zero, where the instantaneous forward rate is given by

$$\mathcal{F}(k; \boldsymbol{\theta})^{NS} = \beta_0 + \beta_1 \exp\left\{-\frac{k}{\tau}\right\} + \beta_2 \left[\frac{k}{\tau} \exp\left\{-\frac{k}{\tau}\right\}\right], \quad (3)$$

where β_0 determines the level of the yield curve, β_1 sets the slope, and β_2 controls the curvature of the curve (the hump of the curve). k is the time to maturity and τ is a positive

¹The spot rate is the yield of a zero-coupon bond at a given maturity.

decay parameter. A small τ makes the exponential term decay slower hence, forcing the short-term and hump components to be concentrated around the shorter maturities. The Svensson specification is found by adding an extra term to (3), giving us

$$\mathcal{F}(k; \boldsymbol{\theta}) = \beta_0 + \beta_1 \exp \left\{ -\frac{k}{\tau} \right\} + \beta_2 \left[\frac{k}{\tau} \exp \left\{ -\frac{k}{\tau} \right\} \right] + \beta_3 \left[\frac{k}{\xi} \exp \left\{ -\frac{k}{\xi} \right\} \right], \quad (4)$$

where β_3 and ξ are new parameters for the new term. The Svensson specification allows for more flexibility by adding a second "hump" to the yield curve which allows for more than one local extremum along the maturity profile which in some cases can improve the fit of the yield curve. One can then derive the implied spot rate by integrating over the forward rate and then dividing by the remaining time to maturity.

3.2 GATSM Framework

Models referred to as Dynamic Term Structure Models (DTSMs) are used extensively in central banks and other financial institutions to analyse the evolution of yield curves and their term premia over time. In short, DTSMs rely on a low-dimensional $F \times 1$ state vector to describe what drives the yield curve, where F is the number of factors.

The model we consider to be the workhorse of this paper is the GATSM where bond yields are affine in the factors and with the factor dynamics being assumed Gaussian. The GATSM can be formulated under two different measures, the physical measure denoted by $\mathbb{P}$, and the risk-neutral probability measure denoted by $\mathbb{Q}$. The physical measure $\mathbb{P}$ governs the actual dynamics of the states hence also how interest rates evolve over time, while the risk-free measure $\mathbb{Q}$ controls the no-arbitrage bond prices of the model. The structure of the GATSM is fairly simple and consists of three equations. To get a clear understanding of the differences under $\mathbb{P}$ and $\mathbb{Q}$ is it first easier to present $\mathbb{P}$ and then derive the $\mathbb{Q}$ subsequently. For notation and structure, we follow the model specification in [Andreasen et al. \(Forthcoming\)](#).

3.3 Physical Measure $\mathbb{P}$

We start by presenting the GATSM under $\mathbb{P}$. Letting the time dimension be denoted by $t = 1, \dots, T$, the first equation specifies the short-rate r_t of the model by some scalar α and the state vector $\mathbf{x}_t$, where the short-rate is affine in the states through the coefficient vector $\boldsymbol{\beta}$

$$r_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t, \quad (5)$$

where $\boldsymbol{\beta}$ is a vector of dimensions $F \times 1$.

The second equation under $\mathbb{P}$ captures the dynamics of the factors $\mathbf{x}_t$ and has a simple VAR(1) representation given by

$$\mathbf{x}_{t+1} = \boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^\mathbb{P}, \quad (6)$$

where the noise $\boldsymbol{\varepsilon}_{t+1}^\mathbb{P}$ is assumed *iid* and is normally distributed with mean $\mathbf{0}$ and covariance matrix $\mathbf{I}$, where $\mathbf{0}$ is the null vector and $\mathbf{I}$ is the identity matrix. Furthermore, both $\boldsymbol{\Phi}^\mathbb{P}$ and $\boldsymbol{\Sigma}$ have dimensions $F \times F$ while $\boldsymbol{\mu}^\mathbb{P}$ is $F \times 1$.

Lastly, the third equation is the pricing kernel, or stochastic discount factor, simply being assumed to have the form

$$M_{t+1} = \exp \left\{ -r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \right\}, \quad (7)$$

where the $F \times 1$ vector $\boldsymbol{\Lambda}_t$ is the market price of risk given each of the factors. Since the differences in yields between different maturities, and their evolution, is largely dependent on the required risk premia accrued by holding bonds, we wish for the price of risk to be flexible in order to better model the risk. We consider the specification of [Duffee \(2002\)](#), a common specification for $\boldsymbol{\Lambda}_t$ in the literature which is given by

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Sigma}^{-1} (\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_x \mathbf{x}_t), \quad (8)$$

with $\boldsymbol{\lambda}_0$ being $F \times 1$ and $\boldsymbol{\lambda}_x$ being a $F \times F$ matrix. This specific form is called the essentially affine specification since it is not completely affine given that if $\boldsymbol{\lambda}_x \neq 0$ then $\boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t$ is not affine in $\mathbf{x}_t$. A reason for using this specification is that it allows the price of risk to vary independently, giving us great flexibility for fitting risk to empirical behaviour of expected excess returns of bonds, thus allowing us to produce model-implied term premia that is consistent with empirical data.

3.4 $\mathbb{P}$ and $\mathbb{Q}$ Dynamics

The price of a bond crucially depends on the required risk compensation of the market participants and given that investors are commonly risk-averse the current price of a bond is generally below that of its expectation to reward investors for holding risk. A somewhat magical result is that under the no-arbitrage assumption one can *once and for all* adjust for the probability of future outcomes such that it incorporates all risk premia of the market participants, you then take the new expectation under this measure, giving us the risk-neutral measure $\mathbb{Q}$. This is preferable to first taking expectations and then adjusting for the risk of each individual investor. Essentially, when risk-neutral probabilities have been found we can then simply price assets by taking the present value of its expected payoff.

We can look at it this way, the $\mathbb{P}$ -measure is also called the 'real-world' measure because under $\mathbb{P}$ is how asset prices actually behave in reality. Stocks in the U.S. tend to drift upward over time and investors earn a return above the theoretical risk-free rate as compensation for holding risk. The prices of risky assets are volatile and unpredictable but will on average reward investors for holding them, so under the $\mathbb{P}$ -measure holding assets is tilted in favour of the risk-taker. Essentially the $\mathbb{P}$ -measure is for when you want to value assets at risk.

The $\mathbb{Q}$ -measure on the other hand is a 'world' where investors are risk-neutral in the way that they do not care about risk at all, and are perfectly indifferent between a safe and a risky asset as long as the expected return is identical. In this world the risk premium is zeroed out and every asset earns exactly the risk-free rate on average no matter how risky.

3.4.1 Real-world Measure to Risk-Free Measure

Mathematically, going from the $\mathbb{P}$ -measure to the $\mathbb{Q}$ -measure means that the risk premium is removed from every asset's drift. For a Gaussian innovation we can move from the $\mathbb{P}$ - to the $\mathbb{Q}$ -measure by using the Radon-Nikodym derivative, given by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left\{ -\frac{1}{2} \mathbf{\Lambda}_t^\top \mathbf{\Lambda}_t - \boldsymbol{\varepsilon}_{t+1}^\top \mathbf{\Lambda}_t \right\}. \quad (9)$$

By computing the density of $\boldsymbol{\varepsilon}$ under both $\mathbb{P}$ and $\mathbb{Q}$ it can be shown² that the Gaussian innovation under the $\mathbb{Q}$ -measure is given by

$$\boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} = \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}} + \mathbf{\Lambda}_t. \quad (10)$$

²See Appendix A3.

And likewise, by inserting this result into the physical dynamics, the law of motion for the states under $\mathbb{Q}$ is given by

$$\begin{aligned} \mathbf{x}_{t+1} &= \underbrace{(\boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\lambda}_0)}_{\boldsymbol{\mu}^{\mathbb{Q}}} + \underbrace{(\boldsymbol{\Phi}^{\mathbb{P}} - \boldsymbol{\lambda}_x)}_{\boldsymbol{\Phi}^{\mathbb{Q}}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} \\ &= \boldsymbol{\mu}^{\mathbb{Q}} + \boldsymbol{\Phi}^{\mathbb{Q}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}}. \end{aligned} \quad (11)$$

We see that the bridge between the $\mathbb{P}$ -measure and the $\mathbb{Q}$ -measure is the market price of risk $\boldsymbol{\Lambda}_t$. The volatility matrix $\boldsymbol{\Sigma}$ is identical under both measures and only the drift changes. $\boldsymbol{\lambda}_0$ is a fixed number for each factor and does not react to shifts in the markets, it can then simply be interpreted as a baseline level of risk compensation and acts as the shift between the unconditional mean of the factors from the real-world measure to the risk-free measure. $\boldsymbol{\lambda}_x$ is the state-dependent component of the market price of risk and each of its elements governs how a particular state affects the price of a particular source of risk. For instance, element i, j of $\boldsymbol{\lambda}_x$ says when factor j increases by one unit how much does the price of risk on factor i change.

3.5 Solving the Model

Pricing zero-coupon bonds is one of the key objectives of our GATSM and it turns out that the solution for bond prices are given by

$$P_t^{(k)} = \exp \{ A_k + \mathbf{B}_k^T \mathbf{x}_t \}, \quad (12)$$

where k denotes maturity, $\log \left(P_t^{(k)} \right)$ is affine in the factors $\mathbf{x}_t$ through the $F \times 1$ vector of coefficients B_k , and the scalar A_k , following [Ang and Piazzesi \(2003\)](#). Earlier we defined our pricing kernel to be strictly positive which guarantees the absence of arbitrage, but to find expressions for A_k and B_k we formally impose a non-arbitrage condition as

$$P_t^{(k)} = \mathbb{E}_t \left[\prod_{i=0}^{k-1} M_{t+1+i} \times 1 \right], \quad (13)$$

which states that the price of a bond yielding one monetary unit in payoff in period t with k periods to maturity, is equal to the expected product of the pricing kernel from t to maturity. This condition follows from

$$P_t^{(k)} = \mathbb{E} \left[M_{t+1} P_{t+1}^{(k-1)} \right],$$

using recursive substitution. For the first iteration $k = 1$ we have

$$A_1 + \mathbf{B}_1^\top \mathbf{x}_t = -(\alpha + \boldsymbol{\beta}^\top \mathbf{x}_t), \quad (14)$$

using equations (13) and (5). For $k > 1$ we can take advantage of equations (12) and (13) to write the bond price as

$$\begin{aligned} P_t^{(k)} &= \mathbb{E}_t \left[M_{t+1} \mathbb{E}_{t+1} \left(\prod_{i=1}^{k-1} M_{t+1+i} \times 1 \right) \right] \\ &= \mathbb{E}_t \left[M_{t+1} P_{t+1}^{(k-1)} \right]. \end{aligned}$$

Combining this expression with (13) we have the equation

$$\begin{aligned} \exp(A_k + \mathbf{B}_k^\top \mathbf{x}_t) &= \mathbb{E}_t \left(M_{t+1} P_{t+1}^{(k-1)} \right) \\ &= \mathbb{E}_t \left(\exp \left\{ -r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \right\} \exp \{ A_{k-1} + \mathbf{B}_{k-1}^\top \mathbf{x}_{t+1} \} \right). \end{aligned}$$

Now by inserting for the moment-generating function of a multivariate random variable

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

and doing a lot of rearranging, we can find that A_k is given by

$$A_k = A_{k-1} - \alpha + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1}, \quad (15)$$

and $\mathbf{B}_k$ is given by

$$\mathbf{B}_k^\top = \mathbf{B}_{k-1}^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top. \quad (16)$$

Given $\mathbf{B}_1^\top = -\boldsymbol{\beta}^\top$ we can also recursively derive the identity for the bond-pricing loadings as³

$$\mathbf{B}_k^\top = -\boldsymbol{\beta}^\top \sum_{i=0}^{k-1} (\boldsymbol{\Phi}^\mathbb{Q})^i \iff \mathbf{B}_k = -\sum_{i=0}^{k-1} [(\boldsymbol{\Phi}^\mathbb{Q})^\top]^i \boldsymbol{\beta}. \quad (17)$$

³See Appendix A7 for full derivation.

Finally, using the closed-form solutions for A_k and $\mathbf{B}_k$ the continuously log-compounded yields are then given by

$$y_t = -\frac{1}{k} \log P_t^{(k)} = -\frac{1}{k} (A_k + \mathbf{B}_k^\top \mathbf{x}_t). \quad (18)$$

The VAR coefficients $\boldsymbol{\mu}^\mathbb{P}$ and $\boldsymbol{\Phi}^\mathbb{P}$ enter with risk-correction in the loadings for A_k and $\mathbf{B}_k$, thus all that matters for pricing bonds is the risk-neutral dynamics of the states.

3.5.1 Yield Decomposition

Now that we have computed the closed-form solution, we can start looking at one of the central outputs of affine term structure models, such as the GATSM, which is its ability to decompose model-implied yields into their expectations and term premium components. The bond recursion loadings A_k and $\mathbf{B}_k$ that we computed earlier, depends both on $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}_x$ which means that the model-implied yield curve is shaped by the market price of risk.

Any bond yield can be decomposed into two distinct terms

$$y_t^{(k)} = \frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}_t^\mathbb{P} [r_{t+i}] + TP_t^{(k)}, \quad (19)$$

where $\mathbb{E}_t^\mathbb{P} [r_{t+i}]$ is the expectation of the short-rate i periods ahead under the $\mathbb{P}$ -measure. The term $TP_t^{(k)}$ is the term premium.

The expected short-rate i periods ahead in our GATSM framework is given by

$$\mathbb{E}_t^\mathbb{P} [r_{t+i}] = \alpha + \boldsymbol{\beta}^\top \mathbb{E}_t^\mathbb{P} [\mathbf{x}_{t+i}], \quad (20)$$

where $\mathbb{E}_t^\mathbb{P} [\mathbf{x}_{t+i}] = \sum_{j=0}^{i-1} (\boldsymbol{\Phi}^\mathbb{P})^j \boldsymbol{\mu}^\mathbb{P} + (\boldsymbol{\Phi}^\mathbb{P})^i \mathbf{x}_t$, and is found by recursive substitution⁴. Equation (19) without the term premium term is also known as the pure expectations hypothesis (EH) yield. The EH is a proposition which states that the long-term rate is determined purely by current and future expected short-rates and is usually rejected based on empirical evidence. The term premium is the extra compensation an investor requires for holding risk, and in the yield curve we see that the market price of risk lives entirely in the term premium component of the yield curve. The term premium is defined as the residual between the bond yield at

⁴See Appendix A5

maturity k and the average of the expected future short-rates over the maturity of the bond

$$TP_t^{(k)} = y_t^{(k)} - \frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}_t^{\mathbb{P}} [r_{t+i}]. \quad (21)$$

To see how $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}_x$ shapes the term premium we first look at the unrealistic case of no market price of risk ($\boldsymbol{\lambda}_0 = \mathbf{0}$, $\boldsymbol{\lambda}_x = \mathbf{0}$), and we see that the bond recursion for $\mathbf{B}_k^{\top}$ would be entirely governed by $\boldsymbol{\Phi}^{\mathbb{P}}$

$$\mathbf{B}_k^{\top} = -\boldsymbol{\beta}^{\top} + \mathbf{B}_{k-1}^{\top} \boldsymbol{\Phi}^{\mathbb{P}},$$

and the term premium would collapse to zero, and where yields would be reduced to be the pure expectations of future short-rates. With $\boldsymbol{\lambda}_0 \neq \mathbf{0}$ and $\boldsymbol{\lambda}_x \neq \mathbf{0}$ the loadings $\mathbf{B}_k$ would diverge from what the pure EH would produce which would open up a wedge between the yield and the expected short-rate and that wedge is exactly the term premium. Given our framework we can alternatively represent the term premium component as the average of the expected one-period excess returns earned while rolling the bond down the yield curve⁵

$$TP_t^{(k)} = \frac{1}{k} \sum_{i=1}^k \mathbb{E}_t^{\mathbb{P}} \left[rx_{t+i}^{(k-i+1)} \right], \quad (22)$$

where $rx_{t+1}^{(k)} \equiv \log P_{t+1}^{(k-1)} - \log P_t^{(k)} - r_t$ is the one-period log holding return on a k -period bond in excess of the short-rate, where under the model the expected excess return is

$$\mathbb{E}_t^{\mathbb{P}} \left[rx_{t+1}^{(k)} \right] = \mathbf{B}_{k-1}^{\top} \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t - \frac{1}{2} \mathbf{B}_{k-1}^{\top} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{\top} \mathbf{B}_{k-1}, \quad (23)$$

where the first term $\mathbf{B}_{k-1}^{\top} (\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_x \mathbf{x}_t) = \mathbf{B}_{k-1}^{\top} \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t$ is the risk premium demanded for holding the bond's exposure to the factor shocks, and the second term is the Jensen convexity correction⁶. We again see that with $\boldsymbol{\lambda}_0 = \mathbf{0}$, $\boldsymbol{\lambda}_x = \mathbf{0}$ the term premium vanishes up to the convexity term.

Now by looking at the result in (11) and this expression, we can conclude that the gap between the physical measure $\mathbb{P}$ and risk-free measure $\mathbb{Q}$, which is governed by $\boldsymbol{\Lambda}_t$, is what drives the model-implied term premium. Where a larger difference between $\boldsymbol{\Phi}^{\mathbb{Q}}$ and $\boldsymbol{\Phi}^{\mathbb{P}}$ results in a larger term premium.

⁵See Appendix A6.

⁶The Jensen convexity correction adjusts for the bias when using expected values with non-linear functions.

4 Two-Factor Model

We will start the analysis of the paper by introducing and analysing the GATSM with two latent factors before specifying and analysing the results of the two-factor latent GATSM. The two-factor GATSM is the GATSM in its simplest specification, thus setting the foundation for further inclusion of macroeconomic factors. The two-factor model consists of equations (5), (6), (7), (11), and (18), with $F = 2$. A more popular approach in the literature is the latent three-factor version which would likely yield a better model fit at the cost being more computationally heavy and more complex in estimation due to more parameters, which is not necessarily favourable in our setting. We do, however, estimate a three-factor GATSM for simple reference alongside the two-factor model, but refrain from using it for further analytical purposes.

4.1 Identification

Before we can estimate the two-factor GATSM we need to impose various restrictions to make it both identifiable and estimable. Firstly, many different parameter vectors can generate identical implied bond yields because the latent factors have no natural scale or orientation. For instance, if we let $\tilde{\mathbf{x}}_t = \mathbf{H}\mathbf{x}_t$ and $\tilde{\mathbf{B}}_t^\top = \mathbf{B}^\top \mathbf{H}^{-1}$ where $\mathbf{H}$ is some 2×2 invertible matrix, and plug these into equation (18) the yield is identical for any invertible $\mathbf{H}$. We cannot tell from bond prices alone whether the true factors are $\mathbf{x}_t$ or $\tilde{\mathbf{x}}_t$. This rotationally indeterminacy is also apparent in the dynamics, hence the need for various model restrictions.

The following restrictions can be grouped into three categories: 1) calibrated restrictions that are fixed before estimation, 2) normalisation restrictions which are required for point identification, and 3) admissibility restrictions which constrains the parameter space during estimation to ensure economically feasible estimates.

The first calibrated restriction is that we impose a zero risk-neutral mean

$$\boldsymbol{\mu}^{\mathbb{Q}} = \mathbf{0},$$

where the long-run level of yields is now controlled entirely by α . If we let $\boldsymbol{\mu}^{\mathbb{Q}} \neq \mathbf{0}$ and with a free α we would be introducing a free level shift that cannot be separately identified. The second calibrated restriction is that we impose unit short-rate loadings

$$\beta_1 = \beta_2 = 1 \Leftrightarrow \boldsymbol{\beta} = \mathbf{1},$$

and is imposed because latent factors have no natural scale. The last calibrated restriction is that we decouple the dynamics of each factor under $\mathbb{Q}$ by setting

$$\Phi_{12}^{\mathbb{Q}} = \Phi_{21}^{\mathbb{Q}} = 0 \Leftrightarrow \mathbf{\Phi}^{\mathbb{Q}} = \begin{bmatrix} \lambda_1^{\mathbb{Q}} & 0 \\ 0 & \lambda_2^{\mathbb{Q}} \end{bmatrix},$$

such that only the eigenvalues of $\mathbf{\Phi}^{\mathbb{Q}}$, denoted by λ , matter for bond pricing, where the diagonal form pins down which factor corresponds to which eigenvalue. Forcing $\mathbf{\Phi}^{\mathbb{Q}}$ to be diagonal resolves its rotation indeterminacy problem. Given this we also impose eigenvalue ordering such that

$$\lambda_1^{\mathbb{Q}} > \lambda_2^{\mathbb{Q}} > 0,$$

to prevent two observationally equivalent representations. The positivity constraint also ensures that the risk-neutral dynamics are stable. Sometimes during optimisation when $\lambda_1^{\mathbb{Q}} \approx \lambda_2^{\mathbb{Q}}$, the diagonal parameterisation of $\mathbf{\Phi}^{\mathbb{Q}}$ approaches a degenerate Jordan case. As a purely numerical safeguard we then relax the diagonal restriction R3 by setting a small off-diagonal

$$\Phi_{12}^{\mathbb{Q}} = (1 - |\lambda_1^{\mathbb{Q}} - \lambda_2^{\mathbb{Q}}|)^{1000},$$

which almost always avoids this degeneracy problem during estimation. This term is effectively zero when the eigenvalues are well-separated and only grows smoothly towards 1 as they converge, so it overrides R3 only in the immediate case of a repeated eigenvalue. For our estimated, well-separated eigenvalues the term is numerically negligible, so the diagonal restriction R3 holds in practice and the bond-pricing recursion is unaffected. Furthermore, we impose a lower Cholesky decomposition on $\mathbf{\Sigma}$ such that

$$\mathbf{\Sigma} = \begin{bmatrix} \Sigma_{11} & 0 \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

which makes the innovation covariance matrix $\mathbf{Q} = \mathbf{\Sigma}\mathbf{\Sigma}^{\top}$ unique. We also impose stationarity under the $\mathbb{P}$ -dynamics by imposing that

$$|\lambda_i^{\mathbb{P}}| < 1, \text{ for } i = 1, 2,$$

where $\lambda_i^{\mathbb{P}}$ is the i th eigenvalue of $\mathbf{\Phi}^{\mathbb{P}}$, not to be confused with the lambda parameters from the market price of risk.

The last three restrictions are those we impose throughout the estimation process to

ensure economically feasible results. They are i) $\alpha \geq 0$, ii) $\lambda_{\min}(\Sigma \Sigma^\top) > 0$, and iii) $\sigma_v > 0$. Restriction i) prevents the intercept from taking an economically implausible sign, since a negative α would mean the baseline short-rate, i.e. when both factors are at zero, is negative which in combination with the factor dynamics could generate an implied yield curve that is systematically negative on average which is inconsistent with the data. The condition ii) simply ensures that we have valid innovation covariance and is also automatically satisfied if $\Sigma_{11}, \Sigma_{22} > 0$. Lastly, condition iii) prevents the Kalman filter innovation covariance becoming singular. In Table 8 a short summary of the restrictions and what they try to resolve is displayed.

Table 8: Two Factor Restrictions

No.	Restriction	Type	Resolves
R1	$\mu^{\mathbb{Q}} = \mathbf{0}$	Calibrated	Level indeterminacy with α
R2	$\beta_1 = \beta_2 = 1$	Calibrated	Scale of latent factors
R3	$\Phi_{12}^{\mathbb{Q}} = \Phi_{21}^{\mathbb{Q}} = 0$	Calibrated	Rotation under $\mathbb{Q}$
R4	$\lambda_1^{\mathbb{Q}} > \lambda_2^{\mathbb{Q}} > 0$	Normalisation	Label switching
R5	Jordan correction	Normalisation	Near-degenerate eigenvalues
R6	Σ lower-triangular, $\Sigma_{ii} > 0$	Normalisation	Sign of factor innovations
R7	$ \lambda_i(\Phi^{\mathbb{P}}) < 1$	Admissibility	$\mathbb{P}$ -stationarity
R8	$\alpha \geq 0$	Admissibility	Sign of short-rate intercept
R9	$\lambda_{\min}(\Sigma \Sigma^\top) > 0$	Admissibility	Positive-definite covariance
R10	$\sigma_v > 0$	Admissibility	Non-singular Kalman innovation

4.2 State-Space Representation

With unobserved latent factors we have to estimate the factors and the model parameters simultaneously from a panel of bond yields in a state-space model. A state-space model consists of a measurement equation and a state equation which controls the law of motion of the state variables. We let $\mathcal{I}_y$ contain the indices for our selected maturities used for estimation. Our selected maturities throughout the paper will be the 1-, 2-, 5-, 7-, 10-, and 15-year maturities, giving us a good coverage of both the short-term and long-term horizons and also those in-between. We denote the number of maturities by M and our indices become $\mathcal{I}_y = \{12, 24, 60, 84, 120, 180\}$ in monthly frequency. The measurement equation for our state-space system is equation (18) and an error term where define $g_0^{(k)} \equiv \frac{-12}{k}A_k$ and $g_{x,i}^{(k)} = \frac{-12}{k}B_{k,i}$, where $B_{k,i}$ is the i th element of $\mathbf{B}_k$. By defining this we convert our yields to be in annualised terms, giving us

$$y_t^{(k)} = g_0^{(k)} + g_{x,1}^{(k)}x_{1,t} + g_{x,2}^{(k)}x_{2,t} + v_t^{(k)}, \quad k \in \mathcal{I}_y, \quad (24)$$

where $v_t^{(k)}$ is the measurement error term with mean zero and variance σ_v^2 . For brevity, throughout this paper we will refer to the standard deviation σ_v as the measurement error. It is more reasonable to assume that yields are measured with errors since [Hamilton and Wu \(2014\)](#) show that one can test the assumption that errors are absent in U.S. yields and they generally reject that yields are measured without errors. We impose that each maturity k has identical measurement error. We can then stack this scalar measurement equation for every chosen maturity k

$$\underbrace{\begin{bmatrix} y_t^{(12)} \\ \vdots \\ y_t^{(180)} \end{bmatrix}}_{\mathbf{y}_t} = \underbrace{\begin{bmatrix} g_0^{(12)} \\ \vdots \\ g_0^{(180)} \end{bmatrix}}_{\mathbf{g}_0} + \underbrace{\begin{bmatrix} g_{x,1}^{(12)} & g_{x,2}^{(12)} \\ \vdots & \vdots \\ g_{x,1}^{(180)} & g_{x,2}^{(180)} \end{bmatrix}}_{\mathbf{G}_x} \underbrace{\begin{bmatrix} x_{1,t} \\ x_{2,t} \end{bmatrix}}_{\mathbf{x}_t} + \underbrace{\begin{bmatrix} v_t^{(12)} \\ \vdots \\ v_t^{(180)} \end{bmatrix}}_{\mathbf{v}_t}, \quad (25)$$

with $\mathbf{v}_t \sim \text{n iid}(\mathbf{0}, \sigma_v^2 \mathbf{I})$. The associated state equation is Eq. (6) from our $\mathbb{P}$ -dynamics. Notation-wise, $\mathbf{G}_x^{(k)}$ with the superscript ' (k) ' denotes the row vector for that maturity k . For instance, $\mathbf{G}_x^{(15)} = \begin{bmatrix} g_{x,1}^{(15)} & g_{x,2}^{(15)} \end{bmatrix}$, and without the superscript it is the full matrix of annualised bond pricing loadings.

4.3 Kalman Filter

We never observe the latent factors directly, denoted as $x_{1,t}$ and $x_{2,t}$, hence we use a Kalman filter to infer what the latent factors must have been given our observed panel of bond yields $\mathbf{y}_t$. The filter essentially works backwards to infer the factors and it does this optimally, meaning that it produces the best possible estimate of $\mathbf{x}_t$ given all available information. The filter makes a prediction for time $t + 1$ given the information up until time t , which is then followed by an update step for each time $t = 1, \dots, T$.

For the first period at $t = 1$ we first need to initialise the filter with some starting inputs. For this we let the starting input be the unconditional mean and variance of the stationary $\mathbb{P}$ -VAR(1) given by

$$\begin{aligned}\hat{\mathbf{x}}_{0|0} &= (\mathbf{I} - \Phi^{\mathbb{P}})^{-1} \boldsymbol{\mu}^{\mathbb{P}}, \\ \text{vec}(\hat{\mathbf{P}}_{0|0}) &= (\mathbf{I} - \Phi^{\mathbb{P}} \otimes \Phi^{\mathbb{P}})^{-1} \text{vec}(\mathbf{Q}),\end{aligned}$$

where $\otimes$ is the Kronecker product and $\text{vec}(\cdot)$ stacks the columns of a matrix such that it becomes a vector, where the first column is at the top and the last column is at the bottom.

The filter starts with the prediction step where, given the filtered mean $\hat{\mathbf{x}}_{t-1|t-1}$ and covariance $\hat{\mathbf{P}}_{t-1|t-1}$ from the previous period, the one-step-ahead predictions are

$$\begin{aligned}\bar{\mathbf{x}}_{t|t-1} &= \boldsymbol{\mu}^{\mathbb{P}} + \Phi^{\mathbb{P}} \hat{\mathbf{x}}_{t-1|t-1}, \\ \bar{\mathbf{P}}_{t|t-1} &= \Phi^{\mathbb{P}} \hat{\mathbf{P}}_{t-1|t-1} (\Phi^{\mathbb{P}})^{\top} + \mathbf{Q},\end{aligned}\tag{26}$$

where for $t = 1$ these are the initialisation values. The predicted yields and its conditional variance are

$$\begin{aligned}\bar{\mathbf{y}}_{t|t-1} &= \mathbf{g}_0 + \mathbf{G}_x \bar{\mathbf{x}}_{t|t-1}, \\ \mathbf{F}_t &= \mathbf{G}_x \bar{\mathbf{P}}_{t|t-1} \mathbf{G}_x^{\top} + \sigma_v^2 \mathbf{I},\end{aligned}\tag{27}$$

where $\mathbf{F}_t$ is the $M \times M$ innovation covariance matrix for the yields. $\mathbf{F}_t$ is essentially a measure for how uncertain the model is about the yields before observing them at time t .

Following the prediction step is the update step where we observe the actual yield at time t , $\mathbf{y}_t$ and compares it to the predicted yield from (27), where the prediction error is

$$\boldsymbol{\eta}_t = \mathbf{y}_t - \bar{\mathbf{y}}_{t|t-1}.$$

The filtered state mean and covariance are then updated as

$$\begin{aligned}\hat{\mathbf{x}}_{t|t} &= \bar{\mathbf{x}}_{t|t-1} + \mathbf{K}_t \boldsymbol{\eta}_t, \\ \hat{\mathbf{P}}_{t|t} &= \bar{\mathbf{P}}_{t|t-1} - \mathbf{K}_t \mathbf{F}_t \mathbf{K}_t^\top,\end{aligned}\tag{28}$$

where $\mathbf{K}_t$ is the Kalman gain which determines how much weight to place on the new observation versus the prior prediction, and is given by

$$\mathbf{K}_t = \bar{\mathbf{P}}_{t|t-1} \mathbf{G}_x^\top \mathbf{F}_t^{-1}.\tag{29}$$

The new filtered values from (28) are then plugged back into the prediction step for the next period. This procedure is done for all t up until T .

As the filter goes through $t = 1, \dots, T$, at each t the observed yields conditional on past information are jointly normally distributed with mean $\bar{\mathbf{y}}_{t|t-1}$ and covariance $\mathbf{F}_t$, giving us a way to compute the Gaussian log-likelihood even if $\mathbf{x}_t$ are unobserved. The per-period log-likelihood contribution is

$$\ell_t = -\frac{M}{2} \log(2\pi) - \frac{1}{2} \log(\det(\mathbf{F}_t)) - \frac{1}{2} \boldsymbol{\eta}_t^\top \mathbf{F}_t^{-1} \boldsymbol{\eta}_t,\tag{30}$$

where the total log-likelihood is the sum of the per-period contributions $\mathcal{L} = \sum_{t=1}^T \ell_t$, and can be viewed as a model fit measure with a higher value meaning a better fit. To prevent just simply artificially increasing the log-likelihood by just increasing the observations T , we divide by T to get the average log-likelihood $\bar{\mathcal{L}} = \frac{\mathcal{L}}{T}$, which is exactly the object we are aiming to numerically maximise and thus granting us our maximum likelihood estimation parameters. What happens in practice is that for each $t = 1, \dots, T$ we pass the log-likelihood value alongside its corresponding parameter vector to our optimiser, which then makes an evaluation given the parameters. The optimiser then proposes a new parameter vector in the direction that best minimises the (negative) log-likelihood. If the change in log-likelihood compared to the last iteration is smaller than some given tolerance, the algorithm is converged and it outputs the objective function value alongside the corresponding parameter vector.

In our two-factor model, with the applied restrictions, the vector of parameters to be estimated in the Kalman Filter alongside the states consists of the 13 parameters

$$\boldsymbol{\theta}_{\text{est}} = (\alpha, \boldsymbol{\mu}^{\mathbb{P}}, \boldsymbol{\Phi}^{\mathbb{P}}, \boldsymbol{\Sigma}, \text{diag}(\boldsymbol{\Phi}^{\mathbb{Q}}), \mathbf{v}_t).$$

4.4 Two-Factor Starting Parameters

The maximum likelihood surface of ATSMs are known to be multimodal and hard to optimise over, hence good starting values are essential to ensure that our numerical optimisers convergences properly. We denote starting values by subscript '0'. We start by inferring some starting values for $\Phi^{\mathbb{P}}$ using principal components analysis (PCA), namely the two first PCs. We start by computing the covariance matrix of our demeaned panel of yields given by $\text{Cov}(\mathbf{Y}_{\text{dm}}) = \mathbf{Y}_{\text{dm}} \mathbf{Y}_{\text{dm}}^{\top} T^{-1}$, where $\mathbf{Y}_{\text{dm}}$ is the whole $M \times T$ panel of demeaned yields. We then solve the characteristic equation $\text{Cov}(\mathbf{Y}_{\text{dm}}) \boldsymbol{\nu} = \lambda \boldsymbol{\nu}$, where $\boldsymbol{\nu}$ is an eigenvector and λ the corresponding eigenvalue, to obtain the six eigenvalues and their corresponding eigenvectors where the two eigenvectors with largest variance are concatenated side-by-side into a matrix $\mathbf{V}$. We finally obtain our time series of PC scores by projecting the eigenvectors onto our yields

$$\mathbf{PC}_t = \mathbf{V}^{\top} \mathbf{Y}_{\text{dm}}.$$

Using these series we perform OLS on a VAR(1) without the intercept term since the PC scores are demeaned

$$\mathbf{PC}_t = \Phi_{\text{PC}} \mathbf{PC}_{t-1} + \boldsymbol{\varepsilon}_t, \quad (31)$$

where we denote the coefficients by $\hat{\Phi}_{\text{PC}}$. Applying this to yield data for the U.S. for the maturities in $\mathcal{I}_y$ we obtain the following estimate

$$\hat{\Phi}_{\text{PC}} = \begin{bmatrix} 0.9945 & -0.0260 \\ 0.0016 & 0.9664 \end{bmatrix}.$$

The two largest eigenvalue moduli in descending order of $\hat{\Phi}_{\text{PC}}$ are $\lambda_1 = 0.9930$ and $\lambda_2 = 0.9679$. We let the initial value $\text{diag}(\Phi_0^{\mathbb{Q}})$ be set to the eigenvalues of $\hat{\Phi}_{\text{PC}}$ in descending order plus a small buffer of 0.004 reflecting the fact that the risk-neutral measure is more persistent than the physical measure, due to the market price of risk flattening the term structure. From experience we also set the diagonal of $\Phi_0^{\mathbb{P}}$ be equal to the diagonal of $\Phi_0^{\mathbb{Q}}$. We do not make any assumptions about the off-diagonal dynamics hence we leave them to be zero initially and let the optimiser decide their sign and magnitude later.

The unconditional mean of the short-rate is given by $\mathbb{E}[r_t] = \alpha + \beta^{\top} \mathbb{E}[\mathbf{x}_t]$ with $\mathbb{E}[\mathbf{x}_t] = (\mathbf{I} - \Phi^{\mathbb{P}})^{-1} \boldsymbol{\mu}^{\mathbb{P}}$. We simply let $\boldsymbol{\mu}^{\mathbb{P}} = \mathbf{0}$ for now and let α carry the entire unconditional mean such that $\mathbb{E}[r_t] \approx \alpha$. This is similar to the procedure in [Ang and Piazzesi \(2003\)](#). The shortest maturity in the data is the one-year yield which we will use as an approximation

for the instantaneous short-rate, where we divide by 12 to go from an annualised yield to a monthly yield, giving us a starting value for α of 0.00404. We expect the one-year yield to be slightly higher than the actual short-rate due to the term premium but not enough to significantly alter the optimisation.

Now given starting values for $\Phi^{\mathbb{Q}}$, α , the restrictions $\beta = \mathbf{1}$ and $\mu^{\mathbb{Q}} = \mathbf{0}$ plus some negligible small placeholder $\Sigma = \text{diag}(1 \times 10^{-6}, 1 \times 10^{-6})$, we can compute the initial bond pricing recursion for A_k and B_k using Eqs. (15) and (16) and the fact that $A_1 = -\alpha$ and $B_1 = -\beta^{\top}$. Given the fact that the model is specified at a monthly rate but the LHS of the measurement equation $\mathbf{y}_t$ is in annualised terms, we again use the definition $g_0^{(k)} \equiv \frac{-12}{k} A_k$ and $g_{x,i}^{(k)} = \frac{-12}{k} B_{k,i}$ such that loadings are in annualised terms.

For $\mu^{\mathbb{P}}$ we choose starting values such that the model's unconditional yield matches the sample mean yield for any maturity k . The unconditional yields for each maturity stacked in a 6×1 vector are given by

$$\mathbb{E}[\mathbf{y}] = \mathbf{g}_0 + \mathbf{G}_x (\mathbf{I} - \Phi^{\mathbb{P}})^{-1} \mu^{\mathbb{P}},$$

rearranging to

$$\mathbf{M} \mu^{\mathbb{P}} = \mathbb{E}[\mathbf{y}] - \mathbf{g}_0,$$

where $\mathbf{M} = \mathbf{G}_x (\mathbf{I} - \Phi^{\mathbb{P}})^{-1}$. The above equation is an over-identified system of equations with six equations but only two unknowns, and is solved using least squares

$$\mu^{\mathbb{P}} = (\mathbf{M}^{\top} \mathbf{M})^{-1} \mathbf{M}^{\top} (\bar{\mathbf{y}}_{data} - \mathbf{g}_0),$$

giving us starting values for $\mu^{\mathbb{P}}$ which are $\mu_0^{\mathbb{P}} = (0.0000056, -0.0000626)^{\top}$.

To determine a starting value for the innovation scale Σ we note that a one-period unpredictable change in yields for our measurement equation satisfies

$$\Delta \mathbf{y}_t = \mathbf{G}_x \Delta \mathbf{x}_t = \mathbf{G}_x \Sigma \varepsilon_{t+1}^{\mathbb{P}},$$

where for the purpose of characterising the innovation scale we have omitted the terms $\mu^{\mathbb{P}}$ and $(\Phi^{\mathbb{P}} - \mathbf{I}) \mathbf{x}_{t-1}$. We only determine starting values for the diagonal of Σ and let the off-diagonal entries be set to zero. The model-implied covariance of monthly yield changes is given by

$$\text{Cov}(\Delta \mathbf{y}_t) = \mathbf{G}_x \Sigma \Sigma^{\top} \mathbf{G}_x^{\top},$$

and our goal is then to match the model-implied yield-change covariance with that of the empirical one. $\text{Cov}(\Delta \mathbf{y}_t)$ is a matrix with dimensions 6×6 while the unknown Σ is a 2×2 diagonal matrix giving us only 2 free parameters but 20 unique equations. This is a heavily over-determined system and in general has no unique solution. To sidestep this we apply PCA to $\Delta \mathbf{y}_t$ which decomposes the empirical covariance matrix as

$$\text{Cov}(\Delta \mathbf{y}_t) = \sum_{j=1}^6 \gamma_j v_j v_j^T,$$

where the eigenvalues $\gamma_1 \geq \gamma_2 \geq \dots \geq \gamma_6$, rank the directions of greatest variance in the vector space, and v_i are the associated eigenvector for each eigenvalue i . We now only need to match the unknown 2×2 matrix Σ with 2 scalar variances which are the 2 largest eigenvalues. Now for a diagonal Σ the contribution of factor j in the model to the yield-change variance is $\sigma_{jj}^2 \cdot \|\mathbf{G}_x[:, j]\|^2$, and setting these equal to each other yields

$$\sigma_{jj} = \frac{\sqrt{\gamma_j}}{\|\mathbf{G}_x[:, j]\|},$$

where we divide by $\|\mathbf{G}_x[:, j]\|$ to go from the PCA yield-change space to the GATSM factor space. The starting values are then computed, giving us

$$\text{diag}(\Sigma_0) = (3.1645 \times 10^{-4}, 1.8225 \times 10^{-4}).$$

Lastly, we need to set a starting value for the measurement error in the yields. We simply set it to be equal to the root mean-squared residual after projecting the PCA scores onto our yields

$$\hat{\varrho} = \sqrt{\frac{1}{6T} \sum_{i=1}^6 \sum_{t=1}^T (y_{i,t} - \hat{y}_{i,t}^{PC})^2},$$

which yields $\hat{\varrho} = 0.001$.

4.5 Choice of Optimiser

For our choice of optimiser we consider two different optimisers for better estimation robustness. The first optimiser is the Nelder-Mead algorithm as described in [Lagarias et al. \(1998\)](#) which is initialised using the `fminsearch` function in MATLAB. The Nelder-Mead (NM) algorithm is a numerical method used to find local minima and is a direct search method. This means that for each optimisation iteration it calculates the value of a function given some parameter values and compares them to the same function given some other values, and takes a step towards the direction of the parameter values which yielded the lowest value. To search around the parameter space the NM algorithm uses what is called a simplex which is a shape consisting of $p + 1$ vertices in p dimensions⁷, where p is the number of parameters to be estimated. Since the NM algorithm is a local optimiser it is very reliant on us providing the correct starting values, since the algorithm can only find the nearest local minimum closest to its starting point.

Essentially, we are betting on the fact that our starting positions lands in the basin of attraction of the actual global optimum. We are aware of this shortfall and a multi-start approach would likely yield better results.

Our second choice of optimiser is to apply a modified Covariance Matrix Adaption Evolution Strategy (CMAES) algorithm which is a global optimiser originally by [Hansen et al. \(2003\)](#) but later modified by [Andreassen \(2010\)](#) to optimise Dynamic Stochastic General Equilibrium (DSGE) models, but works well in the maximum likelihood estimation of a GATSM as well. The CMAES is an evolutionary, derivative-free algorithm where the basic idea is that the algorithm maintains a multivariate normal distribution over the parameter space and at each iteration draws a population of λ candidate parameter vectors from it. It then evaluates the log-likelihood at each candidate and keeps the best ones and uses those to update the distribution which shifts the mean towards the promising region in the parameter space. This process repeats until the convergence.

What makes CMAES particularly useful here is that the shape of the distribution adapts automatically. So if the likelihood consistently improves in a certain direction across iterations the algorithm stretches the search distribution along that direction. This allows it to navigate ridges in the likelihood surface without computing derivatives. The step size is also adjusting accordingly where it increases when the algorithm is making consistent progress and decreases when it starts to fluctuate, thus preventing premature convergence and aimless searching

⁷For a 2-dimensional function, the simplex is a triangle, 3-dimensional a tetrahedron. Above three dimensions it becomes hard to visualise.

around.

The important tuning parameter that needs to be chosen by the user, is the population size λ . A larger population size gives the algorithm a larger coverage of the parameter space and improves its ability to find the global optimum but comes at a computational cost. In addition, a very large population can slow down the adaptation of the covariance matrix, since the algorithm needs more iterations before the search distribution has meaningfully scouted a promising region. So in practice we want the population size to be chosen to balance global search capability against computational feasibility. Throughout this paper when mentioning the use of CMAES we will use the 'rule of thumb' lambda value found in the MATLAB Source Code in [Hansen \(2016\)](#) which is

$$\lambda = 4 + \text{floor}(3 \log(N)),$$

where $\text{floor}(\cdot)$ rounds down to the nearest integer and N is the number of parameter inputted.

Two other important tuning parameters that are used in both the NM and the CMAES are the TolX and TolFun parameters. TolX and TolFun are termination tolerances that tell the optimiser when to stop iterating. TolX is the tolerance on the parameter vector and lives in the parameter space, while TolFun is the tolerance on the objective function value which is the negative log-likelihood in this two-factor case.

In the NM case both tolerances needs to be satisfied before the optimiser stops, while on the other hand, the CMAES check the tolerances independently and stops the run if just one of the tolerances are satisfied. The only other way the optimisers stop is if the maximum number of iterations (MaxIter) is reached without flagging any of the tolerances. MaxIter is set to a very large value so if the optimiser reaches these number of iterations without stopping, is it a clear indication that our optimiser found a poor local optimum.

For the two-factor estimation we set $\text{TolFun} = \text{TolX} = 1 \times 10^{-4}$ and set the $\text{MaxIter} = 10,000$ for the NM, where we let MaxIter be unbounded in CMAES and instead let MaxFunEvals, which controls the cap on objective function evaluations, govern the iteration cap with $\text{MaxFunEvals} = 1 \times 10^6$. The reason we set both tolerances to be loose is that the Kalman filter log-likelihood surface is multimodal and a tighter tolerance mostly buys extra function evaluations rather than resulting in a better optimum.

4.6 Two-Factor GATSM Result

In Table 9 is displayed estimation results for the two-factor with the three-factor model for reference. Estimation is on U.S. data using both the CMAES and Nelder-Mead (NM) optimisers to assess robustness to numerical optimisation, with standard errors in parentheses. At first glance for the two-factor approaches we see that CMA-ES and NM both achieve highly similar average log-likelihoods giving them both a similar model fit, but lower average log-likelihoods than those of the three-factor approaches. The actual U.S yields and the two-factor model-implied yields for maturities 1 and 15 are plotted in Figure 4 where we see that both methods generally has a good yield fit, but shows clear discrepancies near the zero lower bound especially the period following the Financial Crisis. Choosing to use two factors instead of three factors naturally yields a looser model fit due to there being one fewer factor leading to less flexibility.

Both optimisers estimate highly persistent and similar factors under $\mathbb{Q}$ -dynamics with the first eigenvalues $\lambda_{1,CMAES}^{\mathbb{Q}} = 0.998022$, $\lambda_{1,NM}^{\mathbb{Q}} = 0.997934$ being very close to one, essentially acting as a level factor being very close to being non-stationary. The second eigenvalues $\lambda_2^{\mathbb{Q}}$ are similarly also nearly identical across methods. Furthermore, α is estimated at around 0.011 (monthly) for both approaches. The model-implied unconditional short-rate, $\mathbb{E}[r_t] = \alpha + \beta^{\top}(\mathbf{I} - \Phi^{\mathbb{P}})^{-1}\boldsymbol{\mu}^{\mathbb{P}}$, is approximately 4.5% in annualised terms under CMA-ES and 4.3% under NM, sitting just below the 4.84% sample-mean one-year yield. All parameters discussed above are estimated with very tight standard errors indicating that they are well-identified which is one of the advantages of using the more parsimonious two-factor specification instead of the standard three-factor model where we see observe larger standard errors. Additionally, the two-factor specification is quicker to optimise over than the three-factor leading to computationally faster simulations for later analyses.

The first notable difference between the two approaches is in $\boldsymbol{\mu}$ under the $\mathbb{P}$ -dynamics where both show similar estimates but with different signs, although the estimates are negligible small and close to zero indicating near-zero unconditional factor means. Also under the $\mathbb{P}$ -dynamics we see that $\Phi_{12}^{\mathbb{P}}$ is estimated to be positive for the CMAES but negative for the NM. To intuitively clarify what this tells us we can represent Eq. (6) as the following system of equations (excluding the last term)

$$\begin{aligned} x_{1,t+1} &= \mu_1^{\mathbb{P}} + \Phi_{11}^{\mathbb{P}}x_{1,t} + \Phi_{12}^{\mathbb{P}}x_{2,t}, \\ x_{2,t+1} &= \mu_2^{\mathbb{P}} + \Phi_{21}^{\mathbb{P}}x_{1,t} + \Phi_{22}^{\mathbb{P}}x_{2,t}. \end{aligned}$$

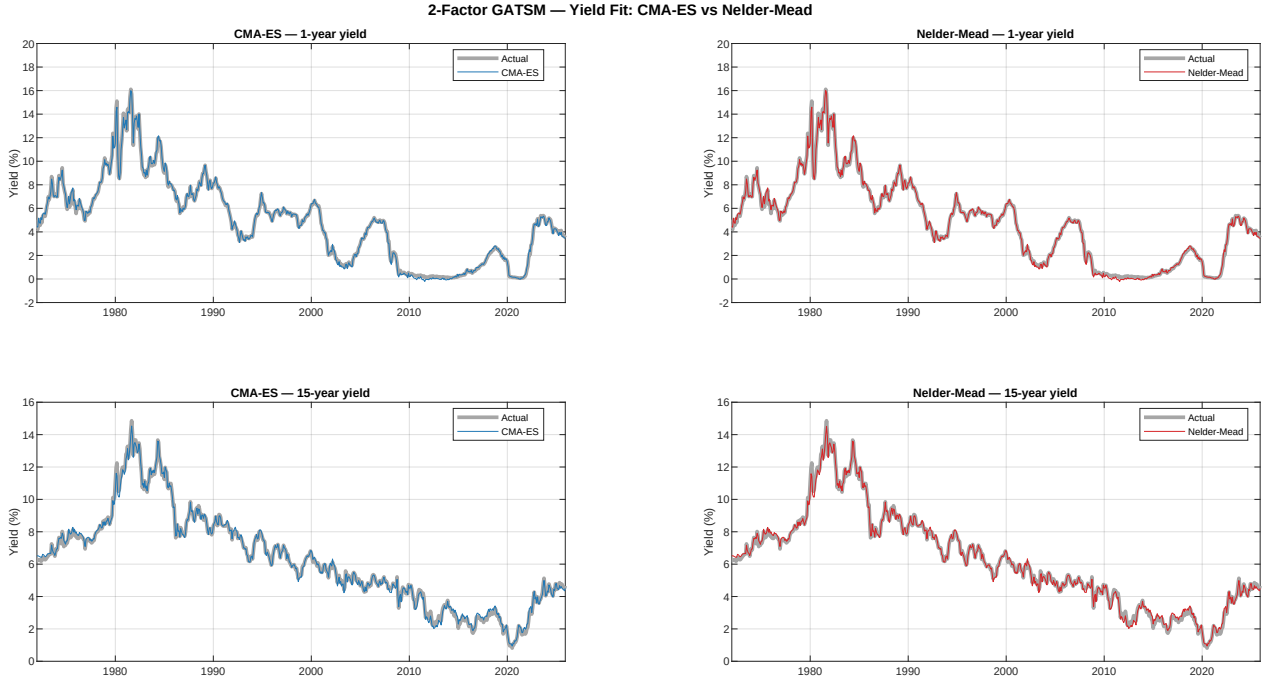


Figure 4: Yield Fit Comparison

From this we can tell that $\Phi_{12}^{\mathbb{P}}$ governs how our second factor, the slope factor $x_{2,t+1}$, feeds into next period's level factor $x_{1,t+1}$. A positive $\Phi_{12}^{\mathbb{P}}$ under CMA-ES of 0.0186 means that a higher slope today predicts a slight increase in the level factor tomorrow, i.e. a steeper yield curve today mildly pulls long-rates up. Under NM where $\Phi_{12}^{\mathbb{P}}$ is -0.0297 this prediction is reversed saying a steep curve today is followed by lower long-rates. Similarly, we notice that the estimates for $\Phi_{21}^{\mathbb{P}}$, too, has different signs between the two optimisers, being negative for CMAES and positive for NM, although both negligible small in magnitude. For the rest of the estimates both optimisers produce near-identical estimates. The differences in the $\mathbb{P}$ -dynamics tells us that CMAES and NM has settled at different local optima, and with very small standard errors the estimates are statistically significant from zero.

What we will consider is to compare the implied term premia from both approaches and choose the optimiser with the most economically sensible term premia. The term premium for a specific maturity k can be computed as

$$\text{TP}_t^{(k)} = y_t^{(k)} - \frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}_t^{\mathbb{P}} [r_{t+i}],$$

where $y_t^{(k)}$ is the model-implied yields from Eq. (18) modified to be in annualised terms by

Table 9: Estimated Parameters – Two-Factor and Three-Factor GATSM

	Two-Factor		Three-Factor	
	CMA-ES (1)	Nelder-Mead (2)	CMA-ES (3)	Nelder-Mead (4)
λ_1^Q	0.998022 (0.000041)	0.997934 (0.000042)	0.999949 (0.000046)	0.999322 (0.000065)
λ_2^Q	0.969681 (0.000297)	0.969775 (0.000296)	0.962525 (0.003778)	0.965422 (0.001791)
λ_3^Q	–	–	0.957003 (0.004380)	0.943403 (0.003241)
α	0.011471 (0.000213)	0.011236 (0.000206)	0.202474 (0.174460)	0.013979 (0.000735)
μ_1^P	-0.000011 (0.000028)	0.000013 (0.000026)	-0.001464 (0.001757)	0.000006 (0.000018)
μ_2^P	-0.000067 (0.000035)	0.000065 (0.000034)	0.007902 (0.269204)	0.000053 (0.000130)
μ_3^P	–	–	-0.010476 (0.268739)	-0.000049 (0.000099)
Φ_{11}^P	0.996704 (0.003815)	1.002951 (0.003455)	0.992501 (0.005114)	1.001205 (0.002497)
Φ_{12}^P	0.018586 (0.006813)	-0.029680 (0.006808)	0.018357 (0.006349)	0.000619 (0.003049)
Φ_{13}^P	–	–	0.018257 (0.006609)	0.001733 (0.004053)
Φ_{21}^P	-0.006655 (0.006063)	0.009240 (0.005778)	0.044233 (1.472351)	0.002320 (0.018710)
Φ_{22}^P	0.969117 (0.009552)	0.990433 (0.006595)	0.826786 (3.369350)	0.998675 (0.026622)
Φ_{23}^P	–	–	-0.119578 (3.340218)	0.000182 (0.026483)
Φ_{31}^P	–	–	-0.057136 (1.471849)	-0.004891 (0.013723)
Φ_{32}^P	–	–	0.135412 (3.364545)	-0.001031 (0.018435)
Φ_{33}^P	–	–	1.080575 (3.365858)	0.994965 (0.018807)
Σ_{11}	0.000317 (0.000007)	0.000320 (0.000008)	0.000265 (0.000006)	0.000207 (0.000004)
Σ_{21}	-0.000179 (0.000016)	-0.000180 (0.000017)	-0.002302 (0.055722)	-0.000443 (0.000073)
Σ_{22}	0.000386 (0.000006)	0.000394 (0.000006)	0.014779 (0.376780)	0.001119 (0.000206)
Σ_{31}	–	–	0.002179 (0.055725)	0.000234 (0.000071)
Σ_{32}	–	–	-0.014825 (0.377046)	-0.000867 (0.000209)
Σ_{33}	–	–	0.000415 (0.000033)	0.000182 (0.000004)
σ_v	0.001295 (0.000009)	0.001297 (0.000009)	0.000484 (0.000004)	0.000634 (0.000006)
$\bar{\mathcal{L}}$	28.652756	28.590724	31.844750	30.893413

multiplying by 12. For this we consider both ends of the maturity spectrum, namely the 1- and 15-year maturities. In Figure 5 is plotted the model-implied term premia for the CMAES and the NM for maturities 1 and 15 over time.

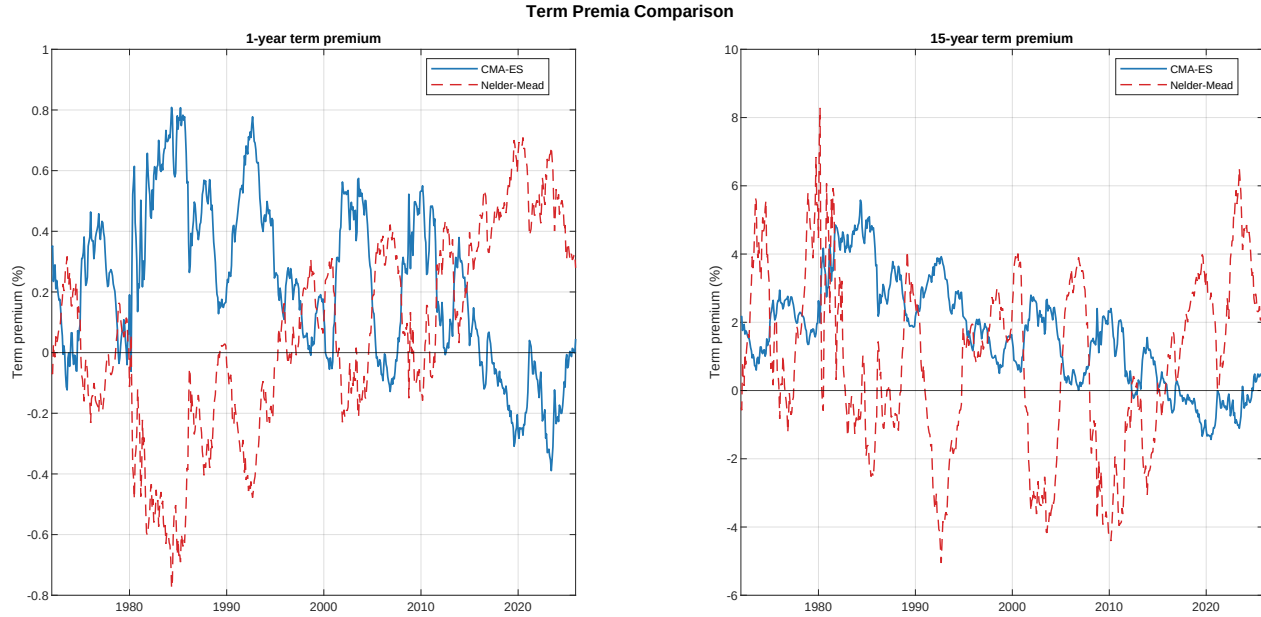


Figure 5: Term Premia Comparison

Generally term premia for the 15 year maturity is of larger magnitude both positively and negatively which we would expect since longer-term bonds carry more risk than short-term bonds for the investor, hence demanding greater compensation. We also clearly see large differences in term premia between the CMAES and the NM. For the 1 year term premium they almost seem completely negatively correlated being on different sides of zero until around 1995. For the 15 year term premium the NM is highly volatile relative to the stable CMAES, with wild oscillations between large negative and positive term premia.

The period of 1979 to 1987 is especially an interesting period for monetary policy in the U.S. due to the then Federal Reserve Chairman Paul Volcker pursuing a drastic monetary tightening, namely suppressing money supply, to combat the high inflation in the U.S. Due to this new monetary policy the federal funds rate rose above 10% during 1979 and later peaked at nearly 20% during 1981 but then quickly swung back again to around the 10% level the following years. Because of this, investors holding long-term bonds naturally faced larger uncertainties about what the short-rates would be in just a few years let alone over a say 10 to 15 year horizon, and we naturally expect that the term premium should be high during this period, since the term premium compensates for this uncertainty about future

short-rates.

The CMAES 1-year and 15-year term premia are largely positive throughout the whole sample period and peaks during the earlier years of Volcker's chairmanship which is consistent with our expectations that during great uncertainty even the very short-term bonds should bear compensation. There is also a visibly negative trend in the term premia over the period, especially for the long 15-year horizon, which could reflect the fact that the U.S. saw a long steady period of moderate business cycles and reduced inflation uncertainty, resulting in a fall in the term premium since the future risks are now lower. Later during the late 2010's the model-implied term premia start becoming mildly negative which could reflect term premium compression arising from large quantitative easing operations by the Federal Reserve. Given all the considerations above we deem the CMAES estimates to be more economically sensible than the NM estimates. This once again shows the advantages in using the CMAES over the NM. Practically what would be even more robust is to first run the CMAES and then use its estimates as starting values for the NM to further refine the optimum. But this goes to show that the Nelder-Mead by itself needs good starting values, and generally estimating over many different starting values would likely yield a better optimum than what we have found here.

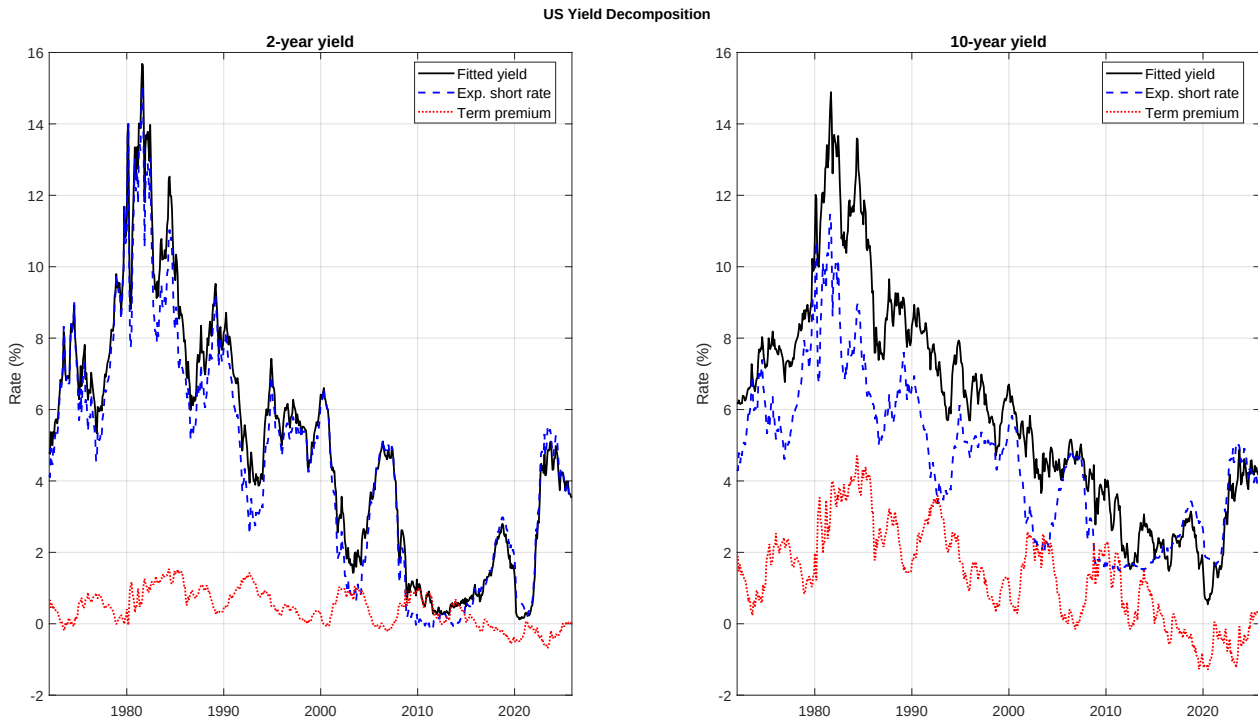


Figure 6: Two-Factor GATSM Yield Decomposition

As noted earlier, one of the main outputs of the GATSM is its yield curve decomposition, and we can compute the model-implied decomposition using Eq. (19). The results for maturities 2 and 10 are plotted in Figure (6). The figure presents the yield curve decomposition from the two-factor GATSM over the full 1972:Q1 - 2025:Q4 sample. Each fitted yield is decomposed into an expected short-rate component, representing what the market expects from rolling over short-term bonds across the life of the bond, and a term premium, capturing the additional compensation investors demand for the uncertainty of holding a long-term bond rather than a sequence of short bonds.

At the 2-year horizon, the expected short-rate dominates. The fitted yield tracks the expected short-rate closely throughout the sample, with the term premium remaining modest and relatively stable. This is expected since the uncertainty about the future path of short-rates at a short horizon is limited, resulting in investors requiring only modest compensation for holding the 2-year bonds.

At the 10-year horizon, the figure changes drastically. A clear and persistent wedge between the fitted yields and the expected short-rate is now present, resulting in the term premium accounting for a substantially larger share of the total yield. This pattern - where the term premium grows with the horizon - is a robust feature of the data and arises naturally from the increased uncertainty that investors face about monetary conditions the further into the future they look to hold their bonds. Committing capital for 10 years exposes the investor to a far wider range of volatilities than a 2-year commitment. This additional risk is reflected in the increased term premium.

The time-series include several distinct crises and economically important events. During the late 1970s and the early 1980s, both yields and the term premium were elevated, consistent with the high macroeconomic uncertainty of the pre-Volcker and Volcker disinflation era. At the time, U.S. inflation had risen to double-digit levels following the oil price shocks in the 1970s. In 1979, Federal Reserve Chairman Paul Volcker responded with a big tightening of the monetary policy, pushing the federal funds rate to extremely high levels to break the inflation expectations [Goodfriend and King \(2005\)](#). This forced the macroeconomic expectations to be highly uncertain and difficult to forecast, resulting in investors demanding substantial compensation for holding long-term bonds. As the Volcker disinflation succeeded and inflation fell back through the early 1980s, the falling inflation volatility, improved central bank credibility, and a more stable macroeconomic environment reduced the uncertainty that investors needed to be compensated for, resulting in the term premium steadily declining.

The post-Financial Crisis period is characterised by a decline in term premium for both

maturities. From around 2009 onward the term premium is persistently declining, reaching near-zero levels by mid 2010s and turning negative from around 2016 onward for the 2-year horizon. A negative term premium should not be seen as irrational investor behaviour - it arises naturally when long-term bonds serve as effective insurance against bad states of the economy. Investors are willing to accept a yield below the expected short-rate path in exchange for a hedge, allowing their bond prices to rise when the economy declines.

From around 2021 onward, both the fitted yields and the expected short-rates rise significantly. The term premium remains relatively stable throughout the end of the sample, suggesting that the yield rise is primarily an expected short-rate increase rather than an increased demand for compensation for holding long-term bonds.

4.6.1 Campell-Shiller

Another way to validate the implied term premium of our GATSM is to investigate whether or not the model matches regression evidence against the expectations hypothesis. This approach was first suggested by [Dai and Singleton \(2002\)](#) where they suggest testing if a GATSM shows the same deviation from the expectation hypothesis (EH) as real data using a standard [Campbell and Shiller \(1991\)](#) predictive regression. The Campbell-Shiller (CS) regression is computed as

$$y_{t+1}^{(k-m)} - y_t^{(k)} = \alpha_{cs} + \beta_{cs} \frac{m}{k-m} \left(y_t^{(k)} - y_t^{(m)} \right) + \varepsilon_t, \quad (32)$$

where m is the short-term maturity set to $m = 1$ years in our case, and the yield spread $\left(y_t^{(k)} - y_t^{(m)} \right)$ is the predictor. Under the EH the slope coefficient β_{cs} should equal to 1, whereas a negative slope coefficients indicates the existence of term premia. The comparison of empirical and model-implied CS coefficients provides a test of whether or not our model generates time-varying term premia that is consistent with the data.

Table 10: Campbell-Schiller Regressions

Maturity	β_{data}	$\beta_{\text{sim.}}$	Sim. median	Sim. std	Sim. 95% CI
2	-0.3973	-0.3907	-0.4025	0.4671	$[-1.2700, 0.5520]$
5	-1.1197	-0.8691	-0.9011	0.7413	$[-2.2397, 0.6792]$
7	-1.5413	-1.1943	-1.2203	0.8518	$[-2.7989, 0.5594]$
10	-2.0815	-1.6614	-1.6790	1.0312	$[-3.6481, 0.4284]$
15	-2.8590	-2.3334	-2.3360	1.3361	$[-4.9612, 0.3266]$

Notes: The actual-data slope β_{cs} is estimated by OLS on observed U.S. yields. Model-implied moments are obtained from $R = 100,000$ simulations. Reported model statistics are computed from the simulated distribution of β_{sim} ; the reported intervals are simulation quantiles rather than analytic confidence intervals.

In Table 10 is reported the Campbell-Shiller slope coefficients estimated on actual U.S. yield data together with statistical moments of the slope distribution computed using $R = 100,000$ replications with our estimated two-factor GATSM being the DGP. We see that the β coefficient of our model follows that of the data in being monotonically declining in maturity. Our simulated CS regressions also rejects the EH beta coefficient of 1 by looking at the 95% confidence intervals.

4.7 Monte Carlo Study Two-Factor

Table 11: Monte Carlo Simulations Two-Factor GATSM

Parameter	True value	Mean bias	Std. dev.	RMSE
Short-rate intercept				
α	0.011471	-0.000010	0.000435	0.000435
$\mathbb{P}$-dynamics: mean				
μ_1	-0.000011	-0.000033	0.000063	0.000071
μ_2	-0.000068	-0.000033	0.000081	0.000087
$\mathbb{P}$-dynamics: AR matrix				
Φ_{11}^P	0.996661	-0.004829	0.007998	0.009343
Φ_{21}^P	-0.006696	-0.004001	0.010512	0.011248
Φ_{12}^P	0.018286	0.002678	0.008857	0.009253
Φ_{22}^P	0.969223	-0.009528	0.012882	0.016023
Volatility (Cholesky)				
Σ_{11}	0.000317	-0.000001	0.000011	0.000011
Σ_{21}	-0.000179	0.000000	0.000021	0.000021
Σ_{22}	0.000386	-0.000001	0.000013	0.000013
$\mathbb{Q}$-dynamics				
λ_1^Q	0.998022	-0.000003	0.000121	0.000121
λ_2^Q	0.969681	-0.000001	0.000719	0.000719
Measurement error				
σ_v	0.000108	-0.000000	0.000002	0.000002

Notes: Mean bias is the mean estimate minus the true DGP value across $R = 1,000$ converged replications. Std. dev. is the standard deviation of the sampling distribution. $\text{RMSE} = \sqrt{\text{Bias}^2 + \text{Var}}$. The DGP uses the CMA-ES maximum likelihood estimates from the full U.S. sample (Jan 1972–Dec 2025, $T = 648$ months). All parameters are in monthly units.

To assess the finite-sample properties of the GATSM, we conduct a Monte Carlo simulation using the CMA-ES estimates from the full U.S. sample (January 1972–December 2025, $T = 648$ months), which serve as the true DGP. Across $R = 1,000$ replications, we simulate yield panels of identical length, re-estimate all parameters using maximum likelihood, and summarise the mean bias, standard deviation and the RMSE of each parameter estimate.

For every replication $\mathcal{R} = 1, \dots, R$ artificial data is generated. We let the CMAES point

estimates be in a vector of parameters $\boldsymbol{\theta}^{\text{true}}$ that are held fixed for every $\mathcal{R}$, with subscript 'true' indicating it represents the true DGP. The unconditional mean and covariance of the states under the true $\mathbb{P}$ -dynamics are computed as

$$\bar{\mathbf{x}}^{\text{true}} = (\mathbf{I} - \boldsymbol{\Phi}^{\mathbb{P},\text{true}})^{-1} \boldsymbol{\mu}^{\mathbb{P},\text{true}}, \quad \mathbf{P}_{\infty}^{\text{true}} = \text{dlyap}(\boldsymbol{\Phi}^{\mathbb{P},\text{true}}, \mathbf{Q}^{\text{true}}), \quad (33)$$

where $\text{dlyap}(\cdot)$ stands for the discrete Lyapunov equation, and $\mathbf{P}_{\infty}^{\text{true}}$ is the covariance of $\mathbf{x}_t^{\text{true}}$ as $t \rightarrow \infty$. The solution to the discrete Lyapunov equation is identical to iterating the covariance recursion forward until it converges, but is much faster computationally. The initial state for replication $\mathcal{R}$ is drawn from the unconditional distribution

$$\mathbf{x}_1^{\mathcal{R}} = \bar{\mathbf{x}}^{\text{true}} + L_{\mathbf{P}} z_{\mathcal{R}} \quad (34)$$

where $z_{\mathcal{R}}$ is normally distributed with mean 0 and covariance $\mathbf{I}$ and is drawn from the replication-specific random stream. The initial drawn state is then propagated forward for $T = 648$ months under the true $\mathbb{P}$ -dynamics

$$\mathbf{x}_{t+1}^{\mathcal{R}} = \boldsymbol{\mu}^{\mathbb{P},\text{true}} + \boldsymbol{\Phi}^{\mathbb{P},\text{true}} \mathbf{x}_t^{\mathcal{R}} + \boldsymbol{\Sigma}^{\text{true}} \mathbf{u}_{t+1}^{\mathcal{R}}, \quad \mathbf{u}_{t+1}^{\mathcal{R}} \sim \text{iid}(\mathbf{0}, \mathbf{I}), \quad (35)$$

giving us one replication of a simulated path of states $\{\mathbf{x}_t^{\mathcal{R}}\}_{t=1}^T$. This path is used to generate an artificial yield panel for replication $\mathcal{R}$. At each date t the model-implied yield for all chosen maturities is computed from the measurement equation and independent noise

$$\tilde{\mathbf{y}}_t^{\mathcal{R}} = \mathbf{g}_0^{\text{true}} + \mathbf{G}_x^{\text{true}} \mathbf{x}_t^{\mathcal{R}} + \sigma_v^{\text{true}} \boldsymbol{\varepsilon}_t^{\mathcal{R}}, \quad \boldsymbol{\varepsilon}_t^{\mathcal{R}} \sim \text{iid}(\mathbf{0}, \mathbf{I}), \quad (36)$$

where $\boldsymbol{\varepsilon}_t^{\mathcal{R}}$ is drawn from the same replication-specific random number generator as the state innovations. The path $\{\mathbf{x}_t^{\mathcal{R}}\}_{t=1}^T$ itself is not provided to the estimator and is only used to construct $\{\tilde{\mathbf{y}}_t^{\mathcal{R}}\}_{t=1}^T$. Every replication $\mathcal{R} = 1, \dots, R$ is then estimated using the same method used to estimate the 'true' values, with one difference which is that we initialise each replication at the true values, in addition to the previous PCA-based starting values. Essentially, each replication runs two independent starts and retains the candidate with highest log-likelihood. For computational ease we choose to use NM over CMAES for every replication, and our argument for this is that we have already reached our 'best' optimum using the global optimiser, and provided that true values are provided as starting points, all that is really left is to refine the optimum.

The results for $R = 1,000$ are reported in Table 11. The findings reveal a pattern where

the bias is almost entirely concentrated in the $\mathbb{P}$ -dynamics AR matrix, $\Phi^{\mathbb{P}}$. The diagonal parameters are systematically underestimated, with mean biases of -0.0048 and -0.0095 for $\Phi_{11}^{\mathbb{P}}$ and $\Phi_{22}^{\mathbb{P}}$ respectively, which is consistent with the well-known downward bias in near-unit-root autoregressive coefficients. All remaining parameters exhibit negligible bias, suggestion that these are identified by the cross-sectional variation in yields rather than the time-series dimension. Although modest in absolute terms, the bias in $\Phi^{\mathbb{P}}$ show meaningful economic consequences. Understated persistence implies that the model mean reverses faster than the true model, causing expected future short-rates to converge too quickly to their long-run mean. This causes the expectations component of long-term yields to be understated, and consequently the term premium to be overstated as a residual.

By estimating a simple latent two-factor GATSM we have showed using simulations that the model estimates contains what has been dubbed finite sample bias in the literature. What we want to proceed with now is to extend the baseline two-factor model by incorporating observable macro factors, turning the model into what is known as a Macro-Finance Term Structure Model (MTSM) in the literature, although we will continue to refer to it as the four-factor model in this paper.

5 Macro Factors

We will in this section describe the four-factor macro-finance GATSM by first briefly outlining the model setup followed by a complete estimation methodology and the results and analyses that follows. Due to surveys, which will be incorporated later, only being available in quarterly frequency, and to prevent going outside of the scope of this paper by estimating a mixed-frequency model, we will in the following work in the quarterly frequency.

5.1 Macro-Finance Model Specification

The general framework is the same as in the two-factor model with equations (5), (6), (7), (11), and (18) being the main equations now with $F = 4$. Composition-wise we follow the approach of [Ang and Piazzesi \(2003\)](#) and partition our factors into a macro- and a latent block. Notationally, we let m_1 and m_2 refer to the inflation and unemployment factors, respectively, while l_1 and l_2 denotes the previous two latent factors, such that the vector of factors is $\mathbf{x}_t = (x_t^{m_1}, x_t^{m_2}, x_t^{l_1}, x_t^{l_2})^\top = (\mathbf{x}_t^m, \mathbf{x}_t^l)^\top$. From equation (6) we partition the companion matrix $\Phi^{\mathbb{P}}$ into

$$\Phi^{\mathbb{P}} = \begin{bmatrix} \Phi_{mm}^{\mathbb{P}} & \Phi_{ml}^{\mathbb{P}} \\ \Phi_{lm}^{\mathbb{P}} & \Phi_{ll}^{\mathbb{P}} \end{bmatrix} = \begin{bmatrix} \Phi_{m_1 m_1}^{\mathbb{P}} & \Phi_{m_1 m_2}^{\mathbb{P}} & \Phi_{m_1 l_1}^{\mathbb{P}} & \Phi_{m_1 l_2}^{\mathbb{P}} \\ \Phi_{m_2 m_1}^{\mathbb{P}} & \Phi_{m_2 m_2}^{\mathbb{P}} & \Phi_{m_2 l_1}^{\mathbb{P}} & \Phi_{m_2 l_2}^{\mathbb{P}} \\ \Phi_{l_1 m_1}^{\mathbb{P}} & \Phi_{l_1 m_2}^{\mathbb{P}} & \Phi_{l_1 l_1}^{\mathbb{P}} & \Phi_{l_1 l_2}^{\mathbb{P}} \\ \Phi_{l_2 m_1}^{\mathbb{P}} & \Phi_{l_2 m_2}^{\mathbb{P}} & \Phi_{l_2 l_1}^{\mathbb{P}} & \Phi_{l_2 l_2}^{\mathbb{P}} \end{bmatrix},$$

and we similarly partition $\mu^{\mathbb{P}}$ and Σ into

$$\mu^{\mathbb{P}} = \begin{bmatrix} \mu_m^{\mathbb{P}} \\ \mu_l^{\mathbb{P}} \end{bmatrix} = \begin{bmatrix} \mu_{m_1}^{\mathbb{P}} \\ \mu_{m_2}^{\mathbb{P}} \\ \mu_{l_1}^{\mathbb{P}} \\ \mu_{l_2}^{\mathbb{P}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{mm} & \Sigma_{ml} \\ \Sigma_{lm} & \Sigma_{ll} \end{bmatrix} = \begin{bmatrix} \Sigma_{m_1 m_1} & \Sigma_{m_1 m_2} & \Sigma_{m_1 l_1} & \Sigma_{m_1 l_2} \\ \Sigma_{m_2 m_1} & \Sigma_{m_2 m_2} & \Sigma_{m_2 l_1} & \Sigma_{m_2 l_2} \\ \Sigma_{l_1 m_1} & \Sigma_{l_1 m_2} & \Sigma_{l_1 l_1} & \Sigma_{l_1 l_2} \\ \Sigma_{l_2 m_1} & \Sigma_{l_2 m_2} & \Sigma_{l_2 l_1} & \Sigma_{l_2 l_2} \end{bmatrix}.$$

We likewise partition the $\mathbb{Q}$ -dynamics in the same manner. The vector of short-rate coefficients is given by $\beta = (\beta_1^m, \beta_2^m, \beta_1^l, \beta_2^l)^\top$ and the loadings in equation (18) are now replaced by

$$g_0^{(k)} \equiv \frac{-4}{k} A_k, \quad \mathbf{G}_x^{(k)} \equiv \frac{-4}{k} \mathbf{B}_k^\top, \quad (37)$$

reflecting the fact that we are now working in a quarterly frequency. Here again $\mathbf{G}_x^k$ is the $1 \times F$ annualised loading row for maturity k , where we write its macro and latent components as $\mathbf{G}_{x,k}^m$ and $\mathbf{G}_{x,k}^l$.

5.2 Four-Factor Estimation Methodology

The estimation proceeds via the Minimum Chi-Square Estimator (MCSE) pipeline proposed in [Hamilton and Wu \(2012\)](#), which takes advantage of the analytical structure of the model to replace a large dimensional optimisation problem, as we saw for the two-factor case, with a sequence of smaller tangible optimisation steps in which a majority of them have closed-form or near-closed-form solutions.

5.2.1 Step 1: First-Stage Reduced VAR OLS

For the four-factor GATSM we similarly need to impose model and parameter restrictions in order to ensure identification. Contrary to the two-factor identification restrictions we do not impose unity on β making the latent factors $\mathbf{x}_t^l$ scale indetermined, so any rescaling $c\mathbf{x}_t^l$ can be absorbed by rescaling the factor loadings β_l/c and the bond-price loadings. To fix the scale of the latent factors we normalise the latent block of the Σ matrix to

$$\Sigma_{ll} = \mathbf{I}. \quad (38)$$

In addition, for the first-stage OLS we impose the independence assumption of [Ang and Piazzesi \(2003\)](#) which states that latent factors do not Granger-cause the macro factors, the macro AR dynamics are independent of latent factors, and that shocks are orthogonal

$$\Phi_{ml}^{\mathbb{P}} = \mathbf{0}, \quad \Phi_{lm}^{\mathbb{P}} = \mathbf{0}, \quad (39)$$

$$\Phi_{ml}^{\mathbb{Q}} = \mathbf{0}, \quad \Phi_{lm}^{\mathbb{Q}} = \mathbf{0}, \quad (40)$$

$$\Sigma_{ml} = \mathbf{0}, \quad \Sigma_{lm} = \mathbf{0}. \quad (41)$$

The restrictions in (39)-(41) imposes block-diagonality on all three of the parameter matrices such that

$$\Phi^{\mathbb{P}} = \begin{bmatrix} \Phi_{mm}^{\mathbb{P}} & \mathbf{0} \\ \mathbf{0} & \Phi_{ll}^{\mathbb{P}} \end{bmatrix}, \quad \Phi^{\mathbb{Q}} = \begin{bmatrix} \Phi_{mm}^{\mathbb{Q}} & \mathbf{0} \\ \mathbf{0} & \Phi_{ll}^{\mathbb{Q}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{mm} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}. \quad (42)$$

The market price of risk λ_x is also block-diagonal and the $\mathbb{Q}$ -dynamics can now be computed as

$$\Phi^{\mathbb{Q}} = \begin{bmatrix} \Phi_{mm}^{\mathbb{P}} & \mathbf{0} \\ \mathbf{0} & \Phi_{ll}^{\mathbb{P}} \end{bmatrix} - \underbrace{\begin{bmatrix} \lambda_{x,mm} & \mathbf{0} \\ \mathbf{0} & \lambda_{x,ll} \end{bmatrix}}_{\lambda_x}. \quad (43)$$

Since our macro factors inflation and unemployment are directly observed they are not

subject to the rotational indeterminacy that is apparent in the latent factors. The identification problem from rotations are then confined entirely to the subspace spanned by $\mathbf{x}_t^l$. The restrictions (39)-(41) separates the rotation problem into two decoupled blocks, macro and latent, meaning we only have to worry about the latent rotational problem.

The macro block follows an autonomous VAR(1) process

$$\mathbf{x}_t^m = \mathbf{A}^{m,\#} + \mathbf{\Phi}_{mm}^{\mathbb{P},\#} \mathbf{x}_{t-1}^m + \mathbf{u}_t^{m,\#}, \quad t = 2, \dots, T, \quad (44)$$

where $(\#)$ denotes reduced-form counterparts, and $\mathbf{u}_t^{m,\#}$ is the Gaussian innovation vector. Let $T_1 = T - 1$ since we lose one observation when running OLS on the VAR(1) process. We can analytically solve for the coefficients using OLS

$$\begin{bmatrix} \hat{\mathbf{A}}^{m,\#} \\ \hat{\mathbf{\Phi}}_{mm}^{\mathbb{P},\#} \end{bmatrix} = (\mathbf{X}_m^\top \mathbf{X}_m)^{-1} \mathbf{X}_m^\top \mathbf{Y}_m, \quad (45)$$

where $\mathbf{Y}_m \in \mathbb{R}^{T_1 \times F_m}$ stacks $\mathbf{x}_t^m$ for $t = 2, \dots, T$ and $\mathbf{X}_m = \begin{bmatrix} \mathbf{1} & \mathbf{x}_{t-1}^m \end{bmatrix}$ is the $T_1 \times (1 + F_m)$ regressor matrix. Taking this to the data we get

$$\hat{\mathbf{A}}^{m,\#} = [-0.0046, -0.0031], \quad \hat{\mathbf{\Phi}}_{mm}^{\mathbb{P},\#} = \begin{bmatrix} 0.8808 & -0.0281 \\ 0.0447 & 0.8970 \end{bmatrix}.$$

The macro block is then identified directly from our OLS estimate $\hat{\mathbf{\Phi}}_{mm}^{\mathbb{P}} = \hat{\mathbf{\Phi}}_{mm}^{\mathbb{P},\#}$. The residual covariance is given by

$$\hat{\mathbf{\Omega}}^{m,\#} = T_1^{-1} (\hat{\mathbf{U}}^m)^\top \hat{\mathbf{U}}^m, \quad (46)$$

where $\hat{\mathbf{U}}^m$ is the matrix of OLS residuals. The macro covariance of innovation $\mathbf{\Sigma}_{mm} \mathbf{\Sigma}_{mm}^\top$ is identified from this OLS residual covariance, but $\mathbf{\Sigma}_{mm}$ is only identified up to an arbitrary orthogonal rotation of the columns. For instance, if $\mathbf{\Sigma}_{mm}$ satisfies $\mathbf{\Sigma}_{mm} \mathbf{\Sigma}_{mm}^\top = \hat{\mathbf{\Omega}}^{m,\#}$ then so does $\mathbf{\Sigma}_{mm} \mathbf{Q}$ for any orthogonal $\mathbf{Q}$. To resolve this we impose that $\mathbf{\Sigma}_{mm}$ is the lower-triangular Cholesky of $\hat{\mathbf{\Omega}}^{m,\#}$

$$\mathbf{\Sigma}_{mm} = \text{chol}(\hat{\mathbf{\Omega}}^{m,\#}). \quad (47)$$

The specific ordering of our macro variables (inflation first, unemployment second) is an identifying normalisation that pins down a unique $\mathbf{\Sigma}_{mm}$ in (47). It imposes a recursive contemporaneous structure in which an inflation innovation can affect unemployment within the same quarter, while an unemployment innovation affects inflation only with a lag, consistent with price stickiness under which real activity responds to nominal shocks faster than

inflation responds to labour-market shocks within a single quarter. Do note later that the estimated off-diagonal of the Cholesky decomposition ($\Sigma_{21} = 0.018$) is very small relative to the diagonals, so the recursive ordering has only a minor effect on the results and primarily matters for the labelling of the structural macro shocks that will be used later in an impulse response analysis.

Now following [Ang and Piazzesi \(2003\)](#), [Hamilton and Wu \(2012\)](#), and [Chen and Scott \(1993\)](#) we partition the yield block into two groups. Group $\mathcal{Y}^1$ which consists of $n_{\text{obs}} = 2$ maturities that are priced exactly and group $\mathcal{Y}^2$ that consists of $n_y = 4$ yields that are measured with error σ_v as in the two-factor model. Furthermore, let $n_m = 2$ denote the number of macro factors and $n_l = 2$ the number of latent factors. The yields in group $\mathcal{Y}^1$ are used to identify the latent factors. In our case we choose the 1-year and 10-year maturity yields for $\mathcal{Y}_t^1$, and are chosen as they span the short and long ends of the curve. Alternatively, we could have chosen the 15-year maturity instead of the 10-year based on our argument but we experience better fitted yields using the 10-year maturity as the long-end representative. The remaining 2, 5, 7 and 15-year yields are then put into $\mathcal{Y}_t^{(2)}$.

By applying the maturities in group $\mathcal{Y}_t^1$ to the affine yield equation in (18) and stacking them gives us

$$\mathbf{y}_t^1 = \begin{bmatrix} y_t^{(1)} \\ y_t^{(10)} \end{bmatrix} = \begin{bmatrix} -A_1/1 \\ -A_{10}/10 \end{bmatrix} + \begin{bmatrix} -\mathbf{B}_1^\top/1 \\ -\mathbf{B}_{10}^\top/10 \end{bmatrix} \mathbf{x}_t, \quad (48)$$

which, after the macro and latent partition, and in annualised units is equal to

$$\mathbf{y}_t^1 = \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \mathbf{x}_t^l, \quad (49)$$

where $\mathbf{G}_x^{1,m}$ and $\mathbf{G}_x^{1,l}$ are the macro and latent column blocks of the loadings evaluated at the $\mathcal{Y}_t^1$ maturities

$$\mathbf{G}_x^{1,m} = \begin{bmatrix} \mathbf{G}_{x,4}^m \\ \mathbf{G}_{x,40}^m \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad \mathbf{G}_x^{1,l} = \begin{bmatrix} \mathbf{G}_{x,4}^l \\ \mathbf{G}_{x,40}^l \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad k = 4, 40 \text{ quarters} = 1, 10 \text{ years}. \quad (50)$$

Doing the same for group $\mathcal{Y}^2$ gives us

$$\mathbf{y}_t^2 = \mathbf{g}_0^2 + \mathbf{G}_x^{2,m} \mathbf{x}_t^m + \mathbf{G}_x^{2,l} \mathbf{x}_t^l, \quad (51)$$

where the loadings are given by

$$\mathbf{G}_x^{2,m} = \begin{bmatrix} \mathbf{G}_{x,8}^m \\ \mathbf{G}_{x,20}^m \\ \mathbf{G}_{x,28}^m \\ \mathbf{G}_{x,60}^m \end{bmatrix} \in \mathbb{R}^{4 \times 2}, \quad \mathbf{G}_x^{2,l} = \begin{bmatrix} \mathbf{G}_{x,8}^l \\ \mathbf{G}_{x,20}^l \\ \mathbf{G}_{x,28}^l \\ \mathbf{G}_{x,60}^l \end{bmatrix} \in \mathbb{R}^{4 \times 2}, k = 8, 20, 28, 60 \text{ quarters} = 2, 5, 7, 15 \text{ years.} \quad (52)$$

Under the independence assumption, yield block $\mathcal{Y}_t^1$ follows a process that is driven by lagged $\mathcal{Y}_{t-1}^1$ and contemporaneous macro factors due to our independence assumption

$$\mathcal{Y}_t^1 = \mathbf{A}^{1,\#} + \Phi_{11}^{\mathbb{P},\#} \mathcal{Y}_{t-1}^1 + \Psi_{1m}^{\mathbb{P},\#} \mathbf{x}_t^m + \mathbf{u}_t^{1,\#}, \quad (53)$$

which is found by inserting for the latent $\mathbb{P}$ -dynamics and rearranging. The substitution leaves a lagged-macro term that we exclude from (53). Under the independence restriction $\Phi_{lm}^{\mathbb{P}} = \mathbf{0}$ this coefficient carries no structural parameter to be recovered, and retaining it would add freely-estimated reduced-form coefficients with no structural counterpart and only inflate the finite-sample bias.

This exclusion does not affect the macro persistence parameters $\Phi_{mm}^{\mathbb{P}}$, which are the main object of this paper, since these are obtained from the autonomous macro block alone. Its only potential direct effect is on the latent block, where our estimation have showed that the term is negligible, although we acknowledge that there will be omitted variable bias arising from this exclusion even though its magnitude should be inconsequential⁸.

Block $\mathcal{Y}_t^2$ is regressed on contemporaneous macro factors and $\mathcal{Y}_t^1$

$$\mathcal{Y}_t^2 = \mathbf{A}^{2,\#} + \Phi_{2m}^{\mathbb{P},\#} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathcal{Y}_t^1 + \mathbf{u}_t^{2,\#}, \quad (54)$$

where this expression is derived by solving for $\mathbf{x}_t^l$ in (49) and inserting into (51)⁹. Both of the regressions are solved using OLS. Given the OLS estimate $\hat{\Phi}_{11}^{\mathbb{P},\#}$ we can identify the latent factor block after performing a similarity transformation given by

$$\hat{\Phi}_{ll}^{\mathbb{P}} = \text{chol} \left(\left(\hat{\Omega}^{1,\#} \right)^\top \hat{\Phi}_{11}^{\mathbb{P},\#} \text{chol} \left(\hat{\Omega}^{1,\#} \right) \right), \quad (55)$$

⁸Re-estimating the Group 1 regression with $\mathbf{x}_{t-1}^m$ shows that the latent persistence eigenvalues of $\hat{\Phi}_{11}^{\mathbb{P},\#}$, which are invariant under the similarity transform recovered in (55), only move from 0.965 to 0.963 and from 0.785 to 0.767. The estimated lagged-macro coefficient is around the magnitude 0.003 and changes the residual covariance $\hat{\Omega}^{1,\#}$ by only about 2%.

⁹See Appendix A1.

which lets $\widehat{\Phi}_{ll}^{\mathbb{P}}$ preserve the eigenvalues of $\widehat{\Phi}_{11}^{\mathbb{P},\#}$, thus inheriting its stationarity. We have now found a solution for $\Phi^{\mathbb{P}}$.

5.2.2 Step 2: Structural Parameters

Now following [Hamilton and Wu \(2012\)](#), the structural parameters $(\Phi^{\mathbb{Q}}, \beta)$ are estimated using the MCSE approach which is an estimation method which involves minimising a quadratic form in the distance between structural model-implied equations and the unrestricted OLS estimates we obtained in Step 1. Asymptotically, the MCSE is equivalent to the full-information maximum likelihood (FIML) but eases the estimation steps since we have already analytically solved for the OLS estimators and the numerical optimisation we are about to approach is confined to a lower-dimensional problem rather than a full non-linear likelihood surface as in the two-factor estimation approach. For macro specific parameters we have $n_m = 2$.

Comparing equations (49) and (53) we see that the OLS slope $\Psi_{1m}^{\mathbb{P},\#}$ must equal the structural macro loading $G_x^{1,m}$, giving us the first MCSE condition

$$G_{x,OLS}^{1,m} = \widehat{\Psi}_{1m}^{\mathbb{P},\#}. \quad (56)$$

Now by substituting the yield recursion expression for $\mathcal{Y}_t^1$ from (49) into the $\mathcal{Y}_t^2$ regression in (54) gives us

$$\mathcal{Y}_t^2 = \left(A^{2,\#} + \Phi_{21}^{\mathbb{P},\#} g_0^1 \right) + \left(\Phi_{2m}^{\mathbb{P},\#} + \Phi_{21}^{\mathbb{P},\#} G_x^{1,m} \right) \mathbf{x}_t^m + \Phi_{21} G_x^{1,l} \mathbf{x}_t^l + \mathbf{u}_t^{2,\#}, \quad (57)$$

and we can see that the second MCSE condition is given by

$$G_{x,OLS}^{2,m} = \Phi_{2m}^{\mathbb{P},\#} + \Phi_{21}^{\mathbb{P},\#} \widehat{\Psi}_{1m}^{\mathbb{P},\#}. \quad (58)$$

After controlling for lagged $\mathcal{Y}_t^1$ and contemporaneous $\mathbf{x}_t^m$, the residuals $\mathbf{u}_t^{1,\#}$ in (53) only absorb the latent factor innovations, so the third MCSE condition is given by

$$\widehat{\Omega}^{1,\#} = G_x^{1,l} (G_x^{1,l})^\top. \quad (59)$$

Now by looking at the coefficients for $\mathbf{x}_t^l$ in (57) the fourth MCSE condition is

$$G_{x,OLS}^{2,l} (G_x^{1,l})^\top = \Phi_{21}^{\mathbb{P},\#} \widehat{\Omega}^{1,\#}, \quad (60)$$

where we have multiplied by $(G_x^{1,l})^\top$ since $G_x^{2,l}$ is unobserved.

The annualised loadings $\mathbf{G}_{x,k}^m$ in (50) and (52) are given by $\frac{4}{k} \sum_{j=0}^{k-1} [(\Phi_{mm}^Q)^\top]^j \beta_m$ and $\frac{4}{k} \sum_{j=0}^{k-1} [(\Phi_{ll}^Q)^\top]^j \beta_l$, respectively. We can then once again decouple the macro and latent blocks into two independent subproblems, where the conditions in (56) and (58) constitutes the macro block, and conditions (59) and (60) controls the latent block. The macro columns of $\mathbf{G}_x$ only depend on Φ_{mm}^Q and β_m while the latent columns only depend on Φ_{ll}^Q and β_l .

In the two-factor model we estimated Φ^Q together with Φ^P , an approach which were largely dependent on starting values to get fine results. Now given that we have analytically solved for Φ^P already we can now find an expression for Φ^Q by anchoring it to Φ^P and solving for the risk corrections λ_{mm} and λ_{ll} instead.

To solve the macro block we let the parameter vector be $\mathbf{p}_m = [\text{vec}(\lambda_{mm}); \beta_m]$, with 6 unknowns to solve for. The residual function $\mathbf{f}_m$ then evaluates the model-implied $\mathbf{G}_x^{1,m}$ and $\mathbf{G}_x^{2,m}$ against the targets in (56) and (58)

$$\mathbf{f}_m(\mathbf{p}_m) = \begin{bmatrix} \text{vec}(\mathbf{G}_x^{1,m}(\mathbf{p}_m) - \mathbf{G}_{x,OLS}^{1,m}) \\ \text{vec}(\mathbf{G}_x^{2,m}(\mathbf{p}_m) - \mathbf{G}_{x,OLS}^{2,m}) \end{bmatrix}. \quad (61)$$

The objective is to minimise

$$\min_{\mathbf{p}_m} \|\mathbf{f}_m(\mathbf{p}_m)\|^2. \quad (62)$$

In the macro subproblem we directly matched $\mathbf{G}_x^{1,m}$ and $\mathbf{G}_x^{2,m}$ against the OLS slopes, because macro factors are observed and appeared as regressors in the $\mathbf{y}_t^1$ and $\mathbf{y}_t^2$ regressions. The latent block is not as straightforward as the macro block due to the latent factors being unobserved.

For the latent block we apply the identification result of Proposition 2 in [Hamilton and Wu \(2012\)](#) where, for $n_l = 2$, the $\mathbb{Q}$ -dynamics matrix Φ_{ll}^Q is parameterised in an equal-diagonal Jordan form

$$\Phi_{ll}^Q = \Phi_{ll}^P - \lambda_{ll}, \quad \lambda_{ll} = \begin{bmatrix} a & b \\ c & a \end{bmatrix}, \quad (63)$$

subject to the constraint $b \leq c$. This restriction achieves identification without restricting the eigenvalues. The parameter vector is $\mathbf{p}_l = [a; b; c; \beta_l]$, with 5 unknowns to solve for. The residual function $\mathbf{f}_l$ is then given by

$$\mathbf{f}_l(\mathbf{p}_l) = \begin{bmatrix} \text{vech}(\mathbf{G}_x^{1,l}(\mathbf{G}_x^{1,l})^\top - \widehat{\Omega}^{1,\#}) \\ \text{vec}(\mathbf{G}_x^{2,l}(\mathbf{G}_x^{1,l})^\top - \widehat{\Phi}_{21}^P \widehat{\Omega}^{1,\#}) \end{bmatrix}, \quad (64)$$

where $\text{vech}(\cdot)$ stacks only the lower-triangular of a symmetric matrix¹⁰. The latent objective is then

$$\min_{\mathbf{p}_l} \|\mathbf{f}_l(\mathbf{p}_l)\|^2. \quad (65)$$

Lastly, before delivering both minimisation problems to our optimiser, we first impose the sign normalisation restriction that the sign of each element of β_l is fixed to be positive, which resolves the observational equivalence between (β_l, Φ_{ll}^Q) and $(-\beta_l, (\Phi_{ll}^Q)^\top)$.

The minimisation problem is solved using the CMAES optimiser, where we set tight absolute bounds $\text{TolFun} = \text{TolX} = 10^{-14}$ with a large function-evaluation budget of 10^6 to ensure convergence. We allow for much tighter tolerance in this case compared to the two-factor maximum likelihood case, and the reason is that what is solved here is a least-squares residual problem where the objective function is smoother than the log-likelihood surface, so extra tolerance is cheap and harmless.

The starting values and box bounds for β_m are derived from the sign and scale of $\mathbf{G}_{x,OLS}^{1,m}$ which provides us with an informative initialisation. Lastly, starting values for β_l are scaled from the diagonal of $\text{chol}(\hat{\Omega}^{1,\#})$.

5.2.3 Step 3: Recovering μ^Q and α Numerically

The intercept of the Q -dynamics in the latent block μ_l^Q is not completely separately identified from the level of the yield curve implied by α and β . Because of this we impose the restriction that

$$\mu_l^Q = \mathbf{0}. \quad (66)$$

The unconditional mean of the latent factors under Q is then $(\mathbf{I} - \Phi_{ll}^Q)^{-1} \mu_l^Q = \mathbf{0}$, and any level contribution from the latent factors is absorbed into α and μ_m^Q .

With the restriction on μ_l^Q the parameter vector to be numerically optimised is $\mathbf{p}_1 = [\alpha; \mu_m^Q]$ with $1 + n_m = 3$ unknowns. The function $\mathbf{f}_1$ then matches the model-implied A^2 loadings against the OLS intercept from $\mathbf{y}^2$

$$\mathbf{f}_1(\mathbf{p}_1) = \mathbf{A}^{2,\#} - \mathbf{A}_{ann}^2(\mathbf{p}_1) + \mathbf{G}_{x,ann}^{2,l} (\mathbf{G}_{x,ann}^{1,l})^{-1} \mathbf{A}^{1,\#}, \quad (67)$$

where $\mathbf{A}_{ann}^i$ denotes the model-implied annualised yield intercepts at the $\mathbf{y}^i$ maturities de-

¹⁰Eq. (59) consists of 2×2 symmetric matrices on both sides so there are only 3 unique scalar restrictions. If we used $\text{vec}(\cdot)$ instead we would write a system of 4 equations where two of them would be identical. Our solver would then be trying to satisfy the same restriction twice, which would waste a degree of freedom.

rived from (15). The optimiser then minimises the following

$$\|\mathbf{f}_1(\mathbf{p}_1)\|^2 + w(a - \bar{y}^{1yr}/ann)^2, \quad (68)$$

where $w(a - \bar{y}^{1yr}/ann)^2$ is a soft penalty term that prevents degenerate solutions where α and $\boldsymbol{\mu}_m^{\mathbb{Q}}$ offset each other while leaving the unconditional yield level far from its empirical value. $\bar{y}^{1yr}$ is the sample mean of the 1-year yield and w is chosen so that the penalty is of comparable magnitude to the $\mathbf{A}^2$ residuals which is a factor of 1×10^6 . CMAES is also the used optimiser here with $\text{TolX} = \text{TolFun} = 1 \times 10^{-10}$. We use looser tolerances here since we have to balance between actual meaningful steps and increases in the soft penalty term.

By experience, during estimation we also apply the hard-bounds $[-0.03 \leq \alpha \leq 0.03]$ which is to prevent the optimiser ending up at a common solution where α drifts to -0.05 , equivalent to a -7.3% annualised short-rate. The α estimate likes to path towards economically infeasible negative regions to keep the estimates of the short-rate loadings $\boldsymbol{\beta}$ empirically plausible, which required us to hard-bound α during estimation to prevent this.

5.2.4 Step 4: Recovering $\boldsymbol{\mu}^{\mathbb{P}}$ Analytically

Now with the parameters $(\alpha, \boldsymbol{\mu}^{\mathbb{Q}}, \boldsymbol{\Phi}^{\mathbb{Q}}, \boldsymbol{\beta}, \boldsymbol{\Sigma})$ all determined we can recover $\boldsymbol{\mu}^{\mathbb{P}}$ analytically from the yield-intercept identification conditions, giving us

$$\boldsymbol{\mu}_m^{\mathbb{P}} = \hat{\mathbf{A}}^{m,\#}, \quad (69)$$

which is the macro OLS intercept. The latent intercept is given by

$$\boldsymbol{\mu}_l^{\mathbb{P}} = \text{chol}(\hat{\boldsymbol{\Omega}}^{1,\#})^{-1} \left(\hat{\mathbf{A}}^{1,\#} - \mathbf{A}_{ann}^1 + \text{chol}(\hat{\boldsymbol{\Omega}}^{1,\#}) \hat{\boldsymbol{\Phi}}_u^{\mathbb{P}} \text{chol}(\hat{\boldsymbol{\Omega}}^{1,\#})^{-1} \mathbf{A}_{ann}^1 \right). \quad (70)$$

5.2.5 Bond Pricing Loadings and Kalman Filter

Now that we have the full structural parameter set, we compute the bond price recursions in (15) and (16) for $k = 1, \dots, 60$ quarters to complete the loadings A_k and $\mathbf{B}_k$ for each maturity. Annualised loadings are then computed using (37).

Lastly, we run a Kalman Filter given our already estimated parameters to recover our latent states and to compute the log-likelihood of our solution.

5.2.6 Identification Test

One of the benefits of using the MCSE approach is that it allows one to test whether the imposed restrictions are consistent with data. As per [Hamilton and Wu \(2012\)](#), the exact Wald statistic is given by

$$\chi^2 = T \left(\mathbf{f}_m^\top \hat{\mathbf{R}}_{mm} \mathbf{f}_m + \mathbf{f}_l^\top \hat{\mathbf{R}}_{ll} \mathbf{f}_l + \mathbf{f}_1^\top \hat{\mathbf{R}}_{11} \mathbf{f}_1 \right),$$

where $\mathbf{f}_m, \mathbf{f}_l, \mathbf{f}_1$ are the gaps between the model-implied and OLS reduced forms. The used weighting matrix here is important since, to make the statistic asymptotically χ^2 , it must be the inverse covariance of these gaps which is the information matrix $\hat{\mathbf{R}}$. The information matrix is block-diagonal in the reduced-form parameters $(\text{vec } \hat{\mathbf{\Pi}}_1, \text{vech } \hat{\mathbf{\Omega}}^{1,\#}, \text{vec } \hat{\mathbf{\Pi}}_2)$, which allows us to split it into three different matrices, with blocks $\hat{\mathbf{\Omega}}^{1,\#,-1} \otimes \hat{\mathbf{Q}}_1$, $\frac{1}{2} \mathbf{D}^\top (\hat{\mathbf{\Omega}}^{1,\#,-1} \otimes \hat{\mathbf{\Omega}}^{1,\#,-1}) \mathbf{D}$, and $\hat{\mathbf{\Omega}}^{2,\#,-1} \otimes \hat{\mathbf{Q}}_2$, where $\hat{\mathbf{Q}}_i = T_1^{-1} \sum_t \mathbf{x}_{it} \mathbf{x}_{it}^\top$ and $\mathbf{D}$ is the duplication matrix¹¹.

For comparison we also report a simpler unweighted version that sets $\hat{\mathbf{R}}_i = \mathbf{I}$,

$$\hat{\chi}^2 = T \left(\|\mathbf{f}_m(\hat{\mathbf{p}}_m)\|^2 + \|\mathbf{f}_l(\hat{\mathbf{p}}_l)\|^2 + \|\mathbf{f}_1(\hat{\mathbf{p}}_1)\|^2 \right). \quad (71)$$

This is not itself $\chi^2(q)$, but it gives a quick, scale-free read on how closely the model reproduces the OLS targets. Only the properly weighted version, evaluated at the parameters that minimise it, has the $\chi^2(q)$ distribution, where q is the number of over-identifying restrictions. The unweighted version simply sums the squared norms of the three blocks; it is not itself $\chi^2(q)$ but indicates how closely the structural restrictions reproduce the reduced-form targets in the loadings metric.

From the macro block we match the model-implied macro factor loadings $\mathbf{G}_{x,OLS}^{1,m}$ and $\mathbf{G}_{x,OLS}^{2,m}$ with their reduced-form OLS counterparts $\hat{\mathbf{\Psi}}_{1m}^{\mathbb{P},\#}$ and $\mathbf{\Phi}_{2m}^{\mathbb{P},\#} + \mathbf{\Phi}_{21}^{\mathbb{P},\#} \hat{\mathbf{\Psi}}_{1m}^{\mathbb{P},\#}$, which is $n_{obs} \times n_m + n_y \times n_m = 2 \times 2 + 4 \times 2 = 12$ conditions to match in $(\boldsymbol{\lambda}_{mm}, \boldsymbol{\beta}_m)$, which is a vector of $n_m^2 + n_m = 4 + 2 = 6$ free parameters, which gives us $12 - 6 = 6$ over-identifying restrictions. The latent MCSE conditions in (59) and (60) provide $\frac{n_l(n_l+1)}{2} + n_y \times n_l = 3 + 4 \times 2 = 11$ equations. The unknowns we need to solve for are the Jordan market price of risk (a, b, c) from Proposition 2 in [Hamilton and Wu \(2012\)](#) and the two latent short-rate loadings $\boldsymbol{\beta}_l$, giving us $3 + n_l = 5$ free parameters. This yields $11 - 5 = 6$ over-identifying parameters from the latent block. Lastly, from the Step 3 we match the model-implied annualised intercepts $\mathbf{A}_{ann}^2$ against their OLS intercept counterparts $\mathbf{A}^{2,\#}$, which provides n_y equations for the

¹¹For a symmetric matrix A , the duplication matrix turns $\text{vech}(A)$ into $\text{vec}(A)$.

$1 + n_m$ unknowns in $(\alpha, \boldsymbol{\mu}_m^{\mathbb{Q}})$, giving us $4 - 3 = 1$ over-identifying restriction.

In total we have $q = 13$, over-identifying restrictions, so we test using a chi-square distribution with 13 degrees of freedom, $\chi^2(13)$, which has a critical value of 22.362 at the 5% significance level.

Evaluated at our estimates, the unweighted statistic is small, $\widehat{\chi}_{\mathbf{I}}^2 = 2.0140 \times 10^{-4} + 2.4227 \times 10^{-10} + 1.9818 \times 10^{-4} = 0.0863$, indicating that the structural loadings reproduce the reduced-form OLS targets to within a small absolute distance. The formal test, however, requires the efficient weighting $\widehat{\mathbf{R}}$. Re-estimating the structural parameters by minimising the efficiently-weighted criterion and evaluating the statistic at that minimiser gives $\widehat{\chi}^2 = 87.18$, exceeding the 1% critical value of $\chi^2(13)$, which is 27.69 with the 5% value being 22.362, so the over-identifying restrictions are formally rejected. This mirrors [Hamilton and Wu \(2012\)](#), who likewise reject the [Ang and Piazzesi \(2003\)](#) specification ($\chi^2 = 462$ against a 1% critical value of 176). The rejection comes from how precisely the model fits the yields. Because the pricing errors are so small, the test holds the model to a very tight standard, which means that even small gaps between the no-arbitrage restrictions and the unrestricted data count as large in this context. Rejections like this are routine for simple affine models and do not weaken our main results who are dependent on the physical persistence parameters, and those are tightly estimated as we will report below. But we should still keep in mind that the over-identifying restrictions in general is not consistent with the data. As [Hamilton and Wu \(2012\)](#) note, in an over-identified structure the sequential block estimates we report differ numerically from the efficiently-weighted estimates, and the statistic above is computed using the efficiently-weighted estimates.

5.3 Macro-Finance Results

Table 12: Estimated Parameters — Four-Factor GATSM

Parameter	Estimate	Parameter	Estimate
Q-dynamics: Eigenvalues		Volatility: Macro Cholesky (Σ_{mm})	
λ_1^Q	1.016943	Σ_{11}	0.472118 (0.023016)
λ_2^Q	0.979789	Σ_{21}	0.018191 (0.030481)
λ_3^Q	0.963959	Σ_{22}	0.433753 (0.020827)
λ_4^Q	0.881890		
Short-rate Intercept		Q-dynamics: Macro AR (Φ_{mm}^Q)	
α	0.012153 (0.002538)	$\Phi_{mm,11}^Q$	0.976306 (0.657334)
		$\Phi_{mm,21}^Q$	0.008719 (0.599120)
		$\Phi_{mm,12}^Q$	0.057549 (0.970701)
		$\Phi_{mm,22}^Q$	1.004596 (0.646081)
P-dynamics: Mean (μ^P)		Q-dynamics: Latent AR (Φ_{ll}^Q)	
μ_1^P (infl)	-0.004645 (0.040571)	$\Phi_{ll,11}^Q$ ($= \Phi_{ll,22}^Q$)	0.925319 (0.035897)
μ_2^P (unemp)	-0.003124 (0.036886)	$\Phi_{ll,12}^Q$	0.027129 (0.058034)
μ_3^P (lat1)	-0.181179 (0.097964)	$\Phi_{ll,21}^Q$	0.087197 (0.052572)
μ_4^P (lat2)	0.182655 (0.180117)		
P-dynamics: Macro AR (Φ_{mm}^P)		Short-rate Loadings (β)	
$\Phi_{mm,11}^P$	0.880789 (0.037173)	β_1 (infl)	0.000746 (0.000336)
$\Phi_{mm,21}^P$	0.044653 (0.035084)	β_2 (unemp)	-0.000771 (0.000407)
$\Phi_{mm,12}^P$	-0.028094 (0.038063)	β_3 (lat1)	0.000109 (0.000675)
$\Phi_{mm,22}^P$	0.897005 (0.035631)	β_4 (lat2)	0.002264 (0.000595)
P-dynamics: Latent AR (Φ_{ll}^P)		Risk-neutral Intercept (μ^Q)	
$\Phi_{ll,11}^P$	0.869830 (0.040274)	μ_1^Q	-0.044488 (0.084402)
$\Phi_{ll,21}^P$	0.080681 (0.040625)	μ_2^Q	-0.354474 (0.074920)
$\Phi_{ll,12}^P$	0.100098 (0.050691)		
$\Phi_{ll,22}^P$	0.880871 (0.052990)		
		Measurement Error	
		σ_v (bps)	59.1130 (0.0000)
		$\frac{\log \mathcal{L}}{T}$	24.216488

Table 12 reports the full set of estimated parameters from the four-factor Macro-finance GATSM. Standard errors in parentheses are full-information maximum-likelihood (FIML)

standard errors evaluated at the (MCSE) point estimates. The observed information matrix is the negative numerical Hessian $-\mathbf{H}$ of the total Kalman-filter log-likelihood $\mathcal{L}(\boldsymbol{\theta}) = \sum_{t=1}^T \ell_t(\boldsymbol{\theta})$, computed by central finite differences at $\hat{\boldsymbol{\theta}}$. The asymptotic covariance is $\widehat{\text{Var}}(\hat{\boldsymbol{\theta}}) = (-\mathbf{H})^{-1}$.

The model combines two observable macro factors, inflation and unemployment, with two latent yield factors, and is estimated on quarterly U.S. data from 1972:Q1 to 2025:Q4.

For the $\mathbb{Q}$ -dynamics the four eigenvalues of the $\mathbb{Q}$ -measure companion matrix follow the pattern typically found in term structure models. The dominant eigenvalue $\lambda_1^{\mathbb{Q}} = 1.017$ is marginally above unity, implying that the dominant pricing factor is effectively permanent under the risk-neutral measure. An eigenvalue above 1 in a stationary model can be somewhat problematic if taken literally as it implies explosive risk-neutral dynamics. Although in our case, this means that a shock to the overall level of interest rates does not die out when pricing bonds, which is what allows the model to match the highly persistent level of long-term yields observed in the data. This is consistent with the high persistence of long-term yields in the data, and is needed to generate an upward-sloping average yield curve. The remaining eigenvalues are all below unity, meaning the variation they capture will eventually die out. However, the next two largest at 0.970 and 0.964 are quite persistent, whereas the smallest one at 0.882 reverts noticeably faster.

The $\mathbb{Q}$ -measure latent block $\Phi_{ll}^{\mathbb{Q}}$ is estimated in the equal-diagonal Jordan form with both diagonal entries at 0.925. Since the diagonal is equal by construction, the off-diagonal elements carry the relevant information about the latent factor dynamics. With $\Phi_{ll,12}^{\mathbb{Q}} = 0.027$ and $\Phi_{ll,21}^{\mathbb{Q}} = 0.087$, the two latent factors are only weakly coupled under the risk-neutral measure, but the asymmetry, with $\Phi_{ll,21}^{\mathbb{Q}}$ being roughly three times larger than $\Phi_{ll,12}^{\mathbb{Q}}$, suggests that one factor responds more to the other than vice versa. The asymmetry in the off-diagonal means that the two latent factors do not influence each other equally. When the first latent factor moves, it has a strong influence on the second factor, but a movement in the second factor has a much weaker influence on the first. This means that hidden forces driving the yield curve are one-directional rather than symmetric.

The $\mathbb{Q}$ -measure macro block $\Phi_{mm}^{\mathbb{Q}}$ has diagonal estimates near unity of 0.976 for inflation and 1.005 for unemployment. These estimates suggest that investors price both inflation and unemployment as nearly permanent forces when valuing bonds. This means that a rise in inflation or unemployment today is expected to persist for a very long time. This reflects how markets price these risks rather than how the variables actually behave. These do however have very large standard errors, with the off-diagonals having very large standard errors rela-

tive to their point estimates. For example $\Phi_{mm,21}^{\mathbb{Q}} = 0.009$ with standard error of 0.599. This reflects the limited information the yield cross-section carries about the risk-neutral dynamics ($\mathbb{Q}$) of inflation and unemployment, as these only enter bond pricing indirectly through the yield equation intercepts. The large standard errors on $\Phi_{mm}^{\mathbb{Q}}$ makes these parameters difficult to interpret. It is unclear whether this reflects a fundamental identification problem, meaning that the the yield cross-section may simply carry too little information about the risk-neutral dynamics of inflation and unemployment, or whether it points to some degree of model misspecification. The imprecision in these parameters does not affect the main results of the paper, since the central findings concern the finite-sample bias in the physical dynamics $\Phi_{mm}^{\mathbb{P}}$, which are well-identified with small standard errors. However, it warrants some caution when interpreting quantities that depend directly on the macro $\mathbb{Q}$ -dynamics, such as term premium decompositions.

The $\mathbb{P}$ -measure macro AR parameters are better identified than the $\mathbb{Q}$ -dynamics. The diagonal of $\Phi_{mm}^{\mathbb{P}}$ is estimated at 0.881 for inflation and 0.897 for unemployment, both with small standard errors. This indicates high but stationary persistence in both series. Unlike the $\mathbb{Q}$ -measure persistence, these estimates capture how inflation and unemployment actually behave. Both are slow-moving and once they experience a shock they tend to revert back to their average very slowly. The off-diagonals are relatively small, suggesting that inflation and unemployment dynamics are largely driven by their own lags but is still affected by the other at a smaller magnitude.

The $\mathbb{P}$ -measure latent AR block shows persistence estimates of 0.870 for the first latent factor and 0.881 for the second. Both factors are persistent, reflecting the slow-moving nature of yield curve dynamics under the physical measure. The two factors are similarly persistent, with only a minor difference between them, suggesting that they both capture long-run variation in the yield curve, with shocks to either resulting in them taking a considerable amount of time to revert back to their long-run means.

The unconditional means of inflation and unemployment are small and economically insignificant, which is expected given that our macro factors are normalised to zero mean.

The diagonal elements of Σ_{mm} are 0.472 for inflation and 0.434 for unemployment. This indicates that both macro series are subject to smaller sizeable unpredictable shocks relative to the normalised latent shocks $\Sigma = \mathbf{I}$. Within the macro block, inflation is slightly more volatile than unemployment. Positive diagonal entries in Σ means that a meaningful part of the movement in inflation and unemployment each quarter comes from genuinely unexpected shocks rather than predictable trends in the data. The small off-diagonal of 0.018 suggests

that unexpected movement in inflation and unemployment are independent of each other in the model. This is consistent with off-diagonal elements of $\Phi_{mm}^{\mathbb{P}}$ which were also insignificant, suggesting a pattern of independence between inflation and unemployment.

The short-rate loading vector β is positive in all entries except the unemployment loading β_2 . The inflation loading is small and positive ($\beta_1 = 0.00075$), while the unemployment loading is small and negative ($\beta_2 = -0.00077$), meaning that higher unemployment is associated with a slightly lower short-rate, consistent with the typical monetary policy response to weak economic conditions of lowering interest rates to stimulate the economy. The second latent factor has the largest loading at $\beta_4 = 0.00226$, suggesting it is the factor most directly tied to short-rate variation. The small size of both macro loadings suggests that while inflation and unemployment do move the short-rate, their direct impact in a given quarter is modest. Much of the short-rate variation is driven by the latent factors rather than the macro variables directly.

The risk-neutral intercepts $\mu_1^{\mathbb{Q}} = -0.045$ and $\mu_2^{\mathbb{Q}} = -0.354$ with standard errors 0.084 and 0.075 respectively are both negative, with $\mu_2^{\mathbb{Q}}$ being the most significant intercept of the two. Since the corresponding $\mathbb{P}$ -measure means are small and close to zero, the $\mathbb{Q}$ -measure mean sits below the $\mathbb{P}$ -measure mean. This gap is what generates the positive average term premium in the model. The lower $\mathbb{Q}$ -measure mean relative to the $\mathbb{P}$ -measure mean implies that investors price interest rates in a more pessimistic way compared to how they actually turn out on average, effectively generating a positive term premium.

The measurement error standard deviation is around 59 basis points annualised, meaning that the model's predicted yields at the intermediate maturities (2, 5, 7, and 15 years) are on average about 0.59 percentage points away from the observed yields. Given that the model is using four factors to fit the entire U.S. yield curve across more than 50 years, this is a reasonable fit.

5.3.1 Yield Curve Decomposition

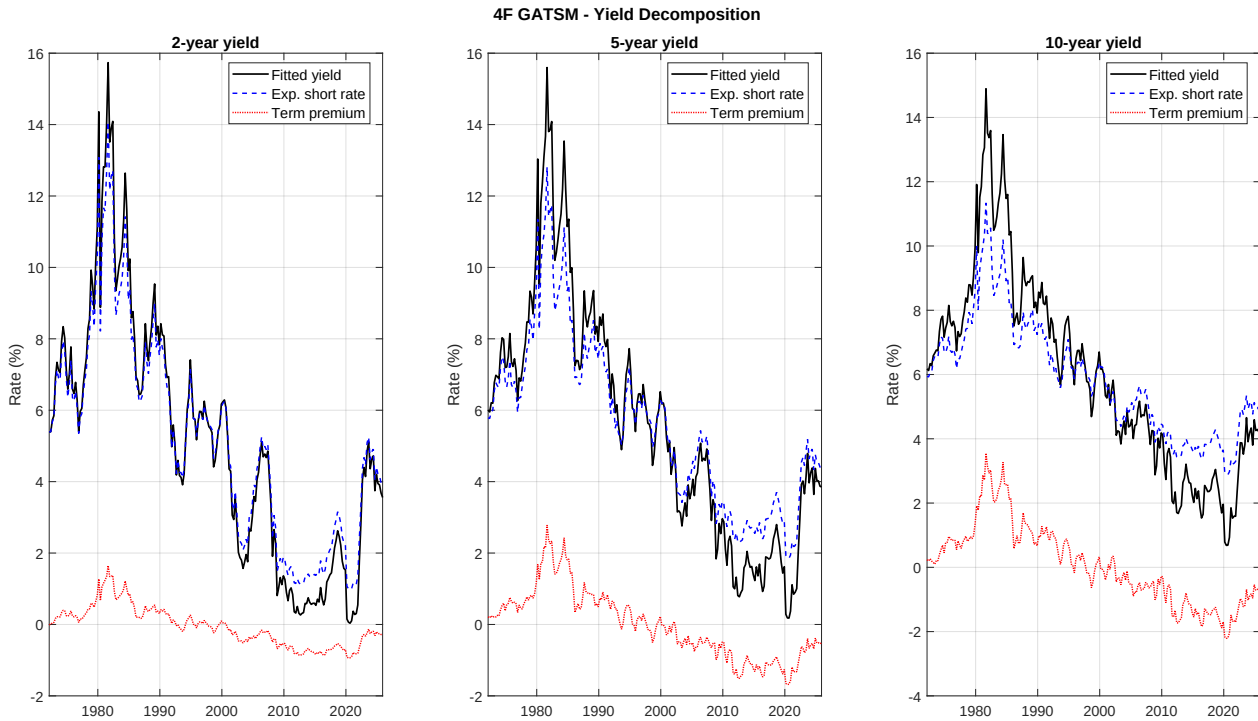


Figure 7: Yield Curve Decomposition for the Four-Factor GATSM

Figure 7 presents the yield curve decomposition for the four-factor GATSM for the 2-, 5-, and 10-year maturities over the full 1972:Q1 - 2025:Q4 sample. The decomposition is set up in the same way as the two-factor decomposition in Figure 6, with the addition of the 5-year maturity and the inclusion of inflation and unemployment as observable state variables.

Compared to the two-factor decomposition, the term premium in the four-factor decomposition is considerably smoother across the sample. At the 10-year maturity in particular, the two-factor model produced a volatile term premium that fluctuated substantially through most of the sample, whereas the four-factor keeps the term-structure on a cleaner path. The addition of the inflation and unemployment macroeconomic variables helps the model absorb some of the variation that the latent state variables were forced to push into the term premium in the two-factor model. This leaves a smoother term premium series, making patterns more visual. The pattern of the term premium is mostly flat with an early peak for the 2-year yields in both models, and the 10-year yields are both peaking around the 1980s, declining steadily with a rebound around 2020. The more substantial differences are seen in the level of the term premium. While the two-factor, 10-year term premium remained

volatile but broadly positive throughout most of the sample, the four-factor, 10-year crosses zero around the 2000s and continues into the negative term premiums, reaching a negative peak of around 3 percent. The 5-year follows a similar pattern, also turning negative around 2000. This means that the four-factor model implies that investors were willing to pay to hold long-term bonds rather than receive compensation for it, effectively hedging themselves against bad states of the economy, as described under the two-factor model.

In terms of the overall levels, the term premium in the four-factor model ranges approximately from -1% to 1.5% at the 2-year horizon, while the 10-year spans wider at roughly -2% to 3.5%. This is consistent with the observation from earlier that longer horizons are exposed to more uncertainty. This is virtually unchanged from the two-factor model, where a general pattern of a greater span in term premium levels is also seen for the longer-term bonds, which is a construct of how our model computes the term premium.

6 Survey Expectations

6.1 Motivation for Surveys

As noted earlier, a well-documented challenge in ATSMs is the finite-sample bias in the estimated physical dynamics $\Phi^{\mathbb{P}}$, see e.g. [Bauer et al. \(2012\)](#), or [Duffee \(2011\)](#) that show that $\mathbb{P}$ -dynamics estimated from yields alone have poor out-of-sample forecasting performance due to this persistent bias. As we know, one of the main features of ATSMs is its ability to model yields and their decomposition, so a skewed estimated term premia is undesirable.

The main motivation for our idea to incorporate surveys as a way to mitigate some of the finite sample bias is [Kim and Orphanides \(2012\)](#), where they in their paper showed that by incorporating survey data of interest rates into a latent three-factor GATSM specification one can reduce this finite sample bias. Their estimation approach differs from ours in the fact that they 'back-out' the states together with the parameter estimates using maximum likelihood with a Kalman filter, a common approach when all states are latent as we did in our two-factor model. We, on the other hand, use surveys on macro variables, namely inflation and unemployment, and use the MCSE approach where we first estimate all parameters and then back-out the latent states based on those estimates through a Kalman filter. This approach was easier to implement given our two extra observable macro factors. Due to this we will have to implement our surveys on our macro factors differently, but we still retain the concept that surveys act as noisy observations of the $\mathbb{P}$ -dynamics conditional expectation as in [Kim and Orphanides \(2012\)](#). Another important distinction is that we do not explicitly model a survey measurement error, where [Kim and Orphanides \(2012\)](#) directly model and estimate an survey measurement error covariance matrix. Although we will later directly add noise to a Monte Carlo study to assess how our estimator would react to noisy surveys, and how this would affect $\hat{\Phi}_{mm}^{\mathbb{P}}$ and the term premia as the noise increases.

The following section will describe our incorporation of surveys to the MCSE framework of [Hamilton and Wu \(2012\)](#) via an OLS-stacking approach based on [Theil and Goldberger \(1992\)](#).

6.2 Survey Incorporation

6.2.1 Surveys and Data

The surveys that we use are the one-quarter-ahead predictions of inflation and unemployment rate taken from the SPF. At each quarter the SPF publishes

$$\mathbf{s}_{t+1|t} = \mathbb{E}_t^{svy} [\mathbf{x}_{t+1}^m], \quad \mathbf{x}_{t+1}^m = \begin{pmatrix} x_{t+1}^{m_1} & x_{t+1}^{m_2} \end{pmatrix}^\top, \quad (72)$$

where the forecast of inflation, $x_{t+1}^{m_1}$ is the SPF `dpgdp3` series (one-quarter-ahead GDP deflator growth forecast made at times t for $t+1$), and the forecast of unemployment, $x_{t+1}^{m_2}$, which is the SPF `UNEMP3` series (one-quarter-ahead unemployment rate forecast made at t for $t+1$). In our four-factor model the one-quarter-ahead conditional expectation of the macro state is

$$\mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+1}^m] = \boldsymbol{\mu}_m^{\mathbb{P}} + \boldsymbol{\Phi}_{mm}^{\mathbb{P}} \mathbf{x}_t^m, \quad (73)$$

which is the quantity estimated by the Block m OLS regression. We treat survey forecasts as noisy but unbiased observations of the conditional expectation in (73), which motivates our inclusion of surveys directly as additional observations in the Block m regression. What we imply is that the SPF form their expectations under the real-world measure $\mathbb{P}$ making survey forecasts direct but imperfect proxies for the conditional expectations of $\boldsymbol{\Phi}_{mm}^{\mathbb{P}}$, which is also the argument used in [Kim and Orphanides \(2012\)](#).

To keep the surveys consistent with the same space as the normalised macro factors used throughout the model, each survey is normalised by its own sample mean and standard deviation, rather than by the statistics given by the actual macro factors

$$\tilde{s}_t^{inf} = \frac{s_t^{inf} - \bar{s}^{inf}}{\hat{\sigma}_s^{inf}}, \quad \tilde{s}_t^{unemp} = \frac{s_t^{unemp} - \bar{s}^{unemp}}{\hat{\sigma}_s^{unemp}}, \quad (74)$$

where s_t^{inf} and s_t^{unemp} denotes the unaltered SPF survey forecasts for inflation and unemployment, respectively. The 'bars' denote the sample mean and $\hat{\sigma}_s$ is the standard deviation of the samples. The stacked vector of normalised surveys is then

$$\mathbf{s}_t^{norm} = \begin{bmatrix} \tilde{s}_t^{inf} \\ \tilde{s}_t^{unemp} \end{bmatrix}, \quad (75)$$

which is in the same units as the used macro factors. The reason we normalise the surveys by their own moments rather than using the moments from the macro factors is that we avoid

imposing that forecasters predict with the same unconditional distribution as the realised macro factors. The small empirical difference between the two distributions is then absorbed into the OLS residuals.

6.2.2 OLS Stacking

The survey forecasts are incorporated as stochastic prior information on the physical transition matrix $\Phi_{mm}^{\mathbb{P}}$, following the mixed estimation framework of [Theil and Goldberger \(1992\)](#). Specifically, the survey observations are stacked as additional weighted rows in the Block m OLS regression, which is the standard implementation of the Theil–Goldberger mixed estimator. The standard Block m regression in (44) uses only the actual macro realisations

$$\mathbf{x}_{t+1}^m = \mathbf{A}^{m,\#} + \Phi_{mm}^{\mathbb{P},\#} \mathbf{x}_t^m + \mathbf{u}_{t+1}^{m,\#}. \quad (76)$$

Under our survey augmentation we implement that the normalised one-quarter-ahead survey forecast $\mathbf{s}_{t+1|t}^{norm}$ is a noisy observation of the same conditional expectation and is given by

$$\mathbf{s}_{t+1|t}^{norm} = \mathbf{A}^{m,\#} + \Phi_{mm}^{\mathbb{P},\#} \mathbf{x}_t^m + \boldsymbol{\varepsilon}_t^{svy}, \quad (77)$$

where $\boldsymbol{\varepsilon}_t^{svy}$ is the survey forecast error assumed to be mean zero. The regressors of (76) and (77) are identical, where both are conditioned on the states $\mathbf{x}_t^m$ at times t . Intuitively this makes sense as the best information that survey forecasts can use in trying to predict the states in the next period $t+1$ is the current states at times t . We again let $T_1 = T - 1$ since we lose one observation because we have $\mathbf{x}_{t+1}^m$ on the left-hand side while $\mathbf{x}_t^m$ is on the right-hand side of (76). $\mathbf{x}_1^m$ can only be used as a regressor since it has no lags itself.

The two sets of observations are combined into a single weighted least-squares system by stacking the actual macro factors with the survey rows. Let $\mathbf{Y}_{svy}^m$ denote the stacked survey observations and let $\mathbf{Y}_{act}^m$ be the vector of stacked actual macro observations. They are then given by

$$\mathbf{Y}_{svy}^m = \begin{bmatrix} \mathbf{s}_{2|1}^{norm} \\ \vdots \\ \mathbf{s}_{T|T_1}^{norm} \end{bmatrix}, \quad \mathbf{Y}_{act}^m = \begin{bmatrix} \mathbf{x}_2^m \\ \vdots \\ \mathbf{x}_T^m \end{bmatrix}.$$

Both of these are denoted by $\mathbf{Y}$ since they act as the left-hand side regressand, where in OLS we usually denote the dependent variable by y . Both $\mathbf{Y}_{svy}^m$ and $\mathbf{Y}_{act}^m$ share the same

regressors which we stack into a regressor matrix given by

$$\mathbf{X}_{svy}^m = \mathbf{X}_{act}^m = \begin{bmatrix} \mathbf{1} & \mathbf{x}_t^m \\ \vdots & \vdots \\ \mathbf{1} & \mathbf{x}_{T_1}^m \end{bmatrix}.$$

Given this we can write up the full weighted least-squares system as

$$\underbrace{\begin{bmatrix} \mathbf{Y}_{act}^m \\ w_{svy} \mathbf{Y}_{svy}^m \end{bmatrix}}_{\mathbf{Y}_{aug}^m} = \underbrace{\begin{bmatrix} \mathbf{X}_{act}^m \\ w_{svy} \mathbf{X}_{svy}^m \end{bmatrix}}_{\mathbf{X}_{aug}^m} \begin{bmatrix} \mathbf{A}^{m,\#} \\ \Phi_{mm}^{\mathbb{P},\#} \end{bmatrix} + \text{residuals}, \quad (78)$$

with the relative weight $w_{svy} \geq 0$ indicating how much weight we put on the surveys relative to the actual observations. The vector $\mathbf{Y}_{aug}^m$ stacks row-on-row wise, in the sense that the first row of $\mathbf{Y}_{svy}^m$ follows directly after the last row of $\mathbf{Y}_{act}^m$. The same is apparent for the regressor matrix $\mathbf{X}_{aug}^m$ which consists of two columns. The augmented OLS estimator in matrix form is then given by

$$\hat{\Pi}_{aug}^m = ((\mathbf{X}_{aug}^m)^\top \mathbf{X}_{aug}^m)^{-1} (\mathbf{X}_{aug}^m)^\top \mathbf{Y}_{aug}^m \in \mathbb{R}^{(1+n_m) \times n_m}. \quad (79)$$

Given that the system is written in the row-stacked layout $\mathbf{Y} = \mathbf{X}\Pi$, where each row of $\mathbf{Y}_{aug}^m$ is a transposed observation $\mathbf{x}_{t+1}^\top$, the blocks of $\hat{\Pi}_{aug}^m$ are the transposes of the structural coefficients. Our intercept and the macro transition matrix are therefore recovered by transposing the corresponding blocks

$$\hat{\mathbf{A}}^{m,\#} = \left(\hat{\Pi}_{aug}^m [1, :] \right)^\top, \quad \hat{\Phi}_{mm}^{\mathbb{P},\#} = \left(\hat{\Pi}_{aug}^m [2 : \text{end}, :] \right)^\top = \begin{bmatrix} \hat{\Phi}_{m_1 m_1}^{\mathbb{P}} & \hat{\Phi}_{m_1 m_2}^{\mathbb{P}} \\ \hat{\Phi}_{m_2 m_1}^{\mathbb{P}} & \hat{\Phi}_{m_2 m_2}^{\mathbb{P}} \end{bmatrix},$$

where $\hat{\Pi}_{aug}^m [2 : \text{end}, :]$ denotes the lower $n_m \times n_m$ block. The OLS estimator minimises exactly the combined weighted objective given by

$$\sum_{t=1}^{T_1} \|\mathbf{x}_{t+1}^m - \mathbf{A}^{m,\#} - \Phi_{mm}^{\mathbb{P},\#} \mathbf{x}_t^m\|^2 + w_{svy}^2 \sum_{t=1}^{T_1} \|\mathbf{s}_{t+1|t}^{norm} - \mathbf{A}^{m,\#} - \Phi_{mm}^{\mathbb{P},\#} \mathbf{x}_t^m\|^2. \quad (80)$$

The survey rows contribute w_{svy}^2 times the weight of each actual macro observation. In the case of [Kim and Orphanides \(2012\)](#) their analogous weighting is determined endogenously by the ratio of the structural innovation variance to the survey measurement variance, both

of which are estimated jointly. Our scalar w_{svy} replaces this ratio with a calibrated constant which is a simpler alternative and is easier implemented in our MCSE framework. If we let $w_{svy} = 1$ this corresponds to the symmetric special case of the [Theil and Goldberger \(1992\)](#) mixed estimator in which the prior and sample information are assigned equal weight. If we let $w_{svy} = 0$ we recover the standard survey-free estimator in (45).

6.2.3 Downstream Estimation Effects

As the Block m is the first step in the MCSE pipeline, we will in this section describe what we expect the survey augmentation will affect in the subsequent MCSE estimation steps, i.e. how we expect the estimates to change. The estimation steps themselves, however, are unchanged. Firstly, the residual covariance matrix $\hat{\Omega}^{m,\#}$ is used further downstream to recover the lower-triangular Σ_{mm} matrix via Cholesky decomposition and is used from the residuals of the actual macro factors only

$$\hat{\Omega}^{m,\#} = T_1^{-1} \left(\hat{\mathbf{U}}_{act}^m \right)^\top \hat{\mathbf{U}}_{act}^m, \quad (81)$$

where actual residuals are computed as

$$\hat{\mathbf{U}}_{act}^m = \mathbf{Y}_{act}^m - \mathbf{X}_{act}^m \hat{\boldsymbol{\Pi}}_{aug}^m. \quad (82)$$

Using the survey-augmented residuals too would blend the structural innovation covariance with the survey forecast error variance and could lead to a skewed estimate of Σ_{mm} , since Σ_{mm} captures only the structural macro shocks to the $\mathbb{P}$ -dynamics macro block VAR(1). As a recap Σ_{mm} governs the covariance of the structural innovation $\mathbf{u}_t^m$ in the macro block

$$\mathbf{x}_{t+1}^m = \boldsymbol{\mu}_m^{\mathbb{P}} + \boldsymbol{\Phi}_{mm}^{\mathbb{P}} \mathbf{x}_t^m + \Sigma_{mm} \mathbf{u}_{t+1}^m,$$

where $\mathbf{u}_{t+1}^m$ is the shock to the actual macro realisation at $t + 1$. The survey forecast error $\boldsymbol{\epsilon}_t^{svy} = \mathbf{s}_{t+1|t}^{norm} - \boldsymbol{\mu}_m^{\mathbb{P}} - \boldsymbol{\Phi}_{mm}^{\mathbb{P}} \mathbf{x}_t^m$ is a different random variable entirely as it reflects the SPF forecast error rather than a macro shock. This approach is also consistent with [Kim and Orphanides \(2012\)](#) where the structural innovation covariance and survey measurement error are treated as two distinct things. Although, in our approach, the survey measurement error is not explicitly modelled. In this paper we will only focus on the case of $w_{svy} = 1$, which assigns each survey row the same least-squares weight as an actual macro row. Because the same scalar is applied to both macro equations under unweighted OLS, $w_{svy} = 1$ imposes

a common survey-to-innovation precision ratio of one for both inflation and unemployment. The Generalised Least Squares efficient choice would instead use variable-specific weights $w_m = \Sigma_{mm}/\hat{\sigma}_m^{svy}$, which would give the unemployment survey a weight close to 3.7 rather than 1. We retain the single scalar $w_{svy} = 1$ as a deliberate simplification.

Going through the steps chronologically, the survey augmentation has no effect on the either Block 1 or Block 2 in Step 1 only the Block m given the independence assumptions, hence the augmentation will not affect the estimates of Φ^Q nor β as they are computed using Block 1 and Block 2 from Step 1. We do expect a small change in the Σ_{mm} estimate since the estimate $\Phi_{mm}^{p,\#}$ will now differ which affects the fitted values $\mathbf{X}_{act}^m \hat{\Pi}_{aug}^m$ and then further affect the actual residuals $\hat{\mathbf{U}}_{act}^m$, which finally enters $\hat{\Omega}^{m,\#}$. We expect a change in α and in μ_m^Q , where the targets $\hat{\mathbf{A}}^{1,\#}$ and $\hat{\mathbf{A}}^{2,\#}$ are unaffected so the only change that could happen for α and μ_m^Q is through the Σ_{mm} channel. $\hat{\mu}_m^p$, on the other hand, is directly taken from the Block m OLS intercept vector $\hat{\mathbf{A}}^{m,\#}$ so we expect a change.

Given this we likely will not see a large difference in estimates other than for the macro block.

6.3 Survey Results

Table 13: Estimated Parameters — Survey-Augmented Four-Factor MF-GATSM

Parameter	Estimate	Parameter	Estimate
Q-dynamics: Eigenvalues		Volatility: Macro Cholesky (Σ_{mm})	
λ_1^Q	1.016943	Σ_{11}	0.482974 (0.023287)
λ_2^Q	0.994510	Σ_{21}	0.017638 (0.037468)
λ_3^Q	0.963959	Σ_{22}	0.434479 (0.077188)
λ_4^Q	0.893979		
Short-rate Intercept		Q-dynamics: Macro AR (Φ_{mm}^Q)	
α	0.012598 (0.001384)	$\Phi_{mm,11}^Q$	0.976306 (0.005810)
		$\Phi_{mm,21}^Q$	0.008719 (0.010455)
		$\Phi_{mm,12}^Q$	0.057549 (0.006968)
		$\Phi_{mm,22}^Q$	1.004596 (0.015741)
P-dynamics: Mean (μ^P)		Q-dynamics: Latent AR (Φ_{ll}^Q)	
μ_1^P (infl)	-0.001801 (0.018714)	$\Phi_{ll,11}^Q (= \Phi_{ll,22}^Q)$	0.938724 (0.015106)
μ_2^P (unemp)	-0.001392 (0.069285)	$\Phi_{ll,12}^Q$	-0.071349 (0.010203)
μ_3^P (lat1)	-0.036919 (0.168162)	$\Phi_{ll,21}^Q$	-0.034985 (0.018021)
μ_4^P (lat2)	-0.006529 (0.141707)		
P-dynamics: Macro AR (Φ_{mm}^P)		Short-rate Loadings (β)	
$\Phi_{mm,11}^P$	0.862767 (0.034055)	β_1 (infl)	0.000746 (0.000010)
$\Phi_{mm,21}^P$	0.020390 (0.027424)	β_2 (unemp)	-0.000771 (0.000059)
$\Phi_{mm,12}^P$	0.074216 (0.054069)	β_3 (lat1)	0.002315 (0.000093)
$\Phi_{mm,22}^P$	0.894481 (0.036363)	β_4 (lat2)	0.000382 (0.000113)
P-dynamics: Latent AR (Φ_{ll}^P)		Risk-neutral Intercept (μ^Q)	
$\Phi_{ll,11}^P$	0.869830 (0.012814)	μ_1^Q	0.310906 (0.023437)
$\Phi_{ll,21}^P$	0.080681 (0.017767)	μ_2^Q	-0.050020 (0.046914)
$\Phi_{ll,12}^P$	0.100098 (0.010730)		
$\Phi_{ll,22}^P$	0.880871 (0.015968)		
		Measurement Error	
		σ_v (bps)	59.1130 (5.1576)
		$\bar{L}$	27.395694

Notes: Same as in Table (12).

Table 13 reports the estimated parameters from the survey-augmented four-factor GATSM, which incorporates one-quarter-ahead SPF forecasts of inflation and unemployment into the macro AR block of the estimation with equal weight alongside actual observations.

For the $\mathbb{Q}$ -dynamics, the eigenvalue structure is broadly similar to the baseline model, with the dominant eigenvalue at $\lambda_1^{\mathbb{Q}} = 1.017$, again marginally above unity. The remaining three are 0.995, 0.964 and 0.894, which suggest the same progressively faster mean-reversion as the baseline model. The near-identical eigenvalues suggest that incorporating survey data does not meaningfully change how investors price the persistence of interest rate shocks when valuing bonds.

The $\mathbb{Q}$ -measure latent block $\Phi_{ll}^{\mathbb{Q}}$ again uses the equal-diagonal Jordan form, with both diagonal entries now at 0.939, which is slightly higher than the baseline estimate of 0.925. These diagonals are still equal by construction, so the off-diagonals carry the relevant information. Unlike the $\mathbb{P}$ -latent block, the $\mathbb{Q}$ -latent block is solved numerically and depends indirectly on the macro dynamics. This means that the survey augmentation can affect the $\mathbb{Q}$ -latent estimates even though it only directly enters through the macro AR block. The off-diagonals are $\Phi_{12}^{\mathbb{Q}} = -0.071$ and $\Phi_{21}^{\mathbb{Q}} = -0.035$. These are both negative, which is a notable difference from the baseline, where both were positive. This suggests a somewhat different dynamic relationship between the two latent factors once survey information is incorporated into the estimation. In contrast to the baseline, where the two latent factors had a one-directional influence on each other, the now negative off-diagonals suggest that the two hidden drivers of the yield curve somewhat offset each other.

The $\mathbb{Q}$ -measure macro block $\Phi_{mm}^{\mathbb{Q}}$ again has diagonal estimates near unity, 0.976 for inflation and 1.005 for unemployment, suggesting that investors still price both macro variables as highly persistent when valuing bonds. The standard errors are considerably smaller than in the baseline model, dropping from 0.657 and 0.646 to 0.006 and 0.016 respectively. These are a considerable improvement in precision of the macro block and suggests that incorporating survey data helps identify the macro $\mathbb{Q}$ -dynamics better. This makes the macro $\mathbb{Q}$ -dynamics more reliable when discussing term premium decompositions. The off-diagonal elements remain small and insignificant.

$\mathbb{P}$ -measure dynamics show that the macro AR diagonal is estimated at 0.863 for inflation and 0.894 for unemployment with standard deviations 0.034 and 0.036 respectively. Both estimates are slightly lower than the baseline estimates of 0.881 and 0.897. This suggests that once survey data is incorporated, inflation and unemployment are reverting slightly faster to their averages in reality, although the difference is small and both series remain highly per-

sistent overall. The off-diagonal remains small, suggesting that inflation and unemployment in the model are partially independent of each other once their own lags are accounted for. Importantly, since the $\mathbb{Q}$ -measure macro block $\Phi_{mm}^{\mathbb{Q}}$ is unchanged from the no-survey (NS) four-factor while the $\Phi_{mm}^{\mathbb{P}}$ declines slightly, the gap between the two widens. A larger gap between $\mathbb{P}$ and $\mathbb{Q}$ indicates an upward pressure on the implied term premium.

The latent AR ($\Phi_{ll}^{\mathbb{P}}$) shows persistence of 0.870 for the first latent factor and 0.881 for the second latent factor. These are both identical to the no-survey model, which is expected since the $\mathbb{P}$ -block is calculated analytically from the yield data alone and therefore unaffected by the survey augmentation. The standard errors change slightly because these depend on the full model. While the coefficients remain the same, the change in standard errors suggests that both latent factors become more significant when including survey data, since the standard deviations drop from 0.04 and 0.05 respectively to around 0.01 for both factors. This means that while the estimated persistence of the two latent factors is unchanged, they are more precisely estimated in the survey-augmented model, suggesting that incorporating survey data indirectly strengthens the inference about latent factor dynamics.

The unconditional means of the macro factors in the $\mathbb{P}$ -dynamics are small and insignificant, which is again expected since they are normalised to zero mean.

The diagonal elements of Σ_{mm} are 0.483 for inflation and 0.434 for unemployment. The off-diagonal is 0.018. Both unemployment and off-diagonal are identical to the baseline model, while inflation is slightly higher. This suggests that incorporating survey data does not meaningfully change the size or independence of the unpredictable shocks to inflation and unemployment as we expected.

The short-rate loadings are similar to the no-survey model. The inflation and unemployment loadings are identical to the baseline model at $\beta_1 = 0.00075$ and $\beta_2 = -0.00077$. This is expected since the survey augmentation only affects the macro AR block and has no direct effect on the short-rate loadings. The latent factor loadings do however differ slightly. $\beta_3 = 0.00232$ and $\beta_4 = 0.00038$ compared to 0.00011 and 0.00226 in the NS model. This shift in the latent loadings means that the first factor plays a somewhat larger role in driving short-rate variation when incorporating survey data. The second latent factor plays a smaller role compared to the NS model. Another likely culprit leading to the difference is that the loadings for the latent factors have been rotated relative to the NS model.

The risk-neutral intercepts are $\mu_1^{\mathbb{Q}} = 0.311$ and $\mu_2^{\mathbb{Q}} = -0.05$ with standard errors of 0.023 and 0.047 respectively. Compared to the NS model, $\mu_1^{\mathbb{Q}}$ is now positive instead of negative and is significant, while $\mu_2^{\mathbb{Q}}$ is considerably closer to zero and is no longer significant. This

means that the $\mathbb{Q}$ -measure mean moves closer to the $\mathbb{P}$ -measure mean, narrowing the intercept gap and constituting a downward pressure on the term premium compared to the no-survey model. However, as noted above, the widening of the $\Phi^{\mathbb{P}} - \Phi^{\mathbb{Q}}$ gap works in the opposite direction, pushing the term premium upward. The effect from the persistence parameter and the effect from the risk-neutral intercepts pull the term premium in opposite directions, and from the parameter estimates alone it is therefore not possible to determine which one dominates. The net effect is ambiguous at this stage, and will be further examined in the following section, where the yield curve decomposition shows directly how the term premium changes when survey data is incorporated.

The measurement error standard deviation is again roughly 59 basis points annualised, which is identical to the NS model. This is again expected since the yield blocks are unchanged between the two models.

6.3.1 Yield Decomposition Surveys

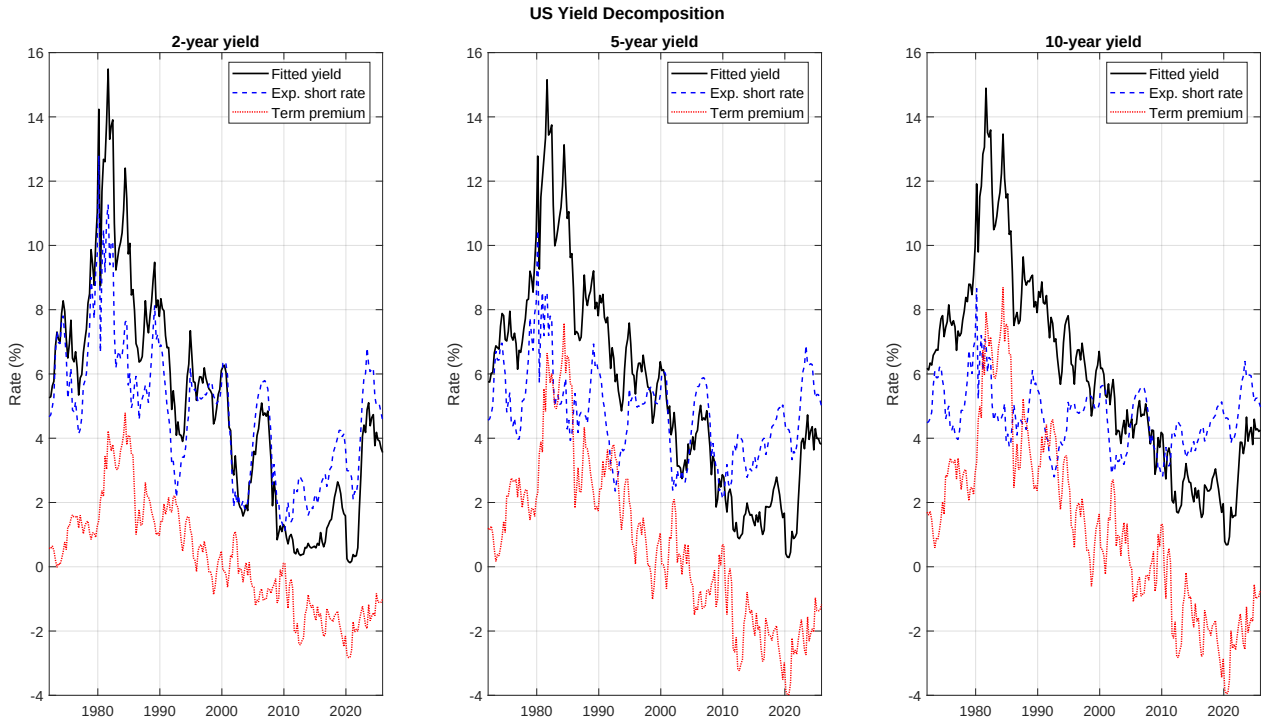


Figure 8: Yield Decomposition - One-Quarter-Ahead Surveys

Figure 8 presents the yield curve decomposition from the survey-augmented four-factor GATSM. Incorporating one-quarter-ahead survey forecasts produces a decomposition that

differs substantially from the no-survey model in figure 7, with the term premium being considerably more elevated across all maturities and throughout most of the sample. This confirms that the widening of the $\Phi^{\mathbb{P}} - \Phi^{\mathbb{Q}}$ gap dominates the narrowing of the risk-neutral intercept gap when survey data is incorporated, resolving the ambiguity noted earlier.

The difference is most striking at the 5- and 10-year maturities. In the no-survey model, both 5- and 10-year term premia crosses zero around year 2000 and remained negative for most of the remaining sample. In the survey-augmented model however, the term premia is considerably more volatile throughout the sample. There, the term premia also crosses zero just before year 2000, but then spike to just over 2% for the 5- and 10-year before dropping again.

All three maturities follow the same overall pattern of an early peak in the 1980s, following by a downward trend until year 2020, where a small rebound is seen. The range of the survey data is considerably larger than the one seen in the no-survey model - the 10-year term premium peaks at just over 8% in the 1980s, while the 5-year peaks at just under 8% at the same time. The 2-year also varies noticeable more than the no-survey model. In the survey-augmented model we see a range from around -3% to just over 4%, whereas the no-survey model only showed term premia ranging from -1% to 1.5%. The overall pattern across all three maturities is showing a larger, but more volatile term premia, with higher peaks but also lower lows.

The mechanism behind this shift is likely that survey forecasts provide additional information about the expected short-rate component that is not contained in the cross-section alone. Rather than inferring expectations solely from the yield cross-section and realised macroeconomic data, the model can incorporate actual investor beliefs, which pulls the expected short-rate down by approximately 2 percent relative to the no-survey model during the earlier part of the sample. Since the fitted yields are almost unchanged, the term premium rises because the gap between fitted yields and expected short-rate is increased. The increased gap is mostly seen in the early part of the sample. In the end of the sample, when the term premium turns negative, the expected short-rate of the survey-augmented and the NS model are more similar, resulting in an increasingly similar term premium between the two models.

In summary, the parameter estimates and yield decomposition show that the primary effect of survey augmentation is regarding the expected short-rate path by lowering it, resulting in an elevation in term premium.

6.3.2 Four-Factor Campbell-Shiller

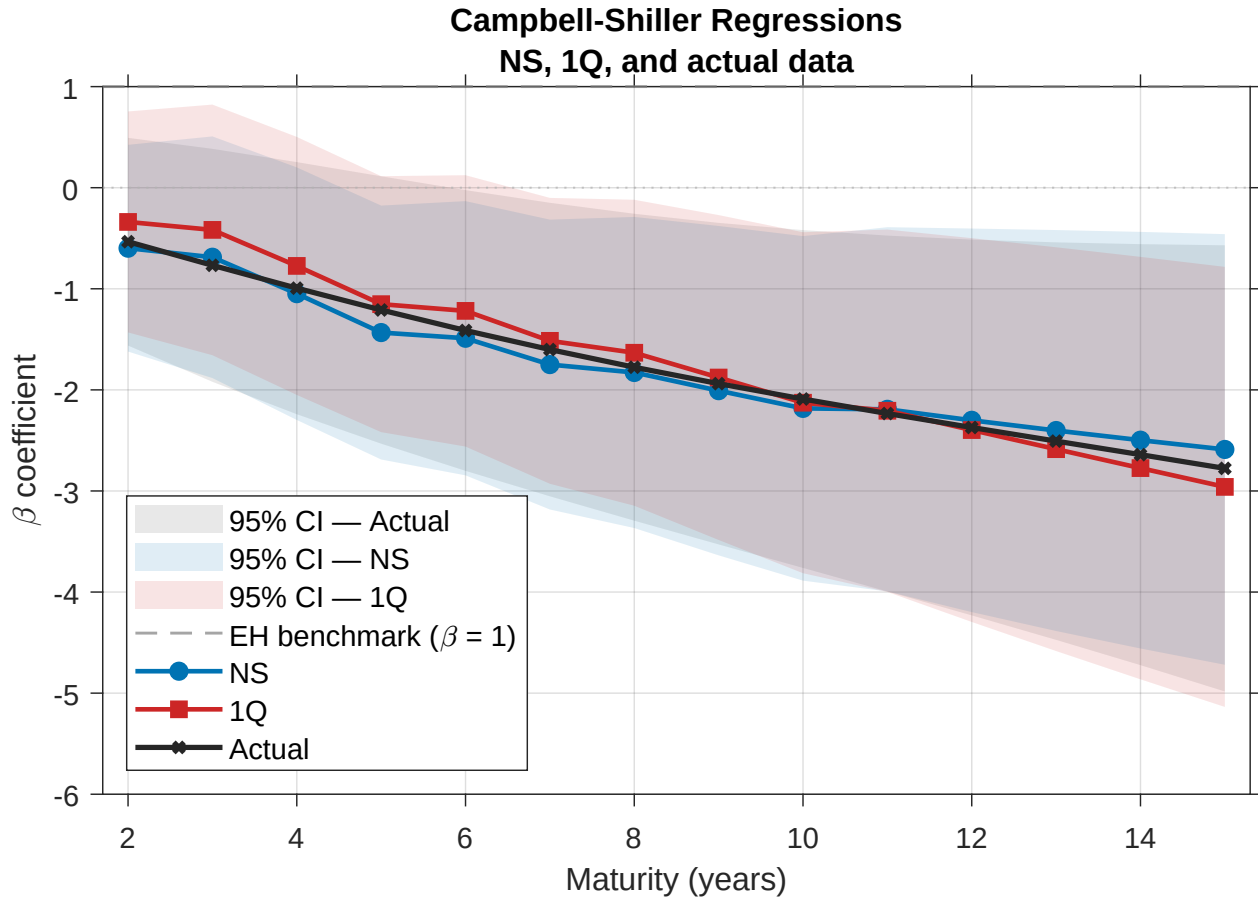


Figure 9: CS Regression Comparison for the Four-Factor Model

Similarly to the two-factor baseline model we can also try to validate the implied term premium of our four-factor models by running Campbell-Shiller regressions using model-implied yields against actual data. Figure 9 plots the CS regressions for the NS and 1Q models up against actual data with the 1-year maturity used as the short-rate.

Both estimators closely follow the same pattern as the actual data which is a negative decline in maturities. The confidence bands are [Newey and West \(1986\)](#) bands on the regression coefficient itself, not a simulation-based band which would be stronger evidence, nonetheless we still see that all three confidence bands exclude the notion of a slope coefficient at 1, once again rejecting the expectation hypothesis.

7 Monte Carlo Finite Sample Test

7.1 Survey-Augmented Monte Carlo Study

To assess the finite-sample properties of the survey-augmented four-factor GATSM and to study how various choices affect our estimators' ability to combat the small-sample bias in its estimate we conduct an extensive Monte Carlo study. The crucial point in the experiment is the fact that the true generating process (DGP) is completely fixed. The three estimators that we consider is the estimator from the four-factor GATSM without surveys (Estimator NS), the four-factor GATSM augmented with one-quarter-ahead survey forecasts (Estimator 1Q), and the four-factor GATSM augmented with four-quarter-ahead survey forecasts (Estimator 4Q).

The key questions of the Monte Carlo study are first: does survey augmentation reduce the persistence bias in $\hat{\Phi}_{mm}^{\mathbb{P}}$ relative to estimator NS, and second: does the one-quarter-ahead 1Q estimator perform better than the four-quarter-ahead 4Q estimator?

7.1.1 Data-Generating Process

The fixed DGP is the estimates from the four-factor GATSM using one-quarter-ahead SPF survey forecasts on inflation and unemployment with the estimates given in Table 12. The reason for specifically using the one-quarter-ahead four-factor GATSM for the DGP is because $\Phi_{mm}^{\mathbb{P}}$ is now already informed by the survey data and is more persistent than the survey-free four-factor version, so the change in finite-sample bias studied here will hopefully be more pronounced. This choice is in parallel to what is conducted in Kim and Orphanides (2012) for their Monte Carlo study, where they use their survey-augmented estimate as the fixed DGP.

The full parameter vector from the 1Q estimator θ^{1Q} will from now on be denoted with the superscript 'true', $\theta^{\text{true}} = \{\Phi^{\mathbb{P},\text{true}}, \Phi^{\mathbb{Q},\text{true}}, \beta^{\text{true}}, \alpha^{\text{true}}, \mu^{\mathbb{P},\text{true}}, \Sigma^{\text{true}}, \sigma_v^{\text{true}}, \mathbf{g}_0^{\text{true}}, \mathbf{G}_x^{\text{true}}\}$.

7.1.2 Survey Measurement and Calibration

Given our MCSE approach for estimation our surveys are not directly observed the estimator itself, and we will generate synthetic samples from an independent measurement model that aims to capture the relationship between the professional forecasts and the true macro states. For the macro variables $m \in \{m_1, m_2\}$ (inflation and unemployment) and the forecast horizons $h \in \{1, 4\}$ measured in quarters, the synthetic survey forecast at time t is generated

via the measurement equation

$$s_{m,t+h|t}^{norm} = \mu_{m,h}^{svy} + \phi_{m,h}^{svy} x_{m,t}^{norm} + \eta_{m,t}, \quad (83)$$

where $x_{m,t}^{norm}$ is the normalised macro state at time t , $s_{m,t+h|t}^{norm}$ is the normalised survey forecast of macro factor m for period $t + h$ made in period t , and $\eta_{m,t}$ is the survey noise and is independent across h and t . Let σ_m^{svy} denote the standard deviation of the OLS residuals. We calibrate the parameters $\{\mu_m^{svy}, \phi_m^{svy}, \sigma_m^{svy}\}$ separately for each macro variable $m \in \{m_1, m_2\}$ and each horizon $h \in \{1, 4\}$ by regressing the normalised historical SPF forecasts $s_{m,t+h|t}^{norm}$ on the normalised macro factor over the full sample from Q1 1972 to Q4 2025. In Table 14 is displayed the regression results for all horizons and all macro factors. Some features that are worth noting are that unemployment is very precisely forecasted, in the sense that the forecasters almost predict next periods unemployment to be identical to the current unemployment, at the one-quarter-horizon ($h = 1$) with an R^2 of 0.987 and relatively low residuals σ_m^{svy} of 0.1161, and if we compare it with the unemployment forecast four-quarters-ahead we see that the longer forecasting horizon results in a worsening precision with an R^2 of 0.914 and much larger relative residual standard deviation σ_m^{svy} . This is within our

Table 14: SPF Survey Calibration

Horizon	Variable	$\mu_{m,h}^{svy}$	$\phi_{m,h}^{svy}$	$\sigma_{m,h}^{svy}$	R^2
$h = 1$ (dpgdp3)	Inflation	0.0003	0.8812	0.4749	0.777
$h = 1$ (UNEMP3)	Unemployment	-0.0035	0.9955	0.1161	0.987
$h = 4$ (dpgdp6)	Inflation	-0.0009	0.7879	0.6184	0.621
$h = 4$ (UNEMP6)	Unemployment	-0.0020	0.9578	0.2938	0.914

expectation given that the four-quarter-ahead forecasts are simply a noisier version of the current macro state than the one-quarter-ahead surveys. Inflation on the other hand is substantially noisier than unemployment at the short horizon ($R^2 = 0.777$, $\sigma_m^{svy} = 0.475$) and at the four-quarter-ahead horizon ($R^2 = 0.621$, σ_m^{svy}), but still captures a significant amount of information about the future of the states. Inflation also sees a drop in precision from the short horizon to the long horizon. The estimated intercepts in the regression are all very close to zero and are economically negligible, and is to be expected given that both SPF forecasts and the macro variables are normalised by their own mean, because if the surveys are unbiased forecasts, in the sense of the [Mincer and Zarnowitz \(1969\)](#), intercepts should be statistically indistinguishable from zero given that the long-run mean of the normalised forecasts are equal to the long-run mean of the normalised factors.

To get a visual impression of the calibrations we have plotted a scatterplot alongside the OLS fitted line for inflation and unemployment in Figures 10 and 11, respectively. The most

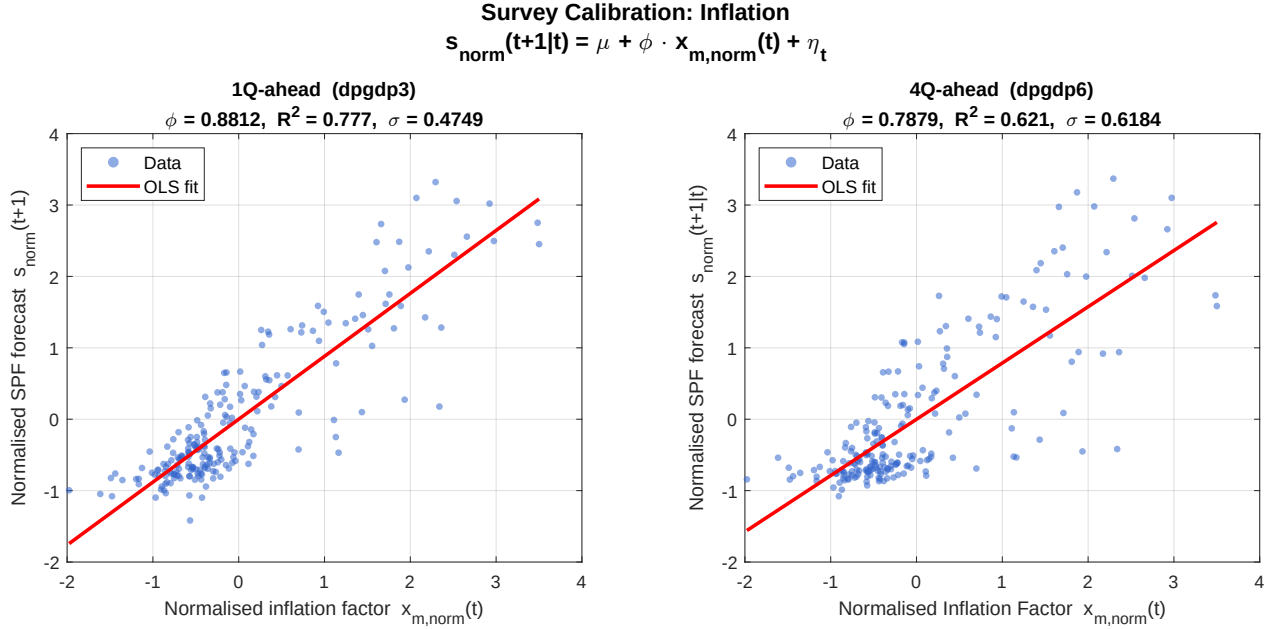
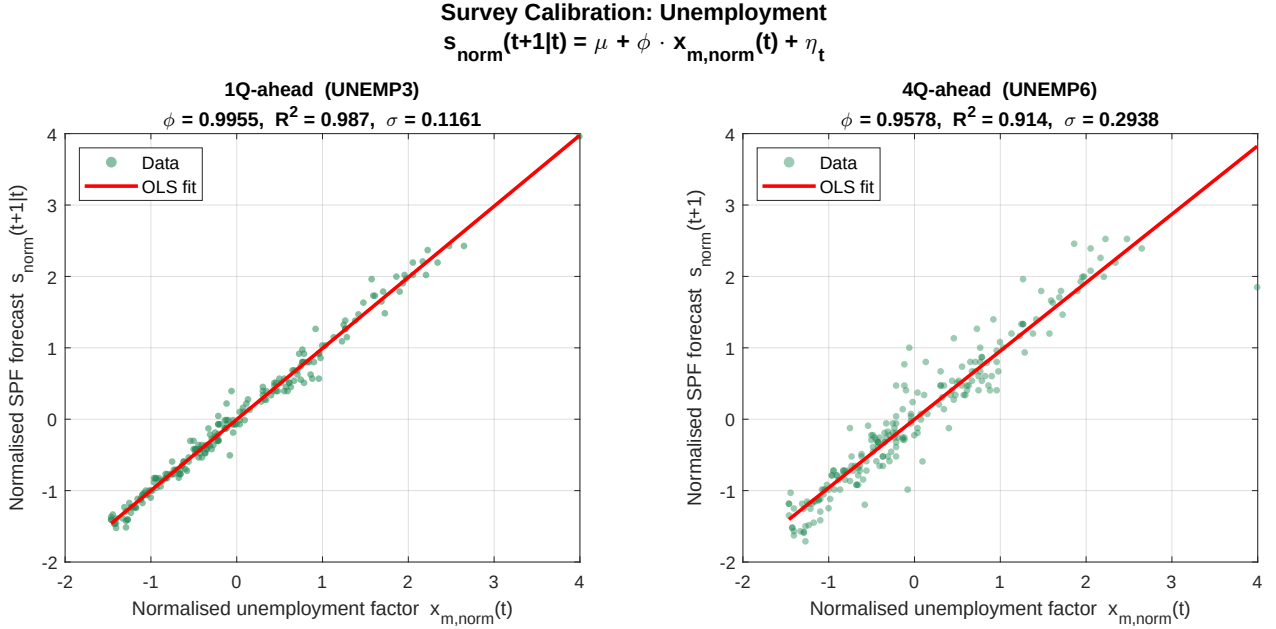


Figure 10: SPF Inflation Calibration

influential parameter for the survey augmentation analysis is the slope coefficient ϕ_m^{svy} , and with our regression setup it tracks how strongly the SPF forecasts co-moves with the macro state, or in other words, how much weight the SPF forecasters put on the current macro state for their forecast of that state in the future. For the short horizon inflation forecast the estimate is $\hat{\phi}_{m1,1Q} = 0.8812$ and falls to $\hat{\phi}_{m1,4Q} = 0.7879$ as the forecast horizon increases to four quarters which in turn hints at the substantial but bounded predictability of inflation suggested in [Stock and Watson \(2007\)](#), although they focus on the difficulty of forecasting changes in inflation rather than the level of informativeness of forecasts which is what we study here. Both the short and long horizon slope estimates of inflation being markedly below unity hints at the fact that forecasters firstly expect inflation to not remain constant and secondly they use information beyond what is embedded in the inflation factor to make their predictions.

For the estimated slope parameter of unemployment the results are noticeably different. The one-quarter slope sits at $\hat{\phi}_{m2,1Q} = 0.996$ barely below unity and still remains close to one at the four-quarter horizon with $\hat{\phi}_{m2,4Q} = 0.958$. The scatterplots for unemployment are also noticeably tighter centered around the fitted line than for inflation. These estimates tells us that the SPF forecasters anchors their forecasts almost one to one with the current

unemployment level. This is consistent with the economic theory of high persistence in U.S. unemployment.



Lastly to conclude this brief survey analysis we will discuss how we expect the estimates of σ_m^{svy} affect the coming Monte Carlo study. The estimated residual standard deviations for inflation ($\hat{\sigma}_{m1,1Q}^{svy} = 0.475$, $\hat{\sigma}_{m1,4Q}^{svy} = 0.618$) are far noisier than those of the unemployment estimate with $\hat{\sigma}_{m2,1Q}^{svy} = 0.116$ for the short horizon and $\hat{\sigma}_{m1,4Q}^{svy} = 0.294$ in the long horizon. The larger residual found in inflation would hint at the relatively modest signal about the current macro state in the survey forecasts, and the benefit of survey augmentation for the four-factor GATSM inflation identification would also be relatively modest. For unemployment we would expect that, given the small residual standard deviations, they would better help with unemployment identification in the GATSM and should help in reducing the finite-sample bias since each synthetic survey observation for unemployment are a high-quality signal of the underlying macro states and therefore almost acts as an equally-good extra unemployment observation.

Now in context of us setting $w_{svy} = 1$, the ratio between the inflation survey residual is accurate for inflation, where the survey residual scale $\hat{\sigma}_{m1,1}^{svy} = 0.475$ is almost close to the structural innovation scale $\Sigma_{11} = 0.483$. For unemployment it understates the precision of the unemployment survey residual since $\hat{\sigma}_{m2,1}^{svy} = 0.116$ lies far below unemployment innovation scale of $\Sigma_{22} = 0.434$. Essentially setting $w_{svy} = 1$ therefore down-weights the unemployment

survey relative to its actual informativeness and what consequences this will be shown in the Monte Carlo results.

7.1.3 Simulation Sampling

The simulation procedure for the survey-augmented Monte Carlo study is similar to the one described in Section 4.7 with one noticeable difference in how we simulate the states. Unlike in the two-factor simulation study which employed a discrete Lyapunov to initialise the state vector from the exact unconditional distribution, our approach for this Monte Carlo study is to employ a burn-in procedure with $T_{burn} = 200$ quarters. For each replication $\mathcal{R} = 1, \dots, R$ starting from $\mathbf{x}_1 = \mathbf{0}$ the state is propagated forward for $T_{burn} + T$ quarters under the true $\mathbb{P}$ -dynamics

$$\mathbf{x}_{t+1} = \boldsymbol{\mu}^{\mathbb{P}, \text{true}} + \boldsymbol{\Phi}^{\mathbb{P}, \text{true}} \mathbf{x}_t + \boldsymbol{\Sigma}^{\text{true}} \mathbf{u}_{t+1}, \quad \mathbf{u}_{t+1} \sim \text{iid}(0, \mathbf{I}). \quad (84)$$

The first T_{burn} quarters are discarded and the following T quarters constitute the simulated panel $\{\mathbf{x}_t\}_{t=1}^T$. The macro path for each replication $\mathcal{R}$ is read directly off of $\{\mathbf{x}_t\}_{t=1}^T$ to mimic the situation where macro factors are directly observable, whereas the two latent factors from the path are only used to compute the yield path and are lastly computed using the Kalman filter once all parameters are estimated. For each period t the simulated yield for maturity k is

$$\tilde{y}_t^{(k)} = g_0^{(k), \text{true}} + \mathbf{G}_x^{(k), \text{true}} \mathbf{x}_t + \boldsymbol{\varepsilon}_t^{(k)},$$

where $\boldsymbol{\varepsilon}_t^{(k)} = 0$ for the maturities in $\mathcal{Y}^1 = \{1, 10\}$ (1-year and the 10-year maturities are priced exactly) and the maturities in $\mathcal{Y}^2 = \{2, 5, 7, 15\}$ are priced with error with $\boldsymbol{\varepsilon}_t \sim \text{iid}(\mathbf{0}, (\sigma_v^{\text{true}})^2)$. To generate synthetic survey forecasts we use equation (83) calibrated using the numbers in Table 14 with $\eta_{m,t} \sim \text{iid}(0, (\sigma_{m,h}^{svy})^2)$. Two independent panels of surveys are generated, one for each forecast horizon $h \in \{1, 4\}$. The $h = 1$ forecast enters the Block m OLS framework as an additional left-hand-side row sharing the regressor $\mathbf{x}_t^m$. We emphasise that the synthetic survey is generated from the empirically calibrated SPF-macro relationship in (83) with slope $\phi_{m,1}^{svy}$, which is not restricted to equal the true conditional-expectation slope $\boldsymbol{\Phi}_{mm}^{\mathbb{P}, \text{true}}$ of the fixed DGP. The two coincide only if forecasters' co-movement with the current state exactly matched the model-implied persistence, which our calibration shows it generally does not.

For inflation we have $\hat{\phi}_{m_1, 1Q}^{svy} = 0.8812$ against $\Phi_{mm, 11}^{\mathbb{P}, \text{true}} = 0.8628$, a negligible wedge, but for unemployment $\hat{\phi}_{m_2, 1Q}^{svy} = 0.9955$ against $\Phi_{mm, 22}^{\mathbb{P}, \text{true}} = 0.8945$, a wedge of roughly 0.10. Since the estimator attributes the survey's co-movement with $\mathbf{x}_t^m$ to $\boldsymbol{\Phi}_{mm}^{\mathbb{P}}$, this wedge means the

unemployment survey should pull the persistence estimate upward toward $\hat{\phi}_{m_2,1}^{svy}$. We retain the empirically calibrated slope deliberately, as it reflects how the actual SPF forecasters incorporate the current macro states when making a prediction about next periods macro states, and we therefore interpret the survey rows as noisy observations of this empirical forecasting relationship rather than as unbiased signals of the model-implied conditional expectation $\mathbb{E}_t^{\mathbb{P}}[\mathbf{x}_{t+1}^m] = \boldsymbol{\mu}_m^{\mathbb{P}} + \boldsymbol{\Phi}_{mm}^{\mathbb{P}} \mathbf{x}_t^m$. The degree to which this injected wedge helps or hurts the finite-sample bias is itself one of the objects the Monte Carlo experiment is designed to measure.

Incorporating four-quarter-ahead survey forecasts is not as easily implemented in the Block m framework given the VAR(1) specification. By iterating the VAR(1) four steps ahead and taking expectations at time t we get

$$\mathbb{E}_t^{\mathbb{P}}[\mathbf{x}_{t+4}^m] = \sum_{j=0}^3 (\boldsymbol{\Phi}_{mm}^{\mathbb{P}})^j \boldsymbol{\mu}_m^{\mathbb{P}} + (\boldsymbol{\Phi}_{mm}^{\mathbb{P}})^4 \mathbf{x}_t^m,$$

which is a non-linear problem and to use this in our OLS stacking approach would require us to know $\boldsymbol{\Phi}_{mm}^{\mathbb{P}}$ before estimating it which is incompatible the closed-form OLS structure of our Block m . To preserve linearity of the system we instead match the four-quarter-ahead survey with the conditional expectation at time $t + 3$

$$\mathbf{s}_{t+4|t}^{norm} \approx \mathbb{E}_{t+3}^{\mathbb{P}}[\mathbf{x}_{t+4}^m] + \boldsymbol{\varepsilon}_t^{svy} = \boldsymbol{\mu}_m^{\mathbb{P}} + \boldsymbol{\Phi}_{mm}^{\mathbb{P}} \mathbf{x}_{t+3}^m + \boldsymbol{\varepsilon}_t^{svy},$$

where $\boldsymbol{\varepsilon}_t^{svy}$ is the residual of the approximated regression. Since $\mathbf{x}_{t+3}^m$ is observed and has a linear relationship with $\boldsymbol{\Phi}_{mm}^{\mathbb{P}}$ it is compatible with the OLS stacking approach. The first three survey observations are dropped since their corresponding $\mathbf{x}_{t+3}^m$ regressors fall outside the available sample, where the amount of usable survey rows is now $T_1 - 3$. Our approach here is an approximation because the two conditional expectations $\mathbb{E}_t^{\mathbb{P}}[\mathbf{x}_{t+4}^m]$ and $\mathbb{E}_{t+3}^{\mathbb{P}}[\mathbf{x}_{t+4}^m]$ are not the same. The surveys $\mathbf{s}_{t+4|t}^{norm}$ were formed using time t information so it is a noisy signal of $\mathbb{E}_t^{\mathbb{P}}[\mathbf{x}_{t+4}^m]$ which conditions on $\mathbf{x}_t^m$. Our approximation instead conditions on $\mathbf{x}_{t+3}^m$ which is a realisation of the state that was unknown when the survey was made. The two conditional expectations are related by the law of iterated expectations

$$\mathbb{E}_t^{\mathbb{P}}[\mathbb{E}_{t+3}^{\mathbb{P}}[\mathbf{x}_{t+4}^m]] = \mathbb{E}_t^{\mathbb{P}}[\mathbf{x}_{t+4}^m],$$

so the one-step expectation at time $t + 3$ is an unbiased predictor of the four-step expectation at time t in the population sense. However, $\mathbf{x}_{t+3}^m$ is viewed as a random variable at time

t meaning that it can be correlated with ε_t^{svy} in the approximated regression through the intermediate innovations $\mathbf{u}_{t+1}^m, \mathbf{u}_{t+2}^m, \mathbf{u}_{t+3}^m$ from (84). This would form an endogeneity bias for the $h = 4$ that is absent in the $h = 1$ case. Nevertheless, this is the most simple approach of incorporating four-quarter-ahead surveys into our Block m framework without making too many changes, and the approximation is expected to be mild in practice because of the high persistence of the macro factor, meaning that $\mathbf{x}_{t+3}^m$ is reasonably predictable given $\mathbf{x}_t^m$. And for the consequences of the approximation in practice for the finite-sample properties of Estimator 4Q relative to Estimator 1Q, is precisely what the upcoming Monte Carlo experiment is designed to assess.

7.1.4 Three Estimators and Bias Metrics

For each replication $\mathcal{R} = 1, \dots, R$ we now have a sample yield path $\left\{y_t^{(k)}\right\}_{t=1}^T$ for maturities $k = \{4, 8, 20, 28, 40, 60\}$ in quarters, a path of observable macro factors $\left\{\mathbf{x}_t^m\right\}_{t=1}^T$, and a path of synthetic survey forecasts $\left\{\mathbf{s}_{t+h|t}^{norm}\right\}_{t=1}^T$ for $h = \{1, 4\}$. For each replication we apply this to the three estimators that apply the same MCSE pipeline as described in Section 5.2 and 6.2 and they only differ in their composition of the Block m OLS regression. One small difference

Table 15: Estimator Overview

Estimator	Block m OLS	Survey horizon
No-Survey (NS)	Actual macro rows only	—
One-quarter-ahead Surveys (1Q)	Actual + h=1 survey rows	$h = 1$
Four-quarter-ahead Surveys (4Q)	Actual + h=4 survey rows	$h = 4$

in the Monte Carlo estimation compared to the empirical estimation is that we let $\hat{\mu}_i^{\mathbb{P}}$ be zero for all replications as a computational simplification. Reasons for this is that firstly the DGP $\mu_i^{\mathbb{P}, \text{true}}$ is small and close to zero but not exactly zero which introduces a small bias, which brings us to the second reason which is that our primary object of interest in the Monte Carlo study is the finite-sample properties of the parameters in the macro block, $\mu_m^{\mathbb{P}}, \Phi_{mm}^{\mathbb{P}}$, and its downstream effect on the model-implied term premium.

The latent $\mu_i^{\mathbb{P}}$ affects the unconditional mean of the latent factors under $\mathbb{P}$ but has no effect on the persistence parameters, the market price of risk, or the yield loadings themselves. So for the sake of the experiment we can view it as a nuisance parameter. Lastly, the analytical recovery of $\hat{\mu}_i^{\mathbb{P}}$ via equation (70) involves an inversion of $\text{chol}\left(\hat{\Omega}^{1, \#}\right)$ which requires that $\hat{\Omega}^{1, \#}$ is well-conditioned which was not a problem during the empirical estimation with $T = 216$ but it could pose a problem when we test using the smaller sample of $T = 100$, where e.g.

the two $\mathcal{Y}^1$ yields happen to be nearly collinear thus making $\hat{\Omega}^{1,\#}$ nearly singular giving us extreme values for $\hat{\mu}_l^{\mathbb{P}}$ which would distort the results. Apart from this one change all of the subsequent steps to the Block m OLS regression are identical across estimators.

The primary objects of interest are the diagonals of $\hat{\Phi}_{mm}^{\mathbb{P}}$ which governs the persistence in the macro dynamics. Given this we will report the results of simulations centered around the macro block. The key question we investigate are whether or not Estimators Q1 and Q4 produce smaller bias results relative to Estimator NS, and whether the using the one-quarter-ahead SPF surveys are preferable to using the four-quarter-ahead surveys. Lastly we will also study how a smaller sample affects the finite-sample properties.

7.2 1Q Monte Carlo Result

Table 16: Monte Carlo Result for the 1Q and NS ($T = 216$)

Parameter	True	Estimator NS			Estimator 1Q ($h = 1$)		
		Mean	Bias	Std.	Mean	Bias	Std.
$R = 2000, T = 216$							
$\mathbb{P}$-dynamics: Macro AR ($\Phi_{mm}^{\mathbb{P}}$)							
$\Phi_{mm,11}^{\mathbb{P}}$	0.86277	0.84068	-0.02208	0.03779	0.86015	-0.00262	0.02607
$\Phi_{mm,21}^{\mathbb{P}}$	0.02039	0.01819	-0.00220	0.03481	0.00905	-0.01134	0.01799
$\Phi_{mm,12}^{\mathbb{P}}$	0.07422	0.07873	+0.00452	0.04045	0.03971	-0.03451	0.02798
$\Phi_{mm,22}^{\mathbb{P}}$	0.89448	0.87179	-0.02269	0.03751	0.93361	+0.03913	0.01925
Short-rate intercept							
α	0.01260	0.01191	-0.00068	0.00266	0.01190	-0.00070	0.00271
Risk-neutral intercept ($\mu_m^{\mathbb{Q}}$)							
$\mu_1^{\mathbb{Q}}$	0.31091	0.04318	-0.26772	0.08677	0.04099	-0.26992	0.08769
$\mu_2^{\mathbb{Q}}$	-0.05002	-0.02435	+0.02567	0.08220	-0.02569	+0.02433	0.08288
Volatility: Cholesky (Σ_{mm})							
Σ_{11}	0.48297	0.47818	-0.00479	0.02281	0.48117	-0.00181	0.02280
Σ_{21}	0.01764	0.01620	-0.00144	0.02912	0.01268	-0.00496	0.02913
Σ_{22}	0.43448	0.43026	-0.00422	0.02132	0.43397	-0.00051	0.02146
Measurement error std.							
σ_v (bps)	59.11305	58.52066	-0.59239	1.44008	58.52066	-0.59239	1.44008

Notes: DGP: 1Q-ahead survey-augmented estimates. Estimator NS = standard MCSE ($w_{\text{svy}} = 0$). Estimator 1Q = survey-augmented, $h = 1$ (dpgdp3/UNEMP3; $\phi_{\text{infl}} = 0.881$, $\phi_{\text{unemp}} = 0.995$). Bias = $\hat{\theta} - \theta_0$. Std = standard deviation of estimates across replications.

Table 16 reports the Monte Carlo results of the no-survey estimator (NS) and the 1-quarter-ahead survey-augmented estimator (1Q) for a sample of $T = 216$ quarters, based on $R = 2,000$ Monte Carlo replications, where each replication uses CMAES in the numerical steps, with tolerances set to 1×10^{-10} for computational ease. This table represents the full sample

available in the paper, providing a baseline for how finite-sample bias arises and whether survey expectations can help correct it.

Looking at the $\mathbb{P}$ -dynamics, the diagonal elements of $\Phi_{mm}^{\mathbb{P}}$ show the mean persistence of inflation and unemployment under the physical measure, meaning how slowly they revert to their long-run average after a shock. Under the NS estimator, the persistence of inflation, $\Phi_{mm,11}^{\mathbb{P}}$, is underestimated by -0.022 relative to its true value of 0.863, whereas the persistence of unemployment, $\Phi_{mm,22}^{\mathbb{P}}$, is underestimated by -0.023 relative to its true value of 0.894, with standard deviations of around 0.038 for both.

Turning to the 1Q estimator, incorporating one-quarter-ahead survey data of inflation and unemployment corrects the negative bias on both diagonals, although not in the same way. For inflation, the bias on $\hat{\Phi}_{mm}^{\mathbb{P}}$ falls from -0.022 to -0.003, a reduction of almost 90 percent. The standard deviation also falls from 0.038 to 0.026, indicating a considerable improvement in bias and precision. For unemployment, the 1Q estimator does not only correct the negative bias of -0.023, but overcorrects it, producing a positive bias of 0.039, meaning that while the NS estimator underestimates the persistence of unemployment, the 1Q estimator overestimates it.

The standard deviation falls by 50%, from 0.038 to 0.019, indicating that the estimator becomes more precise even though it introduces a positive bias. While survey augmentation helps reduce finite-sample bias, the overcorrection in $\hat{\Phi}_{mm,22}^{\mathbb{P}}$ hints at a potential trade-off: if forecasters are too optimistic about the recovery after a crisis or shock in general, meaning that they expect unemployment to revert to its mean faster than it actually does, incorporating the survey data may reduce one source of bias while introducing another source.

The off-diagonal elements $\Phi_{mm,21}^{\mathbb{P}}$ and $\Phi_{mm,12}^{\mathbb{P}}$ capture how much of the movement in one variable is driven by the other. Under the NS estimator, both elements show small biases of -0.002 and 0.005 with standard deviations of 0.035 and 0.040. The true values for these are small, so these biases are modest and do not point to a systematic distortion. This is expected since the finite-sample bias in persistence is specific to own-dynamics.

Under the 1Q estimator, the off-diagonal biases grow with $\Phi_{mm,21}^{\mathbb{P}}$ moving to -0.011 and $\Phi_{mm,12}^{\mathbb{P}}$ to -0.035, while their standard deviations fall noticeably. This suggests that while survey data may help correct the diagonal persistence, the off-diagonals become more biased. However, since the primary motivation for survey augmentation is to reduce finite-sample bias in the persistence of the macro factors, a reduction bias in diagonal persistent at the cost of a modest increase in off-diagonal bias is still considered a net improvement.

The short-rate intercept α has a true value of 0.013 and have a small bias of -0.001 under

both the NS and the 1Q estimator, with almost identical standard deviations around 0.003. This confirms that the level of the short-rate is well-identified by the yield cross-section and is unaffected by the survey augmentation. This is expected since the short-rate intercept is identified from the average level of the yield curve across maturities, which is a feature of the cross-section rather than the time-series.

The risk-neutral intercepts μ_1^Q and μ_2^Q show the largest biases in the table compared to the true values. Under estimator NS they are -0.268 and 0.026 respectively and under estimator 1Q they are -0.270 and 0.024, so virtually the same, meaning that survey augmentation has no effect. This is expected for the same reason as with the short-rate intercept α - these parameters are identified from the cross-section rather than the time-series.

The diagonal elements of the Cholesky volatility Σ_{mm} , which capture the volatility of the unpredictable shocks to inflation and unemployment, show small negative biases under NS of -0.005 for Σ_{11} and -0.004 for Σ_{22} with standard deviations around 0.022. The 1Q estimator reduces these biases to near zero for both elements, with Σ_{11} at -0.002 and Σ_{22} with -0.001, so essentially unbiased. Their standard deviations are near identical to the NS estimator. The off-diagonal Σ_{21} is small and shows similar modest bias under both estimators.

The measurement error standard deviation σ_v is by construction identical across both estimators, at approximately 58.5 basis points against the true value of 59.1 basis points, resulting in a bias of -0.592 basis points.

Overall, the $T = 216$ finite-sample test shows that the NS estimator produces a systematic and expected negative bias in the own-persistence parameters of the macro factors under the physical measure. Incorporating one-quarter-ahead survey data largely corrects this bias, reducing it by nearly 90 percent for inflation persistence and eliminating the negative bias completely for unemployment persistence, although with an overcorrection. Precision is improved across the board for all physical dynamics under 1Q, with standard deviations falling noticeably. The improvements are concentrated in the $\mathbb{P}$ -measure dynamics, where finite-sample bias is theoretically expected, while parameters identified from the yield cross-section are largely unaffected. The bias under NS is present but modest, and the gains from survey augmentation are clear but not major. In the following, the sample is shortened to 100 quarters to examine whether these patterns become more pronounced at reduced samples.

7.3 Monte Carlo Results Small Sample

Table 17: Monte Carlo Result for the 1Q and NS ($T = 100$)

Parameter	True	Estimator NS			Estimator 1Q ($h = 1$)		
		Mean	Bias	Std.	Mean	Bias	Std.
$R = 2000, T = 100$							
$\mathbb{P}$-dynamics: Macro AR ($\Phi^{\mathbb{P}}_{mm}$)							
$\Phi^{\mathbb{P}}_{mm,11}$	0.86277	0.81465	-0.04812	0.06300	0.84779	-0.01497	0.04299
$\Phi^{\mathbb{P}}_{mm,21}$	0.02039	0.01424	-0.00615	0.06023	0.00700	-0.01339	0.03095
$\Phi^{\mathbb{P}}_{mm,12}$	0.07422	0.08316	+0.00895	0.06901	0.04169	-0.03253	0.04654
$\Phi^{\mathbb{P}}_{mm,22}$	0.89448	0.84403	-0.05045	0.06504	0.91980	+0.02532	0.03344
Short-rate intercept							
α	0.01260	0.01172	-0.00087	0.00356	0.01170	-0.00089	0.00358
Risk-neutral intercept ($\mu^{\mathbb{Q}}_m$)							
$\mu^{\mathbb{Q}}_1$	0.31091	0.02445	-0.28646	0.09853	0.02382	-0.28709	0.09644
$\mu^{\mathbb{Q}}_2$	-0.05002	-0.01562	+0.03440	0.09565	-0.01679	+0.03323	0.09704
Volatility: Cholesky (Σ_{mm})							
Σ_{11}	0.48297	0.47290	-0.01007	0.03447	0.47818	-0.00480	0.03433
Σ_{21}	0.01764	0.01578	-0.00185	0.04340	0.01234	-0.00529	0.04324
Σ_{22}	0.43448	0.42382	-0.01066	0.03173	0.42882	-0.00566	0.03187
Measurement error std.							
σ_v (bps)	59.11305	57.77472	-1.33833	2.13144	57.77472	-1.33833	2.13144

Notes: Same as in Table 16 but with $T = 100$.

Table 17 repeats the Monte Carlo simulation of Table 16 for a smaller sample of $T = 100$ quarters, retaining the same DGP and $R = 2,000$ replications. The motivation for examining this shorter sample is to assess whether the finite-sample bias observed at $T = 216$ is amplified when available data is limited, a setting that resembles typical constraints in many term structure applications, especially those using subsamples. Given the name 'finite-sample bias' we expect the bias to worsen as we shorten the number of observations.

Before turning to the results, it is worth noting why the biases documented here may be interpreted as finite-sample bias specifically, rather than other forms of misspecification. Three features of the Monte Carlo design makes this identification possible. First, the data is simulated directly from the estimated model, so by construction there is no misspecification, the model used to generate the data and the model used to estimate the data are identical, ruling out functional error or omitted variables as sources of bias. Second, the true parameter values are known, meaning that bias can be measured directly as the difference between the average estimate across replications and the true value. Third, if the bias is genuinely a finite-sample bias it should be amplified as the sample shrinks. This is the clearest indication of finite-sample bias and is what comparing Table 17 to 16 will help visualise.

Looking at Table 17, the own-persistence parameters of the macro factors under the NS

estimator still show negative bias. Compared to the $T = 216$ estimates, the bias becomes substantially more pronounced at $T = 100$. The bias on inflation $\Phi_{mm,11}^{\mathbb{P}}$ more than doubles from -0.022 to -0.048 and the bias on unemployment $\Phi_{mm,22}^{\mathbb{P}}$ similarly increases from -0.023 to -0.050. With only 100 quarters of data, the estimator observes even fewer mean-reversions in the macro factors and consequently makes even more mistaken conclusions about how fast they revert to their means. The standard deviations also increase considerably, from around 0.038 at $T = 216$ to 0.063 and 0.065 at $T = 100$ respectively. This indicates the greater estimation uncertainty when analysing a shorter sample. Both the larger bias and the increased dispersion are consistent with the expectations of a more substantial finite-sample problem as the sample size shrinks.

Like the $T = 216$ model, the incorporation of surveys provides a meaningful correction. The bias on $\Phi_{mm,11}^{\mathbb{P}}$ falls from -0.048 under NS to -0.015 under 1Q and the standard deviation falls from 0.063 to 0.043, providing smaller bias and a more precise estimation of inflation persistence. The correction for $\Phi_{mm,22}^{\mathbb{P}}$ is again an overcorrection, changing the sign of the bias: the NS bias of -0.050 is now a positive bias of 0.025, with the standard deviation falling from 0.065 to 0.033. Interestingly, when comparing the $T = 216$ and the $T = 100$ models, even though the bias on own-persistence in unemployment $\Phi_{mm,22}^{\mathbb{P}}$ is larger, as one would expect for the smaller $T = 100$ model, the overcorrection from incorporating surveys is of smaller magnitude (0.025 in $T = 100$ versus 0.039 in $T = 216$). The 1Q $\Phi_{mm,22}^{\mathbb{P}}$ estimate moves slightly closer to the true value as the sample shrinks. The reason for this is not immediately clear from the Monte Carlo results.

As noted in Table 16, this positive bias hints to a trade-off when incorporating surveys: while survey data may help reduce finite-sample bias, they may introduce a bias of their own if forecasters are systematically too optimistic about recoveries, expecting factors (unemployment in this case) to revert to their means faster than they actually do, even in the case where we expected the surveys to pull the persistences downwards. Interestingly, the smaller overcorrection at $T = 100$ suggests that the forecast bias is less of a concern when assessing the trade-off at smaller samples.

The off-diagonal elements under NS show small and unsystematic biases of -0.006 for $\Phi_{mm,21}^{\mathbb{P}}$ and 0.009 for $\Phi_{mm,12}^{\mathbb{P}}$ with standard deviations growing to 0.060 and 0.069 respectively. Both the biases and the standard deviations have increases compared to the $T = 216$ model. As in Table 16, the cross-dynamics between inflation and unemployment are not expected to exhibit the same finite-sample distortion as the own-persistent parameters and remain small and unbiased.

Under the 1Q estimator, the off-diagonal biases again become somewhat larger compared to the NS model, but more precise. It is worth noting that overall the bias of the off-diagonal elements under 1Q is largely unchanged when sample size shrinks. The bias of $\Phi_{mm,21}^{\mathbb{P}}$ is slightly larger and the bias of $\Phi_{mm,12}^{\mathbb{P}}$ is slightly lower in $T = 100$ compared to $T = 216$.

As in the $T = 216$ model, the overall conclusion regarding the $\mathbb{P}$ -dynamics is that a reduction in diagonal bias at the cost of a modest increase in off-diagonal bias remains an overall improvement, given that the own-persistent seen in the diagonal elements is the primary target of survey augmentation.

The short-rate intercept α again shows a negligible bias of -0.001 under both estimators, with standard deviations of around 0.004, which is slightly higher than at $T = 216$. This confirms that the level of the short-rate remains well-identified by the yield cross-section even at a smaller sample sizes. Furthermore, the short-rate is still unaffected by survey augmentation, consistent with its cross-section rather than time-series identification.

The risk-neutral intercepts $\mu_1^{\mathbb{Q}}$ and $\mu_2^{\mathbb{Q}}$ show large biases of -0.286 and 0.034 under NS, and these are, like the $T = 216$ model virtually unchanged under 1Q at -0.287 and 0.033. The biases are slightly larger than those at $T = 216$, and the standard deviations have increased slightly as well, suggesting a general increase in estimation uncertainty at shorter samples. Even with the increased uncertainty, the pattern is the same as previously - survey augmentation has no effect on these parameters, since they are identified from the yield cross-section rather than the time-series dynamics of the macro factors, just like the short-rate intercept α .

The diagonal Cholesky elements Σ_{11} and Σ_{22} show increased biases under both NS and 1Q, with slightly increased standard deviations for both as well. This once again highlights the estimation uncertainty as the sample shortens. The 1Q estimator does however still offer marginal improvements, reducing biases under NS from -0.010 and -0.011 to -0.005 and -0.006 respectively for Σ_{11} and Σ_{22} . Despite the increase in bias relative to $T = 216$, the biases remain small in absolute terms - unlike persistence, volatility can be estimated reliably from the variability of the data even at a short sample and does not require observing many mean-reversions to be estimated correctly.

The measurement error standard deviation σ_v is again identical across both estimators at 57.77 basis points, with a slightly larger bias and wider standard deviation than at $T = 216$. This is consistent with the increased estimation uncertainty at shorter samples.

Overall, the $T = 100$ results show the amplification of the finite-sample bias and the correction effect of survey augmentation as expected. Shortening the sample from 216 to 100

quarters roughly doubles the negative bias in the own-persistence of the macro factors under the NS estimator, while increasing standard deviations substantially. This amplification of bias as the sample shortens, combined with the Monte Carlo design that rules out misspecification, provides strong evidence that the biases documented in the tables are genuinely driven by sample size rather than other sources of bias. Incorporating one-quarter-ahead survey expectations corrects the negative bias substantially, with precision improvements across all physical dynamics. The results across both sample sizes reinforce the case that the gains from incorporating survey expectations are consistent even as sample size shrinks.

7.4 Noisy Surveys

The results in Tables 16 and 17 are derived under the assumption that the SPF survey forecasts enter the estimator without additional measurement error beyond their forecast uncertainty. It is worth distinguishing between the two different types of noise that are present in the survey data. The first is forecast error. This is the uncertainty in the forecaster's prediction that exists even for a perfectly rational and well-informed forecaster. This is included in the model as σ_{svy} and is estimated directly from the SPF data. The second type is measurement error. This is noise that can arise from several sources, including differences in respondent information sets or timing mismatches in when forecasts are recorded relative to data releases. The previous results assumed that the forecasts have no measurement errors and only allowed for forecasting errors.

Table 18: Finite-Sample Properties: Robustness to Survey Measurement Error

		$T = 216$			$T = 100$		
Parameter	True	$\kappa = 0$	$\kappa = 0.5$	$\kappa = 1.0$	$\kappa = 0$	$\kappa = 0.5$	$\kappa = 1.0$
Convergence: $T = 216$: 2000/2000; $T = 100$: 1998/2000							
P-dynamics: Macro AR (Φ_{mm}^P)							
$\Phi_{mm,11}^P$	0.86277	-0.0026	-0.0027	-0.0028	-0.0150	-0.0153	-0.0155
$\Phi_{mm,21}^P$	0.07422	-0.0652	-0.0651	-0.0651	-0.0134	-0.0671	-0.0670
$\Phi_{mm,12}^P$	0.02039	+0.0193	+0.0195	+0.0196	-0.0325	+0.0212	+0.0212
$\Phi_{mm,22}^P$	0.89448	+0.0391	+0.0391	+0.0391	+0.0253	+0.0253	+0.0252
Short-rate intercept							
α	0.01260	-0.0007	-0.0007	-0.0007	-0.0009	-0.0009	-0.0009
Risk-neutral intercept (μ_m^Q)							
μ_1^Q	0.31091	-0.2699	-0.2653	-0.2672	-0.2871	-0.2881	-0.2868
μ_2^Q	-0.05002	+0.0243	+0.0247	+0.0238	+0.0332	+0.0336	+0.0364
Volatility: Cholesky (Σ_{mm})							
Σ_{11}	0.48297	-0.0018	-0.0016	-0.0010	-0.0048	-0.0044	-0.0031
Σ_{21}	0.00000	+0.0127	+0.0127	+0.0127	-0.0053	+0.0123	+0.0123
Σ_{22}	0.43448	-0.0005	-0.0005	-0.0005	-0.0057	-0.0056	-0.0055
Measurement error std.							
σ_v (bps)	14.778	-0.1481	-0.1481	-0.1481	-1.3383	-0.3346	-0.3346

Notes: DGP: 1Q-ahead survey-augmented. All columns report 1Q survey-augmented estimates ($h = 1$; dpgdp3/UNEMP3; $\phi_{\text{infl}} = 0.881$, $\phi_{\text{unemp}} = 0.995$). κ = measurement error scale: $\sigma_\omega = \kappa \cdot \sigma_{\text{svy}}$, where σ_{svy} is the baseline SPF forecast error standard deviation calibrated from actual data. The total noise variance becomes $(1 + \kappa^2)\sigma_{\text{svy}}^2$, so $\kappa = 0.5$ increases noise variance by 25% and $\kappa = 1.0$ doubles it relative to baseline. Bias = $\bar{\hat{\theta}} - \theta_0$.

To assess whether the previous results are sensitive to this assumption, table 18 repeats the Monte Carlo exercise while introducing additional measurement noise on top of the forecast uncertainty on the one-quarter-ahead forecast horizon. The extra noise is added through a scaling factor κ , where the additional noise standard deviation is $\sigma_\omega = \kappa * \sigma_{\text{svy}}$. Since this noise is independent of the existing forecast uncertainty, the variances together add a total noise variance of $(1 + \kappa^2)\sigma_{\text{svy}}^2$, this means that $\kappa = 0.5$ increases total noise variance by 25 percent and $\kappa = 1.0$ doubles it. This represents a demanding stress test that likely exceeds what realistic survey measurement errors would be expected to introduce. Since the NS estimator does not use survey data, it is unaffected by κ by construction and is therefore excluded from Table 18 to focus solely on the stability of the 1Q estimator as noise increases. The procedure follows the methodology described in Section 7.1 and only requires a small

change. Synthetic survey forecasts are now sampled for each replication $\mathcal{R}$ using

$$s_{m,t+1|t}^{norm} = \mu_{m,h}^{svy} + \phi_{m,h}^{svy} x_{m,t}^{norm} + \eta_{m,t} + \omega_{m,t}, \quad (85)$$

where $\eta_{m,t} \sim niid(0, (\sigma_m^{svy})^2)$ and $\omega_{m,t} \sim niid(0, (\kappa \cdot \sigma_m^{svy})^2)$. The estimates for $h = 1$ from Table 14 are used for calibration.

The results strongly support the assumption introduced previously. Across both sample sizes and all parameter groups, the biases of the 1Q estimator are extremely stable as κ increases from 0 to 1.0. For the diagonal persistence parameters, the bias on $\Phi_{mm,11}^{\mathbb{P}}$ moves from -0.0026 at $\kappa = 0$ to -0.0028 at $\kappa = 1.0$ for $T = 216$, and from -0.0150 to -0.0155 for $T = 100$. These are negligible changes relative to the magnitude of the bias correction documented earlier. The positive bias on $\Phi_{mm,22}^{\mathbb{P}}$ is similarly stable at 0.039 across all noise levels for $T = 216$ and +0.025 for $T = 100$. The same pattern is true for the short-rate intercept, the risk-neutral parameters, and the Cholesky volatility elements - none show meaningful sensitivity to the introduction of additional measurement noise.

A minor note regarding the convergence rates at $T = 100$, where 2 out of 2,000 replications failed to converge. This is likely caused by one of two conditions imposed on each replication, or a combination of the two. The first one being the maximum iteration limit of 3,000 imposed on each replication and the second one being the tolerance. The convergence failure at $T = 100$ is consistent with the increased numerical difficulty at shorter sample sizes and does not affect the conclusions.

Overall, Table 18 helps validate the assumption used in the analysis regarding no additional measurement error. The bias-correction of the survey augmentation shown in Tables 16 and 17 survive even under stress that doubles the total noise variance of the surveys, providing evidence that the findings cannot be a consequence of survey data being treated as perfectly measured.

8 Short Versus Long Horizon Surveys

8.1 Monte Carlo Results of Differing Forecast Horizons

This section examines whether using a longer forecast horizon will affect the ability of surveys to combat finite-sample bias. The motivation for this examination originates from the data section, where Tables 3 and 6 showed that forecast precision weakens as horizon increases combined with findings from Table 14. The expectation is that the four-quarter-ahead forecasts should carry a weaker signal about the physical dynamics of inflation and unemployment, and should consequently be less effective in terms of reducing finite-sample bias compared to the one-quarter-ahead forecasts.

A way to initially test this is through the signal-to-noise ratio (SNR). This is defined as the ratio between the signal variance (how much survey forecasts reflect a prediction of where the macro variables are heading) and the total forecast noise variance

$$SNR = \frac{(\phi_{m,h}^{svy})^2}{(\sigma_{m,h}^{svy})^2}, \quad (86)$$

where $(\phi_{m,h}^{svy})^2$ is the signal variance and $(\sigma_{m,h}^{svy})^2$ is the total forecast noise variance.

A high SNR means the forecast is primarily informative, while a low SNR indicates that noise dominates the survey and that it therefore carries limited useful information about the true parameters. Comparing the two forecast horizons, the SNRs differ substantially. For inflation, the one-quarter-ahead forecast has an SNR of 3.44, while the four-quarter-ahead has an SNR of just 1.62, less than half. For unemployment the difference is even bigger: the one-quarter-ahead SNR is 73.5, falling to 10.6 at the four-quarter-ahead forecast. In both cases, extending the forecast horizon from one to four quarters substantially reduces the survey signal relative to noise. This is expected since macroeconomic uncertainty is increasing over time. This is consistent with the forecast accuracy results in the data section, where RMSE was shown to be increasing monotonically with horizon for both inflation and unemployment. This was further examined in Table 14, which showed that R^2 is decreasing in horizon and variation is increasing in horizon. These results support the expectation that survey augmentation using a longer horizon (4Q) should be less effective than 1Q. This is examined in detail in Table 19.

Table 19: Finite-Sample Properties: Short- vs. Long-Horizon Survey ($T = 216$)

Parameter	True	Estimator 1Q ($h = 1$)			Estimator 4Q ($h = 4$)		
		Mean	Bias	Std.	Mean	Bias	Std.
$R = 2000, T = 216$							
$\mathbb{P}$-dynamics: Macro AR ($\Phi_{mm}^{\mathbb{P}}$)							
$\Phi_{mm,11}^{\mathbb{P}}$	0.86277	0.86015	-0.00262	0.02607	0.66960	-0.19317	0.05734
$\Phi_{mm,21}^{\mathbb{P}}$	0.02039	0.00905	-0.01134	0.01799	0.07828	+0.05789	0.04312
$\Phi_{mm,12}^{\mathbb{P}}$	0.07422	0.03971	-0.03451	0.02798	0.06208	-0.01213	0.05148
$\Phi_{mm,22}^{\mathbb{P}}$	0.89448	0.93361	+0.03913	0.01925	0.73733	-0.15715	0.05803
Short-rate intercept							
α	0.01260	0.01190	-0.00070	0.00271	0.01194	-0.00066	0.00269
Risk-neutral intercept ($\mu_m^{\mathbb{Q}}$)							
$\mu_1^{\mathbb{Q}}$	0.31091	0.04099	-0.26992	0.08769	0.04369	-0.26722	0.08241
$\mu_2^{\mathbb{Q}}$	-0.05002	-0.02569	+0.02433	0.08288	-0.02626	+0.02376	0.08005
Volatility: Cholesky (Σ_{mm})							
Σ_{11}	0.48297	0.48117	-0.00181	0.02280	0.51318	+0.03021	0.02629
Σ_{21}	0.01764	0.01268	-0.00496	0.02913	0.01395	-0.00369	0.03154
Σ_{22}	0.43448	0.43397	-0.00051	0.02146	0.44824	+0.01376	0.02358
Measurement error std.							
σ_v (bps)	59.11305	58.52066	-0.59239	1.44008	58.52066	-0.59239	1.44008

Notes: Single DGP: V6 parameters (1Q-ahead survey-augmented MCSE). Estimator 1Q = survey-augmented, $h = 1$ (dpgdp3/UNEMP3; $\phi_{\text{infl}} = 0.881$, $\phi_{\text{unemp}} = 0.995$). Estimator 4Q = survey-augmented, $h = 4$ (dpgdp6/UNEMP6; $\phi_{\text{infl}} = 0.788$, $\phi_{\text{unemp}} = 0.958$). Bias = $\bar{\hat{\theta}} - \theta_0$. SD = standard deviation of estimates across replications.

Table 19 compares the one-quarter-ahead survey estimator (1Q, $h = 1$) to the four-quarter-ahead survey estimator (1Q, $h = 4$) for $T = 216$ quarters, keeping the same DGP and $R = 2,000$ replications as the earlier results. Both estimators augment the NS estimator with survey data, but differ in forecast horizon.

The $\mathbb{P}$ -dynamics under the 1Q estimator show small biases of -0.003 for $\Phi_{mm,11}^{\mathbb{P}}$ and 0.039 for $\Phi_{mm,22}^{\mathbb{P}}$ established in Table 16. The 4Q estimator produces substantially worse results on both. The bias on $\Phi_{mm,11}^{\mathbb{P}}$ increases to -0.193, meaning that four-quarter-ahead survey estimator performs much worse than the 1Q estimator. It even performs remarkably worse than the NS estimator, producing a nearly 9 times higher bias. This suggests that the four-quarter-ahead inflation forecasts are too noisy to reliably signal where the actual inflation will be in a full year, resulting in the uncertainty of the forecast dominating and the estimated persistence being pulled in the wrong direction.

Regarding unemployment, the 4Q performs even worse than the NS estimator. Instead of reducing the negative bias of the NS, it introduces a large new negative bias. This results in the unemployment persistence, $\Phi_{mm,22}^{\mathbb{P}}$, to have a bias of -0.157. This is a very large increase relative to the bias of -0.023 under NS or 0.039 under 1Q. This suggests a high difficulty of

prediction the labour market conditions four quarters ahead, resulting in a highly increased bias for unemployment. The standard deviations under 4Q are also considerably wider than under 1Q for both diagonal elements, at 0.057 and 0.058 compared to 0.026 and 0.019 under 1Q for inflation and unemployment respectively. The four-quarter-ahead forecasts therefore does not only produce higher biases, but also greater estimation uncertainty. This is consistent with the lower SNR of the 4Q forecasts, where the weaker signal does not only fail to help, but actually worsens the estimator by pushing the persistence further away from the true parameters.

The off-diagonals elements of the $\mathbb{P}$ -dynamics under 1Q showed modest biases of -0.011 and -0.035 as seen earlier. Under 4Q, $\Phi_{mm,21}^{\mathbb{P}}$ moves to 0.058 and $\Phi_{mm,12}^{\mathbb{P}}$ to -0.012, meaning that $\Phi_{mm,21}^{\mathbb{P}}$ becomes larger and $\Phi_{mm,12}^{\mathbb{P}}$ becomes smaller. Their standard deviations are wider for both off-diagonals.

The short-rate intercept α again shows a negligible bias of -0.0001 under both estimators, confirming as expected that this parameter is unaffected by the choice of survey horizon since it is identified from the yield cross-section rather than the time-series dynamics, as discussed earlier.

The risk-neutral intercepts $\mu_1^{\mathbb{Q}}$ and $\mu_2^{\mathbb{Q}}$ show large and persistence biases under both estimators and are nearly identical for both 1Q and 4Q. As with the short-rate intercept and as discussed earlier, these parameters are unaffected by survey augmentation regardless of horizon by construction because of their identification.

The Cholesky elements Σ_{11} and Σ_{22} show small biases under 1Q of -0.002 and -0.01 respectively. Under 4Q, they move to 0.030 and 0.014 respectively, with slightly wider standard deviations. The opposite signs of the bias and their increased magnitude further indicates that the longer-horizon forecasts distorts the estimation more broadly, and not just on the persistence parameters. This means that the forecasts now distort the estimation enough so the model no longer can separate predictable dynamics from unexpected shocks. Movement that should be attributed to predictability are instead attributed to the shock variance, turning the volatility bias positive for 4Q.

The measurement error standard deviation σ_v is still identical across both estimators at 58.32 basis points as expected.

The short versus long horizon comparison for $T = 216$ shows a clear pattern, that using the four-quarter-ahead forecasts instead of the one-quarter-ahead forecasts does not only fail to improve the NS estimator, but actively worsens the estimation of the physical dynamics across almost every parameter. This is directly consistent with the lower SNR of the four-

quarter-ahead forecasts, the results seen in the data section and findings from Table 14. Using the four-quarter-ahead forecasts does not help reduce the finite-sample bias in our case. The one-quarter-ahead estimator on the other hand outperforms both the NS and the 4Q, resulting in considerably less bias and more precise estimates. The results support the conclusion that when incorporating survey expectations into term structure estimation, the forecast horizon matters considerably where it is preferred to use shorter forecast than longer forecasts.

8.2 Short Versus Long Small Sample

Table 20: Finite-Sample Properties: Short- vs. Long-Horizon Survey ($T = 100$)

Parameter	True	Estimator 1Q ($h = 1$)			Estimator 4Q ($h = 4$)		
		Mean	Bias	Std.	Mean	Bias	Std.
$R = 2000, T = 100$							
$\mathbb{P}$-dynamics: Macro AR ($\Phi_{mm}^{\mathbb{P}}$)							
$\Phi_{mm,11}^{\mathbb{P}}$	0.86277	0.84779	-0.01497	0.04299	0.63387	-0.22889	0.09214
$\Phi_{mm,21}^{\mathbb{P}}$	0.02039	0.00700	-0.01339	0.03095	0.07622	+0.05583	0.07001
$\Phi_{mm,12}^{\mathbb{P}}$	0.07422	0.04169	-0.03253	0.04654	0.05914	-0.01507	0.08310
$\Phi_{mm,22}^{\mathbb{P}}$	0.89448	0.91980	+0.02532	0.03344	0.69246	-0.20202	0.09518
Short-rate intercept							
α	0.01260	0.01170	-0.00089	0.00358	0.01170	-0.00090	0.00354
Risk-neutral intercept ($\mu_m^{\mathbb{Q}}$)							
$\mu_1^{\mathbb{Q}}$	0.31091	0.02382	-0.28709	0.09644	0.02512	-0.28579	0.09846
$\mu_2^{\mathbb{Q}}$	-0.05002	-0.01679	+0.03323	0.09704	-0.01509	+0.03493	0.09562
Volatility: Cholesky (Σ_{mm})							
Σ_{11}	0.48297	0.47818	-0.00480	0.03433	0.51258	+0.02960	0.03990
Σ_{21}	0.01764	0.01234	-0.00529	0.04324	0.01560	-0.00204	0.04882
Σ_{22}	0.43448	0.42882	-0.00566	0.03187	0.44667	+0.01219	0.03610
Measurement error std.							
σ_v (bps)	59.11305	57.77472	-1.33833	2.13144	57.77472	-1.33833	2.13144

Notes: Single DGP: V6 parameters (1Q-ahead survey-augmented MCSE). Estimator 1Q = survey-augmented, $h = 1$ (dpgdp3/UNEMP3; $\phi_{\text{infl}} = 0.881$, $\phi_{\text{unemp}} = 0.995$). Estimator 4Q = survey-augmented, $h = 4$ (dpgdp6/UNEMP6; $\phi_{\text{infl}} = 0.788$, $\phi_{\text{unemp}} = 0.958$). Bias = $\bar{\hat{\theta}} - \theta_0$. SD = standard deviation of estimates across replications.

Table 20 repeats the short versus long horizon comparison for the shorter sample of $T = 100$. Since the $T = 216$ results already showed that 4Q performs worse than both 1Q and NS, it is worth ensuring that this pattern is consistent and is not reversed at shorter samples. The expectation is that the results seen in $T = 216$ will persist or worsen, consistent with the finding in the previous section, that finite-sample bias is amplified at shorter samples.

The $\mathbb{P}$ -dynamics results show that the 4Q estimator again produces substantially larger

biases than 1Q on both diagonal parameters. The bias on $\Phi_{mm,11}^{\mathbb{P}}$ is -0.229 under 4Q compared to -0.015 under 1Q. This is of similar magnitude as seen at $T = 216$, where the biases were -0.193 and -0.003. For unemployment, $\Phi_{mm,22}^{\mathbb{P}}$, the bias is -0.202 under 4Q, compared to a minor overcorrection of 0.025 under 1Q. The standard deviations under 4Q are considerably larger than 1Q, at 0.092 and 0.095 compared to 0.043 and 0.033. This confirms that the 4Q estimator is both more biased and less precise, as seen at $T = 216$. Comparing across the two sample sizes, the biases under 4Q are larger in absolute terms at $T = 100$, consistent with the expected amplification of finite-sample problem at shorter sample. The 1Q estimator still outperforms the 4Q on both diagonal elements, supporting the conclusion that short horizon forecasts are preferred, even at smaller sample sizes.

The off-diagonals of the $\mathbb{P}$ -dynamics show the same pattern as at $T = 216$: the bias of $\Phi_{mm,21}^{\mathbb{P}}$ is larger and the bias of $\Phi_{mm,12}^{\mathbb{P}}$ is smaller, both with roughly double standard deviations.

The short-rate intercept α and the risk-neutral intercepts $\mu_1^{\mathbb{Q}}$ and $\mu_2^{\mathbb{Q}}$ remain virtually unaffected by the choice of survey horizon, since these are identified by the cross-section rather than the time-series dimension, as discussed earlier.

The Cholesky elements show the same sign reversal under 4Q as seen at $T = 216$, with Σ_{11} and Σ_{22} moving from -0.005 and -0.006 under 1Q to 0.030 and 0.012 under 4Q respectively. The standard deviations are slightly wider than the larger sample, consistent with the increased estimation uncertainty at shorter samples.

The measurement error standard deviation σ_v remain identical across both 1Q and 4Q at 57.77 basis points, as expected.

The smaller sample at $T = 100$ confirm that the underperformance of the four-quarter-ahead forecasts relative to the one-quarter-ahead forecasts is not a feature of large samples only. This underperformance, however, persists when data is limited and is even slightly amplified. Across both sample sizes 4Q consistently produces larger biases and wider estimation uncertainty compared to 1Q. The results establish that the forecast horizon is an important determinant of survey augmentation effectiveness where shorter horizons deliver meaningful finite-sample bias corrections, while longer horizons introduce too much noise and actively worsen the estimation relative to using no-survey augmentation at all.

9 Impulse Response Analysis

To round off the analysis of this paper we use the estimated models and Monte Carlo simulation results to show the dynamic response of the yield curve to structural macroeconomic shocks. The preceding Monte Carlo study showed that survey-augmented estimation substantially reduces the small-sample bias in the inflation persistence parameter, but over-corrects in the case of unemployment. Our impulse response analysis attempts to turn the model parameters, that are somewhat difficult to interpret economically, into something that is economically interpretable.

9.1 How IRFs Are Computed

We want to compute the impulse response function (IRF) given the model, where an IRF basically states: if exactly one structural positive shock happens at time t how does the dynamics of the model respond at horizon h . In this way one can analyse how the model-implied 'economy' reacts to an economically interpretable shock, e.g. an inflation shock, over period and see how long it takes for each factor to revert back to its initial position. We will refer to positive shocks as just shocks from here on out. Recall that the annualised model-implied yields for maturity k is given by

$$y_t^{(k)} = g_0^{(k)} + \mathbf{G}_x^{(k)} \mathbf{x}_t,$$

the state dynamics under the $\mathbb{P}$ -measure is given by

$$\mathbf{x}_{t+1} = \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}}, \quad \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}} \sim \text{iid}(0, \mathbf{I}),$$

and the short-rate can be found via

$$r_t = \alpha + \boldsymbol{\beta}^{\top} \mathbf{x}_t.$$

Because $y_t^{(k)}$ is linear in $\mathbf{x}_t$, and $\mathbf{x}_t$ is then linear in $\mathbf{x}_{t-1}$ we can compute the response of any yield in the model to a structural shock analytically. Let j be the index of structural shocks where there exists one structural shock per model factor. Recall that the ordering of

the states are

$$\mathbf{x}_t = \begin{bmatrix} x_t^{m_1} \\ x_t^{m_2} \\ x_t^{l_1} \\ x_t^{l_2} \end{bmatrix} = \begin{bmatrix} \text{Inflation} \\ \text{Unemployment} \\ \text{Latent factor 1} \\ \text{Latent factor 2} \end{bmatrix},$$

so $j = \{1, 2, 3, 4\}$ where $j = 1$ is a structural shock to inflation, $j = 2$ is a structural shock to unemployment and so on. Now let $\mathbf{e}_j$ denote the j -th standard basis vector where element j is equal to 1 and the rest of the entries are all 0. The isolated j -th structural shock is then given by

$$\boldsymbol{\epsilon}_j^{\mathbb{P}} = \boldsymbol{\Sigma} \mathbf{e}_j. \quad (87)$$

Given our independence assumption between the macro and the latent block matrix $\boldsymbol{\Sigma}$ is block-diagonal given by

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & 0 & 0 & 0 \\ \Sigma_{21} & \Sigma_{22} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where the upper block is the lower Cholesky factor of the macro block and the lower block is the identity matrix for the latent factors. What this means is that macro shocks never directly hit the latent factors, and latent factor shocks never directly hit the macro factors.

By ordering, inflation shocks propagates through to the unemployment factor but not vice versa. Now to derive the formula for the IRF iterate the VAR(1) dynamics under $\mathbb{P}$ that gives us

In the same period!

$$\mathbf{x}_{t+h} = \sum_{j=0}^{h-1} (\boldsymbol{\Phi}^{\mathbb{P}})^j \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^h \mathbf{x}_t + (\boldsymbol{\Phi}^{\mathbb{P}})^{h-1} \boldsymbol{\Sigma} \mathbf{e}_j.$$

The IRF is defined as the difference between the shocked and 'unshocked' paths

$$\mathbf{x}_{t+h|\mathbf{u}_{t+1}=\mathbf{e}_j} - \mathbf{x}_{t+h|\mathbf{u}_{t+1}=\mathbf{0}} = (\boldsymbol{\Phi}^{\mathbb{P}})^{h-1} \boldsymbol{\Sigma} \mathbf{e}_j, \quad (88)$$

and given the linearity of yields the IRF for yields for a given horizon h , maturity k , and structural shock j is finally

$$\mathbf{IRF}_y^{(k,j)}(h) = \mathbf{G}_x^{(k)} (\boldsymbol{\Phi}^{\mathbb{P}})^{h-1} \boldsymbol{\Sigma} \mathbf{e}_j. \quad (89)$$

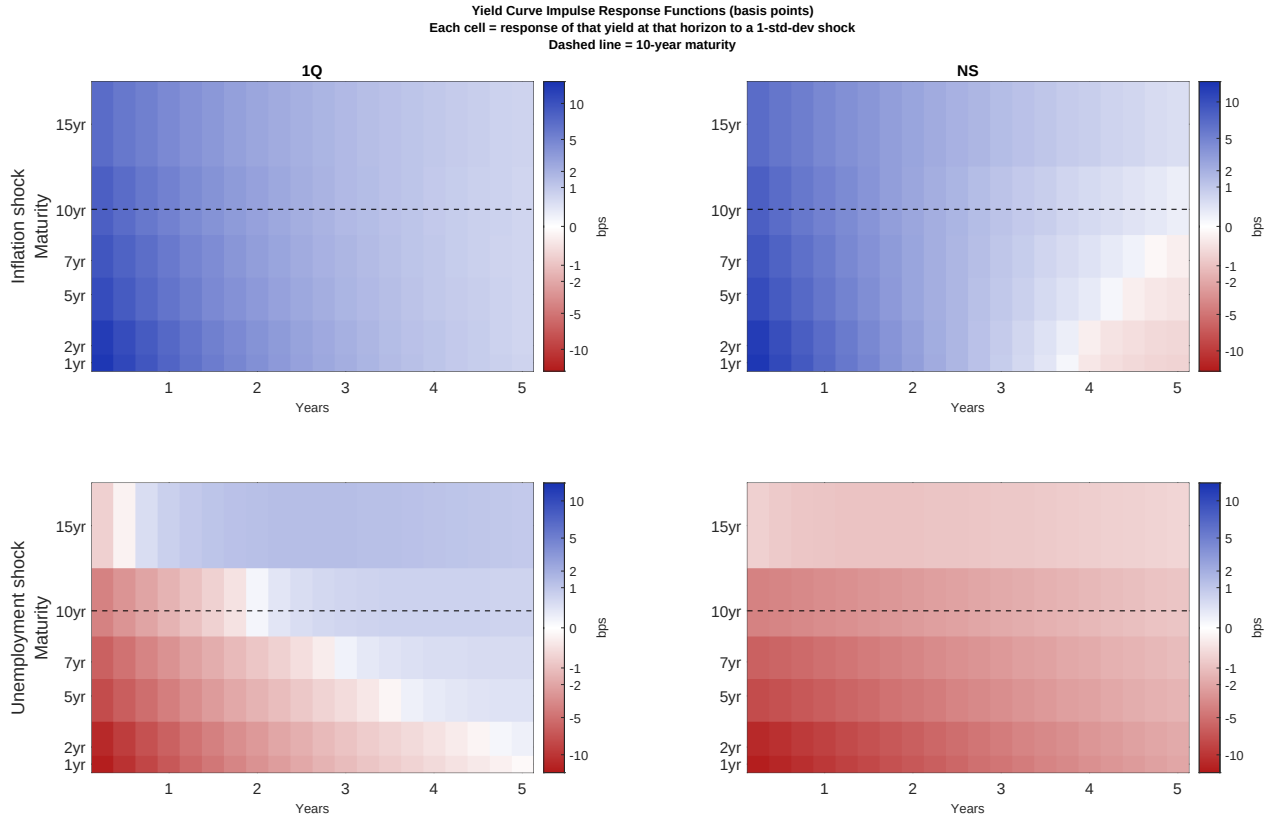


Figure 12: Yield Curve Heatmap of Macro Shocks

Given that we now have the IRF for model-implied yields we can compute the impulse responses of the decomposition using equations (19) and (21). We can similarly compute the impulse response for the short-rate r_t under the $\mathbb{P}$ -measure as

$$\mathbf{IRF}_r^{(k,j)}(h) = \beta^\top (\Phi^{\mathbb{P}})^{h-1} \Sigma \mathbf{e}_j, \quad (90)$$

and for the short-rate impulse response under $\mathbb{Q}$ we replace $\Phi^{\mathbb{P}}$ with $\Phi^{\mathbb{Q}}$. The difference between the short-rate response under $\mathbb{P}$ and $\mathbb{Q}$ then gives us a measure of the risk premium component of the short-rate which tells us how much the market under- or over-values the short-rate path relative to what it is actually expected to do under $\mathbb{P}$.

9.2 Yield Response of Structural Shocks

In Figure 12 is displayed a heatmap for both structural macro shocks for Estimator NS and 1Q. The two quadrants to the left are the response of Estimator 1Q to a inflation and unemployment shock, while the two blocks to the right are for Estimator NS. The way to interpret the figure is that the x-axis shows the time elapsed in years after the structural shock. The y-axis represents the 1, 2, 5, 7, 10, and 15 years maturities. The heatmap is constructed such that there exists a rectangular cell for each maturity for every quarter in time elapsed since the shock. The colour of each cell represents the response of the yield curve at the corresponding maturity on the y-axis to a 1 standard deviation shock, where blue indicates an increase in bps and red a decrease.

Generally, we see that both estimators show the same on-impact response to both the inflation shock but also the unemployment shock. Given an inflation shock both estimators initially react with an increase in the yield curve with the largest increase occurring at the short maturities 1 and 2 where the magnitude of the effect monotonically declines as the maturity increases. This response aligns with the Fisher effect of monetary theory where investors demand higher nominal interest rates following higher inflation to retain their initial purchasing power. Since the short maturities experience a larger relative increase than the longer maturities the slope of the yield curve decreases on impact.

For Estimator 1Q the response of the shock is decreasing but uniformly blue as time passes with no sign-reversals, in short, the survey augmentation model in response to an inflation shock produces a clean upward shift in the yield curve that fades over time consistent with a well-identified physical measure in which inflation is transitory. The drastic difference between the two estimators is that the NS display a sign-reversal starting 4 years after the initial shock basically saying that an inflation shock will eventually lower the yield at the short maturities below what the yields initially were.

What drives this difference is largely due to the difference in persistence between the two estimators. Recall that the 1Q inflation persistence estimate is $\Phi_{mm,11}^{\mathbb{P},1Q} = 0.863$ and the NS estimate is $\Phi_{mm,11}^{\mathbb{P},NS} = 0.88$. Although the difference appears small in absolute terms, repeated uses of $(\Phi)^h$ over longer periods will result in a distinctive difference long-term (for instance, at $h = 20$, $0.881^{20} - 0.863^{20} = 0.024$). An explanation is that the larger persistence for the NS initially pushes the short yields up hard on-impact but the dynamics later overshoots and the model then predicts a mean-reversion that is below that of the baseline. Economically speaking a pure inflation shock should not result in lower shorter yields. In this sense Estimator 1Q is more aligned with economic theory than the NS.

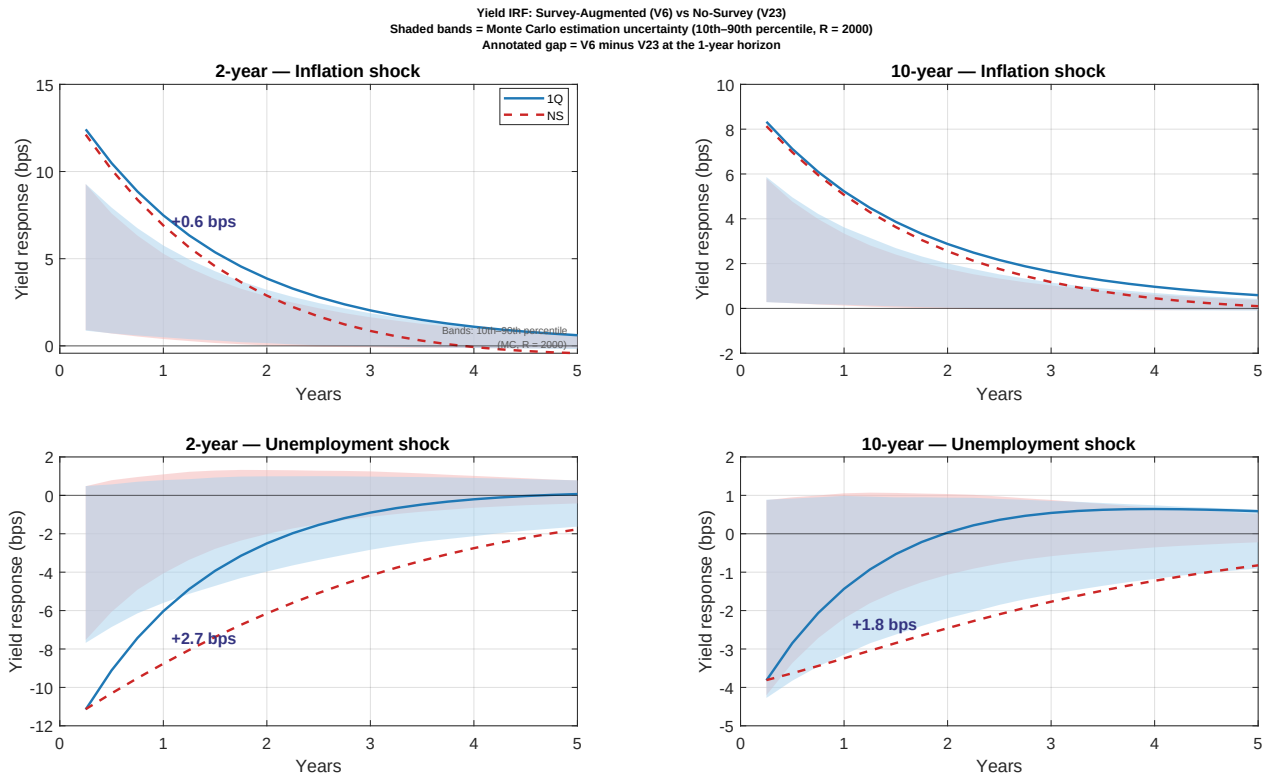


Figure 13: Yield Impulse Responses

Following an unemployment shock both estimators show same responses on impact with a downward shift in the yield curve and a decrease in the yield slope. Economically this is a plausible response since the Federal Reserve would start to perform monetary easing to combat the higher unemployment by lowering interest rates, and investors would then revise their expectations about the short-rate to be lower, where the expected short-rate channel weakens with maturity. The estimators start to differ three quarters after the shock. The NS shows a uniformly red response which declines with time while the 1Q starts to show a sign-reversal for the 15-year maturity three quarters in and a sign-reversal for all other maturities as time passes.

For the unemployment shock we cannot attribute the difference in impulse responses for 1Q and NS based on their persistence parameters since the parameters are near-identical ($\Phi_{mm,22}^{P,1Q} = 0.894$, $\Phi_{mm,22}^{P,NS} = 0.897$). Likewise the unemployment element of β and the Σ_{mm} elements are also barely indistinguishable across the two estimators. The largest parameter difference between the two estimators is the $\Phi_{mm,12}^P$ parameter which governs the feedback from current unemployment into future inflation dynamics. The 1Q estimate for this is $\Phi_{mm,12}^{P,1Q} = 0.074216$, implying that an increase in unemployment will result in a positive

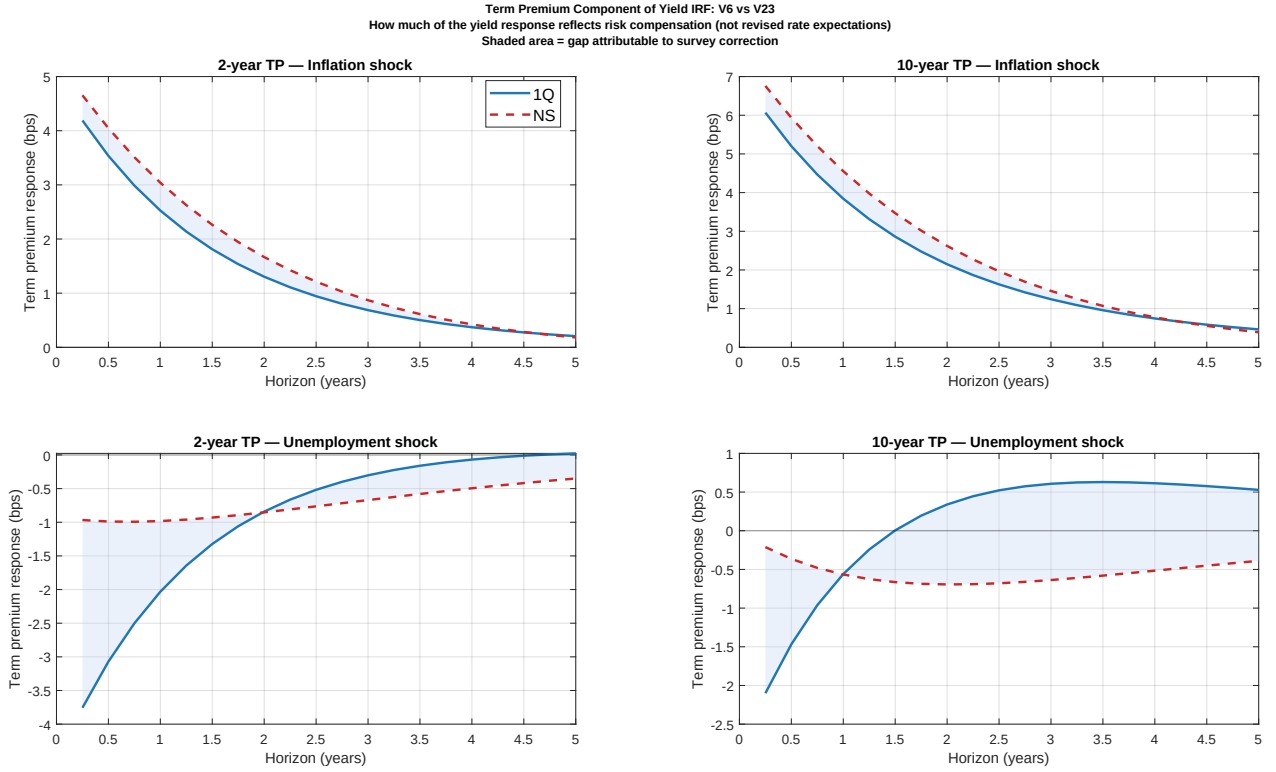


Figure 14: IRF: Term Premium Component

spillover on inflation over the next period.

An economic explanation for this could be that labour market slack first generated disinflationary pressure that the Fed partially accommodates, but where second-round effects eventually push inflation upward. The NS result for the same parameter is $\Phi_{mm,12}^{\mathbb{P},NS} = -0.028094$ which means that a rise in unemployment feeds negatively into inflation. Economically, this is essentially that increased labour market slack causes wage growth and consumer demand to slow down thus lowering inflation. Both estimates have reasonable economic explanations.

Most importantly is that the difference in signs between the two estimators results in fundamentally different joint macro dynamics following an unemployment shock. For Estimator 1Q we see that the unemployment shock partially transmits into the inflation factor over time, thus activating the inflation loading ($\beta_{infl}^{1Q} = 0.0007746$) in the short-rate equation and then reduces some of the yield decline. On the NS side, the unemployment shock essentially suppresses inflation, thus reinforcing the negative short-rate response through $\beta_{infl}^{NS} = 0.000746$ rather than trying to offset it. It is this mechanic that primarily drives the difference in responses to an unemployment between the two estimators, which is shown in Figure 13, where Figure 13 shows the yield impulse responses for maturities 2 and 10 to an inflation

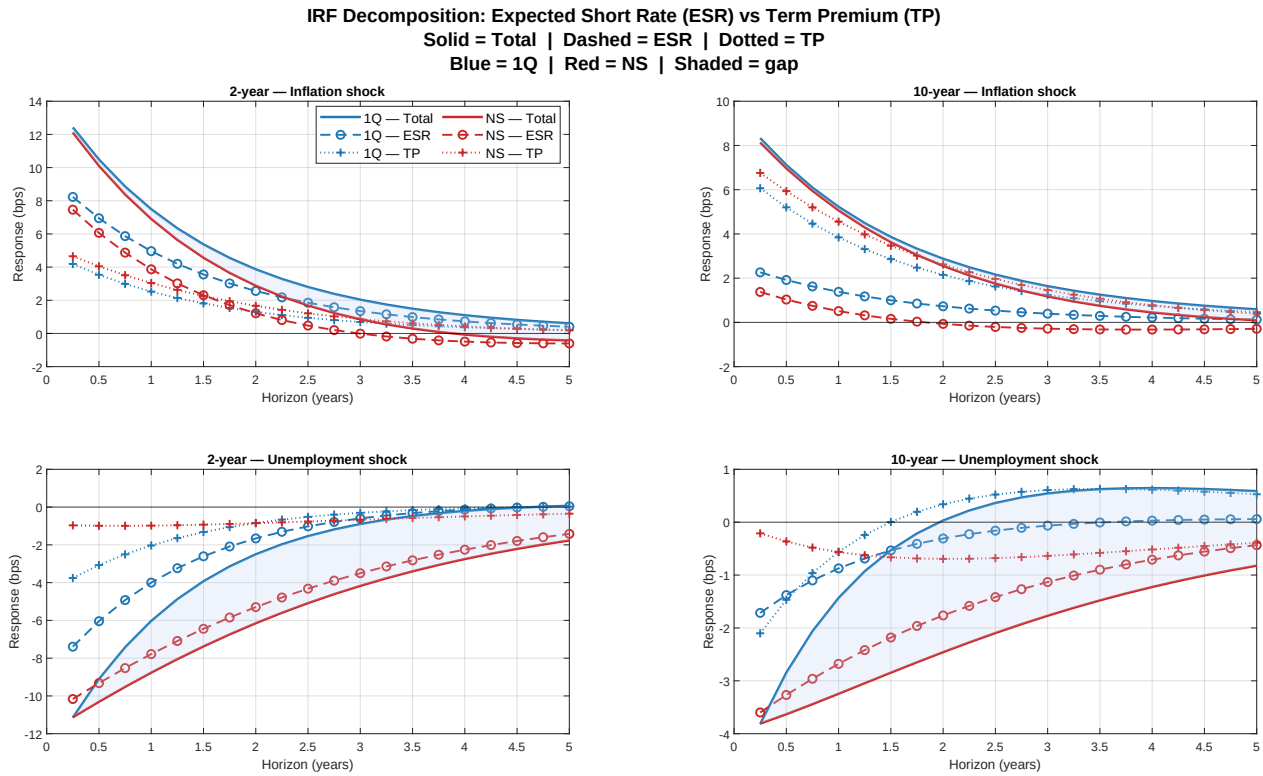


Figure 15: IRF: Yield Decomposition

and unemployment shock.

The 1Q sign-reversal somewhat reflects the economically meaningful operation of the term premium channel. We see that the term premium component of the yield curve starts becoming positive as can be seen in Figure 14. Figure 14 plots the impulse responses of the term premium component of the yield curve following an inflation shock and an unemployment shock for the 2 and 10-year bond yields for estimators 1Q and NS. The term premium component in response to an inflation shock starts becoming positive after 6 quarters and that is around the time where the impulse response of the 10-year yield starts changing signs, and this is where the positive response of the term premium starts to offset and eventually reverse the effect of the negative expected short-rate component at longer maturities. This happens because investors start demanding slightly greater compensation for interest rate risk in an environment of heightened economic uncertainty. For the NS we see that in Figure 14 the impulse responses to an inflation shock results in fall in term premium for the 2 and 10-year maturities, which is an explanation for the yield curve response to show no sign of sign-reversal or mean-reversal over the 5 years following the shock.

A comparison between the response of the expected short-rate and the term premium is

plotted in Figure 15 and makes some of the points discussed above more clear. Looking at the decomposition of the NS 2-year maturity yield following an unemployment shock, the total response is almost entirely due to the expected short-rate, where the response of the term premium is small and reverts back to its initial position almost linearly. Likewise for the 1Q case the expected short-rate makes up a majority of the impulse response, but not with the same large difference as in the NS case.

For the NS estimator's 10-year maturity yield the response of an unemployment shock is almost identical to that of the 2-year yield with the only difference being that the response of the expected short-rate is slightly lower and takes a slight convex form over time.

For the 1Q estimator we see that the initial response of the term premium is slightly more negative than the expected short-rate given that on impact the unemployment shock compresses the term premium, and falling expected short-rates reduce the price of interest rate risk. Later the term premium response turns positive at around 6 quarters in, and from there onwards starts offsetting the negative response of the short-rate, resulting in a net positive yield response to the unemployment shock. The inclusion of surveys therefore has a large impact on the longer maturities primarily because it changes the relative contribution of the two yield curve components. In Figure 15 we see that the responses of the two yield components to an inflation shock remain largely the same in their shape and relative contribution both at the short maturity and long maturity and only differs in magnitude, which is due to symmetric signs in the inflation to unemployment channel ($\Phi_{mm,21}^{\mathbb{P},NS} = 0.045$, $\Phi_{mm,21}^{\mathbb{P},1Q} = 0.0204$).

Our IRF analysis of the yield has translated the results of our survey augmentation into easier interpretable economic terms. Generally, both estimators agree on the on-impact responses to each of the macro shocks, but starts diverging in their persistence, shape, and relative composition at the longer post-shock horizons, all due to differences in $\Phi_{mm}^{\mathbb{P}}$. How macro shocks are transmitted to bond yields are sensitive to whether the physical-measure dynamics are correctly identified. While we make no claim about which estimator recovers the true physical dynamics, the inflation sign-reversal under the NS provides theoretical grounds for preferring the survey-augmented specification.

9.3 State Vector Responses

To get a better understanding of what drives the divergence in the yield impulse responses between the two estimators, it is useful to trace the shock transmission directly at the state level before they can transmit through to the yield loadings. Figure 16 shows the decomposition of the impulse responses down to the level of the state vector itself. For estimators 1Q and NS, the figure plots the within-block responses of each factor to the structural shocks of each factor within its block. Recall that given our independence assumption blocks Φ_{mm} and Φ_{ll} are independent. The plots within the figure are plotted such that they mimic the Φ blocks

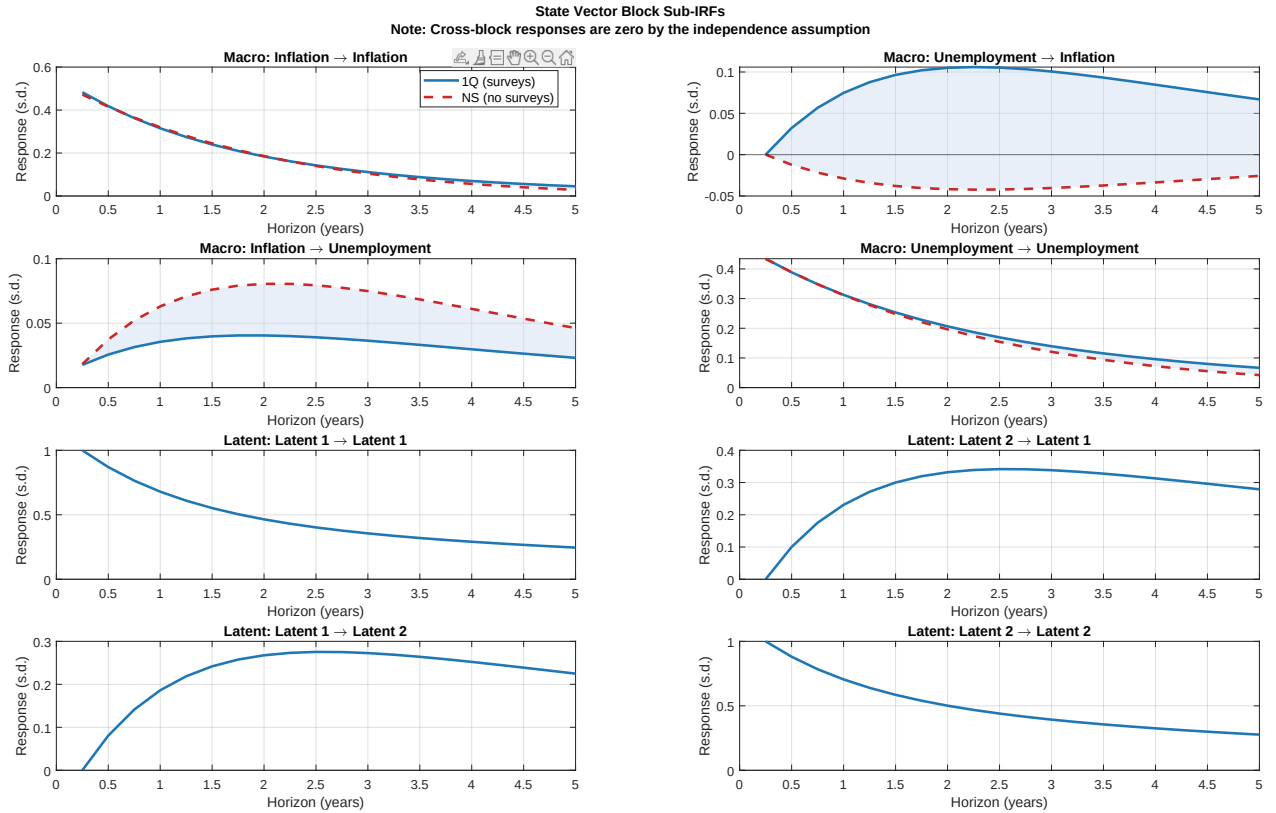


Figure 16: Factor Impulse Responses

where the diagonals show the own-persistence, and the off-diagonals show the intra-macro effects.

The inflation on inflation panel shows that the two responses are nearly indistinguishable at the short horizon and only gradually separates at the longer horizon. At $h = 20$ quarters the propagated responses differ only by around $0.881^{20} - 0.863^{20} \approx 0.0268$ which is roughly an 18 percent larger residual response under NS relative to 1Q but still large enough over long periods to produce the sign-reversal in Figure 12. The unemployment on unemployment

panel shows the same result where the responses follow each other at the start and only starts to differ in the mid to long horizon. The off-diagonal macro effects are the more interesting and informative panels to look at.

The unemployment on inflation panel shows the key difference between the two estimators directly at the state vector level. For the 1Q estimator the response is positive and persistent peaking at around 1 standard deviation, reflecting the spillover from unemployment into future inflation as we mentioned in the yield impulse response section. Under the NS the response is negative and reflects the estimated negative spillover.

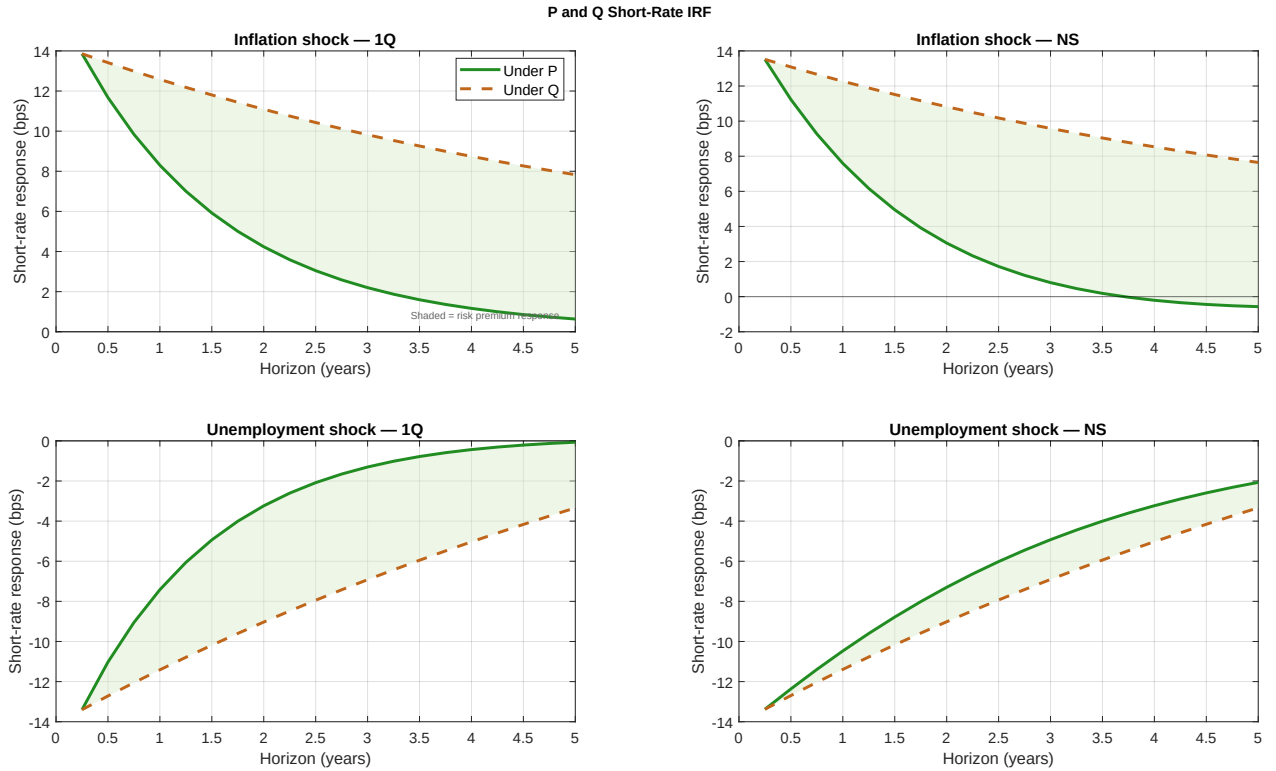


Figure 17: Short-Rate Impulse Response under $\mathbb{P}$ and $\mathbb{Q}$

The inflation on unemployment is positive throughout for both estimators but is larger for the NS which is consistent with our estimates where the NS spillover is larger ($\Phi_{mm,21}^{\mathbb{P},\text{NS}} = 0.045$) than that of the 1Q ($\Phi_{mm,21}^{\mathbb{P},1\text{Q}} = 0.0204$).

Given that the latent block estimates are identical across estimators under $\mathbb{P}$, the ones plotted are for the 1Q. What we can tell is that the impulse response of the latent block is positive for all within structural shocks.

Figure 17 plots the impulse response of the short-rate under $\mathbb{P}$ and $\mathbb{Q}$ and offers a complementary perspective on the shock transmission mechanism. The shaded area between the

two responses is the difference given by

$$\mathbf{IRF}_r^{\mathbb{P},(k,j)}(h) - \mathbf{IRF}_r^{\mathbb{Q},(k,j)}(h),$$

and is the risk premium component of the short-rate IRF. Since $\Phi^{\mathbb{Q}}$ is identical across estimators any differences between them can be contributed to their differences in $\Phi^{\mathbb{P}}$.

Given an inflation shock the $\mathbb{Q}$ path lies above the $\mathbb{P}$ path at all horizons and for both estimators. The market prices inflation as being more persistent than what the $\mathbb{P}$ -dynamics would imply thus investors are demanding a higher risk premium compensation in light of heightened inflation uncertainty over the life of the bond. The gap is slightly larger for the NS because the less persistent $\mathbb{P}$ path under 1Q brings $\mathbb{P}$ further away from $\mathbb{Q}$.

For the unemployment shock the two estimators starts diverging already right at impact. The risk-neutral expected short-rate fall by a larger magnitude than the $\mathbb{Q}$ path across estimators, where the fall for the 1Q estimator becomes drastically larger than that of the NS over time. The large wedge between the two measures for the 1Q implies that investors pricing bonds under $\mathbb{Q}$ expects a more severe and prolonged monetary easing than what the $\mathbb{P}$ -dynamics predicts. For the NS this wedge gradually widens at the 2 year horizon but the gap slowly closes as we approach the 5-year horizon.

9.4 Variance Decomposition

Table 21: Forecast Error Variance Decomposition

Mat.	Estimator 1Q					Estimator NS				
	$h = 1Q$	1Y	2Y	5Y	10Y	$h = 1Q$	1Y	2Y	5Y	10Y
Inflation										
1y	2.4	2.1	1.7	1.1	0.9	2.3	1.8	1.3	0.8	0.7
2y	2.6	2.4	2.1	1.5	1.3	2.3	1.7	1.2	0.8	0.6
5y	2.7	2.9	2.9	2.5	2.2	2.1	1.5	1.1	0.6	0.5
7y	2.4	2.8	2.9	2.8	2.6	2.0	1.4	1.0	0.6	0.5
10y	2.1	2.4	2.7	2.8	2.8	1.8	1.3	1.0	0.6	0.5
15y	1.7	2.0	2.3	2.6	2.6	1.8	1.4	1.1	0.7	0.5
Unemployment										
1y	2.1	1.7	1.3	0.9	0.7	2.1	2.1	2.0	1.5	1.3
2y	2.1	1.8	1.5	1.0	0.8	1.9	1.9	1.7	1.3	1.1
5y	1.5	1.5	1.3	1.1	1.0	1.3	1.2	1.0	0.8	0.6
7y	1.0	1.0	0.9	0.8	0.8	0.9	0.8	0.7	0.5	0.4
10y	0.4	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.3	0.2
15y	0.0	0.0	0.1	0.3	0.3	0.0	0.0	0.0	0.0	0.0
Latent 1										
1y	95.2	91.4	84.2	71.6	66.5	2.9	7.5	13.2	21.8	25.1
2y	94.8	94.0	88.7	76.5	71.0	9.4	14.7	20.0	27.0	29.6
5y	82.0	86.8	88.1	82.8	78.8	29.1	31.8	34.1	36.7	37.6
7y	72.4	77.2	80.1	78.8	76.7	37.6	38.5	39.2	40.1	40.4
10y	61.8	65.1	67.5	68.4	67.8	45.1	44.3	43.7	42.9	42.7
15y	51.8	52.8	53.6	53.9	53.9	51.2	48.9	47.2	45.2	44.5
Latent 2										
1y	0.3	4.9	12.8	26.4	31.8	92.7	88.6	83.5	75.9	72.9
2y	0.5	1.8	7.7	21.0	26.9	86.4	81.7	77.1	71.0	68.7
5y	13.8	8.8	7.7	13.7	18.0	67.5	65.5	63.8	61.9	61.2
7y	24.1	19.0	16.1	17.5	19.9	59.6	59.3	59.1	58.8	58.7
10y	35.7	32.1	29.4	28.4	29.0	52.7	54.0	55.0	56.2	56.6
15y	46.5	45.2	44.0	43.2	43.2	47.0	49.7	51.8	54.1	55.0

The impulse responses gave us an overview of how the factors in our models affect the dynamics of the yield curve following structural shocks. Another way to get meaningful economic interpretations of how our factors affect the yield curve is by computing the forecast error variance decomposition (FEVD). Table 21 reports the FEVD for both estimators across six maturities and five forecast horizons, following the method of [Ang and Piazzesi \(2003\)](#). Formally, the share of total forecast variance attributable to shock j is

$$\Omega_{k,j}(h) = \frac{\text{MSE}_j^{(k)}(h)}{\text{MSE}^{(k)}(h)} = \frac{\sum_{i=0}^{h-1} \Psi_{i,k,j}^2}{\sum_{j=1}^4 \sum_{i=0}^{h-1} \Psi_{i,k,j}^2}, \quad (91)$$

where $\Psi_{i,k,j} \equiv \mathbf{G}_x^{(k)} (\Phi^{\mathbb{P}})^i \Sigma \mathbf{e}_j$, and is closely related to the IRF specification¹²

Across both estimators the combined macro contributions remain below 5% of the total variance at all maturities and horizons. Overall macro variables makes up a very little share of the total forecast error variance and is consistent with the findings in e.g. [Diebold et al. \(2006\)](#) who report macro shares of 8%, 4%, and 2% at the one-month horizon for the 1-, 12-, and 60-month yields, respectively. Our findings are the opposite of what is found in [Ang and Piazzesi \(2003\)](#) where macro factors make up 50% of the share at the $h = 1$ and increases as the forecast period becomes larger.

The big difference between the latent 1 and latent two factors between the two estimators is purely rotational. $\Phi_{ll}^{\mathbb{P}}$ is identical across estimators but the estimates for β for the two latent block are rotated which propagates through $\mathbf{G}_{x,ll}$ during recursion.

¹²See Appendix A4 for derivations.

10 Discussion

This section provides an overview of the findings of this paper, relates them to the broader term structure literature, and reflects on the methodological choices made throughout the paper. In this paper we apply a stepwise model extension, where we start by introducing a two-factor latent GATSM to set a baseline, which we use to show the existence of the finite-sample bias that is prevalent in affine term structure models. After that we extend the specification to a four-factor macro-finance GATSM by incorporating inflation and unemployment alongside the two previous latent factors. We thereafter augment our four-factor model with macro surveys to analyse whether or not survey expectations can help in mitigating finite-sample bias. Each extension is motivated by a distinct research question: whether observable macroeconomic variables contain information beyond latent yield factors, and whether survey expectations can improve the estimation and reduce the finite-sample bias present in the model.

10.1 Overview of Findings

The analysis of this paper includes three model specifications, each stepwise extending a baseline GATSM. The two-factor latent GATSM establishes a baseline and produces a yield decomposition showing a term premium being positive on average, increasing with maturity, and peaking during the high-uncertainty period in the 1980s. The Monte Carlo study for this model reveals a clear and systematic negative bias in the diagonal persistence parameters of $\Phi^{\mathbb{P}}$, consistent with the well-documented underestimation of persistence in near-unit-root autoregressive systems. This provides the central motivation for the model extensions that follows.

The four-factor macro-finance GATSM extends the baseline two-factor model by including inflation and unemployment as observable state variables. Their inclusion produces a smoother term premium series relative to the two-factor model, as the macro factors absorb yield variation that was previously entirely absorbed by the latent factors. Despite the extended model, the finite-sample bias persists, motivating the survey augmentation.

The survey-augmented model is the central focus of the paper. Incorporating one-quarter-ahead SPF forecasts into the estimation of the physical macro dynamics via OLS stacking shows a reduction in the persistence parameters of $\Phi_{mm}^{\mathbb{P}}$, with the reduction becoming more pronounced at shorter sample sizes, where the conventional NS estimator showed higher bias. The survey augmentation elevates the term premium considerably relative to the no-survey

four-factor model, as SPF forecasts reflect real-world expectations about future macro outcomes, providing additional information about the physical dynamics beyond what is seen in realised macro data. However, the bias reduction comes with a trade-off. While the bias in inflation persistence is nearly eliminated, the unemployment persistence estimate over-corrects and turns positive, an indication that unemployment surveys anchor the persistence parameter below the true DGP. This would suggest that SPF forecasters systematically expect unemployment to mean-revert faster than it actually does. Finally, the comparison between the short and long horizon surveys clearly shows that the forecast horizon matters for the survey augmentation effectiveness, with one-quarter-ahead forecasts considerably outperforming four-quarter-ahead forecasts due to their substantially higher signal-to-noise ratio.

10.2 Relation to the Literature

The main motivation for our inclusion of surveys is the paper by [Kim and Orphanides \(2012\)](#). They show by using a three-factor latent GATSM, that they can substantially mitigate finite-sample bias by augmenting their model with survey forecasts of interest rates. Their survey forecasts are from the Blue Chip Economic Indicators, with forecasts of the three-month T-bill rate, which enters directly into the Kalman Filter measurement equation, where we in our paper instead use survey forecasts on inflation and unemployment from the SPF, and augment our model with surveys in a stacking OLS manner. The main mechanism for why surveys can be used to reduce finite sample bias lies in the physical dynamics of the state equation, where surveys provide direct cross-sectional observations of the conditional expectation of the state vector, thus anchoring the physical dynamics to real forward-looking expectations.

[Kim and Orphanides \(2012\)](#) find that by running Monte Carlo simulations their median values of the persistence parameters (the diagonal of Φ^p) using surveys are closer to the true parameters than without surveys. They likewise find that a longer sample without surveys anchors the simulated median closer to the true value than with a shorter sample without surveys. In our paper we make use of this same conditional expectation mechanism, but it is implemented in a different way and in a different setting, which reduces the direct comparability between our result and [Kim and Orphanides \(2012\)](#). Our primary Monte Carlo experiment involved setting the survey-augmented GATSM as the data-generating process as in [Kim and Orphanides \(2012\)](#) and using that we simulate 2000 replications each with a sample observation length of 216 quarters. What we find is that the inflation persistence

parameter of the survey-augmented model is anchored closer to the true value compared to the model without surveys. For the unemployment persistence parameter, on the other hand, we find that the surveys somehow over-correct i.e. the bias for the NS model is negative but the survey-augmented model shows a positive bias which is an interesting result. An unlikely explanation for this could be that the unemployment persistence parameter is badly identified compared to the model without surveys, which would in turn mean that the inclusion of surveys for unemployment worsens the finite sample bias, but the overall lower standard errors for the survey-augmented model would suggest a better identification than the no-survey model. The more likely explanation is that the calibrated survey slope for unemployment $\hat{\phi}_{m_2}^{svy} = 0.9955$ exceeds that of the unemployment persistence thus pulling persistence upward when incorporating surveys.

[Kim and Orphanides \(2012\)](#) use Blue Chip surveys of interest rates, whereas we employ SPF forecasts of the GDP-deflator inflation and unemployment. The Blue Chip surveys target the endogenous short-rate itself which is informative about $\Phi^{\mathbb{P}}$ through the EH and enters directly into the Kalman filter for maximum likelihood estimation. Our SPF forecasts instead enter the physical dynamics during the Block m estimation. The fact that a statistical bias reduction is obtained even from our more indirect use of surveys reinforces the general robustness of the use of surveys to anchor estimations.

One of the more subtle reasons that surveys can help in reducing small-sample bias as noted by [Kim and Orphanides \(2012\)](#) is that it provides actual real-time information at time t , which we will discuss briefly later.

Now in the context of finite-sample bias, which has been a persistent concern in the literature, there exists other methods to combat finite-sample bias that could have been studied instead. For instance, [Hamilton and Wu \(2014\)](#) propose various ways to bias-correct estimates of VAR parameters such as analytical bias approximation, bootstrap bias correction, and indirect inference. Once these bias-corrected estimates are obtained we can simply replace the standard OLS estimates in our MCSE approach with these bias-corrected estimates instead. Another bias-correction approach is that of [Andreasen and Kristensen \(2025\)](#) which avoids repeatedly having to optimise a log-likelihood function to adjust for finite sample bias in state space models specifically such as our two-factor model.

Macro data is exposed to constant revisions many quarters after they are published which could introduce information beyond time t information which would lead to a look-ahead bias. The use of real-time information using surveys could potentially instead be targeting this look-ahead bias and not the finite-sample bias, or it could help alleviate both. What

is shown in [Ghysels et al. \(2018\)](#) is that revised macro data has a high predictive power for excess bond returns, but if you split the macro data into pre- and post-revision components, the post-revision component makes up a sizeable fraction of the overall predictive power, confirming that revised macro data carries information beyond what was available at time t , introducing a look-ahead bias in the process.

The implications for our model-implied yield decomposition after incorporating surveys are that the implied yields themselves remain largely unchanged, but its decomposition is now markedly different. The term premium for the 1Q model is on average larger than that of the NS term premium for all maturities. Comparing this to the findings in [Kim and Orphanides \(2012\)](#), they find that the implied forward term premium of their survey model is on average lower than that of the no-survey model. This difference in findings is likely due to the different surveys applied. [Kim and Orphanides \(2012\)](#)'s surveys of the 3-month T-bill directly pins down the expected short-rate and greatly improves how their model can pin down the short-rate as can be seen in Figure 4 of [Kim and Orphanides \(2012\)](#). We on the other hand use SPF surveys of inflation and unemployment which does not pin down the expected short-rate directly but indirectly, through the $\mathbb{P}$ -dynamics of the macro states. We do note that we cannot make direct comparisons as [Kim and Orphanides \(2012\)](#) report the forward term premium which is closely related to but not identical to our reported term premium.

The estimation strategy of our model relies on the MCSE approach of [Hamilton and Wu \(2012\)](#), who propose it as a more computationally tractable alternative to the classic maximum likelihood estimation for affine term structure models. In addition, they show that it is asymptotically equivalent to a full-information maximum likelihood. [Hamilton and Wu \(2012\)](#) show that some of the commonly used canonical representations of GATSMs are badly identified, or even unidentified, with one of them being the macro-finance GATSM of [Ang and Piazzesi \(2003\)](#) which our four-factor framework is primarily based on. They propose a new parametrisation which is the equal-diagonal Jordan form for the $\mathbb{Q}$ -measure latent block that achieves identification while accommodating both real and complex conjugate eigenvalues without the need to impose a lower-triangular restriction.

We adopt this parameterisation in our analysis, and our results align with [Hamilton and Wu \(2012\)](#)'s findings in that the latent block under the $\mathbb{Q}$ -dynamics are robustly identified with low standard errors, while the macro block under $\mathbb{Q}$ is not with diagonal estimates of 0.978 and 1.005 with standard errors 0.657 and 0.646, respectively. The use of surveys drastically improves the identification of our macro block where the standard errors for the

diagonal under $\mathbb{Q}$ becomes 0.005 and 0.016, once again reinforcing the effectiveness of surveys to pin down estimates.

Hamilton and Wu (2012) further show that the estimates for the macro model in Ang and Piazzesi (2003) is a local minimum instead of the global, where a notable difference between the global minimum and the one found in Ang and Piazzesi (2003) (Local1) is in the signs of the market price of risk matrix for the macro block. The estimates of Local1 imply that an increase in inflation increases the price of inflation risk and decreases the price of output risk, whereas the estimate at the global minimum reverses the sign of this estimate.

We have in our paper refrained from directly discussing the market price of risk and more so indirectly through the talks about the differences in $\Phi^{\mathbb{Q}}$ and $\Phi^{\mathbb{P}}$. The mechanics that are governed by $\lambda_{x,mm}$ are important to discuss, especially in the light of policy evaluation.

The Fed, and other central banks for that matter, care about the term premia because they affect the long-term cost of borrowing independently of the policy rate that they can directly control. When the Fed adjusts the federal funds rate, they control only the short end of the yield curve, but many longer-term loans depend on the term premia which sets the long maturity yields and thus borrowing costs. It is therefore important for the Fed to know how the term premium responds to a change in policy rate. Because of this, a model that can accurately decompose the yield curve is of great importance for policy making. In this context the matrix $\lambda_{x,mm}$ is particularly significant as it can reveal why the term premium varies, specifically how much of the term premium variation that can be attributed to either changes in inflation or unemployment, precisely the dual mandate of the Fed.

Recall in our model that the macro block of the market price of risk is affine in the state

$$\Lambda_t^m = \Sigma_{mm}^{-1} \lambda_{0,mm} + \Sigma_{mm}^{-1} \lambda_{x,mm} \mathbf{x}_t^m,$$

where $\Sigma_{mm}^{-1} \lambda_{x,mm}$ is the 2×2 sensitivity matrix that governs how each macro factor adjusts to the price of each source of macro shock risk. Specifically, element (i, j) measures the change in market price of risk for factor i when factor j increases by one-unit. Table 22 reports the model-implied market price of risk for the NS and 1Q estimates. We wish to discuss the model-implied market price of risk in comparison to the estimates in Hamilton & Wu and in Ang and Piazzesi (2003) but before that we need to review some of the methodological choices made in our model compared to theirs.

Firstly, Ang and Piazzesi (2003) and Hamilton and Wu (2012) employ three latent factors in addition to the two macro factors where we in our model only use two which alters the identification of the yield blocks and in turn the overidentification of the MCSE objective.

Secondly, their two macro factors are composites of inflation and real activity measures derived using PCA, where we in comparison use the GDP deflator and the unemployment rate directly. Our economic scales will therefore differ.

Lastly, and most importantly, [Ang and Piazzesi \(2003\)](#) and [Hamilton and Wu \(2012\)](#) imposes a lower-triangular canonical form for $\Phi_{lt}^{\mathbb{P}}$ with diagonal elements in descending order following [Dai and Singleton \(2000\)](#), the same restriction we employed in our two-factor model, whereas we in the four-factor model apply the propositions in [Hamilton and Wu \(2012\)](#) instead. Given this our comparison of the estimated market price of risk will be qualitative rather than focusing on magnitudes.

Table 22: Implied Market Price of Risk for the Macro Block

	NS		1Q	
	Infl.	Unemp.	Infl.	Unemp.
$\Sigma^{-1}\lambda_{x,mm} = \Sigma_{mm}^{-1}(\Phi_{mm}^{\mathbb{P}} - \Phi_{mm}^{\mathbb{Q}})$				
Infl. risk	-0.2023	-0.1814	-0.2351	0.0345
Unemp. risk	0.0913	-0.2404	0.0364	-0.2548
$\Sigma^{-1}\lambda_{0,mm} = \Sigma_{mm}^{-1}(\mu_m^{\mathbb{P}} - \mu_m^{\mathbb{Q}})$				
$\lambda_{0,mm}$	0.0844	0.8065	-0.6475	0.1382

For element $[1, 1]$ of $\lambda_{x,mm}$ both of our estimators predict a negative effect for the price of inflation risk when inflation increases which is consistent with [Ang and Piazzesi \(2003\)](#) but inconsistent with [Hamilton and Wu \(2012\)](#)'s global minimum where they expect a positive effect. What a negative effect entails is that when inflation is already high an inflation shock will produce a smaller term premium response than if inflation were low. Economically both effects can be argued for. A positive effect would mean that in situations of high inflation an additional inflation shock would further increase the financial instability, and investors would demand even more in compensation for holding bonds. A negative effect would imply that investors see high inflation as transitory since they expect the Fed to combat this high inflation through open market operations, so when the economy is hit by an additional inflation shock, they see it as less persistent, hence the required term premium for this additional shock is lower.

The NS estimator predicts a negative sign for element $[1, 2]$ while the 1Q model produces a positive sign. The positive sign is consistent with both [Ang and Piazzesi \(2003\)](#) and [Hamilton & Wu](#). What a positive estimate means economically is that when unemployment is high and the Fed faces pressure to ease the interest rate investors become more uncertain

about whether the Fed will also be able to simultaneously anchor inflation, thus raising the compensation required for bearing inflation risk. The reverse sign between the NS and 1Q can be traced back to surveys flipping the sign of $\Phi_{mm,12}^{\mathbb{P}}$ which show how surveys can anchor the *mathbbP*-dynamics to achieve economically meaningful estimates.

Both of our models estimate a positive sign for the $[2, 1]$ element which governs how an increase in inflation affects the market price of unemployment risk. Our estimates are consistent with [Ang and Piazzesi \(2003\)](#) but not in-line with the negative estimate found in the Hamilton & Wu's global minimum.

Lastly the $[2, 2]$ element is estimated to be negative for both estimators and is consistent with the findings in both [Ang and Piazzesi \(2003\)](#) and [Hamilton and Wu \(2012\)](#). We do note that the global minimum does not directly mean that its estimates are the correct in terms of economic interpretation, more so that it represents the best model fit based on log-likelihood value.

As mentioned earlier the benefit of using the MCSE as opposed to Kalman filtering MLE is that we can directly test the over-identifying restrictions. As in [Hamilton and Wu \(2012\)](#)'s own application, the efficiently-weighted test formally rejects these restrictions; this reflects the tightness of the yield-pricing fit rather than a failure of the bias analysis, and the well-identified physical persistence parameters on which our central findings rest are unaffected.

10.3 Alternative Data

Several data choices were made in this paper, like the choice of the GDP deflator as the inflation measure rather than CPI or PCE inflation. This choice is primarily driven by the availability of long and consistent SPF forecast data for the GDP price index, which begins in 1968, allowing the sample to start in 1972. CPI-based SPF forecasts are also available but have a shorter history, requiring a shorter final sample. Since the GDP deflator and CPI exhibit broadly similar dynamics, this choice is not expected to have affected the main findings. One concern worth noting is that the SPF transitions from forecasting the GNP implicit deflator to the GDP price index in 1992, while the realised series uses GDP throughout. Since the analysis uses growth rates rather than levels or the GDP deflator, and since GNP and GDP are very similar for the U.S., this transition is not expected to meaningfully affect the results.

The unemployment rate is chosen to capture real activity. This is chosen due to the direct definition and the long history as a SPF forecast variable. An alternative would be to follow [Ang and Piazzesi \(2003\)](#) more closely and use a set of real activity measure, but since

we incorporate survey expectations directly into the estimation, having a single well-defined variable with a corresponding SPF forecast series is preferred, making unemployment the most appropriate choice. GDP growth was also considered as an alternative real activity measure, and similarly has a direct SPF forecast series available. Unemployment is preferred as it captures labour market more directly and together with inflation forms the two variables of the Fed's dual mandate as mentioned earlier, making the combination natural in the context of a model examining monetary policy expectations and term premia.

The sample period of 1972:Q1 to 2025:Q4 is chosen to balance time-series length and maturity coverage in the yield data. An alternative would be to start the sample in 1980 when the Svensson yield curve methodology became available, avoiding the earlier period where Nelson-Siegel specification is used. However, this would substantially reduce the available data for estimating the persistent dynamic and would exclude the Volcker disinflation period. The maturity range of 1 to 15 years is chosen to ensure consistent coverage across the full sample. Using only maturities up to 7 years would allow the sample to start earlier, while extending to 30 years maturity would significantly reduce the sample size. Both extremes include a trade-off, and a balance between the two was chosen instead.

The analysis uses the median rather than the mean SPF forecast series. The median is less sensitive to outliers and extreme individual responses, which is relevant in the earlier part of the sample where the number of respondents is smaller. Using the mean could potentially produce different results in periods where responses were markedly skewed. The two series are however relatively close over most of the sample, so this choice is not expected to have influenced the findings meaningfully.

Finally, the estimation of the four-factor models are conducted at quarter frequency to match the availability of SPF survey data, whereas the two-factor model was estimated on a monthly frequency. We acknowledge this mismatch in frequency which results in the four-factor and four-factor models not being completely comparable.

Using monthly yield data directly alongside quarterly surveys could have improved the overall identification of our estimates due to the increased amount of data points but could in turn have introduced temporal aggregation bias. To prevent this one could adopt the mixed-frequency estimation framework of [Forni and Marcellino \(2016\)](#) which alleviates this bias. Implementing this would introduce a new complexity to the already difficult estimation of affine term structure models, and for this reason we refrained from implementing it.

Across the choices made in this paper, the primary consideration has been consistency between the realised macroeconomic data and the available SPF forecast series, which is a

necessary condition for the survey augmentation to be meaningful.

10.4 Alternative Horizon or Timing Choice

The forecast horizon and timing of the survey data is an important part of the paper, and the choices made are important for the findings. The paper uses one-quarter ahead SPF forecasts as the survey input. They are used over longer-horizon forecasts because the analysis showed that shorter horizons carry substantially higher signal-to-noise ratio and outperform longer horizons in reducing finite-sample bias. An intermediate horizon such as two or three-quarters ahead could have been implemented instead, resulting in a balance between the short and long horizon surveys. This was however not chosen because working with the two extremes of the quarterly surveys (one and four-quarters ahead) provided the opportunity to examine the difference in finite-sample bias reducing abilities between the two horizons.

As discussed in Section 7.1, the four-quarter-ahead estimator uses an approximation that conditions on time $t + 3$ information rather than time t information, which could potentially introduce an endogeneity bias that contributes to its poor finite-sample performance on top of the already poor performance that stems from its lower signal-to-noise ratio.

The use of annual rather than quarterly survey horizons was not explored in the paper. Annual forecasts are available in the SPF for both inflation and unemployment, but have a shorter history than the quarterly forecasts and would require additional complexity in the form of mixed-frequency estimation or frequency conversion to match the quarterly framework. These were not pursued given the increased complexity of incorporation and because the quarterly horizon comparison suggests that longer horizons are less effective in reducing finite-sample bias.

The observed yields and the survey forecasts have a slight timing mismatch. SPF forecasts are typically published in the middle of the quarter, while yields are observed daily and are matched to end-of-month observations in this paper. This means that the information sets available to forecasters when forming their expectations and the information set of the end-of-quarter yields are not perfectly aligned. [Kim and Orphanides \(2012\)](#) face a similar timing challenge when incorporating Blue Chip surveys, noting that the precise timing of forecast formation relative to data releases introduces uncertainty about the exact information set of the forecasters.

10.5 Robustness and Sensitivity

The robustness check and sensitivity analyses conducted in the paper aim to strengthen the reliability of the main findings, though they should not be interpreted as a guarantee of uncertainty-free results.

The noisy survey test shows that the bias-correcting properties of the one-quarter-ahead survey augmentation are stable even when the total noise variance of the surveys is doubled, confirming that the results are not an artefact of treating survey data as perfectly measured. The stability across noise levels is consistent with the high SNRs documented for the one-quarter-ahead survey forecasts, suggesting that the surveys carry a strong enough signal so that increases in measurement error do not materially affect the estimation.

The small sample analysis at $T = 100$ quarters confirms that the finite-sample bias is amplified by sample size, roughly doubling the negative bias as the sample shortens. Survey augmentation continues to provide a meaningful correction at shorter samples. Notably, the overcorrection in unemployment persistence observed at $T = 216$ persists at $T = 100$, suggesting that this is a feature of how SPF forecasters form unemployment expectations. This reinforces the trade-off noted earlier, between reducing finite-sample bias and potentially introducing forecast bias from the surveys themselves.

The comparison between CMAES and Nelder-Mead in the two-factor model reveals sensitivity to the choice of numerical optimiser, with the two approaches settling at different local optima and implying different term premium series. This shows that the choice of optimiser can have meaningful consequences for the economic conclusions drawn from the model. The use of CMAES as the preferred optimiser is motivated by its global search properties, which make it less dependent on starting values than the Nelder-Mead.

Finally, the Campbell-Shiller regression results confirm that the two-factor GATSM produces time-varying term premia consistent with the empirical rejection of the expectations hypothesis in U.S. yield data, with model-implied slope coefficients close to those estimated on actual data.

10.6 Policy Implications

Gaussian affine term structure models have become standard tools in central banks and financial institutions for decomposing yields into expected short-rates and term premia, informing monetary policy decisions and assessing market expectations [Kim and Orphanides \(2012\)](#). The accuracy of these decompositions depends critically on the correct identification of the physical dynamics, making the finite-sample bias problem documented in this paper relevant for empirical term structure modellers. By incorporating survey expectations into the estimation of the physical macro dynamics, the survey-augmented model aims to contribute to more reliable term premium estimates, which in turn could improve the quality of the policy-relevant information extracted from the term structure models.

The elevated term premia implied by the survey-augmented model relative to the baseline four-factor model have direct implications for bond investors. Higher premia reflect greater compensation demanded by investors for holding long-term bonds, meaning long-term yields are higher than what expected future short-rate alone would imply. For central banks and investors, understanding whether movements in long-term yields are driven by expectations of future short-rates or by risk compensation is important for portfolio decisions and monetary policies.

The post-2022 surge in inflation provides a relevant example of this mechanism. As inflation rose sharply following the COVID-19 pandemic, rising inflation expectations caused investors to revise their expectations of future short-rates upward through the Fisher effect, pushing nominal yields higher. At the same time, the increased inflation uncertainty raised the inflation risk premium, reflecting the raised compensation investors demanded for holding long-term bonds under a more uncertain inflation outlook, putting additional upward pressure on term premia beyond what the revised rate expectations alone would imply. The rebound in term premia visible in the survey-augmented model around this period is consistent with both of these channels happening simultaneously, suggesting that the post-2022 rise in long-term yields reflect a combination of higher interest rate expectations and greater risk compensation rather than either alone.

Finally, the impulse response analysis illustrates that the dynamic response of the yield curve to macro shocks differs meaningfully between the survey-augmented and no-survey models. In particular, the survey-augmented model produces more economically sound responses, with inflation shocks generating an upward shift in yields that fades monotonically rather than the sign-reversal observed under the no-survey estimator. This dynamic is important for central banks that use term structure models to assess how macro shocks affect

the bond markets. If the model used misidentifies the persistence of macro shocks it may produce misleading impulse responses, potentially leading to incorrect assessments of how monetary policy shocks affect the yield curve. The survey-augmented model offers a modest step toward a more reliable dynamic inference in this context.

10.7 Limitations and Future Research

The paper includes some limitations that are worth reflecting on. The most fundamental assumption of the model is the Gaussian assumption underlying the GATSM framework. This assumption rules out fat tails, non-linearity, and stochastic volatility. This is a well-known limitation of the Gaussian assumption and becomes particularly relevant during periods of prolonged low-rates which is closely related with the concern of how the model treats periods near the zero lower bound. During the post-Financial-Crisis and post-COVID periods where short-rates were near or at zero, the Gaussian assumption may produce unreliable term premium estimates, since the model can imply negative interest rates.

A shadow rate model or other non-linear extension could potentially provide a more appropriate framework for these periods, though examining this is outside the scope of this paper. For instance, [Andreasen et al. \(2025\)](#) show that the slope coefficient for a given yield spread is of greater magnitude near zero lower bound periods than away from it, and they use this result to test how different term structure models account for the zero lower bound, finding that a linear-rational square-root model performs best at modelling the zero lower bound.

The independence assumption between the macro and latent blocks, imposed following [Ang and Piazzesi \(2003\)](#), restricts feedback between the two blocks in both directions. This means that latent factors cannot Granger-cause macro variables and macro shocks cannot directly affect the latent factors. [Ang and Piazzesi \(2003\)](#) acknowledge this as a limitation, noting that empirical evidence suggests that the term structure can predict movements in macroeconomic activity, which is in contradiction to the strict independence assumption. Relaxing this assumption would allow for a direct interaction between the financial and macroeconomic conditions but would substantially increase the complexity of the estimation. This is left as a potential extension.

The survey augmentation relies on a fixed weight $w_{svy} = 1$, which assigns each survey observation the same least-squares weight as an actual macro observation. As discussed in Section 6, this imposes a common survey-to-innovation precision ratio of one for both macro variables, which is accurate for inflation but understates the precision of the unemployment

survey, where a Generalised Least Squares efficient scheme would weight the surveys variable-by-variable instead of imposing the same weights on both surveys even though they differ in their precision. To this, one of the likely reasons why we see such a stark difference in effect between inflation and unemployment surveys, is that by setting $w_{svy} = 1$ we down-weight the unemployment survey relative to its actual informativeness. [Kim and Orphanides \(2012\)](#) on the other hand estimate the survey measurement error endogenously through the Kalman filter and allow the data to determine the relative weight assigned to surveys. While the noisy survey robustness test shows that the results are stable under additional measurement noise, the fixed weight assumption remains a simplification that can be relaxed as a natural robustness check.

The four-quarter-ahead estimator is implemented using an approximation as discussed in the earlier sections, which may contribute to its poor performance alongside the lower signal-to-noise ratio compared to the short-horizon forecasts. This means that the conclusion favouring short-horizon surveys is based on a comparison where the longer-horizon estimator may include additional approximation bias, but also the endogeneity bias as a result of how we have incorporated the four-quarter-ahead surveys. The performance gap between the two horizons may therefore narrow under a true longer-horizon implementation. It is worth noting, however, that the lower signal-to-noise ratio of four-quarter-ahead forecasts is a fundamental property of the longer horizon itself, independent of the approximation, it is simply harder to predict macroeconomic conditions four quarters ahead than one quarter ahead. This suggests that the preference for shorter horizons is likely to persist even under a true longer-horizon implementation.

The analysis is conducted on U.S. data over a single sample period, which limits the generalisability of the findings. Whether the bias-reducing properties of survey augmentation extend to other countries or monetary policies remains an open question.

Collectively, these limitations point to several natural extensions of the analysis, which are discussed in the remainder of this section.

A potential extension is to apply the survey-augmented macro-finance GATSM to other countries. An extension to the UK, other G7 economies, or a broader panel of countries would provide valuable evidence on whether the bias-reducing properties of survey augmentation are a general phenomenon or a feature specific to the U.S. institutional and monetary policy context. Estimating the model across multiple countries could also potentially allow for the identification of a global volatility factor capturing common variation in term premia across markets, though whether such a factor is identifiable within this framework would require

further investigation.

Relaxing the independence assumption between the macro and the latent blocks would allow for feedback between the two in both directions, enabling the model to capture the empirical observation that yield curves are able to forecast macroeconomic activity. This would allow for more precise estimation to better examine their direct effects, but would substantially increase the complexity of the estimation. Such a relaxation would also be a necessary prerequisite for incorporating additional financial variables as observable factors.

A natural candidate would be stock market returns, following a de-trending procedure to ensure stationarity. Ordering equity prices last in the Cholesky decomposition would allow them to respond to monetary policy shocks while restricting contemporaneous feedback from equity markets to the policy rate, consistent with the so-called Fed put. This could help capture how financial market conditions transmit to the yield curve beyond what inflation and unemployment alone can explain.

The paper uses two latent factors and two macro factors. Extending to three latent factors like [Ang and Piazzesi \(2003\)](#), would provide a richer cross-sectional fit and allow the model to capture more yield curve movements but in turn would introduce more uncertainty and complexity in estimation.

The incorporation of longer-horizon survey information could be extended by following [Kim and Orphanides \(2012\)](#), by including forecasts of short-term interest rate directly into the model, rather than being a state variable. Longer horizons could be incorporated without the approximation bias and endogeneity bias that arises when iterating multi-step-ahead macro conditional expectations through the VAR(1).

The analysis in the paper uses revised macroeconomic data throughout. Using real-time data instead would better reflect the information actually available to investors and forecasters at each point in time, addressing the concern raised by [Ghysels et al. \(2018\)](#) that the use of revised data may overstate the informational content of macro variables, because a substantial amount of predictive power is seen from the additional information in the revised data relative to the real-time data.

Every examination and model extension of the paper is done in-sample. An out-of-sample forecasting analysis would provide a further insight into the empirical application of the model. Using a rolling or expanding window approach to forecast yields at various horizons and comparing the survey-augmented model against the no-survey baseline would assess whether the bias correction translates into improved forecast accuracy, complementing the in-sample Monte Carlo findings of the paper.

11 Conclusion

This paper investigates whether survey expectations of macroeconomic variables, namely inflation and unemployment, can help mitigate the finite-sample bias that is prevalent in affine term structure models. The analysis is done through stepwise model extensions, each motivated by a distinct question.

A baseline two-factor latent GATSM is first estimated via Kalman filtering maximum likelihood using U.S. yield data from 1972 to 2025. A Monte Carlo study of the two-factor latent GATSM reveals the existence of the finite-sample bias in the estimates. Extending the model to a four-factor macro-finance GATSM, estimated using the MCSE pipeline, by including inflation and unemployment as additional factors results in a smoother model-implied term premium series as the macro factors absorb a share of the yield variation previously concentrated in the latent block. Despite all of this, the finite-sample bias persists, motivating the survey augmentation extension.

The model is extended by including one-quarter ahead SPF forecasts of inflation and unemployment through the OLS stacking approach. Running extensive Monte Carlo studies shows that the incorporation of surveys is firstly able to mitigate some of the finite-sample bias, and secondly the size of the finite-sample bias is larger when using a shorter than a longer sample size. Thirdly, the use of surveys markedly improves the identification of the estimates under the risk-neutral measure. Lastly, the survey-augmented model implies a substantially elevated term premium relative to the no-survey baseline model.

The results also reveal a meaningful trade-off. While the bias in inflation persistence is greatly mitigated, the bias of the unemployment persistence parameter is over-corrected going from negative to positive, suggesting that forecasters systematically expect unemployment to mean-revert faster than historical data would imply. This finding reinforces the general idea that survey augmentation introduces a new source of potential bias namely the forecast bias of the forecasters themselves.

Taken together, the results support the general idea of incorporating survey expectations as additional real-time information to pin down estimates in a macro-finance GATSM setting.

References

- Andreasen, M. M. (2010), ‘How to maximize the likelihood function for a dsge model’, *Computational Economics* **35**(2), 127–154.
- Andreasen, M. M., Engsted, T. and Møller, S. V. (Forthcoming), *Book Draft for Empirical Asset Pricing - Chapter 8: Interest Rates and Dynamic Term Structure Models*.
- Andreasen, M. M., Jørgensen, K. and Meldrum, A. (2025), ‘Bond risk premiums at the zero lower bound’, *Journal of Econometrics* **247**, 105939.
- Andreasen, M. M. and Kristensen, D. (2025), ‘Estimating state space models: Simple corrections for finite sample bias’, *SSRN Working Paper No. 5382755*.
- Ang, A. and Piazzesi, M. (2003), ‘A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables’, *Journal of Monetary economics* **50**(4), 745–787.
- Bauer, M. D., Rudebusch, G. D. and Wu, J. C. (2012), ‘Correcting estimation bias in dynamic term structure models’, *Journal of Business & Economic Statistics* **30**(3), 454–467.
- Campbell, J. Y. and Shiller, R. J. (1991), ‘Yield spreads and interest rate movements: A bird’s eye view’, *The Review of Economic Studies* **58**(3), 495–514.
- Chen, R.-R. and Scott, L. (1993), ‘Maximum likelihood estimation for a multifactor equilibrium model of the term structure of interest rates’, *The Journal of Fixed Income* **3**(3), 14–31.
- Dai, Q. and Singleton, K. J. (2000), ‘Specification analysis of affine term structure models’, *The journal of finance* **55**(5), 1943–1978.
- Dai, Q. and Singleton, K. J. (2002), ‘Expectation puzzles, time-varying risk premia, and affine models of the term structure’, *Journal of financial Economics* **63**(3), 415–441.
- Diebold, F. X., Rudebusch, G. D. and Aruoba, S. B. (2006), ‘The macroeconomy and the yield curve: a dynamic latent factor approach’, *Journal of econometrics* **131**(1-2), 309–338.
- Duffee, G. R. (2002), ‘Term premia and interest rate forecasts in affine models’, *The Journal of Finance* **57**(1), 405–443.

- Duffee, G. R. (2011), ‘Information in (and not in) the term structure’, *The Review of Financial Studies* **24**(9), 2895–2934.
- Fed (2019), ‘Yield curve models and data’.
URL: <https://www.federalreserve.gov/data/nominal-yield-curve.htm>
- Fed (2026), ‘The u.s. treasury yield curve: 1961 to the present’.
URL: <https://www.federalreserve.gov/econres/feds/the-us-treasury-yield-curve-1961-to-the-present.htm>
- Fed-Philadelphia (2026), ‘Data files - survey of professional forecasters (rgdp)’.
URL: <https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/median-forecasts>
- Fed-St.Louis (2026a), ‘Gross domestic product: Implicit price deflator’.
URL: <https://fred.stlouisfed.org/series/GDPDEF>
- Fed-St.Louis (2026b), ‘Unemployment rate’.
URL: <https://fred.stlouisfed.org/series/UNRATE>
- Foroni, C. and Marcellino, M. (2016), ‘Mixed frequency structural vector auto-regressive models’.
- Ghysels, E., Horan, C. and Moench, E. (2018), ‘Forecasting through the rearview mirror: Data revisions and bond return predictability’, *The Review of Financial Studies* **31**(2), 678–714.
- Goodfriend, M. and King, R. G. (2005), ‘The incredible volcker disinflation’, *Journal of Monetary Economics* **52**(5), 981–1015.
- Hamilton, J. D. and Wu, J. C. (2012), ‘Identification and estimation of gaussian affine term structure models’, *Journal of Econometrics* **168**(2), 315–331.
- Hamilton, J. D. and Wu, J. C. (2014), ‘Testable implications of affine term structure models’, *Journal of Econometrics* **178**, 231–242.
- Hansen, N. (2016), ‘The cma evolution strategy: A tutorial’, *arXiv preprint arXiv:1604.00772*.

- Hansen, N., Müller, S. D. and Koumoutsakos, P. (2003), ‘Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (cma-es)’, *Evolutionary computation* **11**(1), 1–18.
- Kim, D. H. and Orphanides, A. (2012), ‘Term structure estimation with survey data on interest rate forecasts’, *Journal of Financial and Quantitative Analysis* **47**(1), 241–272.
- Lagarias, J. C., Reeds, J. A., Wright, M. H. and Wright, P. E. (1998), ‘Convergence properties of the nelder–mead simplex method in low dimensions’, *SIAM Journal on optimization* **9**(1), 112–147.
- Mincer, J. A. and Zarnowitz, V. (1969), The evaluation of economic forecasts, in ‘Economic forecasts and expectations: Analysis of forecasting behavior and performance’, NBER, pp. 3–46.
- Nelson, C. R. and Siegel, A. F. (1987), ‘Parsimonious modeling of yield curves’, *Journal of business* pp. 473–489.
- Newey, W. K. and West, K. D. (1986), ‘A simple, positive semi-definite, heteroskedasticity and autocorrelationconsistent covariance matrix’.
- Stock, J. H. and Watson, M. W. (2007), ‘Why has us inflation become harder to forecast?’, *Journal of Money, Credit and banking* **39**, 3–33.
- Svensson, L. E. (1994), ‘Estimating and interpreting forward interest rates: Sweden 1992–1994’.
- Theil, H. and Goldberger, A. S. (1992), On pure and mixed statistical estimation in economics, in ‘Henri Theil’s Contributions to Economics and Econometrics: Econometric Theory and Methodology’, Springer, pp. 317–332.

12 Appendices

12.1 Appendix A1: MCSE Yield Partition

Group 1: Model says

$$\mathbf{y}_t^1 = \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \mathbf{x}_t^l,$$

and its lagged expression is

$$\mathbf{y}_{t-1}^1 = \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m + \mathbf{G}_x^{1,l} \mathbf{x}_{t-1}^l,$$

which we can invert to get

$$\mathbf{x}_{t-1}^l = (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_{t-1}^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m).$$

Under the independence assumption the latent block under $\mathbb{P}$ has the law of motion

$$\mathbf{x}_t^l = \boldsymbol{\mu}_l^{\mathbb{P}} + \boldsymbol{\Phi}_{ll}^{\mathbb{P}} \mathbf{x}_{t-1}^l + \mathbf{u}_t^l.$$

Now substitute in for the inverted $\mathbf{x}_{t-1}^l$ expression

$$\mathbf{x}_t^l = \boldsymbol{\mu}_l^{\mathbb{P}} + \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_{t-1}^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m) + \mathbf{u}_t^l.$$

Now plug in this expression for $\mathbf{x}_t^l$ into the model equation for $\mathbf{y}_t^1$

$$\mathbf{y}_t^1 = \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \left[\boldsymbol{\mu}_l^{\mathbb{P}} + \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_{t-1}^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m) + \mathbf{u}_t^l \right]$$

Now expand and collect terms

$$\begin{aligned}
\mathbf{y}_t^1 &= \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \boldsymbol{\mu}_l^{\mathbb{P}} + \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_{t-1}^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m) + \mathbf{G}_x^{1,l} \mathbf{u}_t^l \\
&= \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \boldsymbol{\mu}_l^{\mathbb{P}} + \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{y}_{t-1}^1 - \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{g}_0^1 - \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m \\
&\quad + \mathbf{G}_x^{1,l} \mathbf{u}_t^l \\
&= \underbrace{\left(\mathbf{g}_0^1 + \mathbf{G}_x^{1,l} \boldsymbol{\mu}_l^{\mathbb{P}} - \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{g}_0^1 \right)}_{\mathbf{A}^{1,\#}} + \underbrace{\mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{y}_{t-1}^1}_{\boldsymbol{\Phi}_{11}^{\mathbb{P},\#}} + \underbrace{\mathbf{G}_x^{1,m} \mathbf{x}_t^m - \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m}_{\boldsymbol{\Psi}_{1m}^{\mathbb{P},\#}} \\
&\quad + \mathbf{G}_x^{1,l} \mathbf{u}_t^l \\
&= \mathbf{A}^{1,\#} + \boldsymbol{\Phi}_{11}^{\mathbb{P},\#} \mathbf{y}_{t-1}^1 + \boldsymbol{\Psi}_{1m}^{\mathbb{P},\#} \mathbf{x}_t^m - \mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m + \mathbf{G}_x^{1,l} \mathbf{u}_t^l.
\end{aligned}$$

The term $\mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{ll}^{\mathbb{P}} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m} \mathbf{x}_{t-1}^m$ is a lagged-macro loading. It coincides with ϕ_{m1}^* in the exactly-priced Eq. (36) in [Hamilton and Wu \(2012\)](#). The block-independence assumption $\boldsymbol{\Phi}_{lm}^{\mathbb{P}} = \boldsymbol{\Phi}_{ml}^{\mathbb{P}} = \mathbf{0}$ removes the direct feedback channel given by $\mathbf{G}_x^{1,l} \boldsymbol{\Phi}_{lm}^{\mathbb{P}}$, but the inversion channel remains, and the term is generally non-zero whenever the exactly-priced yields load on the macro factor $(\mathbf{G}_x^{1,m}) \neq \mathbf{0}$. Because the exactly-priced yields load only very weakly on the macro factor in our estimates ($\beta_m \approx \pm 0.0007$, so $\mathbf{G}_x^{1,m}$ is small), the coefficient on this lagged-macro term, $\boldsymbol{\Phi}_{11}^{\mathbb{P},\#} \mathbf{G}_x^{1,m}$, is negligible. We therefore drop it and work with the approximate Group 1 reduced form. We are then left with

$$\mathbf{y}_t^1 = \mathbf{A}^{1,\#} + \boldsymbol{\Phi}_{11}^{\mathbb{P},\#} \mathbf{y}_{t-1}^1 + \boldsymbol{\Psi}_{1m}^{\mathbb{P},\#} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \mathbf{u}_t^l.$$

Group 2: Model says

$$\mathbf{y}_t^2 = \mathbf{g}_0^2 + \mathbf{G}_x^{2,m} \mathbf{x}_t^m + \mathbf{G}_x^{2,l} \mathbf{x}_t^l.$$

The yield equation for group 1 was given by

$$\mathbf{y}_t^1 = \mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \mathbf{x}_t^l.$$

Since $\mathbf{G}_x^{1,l}$ is an 2×2 invertible matrix by construction (two maturities are chosen precisely to span the latent factor space), we can solve for the latent factors

$$\mathbf{x}_t^l = (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_t^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_t^m).$$

Inserting this expression into $\mathbf{y}_t^2$ yields

$$\mathbf{y}_t^2 = \mathbf{g}_0^2 + \mathbf{G}_x^{2,m} \mathbf{x}_t^m + \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} (\mathbf{y}_t^1 - \mathbf{g}_0^1 - \mathbf{G}_x^{1,m} \mathbf{x}_t^m).$$

Now expand and collect terms

$$\begin{aligned} \mathbf{y}_t^2 &= \mathbf{g}_0^2 + \mathbf{G}_x^{2,m} \mathbf{x}_t^m + \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} \mathbf{y}_t^1 - \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} \mathbf{g}_0^1 - \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m} \mathbf{x}_t^m \\ &= \underbrace{(\mathbf{g}_0^2 - \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} \mathbf{g}_0^1)}_{\mathbf{A}^{2,\#}} + \underbrace{(\mathbf{G}_x^{2,m} - \mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1} \mathbf{G}_x^{1,m})}_{\Phi_{2m}^{\mathbb{P},\#}} \mathbf{x}_t^m + \underbrace{\mathbf{G}_x^{2,l} (\mathbf{G}_x^{1,l})^{-1}}_{\Phi_{21}^{\mathbb{P},\#}} \mathbf{y}_t^1 \\ &= \mathbf{A}^{2,\#} + \Phi_{2m}^{\mathbb{P},\#} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{y}_t^1. \end{aligned}$$

12.2 Appendix A2: MCSE Conditions

Condition 2:

Take

$$\mathbf{y}_t^2 = \mathbf{A}^{2,\#} + \Phi_{2m}^{\mathbb{P},\#} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{y}_t^1 + \mathbf{u}_t^{2,\#}.$$

Now insert for the yield expression for $\mathbf{y}_t^1$

$$\mathbf{y}_t^2 = \mathbf{A}^{2,\#} + \Phi_{2m}^{\mathbb{P},\#} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} (\mathbf{g}_0^1 + \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \mathbf{G}_x^{1,l} \mathbf{x}_t^l) + \mathbf{u}_t^{2,\#}.$$

Now expand and collect terms

$$\begin{aligned} \mathbf{y}_t^2 &= \mathbf{A}^{2,\#} + \Phi_{2m}^{\mathbb{P},\#} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{g}_0^1 + \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,m} \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,l} \mathbf{x}_t^l + \mathbf{u}_t^{2,\#} \\ &= (\mathbf{A}^{2,\#} + \Phi_{21}^{\mathbb{P},\#} \mathbf{g}_0^1) + (\Phi_{2m}^{\mathbb{P},\#} + \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,m}) \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,l} \mathbf{x}_t^l + \mathbf{u}_t^{2,\#} \\ &= (\mathbf{A}^{2,\#} + \Phi_{21}^{\mathbb{P},\#} \mathbf{g}_0^1) + (\Phi_{2m}^{\mathbb{P},\#} + \Phi_{21}^{\mathbb{P},\#} \widehat{\Psi}_{1m}^{\mathbb{P},\#}) \mathbf{x}_t^m + \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,l} \mathbf{x}_t^l + \mathbf{u}_t^{2,\#}. \end{aligned}$$

The second MCSE condition is then given by

$$\mathbf{G}_{x,OLS}^{2,m} = \Phi_{2m}^{\mathbb{P},\#} + \Phi_{21}^{\mathbb{P},\#} \widehat{\Psi}_{1m}^{\mathbb{P},\#}.$$

Condition 3: With

$$\mathbf{y}_t^1 = \mathbf{A}^{1,\#} + \Phi_{11}^{\mathbb{P},\#} \mathbf{y}_{t-1}^1 + \Psi_{1m}^{\mathbb{P},\#} \mathbf{x}_t^m + \mathbf{u}_t^{1,\#},$$

After controlling for both $\mathbf{y}_{t-1}^1$ and $\mathbf{x}_t^m$, the residuals $\mathbf{u}_t^{1,\#}$ only absorb the latent factor

innovations

$$\widehat{\Omega}^{1,\#} = \frac{1}{T_1} \sum_t \mathbf{u}_t^{1,\#} \left(\mathbf{u}_t^{1,\#} \right)^\top = \mathbf{G}_x^{1,l} \Sigma_{ll} \left(\mathbf{G}_x^{1,l} \right)^\top = \mathbf{G}_x^{1,l} \left(\mathbf{G}_x^{1,l} \right)^\top,$$

where it is used that $\Sigma_{ll} = \mathbf{I}$.

The third MCSE condition is then

$$\widehat{\Omega}^{1,\#} = \mathbf{G}_x^{1,l} \left(\mathbf{G}_x^{1,l} \right)^\top.$$

Condition 4: Looking at the expression used in condition 2 and comparing coefficients, the fourth condition must be given by

$$\mathbf{G}_x^{2,l} = \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,l},$$

but since $\mathbf{G}_x^{1,l}$ is unobserved we first multiply both sides by $\left(\mathbf{G}_x^{1,l} \right)^\top$

$$\mathbf{G}_x^{2,l} \left(\mathbf{G}_x^{1,l} \right)^\top = \Phi_{21}^{\mathbb{P},\#} \mathbf{G}_x^{1,l} \left(\mathbf{G}_x^{1,l} \right)^\top.$$

The fourth condition is then given by

$$\mathbf{G}_{x,OLS}^{2,l} \left(\mathbf{G}_x^{1,l} \right)^\top = \Phi_{21}^{\mathbb{P},\#} \widehat{\Omega}^{1,\#}.$$

12.3 Appendix A3: $\mathbb{P}$ to $\mathbb{Q}$

For a Gaussian innovation we can move from the $\mathbb{P}$ to the $\mathbb{Q}$ -measure by using the Radon-Nikodym derivative, given by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left\{ -\frac{1}{2} \Lambda_t^\top \Lambda_t - \boldsymbol{\varepsilon}_{t+1}^\top \Lambda_t \right\}.$$

The density for $\boldsymbol{\varepsilon}_{t+1}$ under $\mathbb{P}$ is simply the multivariate Gaussian density

$$\begin{aligned} \mathbb{P}(\boldsymbol{\varepsilon}_{t+1}) &= (2\pi)^{-\frac{F}{2}} \det(\mathbf{I})^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\boldsymbol{\varepsilon}_{t+1} - \mathbf{0})^\top \mathbf{I}^{-1} (\boldsymbol{\varepsilon}_{t+1} - \mathbf{0}) \right) \\ &= (2\pi)^{-\frac{F}{2}} \exp \left(-\frac{1}{2} \boldsymbol{\varepsilon}_{t+1}^\top \boldsymbol{\varepsilon}_{t+1} \right). \end{aligned}$$

The density for $\boldsymbol{\varepsilon}_{t+1}$ under $\mathbb{Q}$ can then be found by $\mathbb{Q}(\boldsymbol{\varepsilon}_{t+1}) = \mathbb{P}(\boldsymbol{\varepsilon}_{t+1}) \times d\mathbb{Q}/d\mathbb{P}$, giving us

$$\mathbb{Q}(\boldsymbol{\varepsilon}_{t+1}) = (2\pi)^{-\frac{F}{2}} \exp\left(-\frac{1}{2}\boldsymbol{\varepsilon}_{t+1}^\top \boldsymbol{\varepsilon}_{t+1}\right) \exp\left\{-\frac{1}{2}\boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t - \boldsymbol{\varepsilon}_{t+1}^\top \boldsymbol{\Lambda}_t\right\},$$

which can be rewritten as

$$\mathbb{Q}(\boldsymbol{\varepsilon}_{t+1}) = (2\pi)^{-\frac{F}{2}} \exp\left(-\frac{1}{2}[\boldsymbol{\varepsilon}_{t+1} + \boldsymbol{\Lambda}_t]^\top [\boldsymbol{\varepsilon}_{t+1} + \boldsymbol{\Lambda}_t]\right).$$

By looking at the differences between the densities under $\mathbb{P}$ and $\mathbb{Q}$ it shows that our Gaussian innovation under $\mathbb{Q}$ is given by

$$\boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} = \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}} + \boldsymbol{\Lambda}_t.$$

Furthermore, using the above result we can derive

$$\begin{aligned} \mathbf{x}_{t+1} &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} (\boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} - \boldsymbol{\Lambda}_t) \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} - \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} - \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1} (\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_x \mathbf{x}_t) \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} - \boldsymbol{\lambda}_0 - \boldsymbol{\lambda}_x \mathbf{x}_t \\ &= \underbrace{(\boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\lambda}_0)}_{\boldsymbol{\mu}^{\mathbb{Q}}} + \underbrace{(\boldsymbol{\Phi}^{\mathbb{P}} - \boldsymbol{\lambda}_x)}_{\boldsymbol{\Phi}^{\mathbb{Q}}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}} \\ \mathbf{x}_{t+1} &= \boldsymbol{\mu}^{\mathbb{Q}} + \boldsymbol{\Phi}^{\mathbb{Q}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{Q}}. \end{aligned}$$

12.4 Appendix A4: FEVD

Yields are given by

$$y_t^{(k)} = g_0^{(k)} + \mathbf{G}_x^{(k)} \mathbf{x}_t,$$

and the state dynamics are

$$\mathbf{x}_{t+1} = \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}}.$$

The IRF for yields given horizon h , maturity k , and structural shock j is

$$\text{IRF}_y^{(k,j)}(h) = \mathbf{G}_x^{(k)} (\boldsymbol{\Phi}^{\mathbb{P}})^{h-1} \boldsymbol{\Sigma} \mathbf{e}_j.$$

Use the fact that a stationary AR(1) process can be represented by a MA(∞) process, so that the state vector has the following MA representation

$$\mathbf{x}_t - \underbrace{\bar{\mathbf{x}}}_{(\mathbf{I} - \Phi^{\mathbb{P}})^{-1} \mu^{\mathbb{P}}} = \sum_{i=0}^{\infty} (\Phi^{\mathbb{P}})^i \Sigma \varepsilon_{t-i}^{\mathbb{P}}.$$

Since yields are affine in $\mathbf{x}_t$ we can write

$$y_t^{(k)} - \bar{y}^{(k)} = \mathbf{G}_x^{(k)} (\mathbf{x}_t - \bar{\mathbf{x}}) = \sum_{i=0}^{\infty} \mathbf{G}_x^{(k)} (\Phi^{\mathbb{P}})^i \Sigma \varepsilon_{t-i}^{\mathbb{P}}.$$

Now define $\Psi_{i,k,j} \equiv \mathbf{G}_x^{(k)} (\Phi^{\mathbb{P}})^i \Sigma \mathbf{e}_j$ as the (k, j) element of the MA coefficient matrix $\Psi_i = \mathbf{G}_x (\Phi^{\mathbb{P}})^i \Sigma$ at i . The h -step ahead forecast error for yield $y_t^{(k)}$ is

$$\begin{aligned} y_{t+h}^{(k)} - y_{t+h|t}^{(k)} &= \sum_{i=0}^{h-1} \mathbf{G}_x^{(k)} (\Phi^{\mathbb{P}})^i \Sigma \varepsilon_{t+h-i}^{\mathbb{P}} \\ &= \sum_{i=0}^{h-1} \sum_{j=1}^4 \Psi_{i,k,j} \varepsilon_{t+h-i,j}^{\mathbb{P}}. \end{aligned}$$

Since $\varepsilon^{\mathbb{P}} \sim \text{iid}(\mathbf{0}, \mathbf{I})$, all shocks are serially uncorrelated and mutually orthogonal. The mean squared error of the forecast error h steps ahead for the yield at maturity k is then

$$\text{MSE}^{(k)}(h) = \text{Var} \left(\sum_{i=0}^{h-1} \sum_{j=1}^4 \Psi_{i,k,j} \varepsilon_{t+h-i,j}^{\mathbb{P}} \right) = \sum_{j=1}^4 \sum_{i=0}^{h-1} \Psi_{i,k,j}^2.$$

The portion of $\text{MSE}^{(k)}(h)$ that can attributed solely to shock j is

$$\text{MSE}_j^{(k)}(h) = \sum_{i=0}^{h-1} \Psi_{i,k,j}^2.$$

The share of total forecast variance attributable to shock j is

$$\Omega_{k,j}(h) = \frac{\text{MSE}_j^{(k)}(h)}{\text{MSE}^{(k)}(h)} = \frac{\sum_{i=0}^{h-1} \Psi_{i,k,j}^2}{\sum_{j=1}^4 \sum_{i=0}^{h-1} \Psi_{i,k,j}^2}.$$

12.5 Appendix A5: $\mathbb{P}$ Recursion

The law of motion under the $\mathbb{P}$ -measure is given by

$$\mathbf{x}_{t+1} = \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^{\mathbb{P}}.$$

The conditional expectation under $\mathbb{P}$ at time t is

$$\mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+1}] = \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t.$$

Now iterate forward. The two period ahead ($j = 2$) is

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+2}] &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+1}] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbb{E}_t^{\mathbb{P}} [\boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} [\boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 \mathbf{x}_t \\ &= (\mathbf{I} + \boldsymbol{\Phi}^{\mathbb{P}}) \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 \mathbf{x}_t. \end{aligned}$$

For $j = 3$

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+3}] &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+2}] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbb{E}_t^{\mathbb{P}} [(\mathbf{I} + \boldsymbol{\Phi}^{\mathbb{P}}) \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 \mathbf{x}_t] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} [(\mathbf{I} + \boldsymbol{\Phi}^{\mathbb{P}}) \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 \mathbf{x}_t] \\ &= \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^3 \mathbf{x}_t \\ &= (\mathbf{I} + \boldsymbol{\Phi}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2) \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^3 \mathbf{x}_t. \end{aligned}$$

Now the pattern is clear and after j steps we have

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+i}] &= (\mathbf{I} + \boldsymbol{\Phi}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^2 + \dots + (\boldsymbol{\Phi}^{\mathbb{P}})^{i-1}) \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^i \mathbf{x}_t \\ \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+i}] &= \sum_{j=0}^{i-1} (\boldsymbol{\Phi}^{\mathbb{P}})^j \boldsymbol{\mu}^{\mathbb{P}} + (\boldsymbol{\Phi}^{\mathbb{P}})^i \mathbf{x}_t. \end{aligned}$$

12.6 Appendix A6: Alternative Term Premium

The term premium for a given maturity k is given as the wedge between the yield and the average of the expected future short-rates

$$TP_t^{(k)} = y_t^{(k)} - \frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}_t^{\mathbb{P}} [r_{t+i}].$$

One-period expected excess return. If one buys a k -period bond at time t and then sells it as a $(k-1)$ -period bond at $t+1$, the one-period log holding return in excess of the short-rate is

$$rx_{t+1}^{(k)} \equiv \log P_{t+1}^{(k-1)} - \log P_t^{(k)} - r_t = A_{k-1} + \mathbf{B}_{k-1}^\top \mathbf{x}_{t+1} - A_k - \mathbf{B}_k^\top \mathbf{x}_t - r_t.$$

Take the conditional expectation under $\mathbb{P}$ and insert for $\mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+1}] = \boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t$ and $r_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t$.

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} [rx_{t+1}^{(k)}] &= \mathbb{E}_t^{\mathbb{P}} [A_{k-1} + \mathbf{B}_{k-1}^\top \mathbf{x}_{t+1} - A_k - \mathbf{B}_k^\top \mathbf{x}_t - r_t] \\ &= (A_{k-1} - A_k) + \mathbf{B}_{k-1}^\top \mathbb{E}_t^{\mathbb{P}} [\mathbf{x}_{t+1}] - \mathbf{B}_k^\top \mathbf{x}_t - r_t \\ &= (A_{k-1} - A_k) + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t) - \mathbf{B}_k^\top \mathbf{x}_t - \alpha - \boldsymbol{\beta}^\top \mathbf{x}_t. \end{aligned}$$

Now insert (15) and (16)

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} [rx_{t+1}^{(k)}] &= \left(\alpha - \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\lambda}_0) - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} \right) + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t) \\ &\quad - (\mathbf{B}_{k-1}^\top (\boldsymbol{\Phi}^{\mathbb{P}} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top) \mathbf{x}_t - \alpha - \boldsymbol{\beta}^\top \mathbf{x}_t \\ &= -\mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\lambda}_0) - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^{\mathbb{P}} + \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t) - \mathbf{B}_{k-1}^\top (\boldsymbol{\Phi}^{\mathbb{P}} - \boldsymbol{\lambda}_x) \mathbf{x}_t \\ &= -\mathbf{B}_{k-1}^\top \boldsymbol{\mu}^{\mathbb{P}} + \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_0 - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} + \mathbf{B}_{k-1}^\top \boldsymbol{\mu}^{\mathbb{P}} + \mathbf{B}_{k-1}^\top \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t - \mathbf{B}_{k-1}^\top \boldsymbol{\Phi}^{\mathbb{P}} \mathbf{x}_t + \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_x \mathbf{x}_t \\ &= \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_0 - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} + \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_x \mathbf{x}_t \\ &= \mathbf{B}_{k-1}^\top (\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_x \mathbf{x}_t) - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} \\ &= \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t - \frac{1}{2} \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1}. \end{aligned}$$

The first term $\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t$ is the risk premium and the second is the Jensen convexity correction which adjusts for the bias when using expected values with non-linear functions.

Term Premium. Holding the k -period bond to maturity and rolling it down one period at

a time the log holding returns is $-\log P_t^{(k)} = ky_t^{(k)}$ since $P_{t+k}^{(0)} = 1$

$$ky_t^{(k)} = \sum_{i=1}^k \left(\log P_{t+i}^{(k-i)} - \log P_{t+i-1}^{(k-i+1)} \right).$$

Now insert for the expected one-period excess return

$$ky_t^{(k)} = \sum_{i=1}^k \left(rx_{t+i}^{(k-i+1)} + r_{t+i-1} \right).$$

Now that the conditional expectation under the physical measure and rearrange a bit

$$\begin{aligned} \mathbb{E}_t^{\mathbb{P}} \left[ky_t^{(k)} \right] &= \sum_{i=1}^k \left(\mathbb{E}_t^{\mathbb{P}} \left[rx_{t+i}^{(k-i+1)} \right] + \mathbb{E}_t^{\mathbb{P}} \left[r_{t+i-1} \right] \right) \\ \Leftrightarrow y_t^{(k)} &= \frac{1}{k} \sum_{i=1}^k \mathbb{E}_t^{\mathbb{P}} \left[rx_{t+i}^{(k-i+1)} \right] + \frac{1}{k} \sum_{i=1}^k \mathbb{E}_t^{\mathbb{P}} \left[r_{t+i-1} \right] \\ y_t^{(k)} &= \frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}_t^{\mathbb{P}} \left[r_{t+i} \right] + \frac{1}{k} \sum_{i=1}^k \mathbb{E}_t^{\mathbb{P}} \left[rx_{t+i}^{(k-i+1)} \right]. \end{aligned}$$

Now by using the definition of the term premium we see that

$$TP_t^{(k)} = \frac{1}{k} \sum_{i=1}^k \mathbb{E}_t^{\mathbb{P}} \left[rx_{t+i}^{(k-i+1)} \right],$$

must be the term premium in alternative form.

12.7 Appendix A7: Solving the Model

Non-arbitrage condition.

For $k = 1$

$$\begin{aligned} P_t^{(1)} &= \mathbb{E} \left[M_{t+1} P_{t+1}^{(0)} \right] \\ &= \mathbb{E} [M_{t+1}] . \end{aligned}$$

For $k = 2$

$$\begin{aligned} P_t^{(2)} &= \mathbb{E} \left[M_{t+1} P_{t+1}^{(1)} \right] \\ &= \mathbb{E} [M_{t+1} M_{t+2}] . \end{aligned}$$

For $k = 3$

$$\begin{aligned} P_t^{(3)} &= \mathbb{E} \left[M_{t+1} P_{t+1}^{(2)} \right] \\ &= \mathbb{E} [M_{t+1} M_{t+2} M_{t+3}] . \end{aligned} \tag{92}$$

We see the pattern and generally for a given k the price of a bond is

$$P_t^{(k)} = \mathbb{E} \left[\prod_{i=0}^{k-1} M_{t+1+i} \times 1 \right] .$$

Solving for bond yields. Assume that the price of zero-coupon bonds is given by the affine equation

$$P_t^{(k)} = \exp \{ A_k + B_k^\top \mathbf{x}_t \} ,$$

Since

$$P_t^{(k)} = \exp (A_k + \mathbf{B}_k^\top \mathbf{x}_t) ,$$

we then have that

$$\begin{aligned} \exp (A_k + \mathbf{B}_k^\top \mathbf{x}_t) &= \mathbb{E}_t \left(M_{t+1} P_{t+1}^{(k-1)} \right) \\ &= \mathbb{E}_t \left(\exp \left\{ -r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t^\top \boldsymbol{\epsilon}_{t+1}^\mathbb{P} \right\} \exp \{ A_{k-1} + \mathbf{B}_{k-1}^\top \mathbf{x}_{t+1} \} \right) . \end{aligned}$$

Now inserting Eq. (6) we get

$$\begin{aligned}
\exp(A_k + \mathbf{B}_k^\top \mathbf{x}_t) &= \mathbb{E}_t \left(\exp \left\{ -r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \right\} \exp \left\{ A_{k-1} + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^\mathbb{P}) \right\} \right) \\
&= \mathbb{E}_t \left(\exp \left\{ A_{k-1} - r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t \right\} \exp \left\{ -\boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t + \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^\mathbb{P}) \right\} \right) \\
&= \exp \left(A_{k-1} - r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t) \right) \mathbb{E}_t \left(\exp \left(-\boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} + \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \right) \right).
\end{aligned}$$

Now note that for any multivariate random variable $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the moment-generation function (MGF) is given by

$$\mathbf{M}_{\mathbf{X}}(\boldsymbol{\tau}) = \exp \left\{ \boldsymbol{\tau}^\top \boldsymbol{\mu} + \frac{1}{2} \boldsymbol{\tau}^\top \boldsymbol{\Sigma} \boldsymbol{\tau} \right\},$$

where $\boldsymbol{\tau}$ is a conformable vector. For $\boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and letting $\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} - \boldsymbol{\Lambda}_t^\top$ be the conformable vector we can rewrite the expectation term as

$$\begin{aligned}
\mathbb{E}_t \left(\exp \left(-\boldsymbol{\Lambda}_t^\top \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} + \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\varepsilon}_{t+1}^\mathbb{P} \right) \right) &= \exp \left\{ \frac{1}{2} (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} - \boldsymbol{\Lambda}_t^\top) (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} - \boldsymbol{\Lambda}_t^\top)^\top \right\} \\
&= \exp \left\{ \frac{1}{2} (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} - \mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t^\top \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} + \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t) \right\} \\
&= \exp \left\{ \frac{1}{2} (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} - 2\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t + \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t) \right\}.
\end{aligned}$$

The full expression is then

$$\begin{aligned}
\exp(A_k + \mathbf{B}_k^\top \mathbf{x}_t) &= \exp \left(A_{k-1} - r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t) \right) \\
&\quad \times \exp \left(\frac{1}{2} (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} - 2\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t + \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t) \right).
\end{aligned}$$

Taking logs then yields

$$A_k + \mathbf{B}_k^\top \mathbf{x}_t = A_{k-1} - r_t - \frac{1}{2} \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t) + \frac{1}{2} (\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Sigma}^\top \mathbf{B}_{k-1} - 2\mathbf{B}_{k-1}^\top \boldsymbol{\Sigma} \boldsymbol{\Lambda}_t + \boldsymbol{\Lambda}_t^\top \boldsymbol{\Lambda}_t).$$

We then simply rewrite a bit

$$\begin{aligned}
&= A_{k-1} - r_t - \frac{1}{2} \mathbf{\Lambda}_t^\top \mathbf{\Lambda}_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} - \mathbf{B}_{k-1}^\top \Sigma \mathbf{\Lambda}_t + \frac{1}{2} \mathbf{\Lambda}_t^\top \mathbf{\Lambda}_t \\
&= A_{k-1} - r_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} + \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} - \mathbf{B}_{k-1}^\top (\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_x \mathbf{x}_t) \\
&= A_{k-1} - r_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} + \mathbf{B}_{k-1}^\top \boldsymbol{\Phi}^\mathbb{P} \mathbf{x}_t - \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_x \mathbf{x}_t \\
&= A_{k-1} - r_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} + (\mathbf{B}_{k-1}^\top \boldsymbol{\Phi}^\mathbb{P} - \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_x) \mathbf{x}_t \\
&= A_{k-1} - \alpha - \boldsymbol{\beta}^\top \mathbf{x}_t + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} + (\mathbf{B}_{k-1}^\top \boldsymbol{\Phi}^\mathbb{P} - \mathbf{B}_{k-1}^\top \boldsymbol{\lambda}_x) \mathbf{x}_t \\
&= A_{k-1} - \alpha + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1} + (\mathbf{B}_{k-1}^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top) \mathbf{x}_t.
\end{aligned}$$

From this we can clearly see that A_k is given by

$$A_k = A_{k-1} - \alpha + \mathbf{B}_{k-1}^\top (\boldsymbol{\mu}^\mathbb{P} - \boldsymbol{\lambda}_0) + \frac{1}{2} \mathbf{B}_{k-1}^\top \Sigma \Sigma^\top \mathbf{B}_{k-1},$$

and $\mathbf{B}_k$ is given by

$$\mathbf{B}_k^\top = \mathbf{B}_{k-1}^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top.$$

We can further rewrite the expression for $\mathbf{B}_k$ by using the fact that $\mathbf{B}_1^\top = -\boldsymbol{\beta}^\top$. For $k = 1$

$$\mathbf{B}_1^\top = -\boldsymbol{\beta}^\top.$$

For $k = 2$

$$\begin{aligned}
\mathbf{B}_2^\top &= \mathbf{B}_1^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top \\
&= -\boldsymbol{\beta}^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top \\
&= -\boldsymbol{\beta}^\top \left((\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) + (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x)^0 \right) \\
&= -\boldsymbol{\beta}^\top \left((\boldsymbol{\Phi}^\mathbb{Q})^1 + (\boldsymbol{\Phi}^\mathbb{Q})^0 \right).
\end{aligned}$$

For $k = 3$

$$\begin{aligned}
\mathbf{B}_3^\top &= \mathbf{B}_2^\top (\boldsymbol{\Phi}^\mathbb{P} - \boldsymbol{\lambda}_x) - \boldsymbol{\beta}^\top \\
&= \mathbf{B}_2^\top \boldsymbol{\Phi}^\mathbb{Q} - \boldsymbol{\beta}^\top \\
&= -\boldsymbol{\beta}^\top \left((\boldsymbol{\Phi}^\mathbb{Q})^1 + (\boldsymbol{\Phi}^\mathbb{Q})^0 \right) \boldsymbol{\Phi}^\mathbb{Q} - \boldsymbol{\beta}^\top \\
&= -\boldsymbol{\beta}^\top \left((\boldsymbol{\Phi}^\mathbb{Q})^2 + (\boldsymbol{\Phi}^\mathbb{Q})^1 \right) - \boldsymbol{\beta}^\top \\
&= -\boldsymbol{\beta}^\top \left((\boldsymbol{\Phi}^\mathbb{Q})^2 + (\boldsymbol{\Phi}^\mathbb{Q})^1 + (\boldsymbol{\Phi}^\mathbb{Q})^0 \right).
\end{aligned}$$

We then see that the general pattern for k is

$$\mathbf{B}_k^\top = -\boldsymbol{\beta}^\top \sum_{i=0}^{k-1} (\boldsymbol{\Phi}^\mathbb{Q})^i.$$

Transposed using $(AB)^\top = B^\top A^\top$ we get

$$\mathbf{B}_k = -\sum_{i=0}^{k-1} [(\boldsymbol{\Phi}^\mathbb{Q})^\top]^i \boldsymbol{\beta}.$$

12.8 Appendix B1: Extra Figures

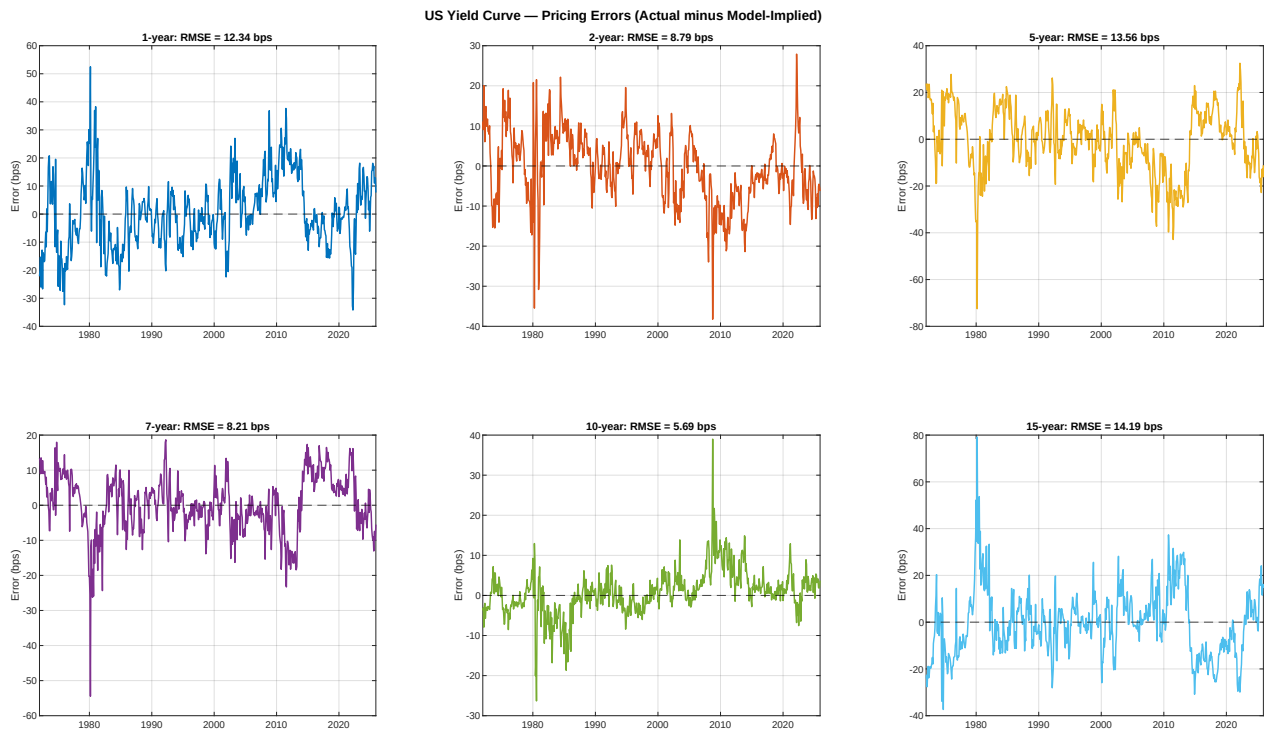


Figure 18: Pricing Errors for the Two-Factor GATSM (CMAES)

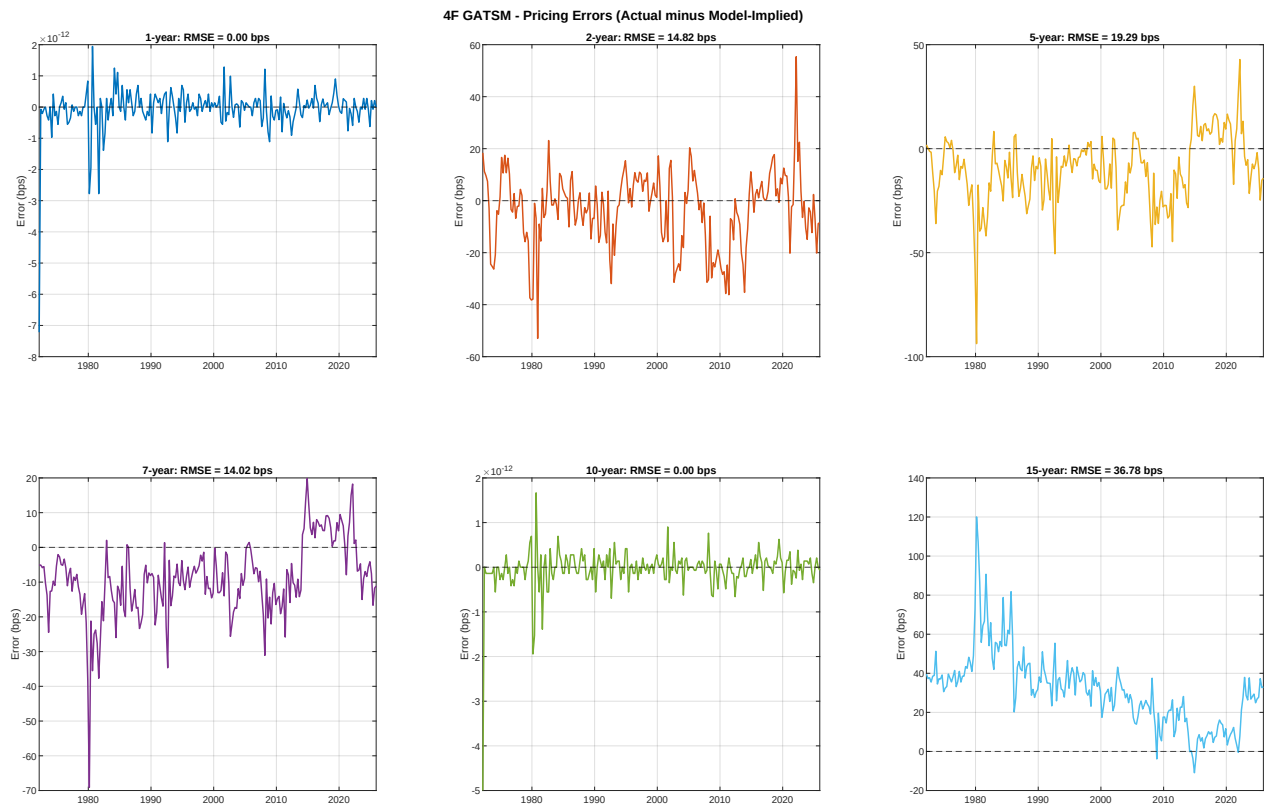


Figure 19: Pricing Errors for the Four-Factor GATSM (NS)

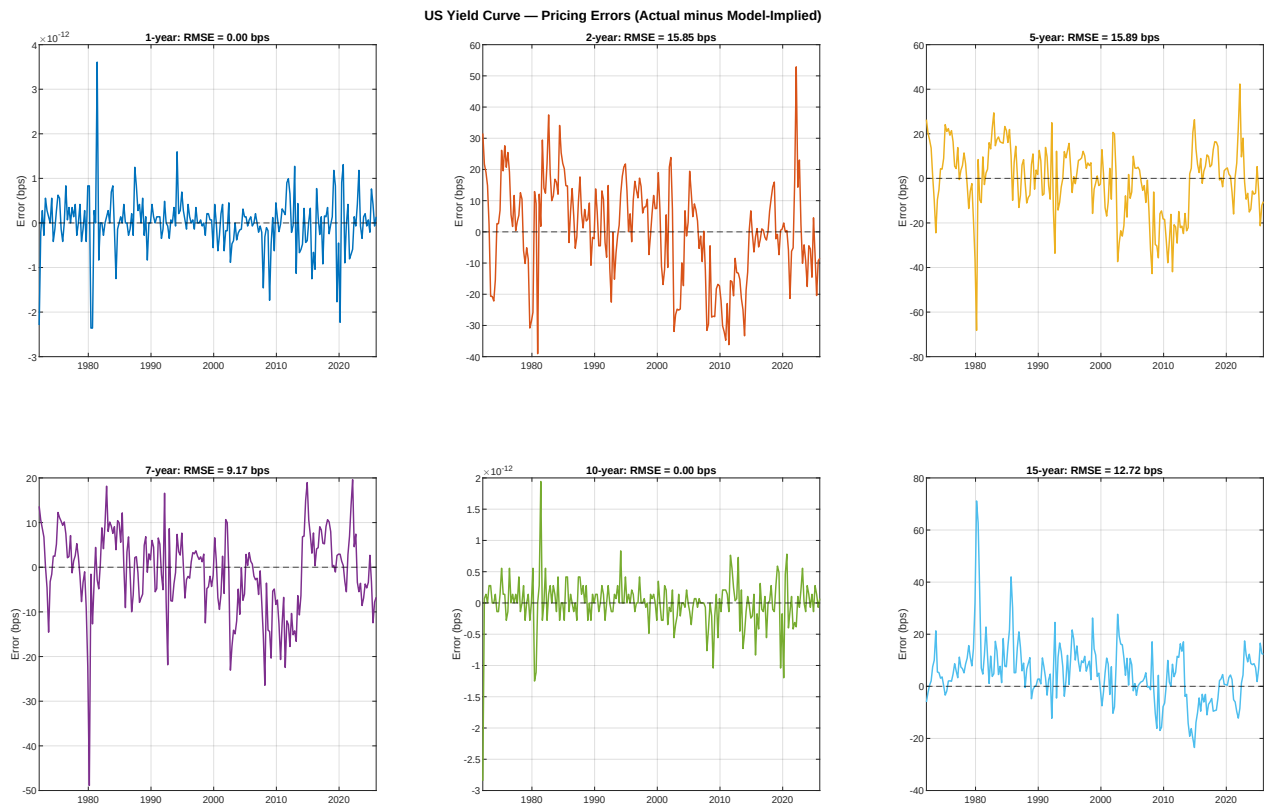


Figure 20: Pricing Errors for the Four-Factor GATSM with surveys (1Q)

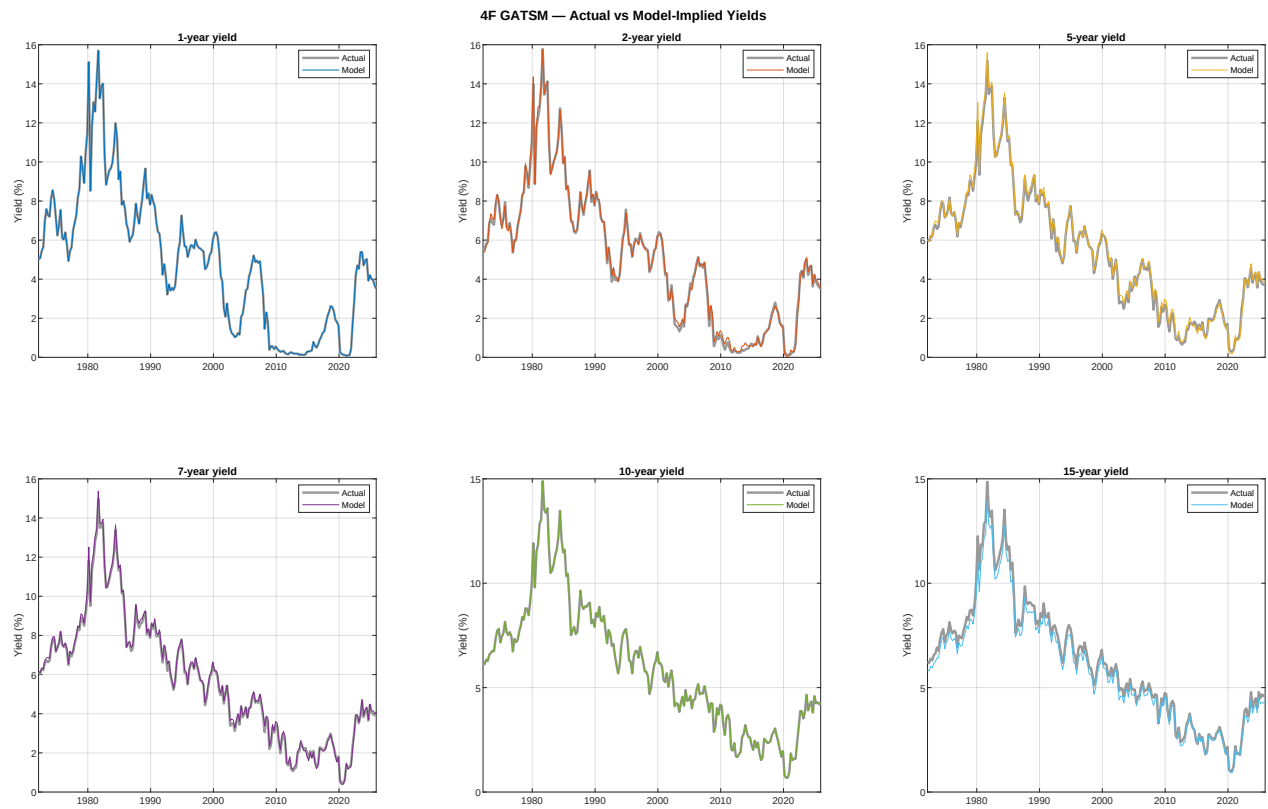


Figure 21: Yield Fit Plot for the Four-Factor GATSM (NS)

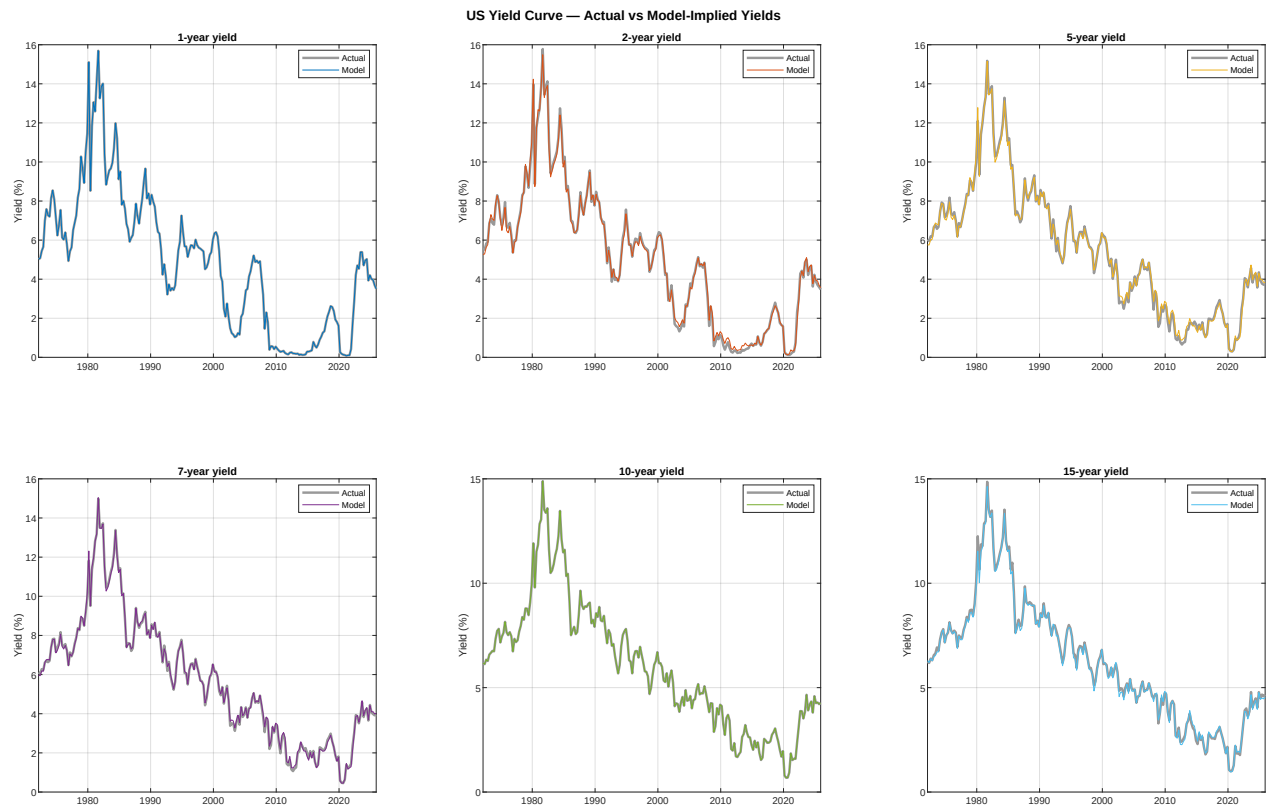


Figure 22: Yield Fit Plot for the Four-Factor GATSM with surveys (1Q)

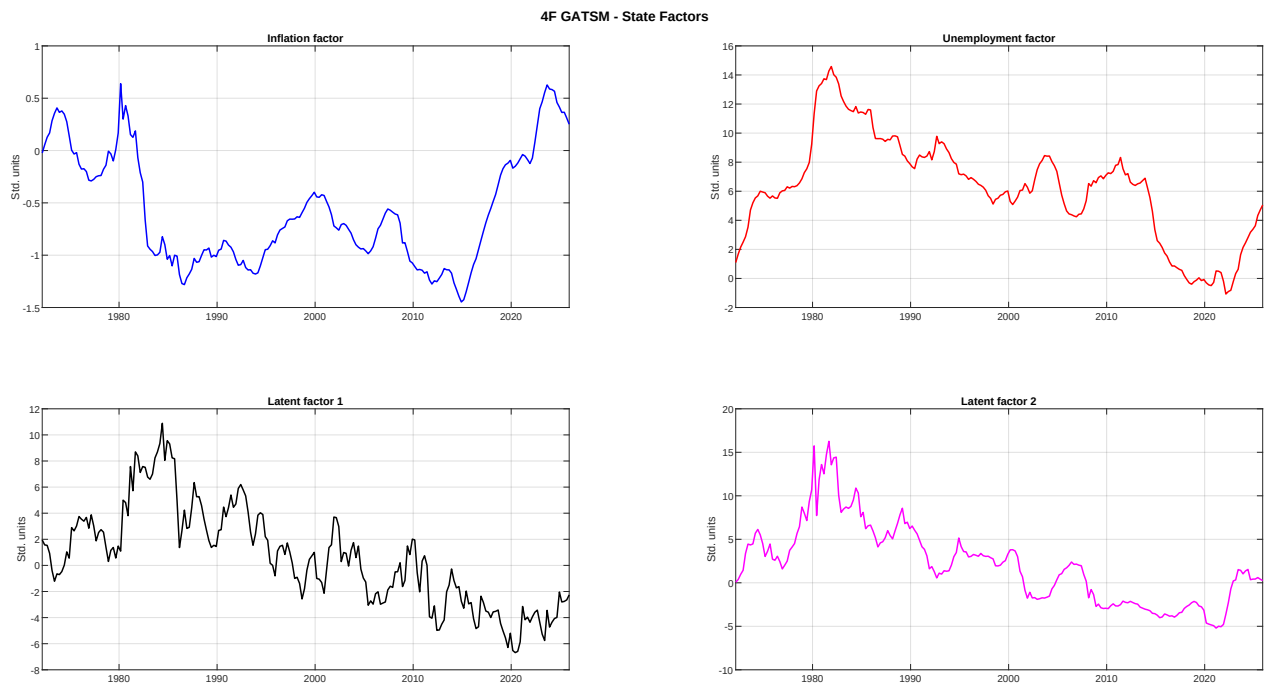


Figure 23: Figure of Factors in the Four-Factor GATSM (NS)

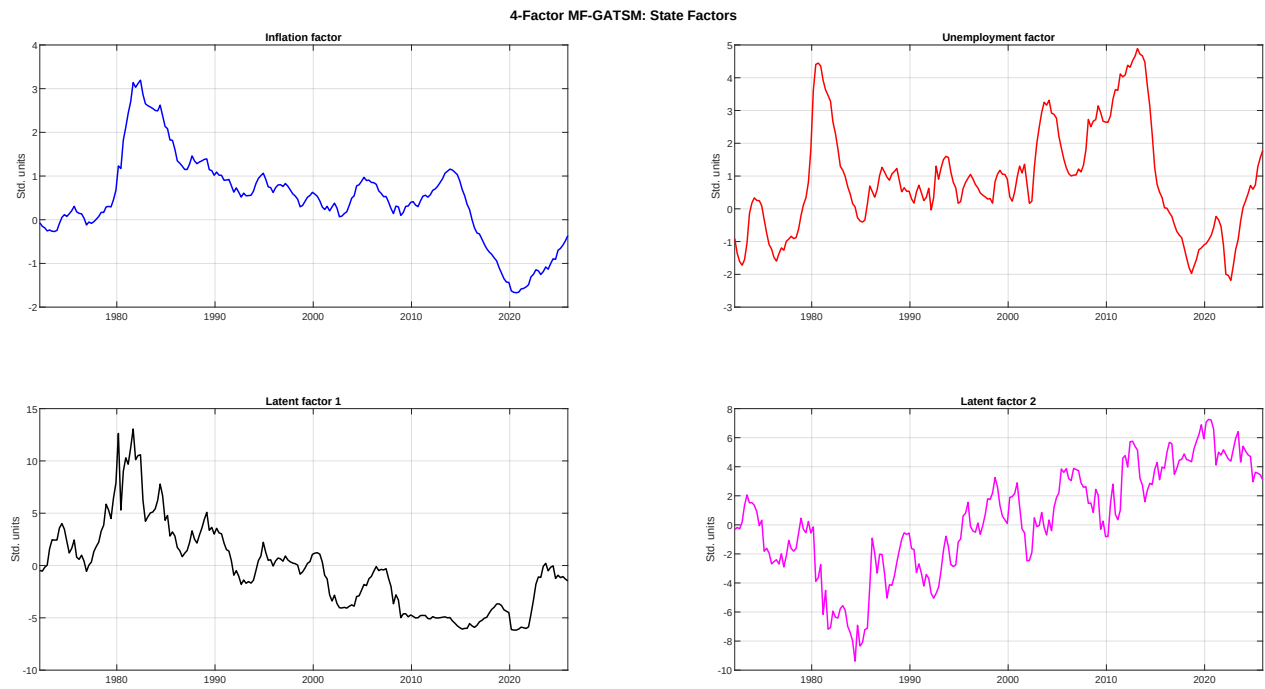


Figure 24: Figure of Factors in the Four-Factor GATSM with surveys (1Q)

12.9 Appendix C1: Main Survey Script

```
% This script incorporates surveys into the four-factor GATSM
%
% Files needed in dir
%   US_monthly_yields_Jan1972_Dec2025.csv
%   US_monthly_yields_Jan1972_Dec2025_maturities.csv
%   US_macro_inflation_unemployment_MF.csv

%% User Settings

% Maturities to use in estimation (years).
% Expressed in quarters; must match values in the maturities
  CSV
% when converted from years (multiply years by 4).
matSelect = [1 2 5 7 10 15] * 4;      % Maturities in quarters

% Yields priced exactly (used to invert latent factors); must
  equal nxL = 2.
% Remaining maturities are priced with measurement error.
% We choose the 1-year and 10-year to span the curve.
exactYears = [1 10];                  % years; must be a subset
  of matSelect/4
errorYears = setdiff(matSelect/4, exactYears);

% Factor dimensions
nxM = 2;      % number of observable macro factors (inflation,
  unemployment)
nxL = 2;      % number of latent yield factors
nx  = nxM + nxL; % total state dimension (= 4)

% Annualisation factor: quarterly data -> annual yields
ann = 4;
```

```

w_svy = 1.0;

%% Load data

% 2a: Yield data (monthly CSV -> quarterly sub-sample)
yields_raw = csvread('Data/US_monthly_yields_Jan1972-Dec2025.
    csv');
mats_years = csvread('Data/
    US_monthly_yields_Jan1972-Dec2025_maturities.csv');

% Validate that every requested maturity exists in the CSV
assert(all(ismember(matSelect/ann, mats_years)), ...
    ['One or more maturities in matSelect are not available in
    the CSV. ' ...
    'Available maturities (years): ' num2str(mats_years)]);

% Sub-sample monthly -> quarterly: rows 3, 6, 9, ... (Mar, Jun,
    Sep, Dec)
% Starting Jan 1972, the first quarterly obs is March 1972 =
    row 3.
T_monthly = size(yields_raw, 1);
qIdx      = 3 : 3 : T_monthly;      % row indices of quarterly
    observations
T          = length(qIdx);          % number of quarterly obs

numObs = length(matSelect);          % total number of yield
    maturities used
data    = zeros(numObs, T);          % numObs x T (annualised
    decimal)
for i = 1:numObs
    col      = find(mats_years == matSelect(i)/ann);
    data(i,:) = yields_raw(qIdx, col)' / 100;
end

```

```

% 2b:
% Y1 has nxL rows      one per latent factor (used to invert
    x_latent).
% Y2 has Ne = numObs - nxL rows      priced with common
    measurement error stdY.
idxExact  = ismember(matSelect/ann, exactYears);    % logical,
    length numObs
idxError   = ~idxExact;
Ne         = sum(idxError);

matSelect_Y1 = matSelect(idxExact);    % nxL-element vector (
    quarters)
matSelect_Y2 = matSelect(idxError);    % Ne-element vector (
    quarters)

Y1 = data(idxExact, :);    % nxL x T  yields priced without
    error
Y2 = data(idxError, :);    % Ne  x T  yields priced with
    measurement error

% 2c: Macro data (quarterly, aligned to yield sub-sample)
% The macro CSV has valid values every 3rd row (same cadence as
    qIdx).
macro_raw = csvread('Data/US_macro_inflation_unemployment_MF.
    csv'); % T_monthly x 2
macro_Q    = macro_raw(qIdx, :);    % T x 2  [inflation,
    unemployment]

% Fill any residual NaNs by linear interpolation
for j = 1:2
    col      = macro_Q(:,j);
    nanIdx   = isnan(col);
    if any(nanIdx)

```

```

        col(nanIdx) = interp1(find(~nanIdx), col(~nanIdx), ...
                               find(nanIdx), 'linear', 'extrap')
    ;

    macro_Q(:,j) = col;
end
end

% Normalise to zero mean and unit variance
macro_mean = mean(macro_Q);
macro_std  = std(macro_Q);
xMacro     = ((macro_Q - macro_mean) ./ macro_std)';    % nxM x
T

if w_svy > 0
    fprintf(' Loading KO survey data (w_svy=%.2f)...\\n', w_svy
    );
    T_infl_raw = readtable('US_SurveyData.xlsx', 'Sheet', '
        Inflation', ...
                           'ReadRowNames', true);
    T_unemp_raw = readtable('US_SurveyData.xlsx', 'Sheet', '
        Unemployment', ...
                           'ReadRowNames', true);

    surv_infl_raw = double(T_infl_raw(:, 'dpgdp3'));    % T x
    1 (% ann., 1Q-ahead)
    surv_unemp_raw = double(T_unemp_raw(:, 'UNEMP3'));    % T x
    1 (% , 1Q-ahead)

    % Normalise each survey series by its own mean and std
    svy_mean_infl = mean(surv_infl_raw, 'omitnan');
    svy_std_infl  = std( surv_infl_raw, 'omitnan');
    svy_mean_unemp = mean(surv_unemp_raw, 'omitnan');
    svy_std_unemp  = std( surv_unemp_raw, 'omitnan');

```

```

xSvy_infl = (surv_infl_raw - svy_mean_infl) /
    svy_std_infl;    % T x 1
xSvy_unemp = (surv_unemp_raw - svy_mean_unemp) /
    svy_std_unemp;    % T x 1

% Stack: xSurvey is nxM x T (rows = [infl; unemp])
xSurvey = [xSvy_infl'; xSvy_unemp'];

fprintf('  Survey normalised: infl(mean=%.4f,std=%.4f),
    unemp(mean=%.4f,std=%.4f)\n', ...
    mean(xSvy_infl), std(xSvy_infl), mean(xSvy_unemp), std(
        xSvy_unemp));
else
    xSurvey = [];    % no survey data; Block m OLS uses actual
        only
    fprintf('  w_svy=0: survey augmentation disabled (standard
        V23).\n');
end

dateStart = datenum(1972, 3, 1);
setup_dates = dateStart + (0:T-1)' * 91;    % approximate
    quarterly spacing
dateVec = datetime(1972, 3, 1) + calmonths(3*(0:T-1));

fprintf('  Quarterly obs: T = %d\n', T);
fprintf('  Maturities (years): %s\n', num2str(matSelect/ann));
fprintf('  Priced exactly (Y1): %s yr\n', num2str(exactYears));
fprintf('  Priced with error (Y2): %s yr\n', num2str(errorYears
    ));

%% First-stage OLS (reduced-form VAR)

T1 = T - 1;    % usable observations after one lag

ym_act = xMacro(:, 2:end)';    % T1 x nxM (

```

```

    actual at t+1)
xm_act = [ones(T1,1), xMacro(:,1:end-1)']; % T1 x (1+nxM)

if w_svy > 0 && ~isempty(xSurvey)
    % Survey dependent variable: xSurvey(:,t) = E^svy_{t-1}[x_{macro,t}]
    % We need survey at t matching actual at t (both are t+1 in the OLS).
    % xSurvey(:,t) = dpdgp3/UNEMP3 at t = 1Q-ahead forecast made at t for t+1.
    % For OLS rows t=1..T1:
    %   actual dep = xMacro(:,t+1) indexed as xMacro(:,2:end) -> ym_act
    %   survey dep = xSurvey(:,t) indexed as xSurvey(:,1:end-1) -> ym_svy
    %   regressor = xMacro(:,t) indexed as xMacro(:,1:end-1) -> xm_act/xm_svy
    % Note: dpdgp3_t forecasts x_m(t+1) using info at t, with no look-ahead.
    % For h=1 surveys: dpdgp3_t forecasts x_m(t+1), so for OLS row t
    % we need xSurvey(:,t), i.e. indices 1..T-1 (not 2..T).
    ym_svy = xSurvey(:, 1:end-1)'; % T1 x nxM (survey at t, forecasting t+1)
    xm_svy = [ones(T1,1), xMacro(:,1:end-1)']; % T1 x (1+nxM) (same RHS as actual)

    % Stack with weight w_svy
    ym = [ym_act; w_svy * ym_svy];
    xm = [xm_act; w_svy * xm_svy];
else
    ym = ym_act;
    xm = xm_act;
end

```

```

paramM      = xm \ ym;
Am_star     = paramM(1, :)';                                % nxM x 1
phiP_starmm = paramM(2:end, :)';                            % nxM x
              nxM

% Residuals and Omega_star computed from ACTUAL data only (
  structural noise)
umt_star     = ym_act - xm_act * paramM;                    % T1 x nxM
Omega_star   = (umt_star' * umt_star) / T1;                  % nxM x
              nxM

fprintf('  Block m: %d actual + %d survey obs (w_svy=%.2f).\n',
      ...
      T1, T1 * double(w_svy>0), w_svy);
fprintf('  phiP_starmm:\n');  disp(phiP_starmm);

% Block 1: Y1 on lagged Y1 and contemporaneous macro
y1          = Y1(:, 2:end)';
              % T1 x nxL
x1          = [ones(T1,1), Y1(:,1:end-1)', xMacro(:,2:end)'];
              % T1 x (1+nxL+nxM)
param1      = x1 \ y1;
A1_star     = param1(1, :)';                                % nxL x 1
              (annual)
phiP_star11 = param1(2:1+nxL, :)';                          % nxL x nxL
              (lagged Y1)
psiP_star1m = param1(2+nxL:end, :)';                        % nxL x nxM
              (contemp. macro, annual)
ult_star    = y1 - x1 * param1;                              % T1 x nxL
Omega_star1 = (ult_star' * ult_star) / T1;                  % nxL x nxL
              (annual^2)

% Block 2: Y2 on contemporaneous macro and Y1

```

```

y2                = Y2(:, 2:end)';

                                % T1 x Ne
x2                = [ones(T1,1), xMacro(:,2:end)', Y1(:,2:end)'];
                    % T1 x (1+nxM+nxL)
param2            = x2 \ y2;
A2_star           = param2(1, :)';                                % Ne x 1 (
                    annual)
phiP_star2m       = param2(2:1+nxM, :)';                          % Ne x nxM
phiP_star21       = param2(2+nxM:end, :)';                        % Ne x nxL
u2t_star          = y2 - x2 * param2;                             % T1 x Ne

fprintf('  OLS complete.\n');

%% Recover sigma and phiP analytically
sigma_macro = chol(Omega_starm, 'lower');                        % nxM x nxM
                    lower triangular
sigma        = blkdiag(sigma_macro, eye(nxL)); % nx x nx

B1l_chol      = chol(Omega_star1, 'lower');                        % nxL x nxL (
                    Cholesky of Y1 resid var)
phiP_mm       = phiP_starmm;                                       % nxM x nxM (
                    from macro OLS)
phiP_ll       = B1l_chol \ phiP_star11 * B1l_chol; % nxL x nxL
                    (similarity)
phiP          = blkdiag(phiP_mm, phiP_ll);                        % nx x nx (
                    block diagonal)

eigsP = abs(eig(phiP));
fprintf('  max|eig(phiP)| = %.4f ', max(eigsP));
if max(eigsP) >= 1
    fprintf('  WARNING: P-dynamics non-stationary.\n');
else
    fprintf('  (stationary)\n');
end

```

```

fprintf('  sigma_macro =\n');  disp(sigma_macro);

%% Solve for (Lambda_mm, Lambda_ll, beta) numerically

% OLS targets (all in annual decimal units)
B1m_OLS = psiP_star1m; % nxL x nxM
B2m_OLS = phiP_star2m + phiP_star21 * B1m_OLS; % Ne x nxM
B1B1_OLS = Omega_star1; % nxL x nxL
B2B1_OLS = phiP_star21 * Omega_star1; % Ne x nxL

% Compact maturity struct
mature.exact = matSelect_Y1;
mature.error = matSelect_Y2;

% ---- CMA-ES options (shared by both subproblems)
-----
cmaOpts.Display      = 'off';
cmaOpts.Plotting     = 'off';
cmaOpts.Saving       = 0;
cmaOpts.VerboseModulo = 0;
cmaOpts.TolFun       = 1e-14;
cmaOpts.TolX         = 1e-14;
cmaOpts.MaxFunEvals   = 1e6;
cmaOpts.StopOnWarnings = 'no';

% Subproblem A: macro block (Lambda_mm, beta_m)
% Parameter vector: p = [Lambda_mm(:); beta_m] (nxM^2 + nxM
%   elements)
% SSR = ||local_macroBlock_residuals(p)||^2
%
% SIGN: beta_m[k] sign equals the column-sum sign of B1m_OLS.
% SCALE: |beta_m[k]| ~ max(|B1m_OLS[:,k]|) / (ann * 0.86).

beta_m_sign = sign(sum(B1m_OLS, 1)); %

```

```

    1 x nxM
beta_m_scale = max(abs(B1m_OLS), [], 1)' / (ann * 0.86);    %
    nxM x 1

% Lower/upper bounds for beta_m, derived from the sign and
% expected scale.
% For a positive-sign factor: lb = +scale/10, ub = +5*scale (
% positive range)
% For a negative-sign factor: lb = -5*scale, ub = -scale/10 (
% negative range)
% The floor of scale/10 (away from zero) prevents CMA-ES from
% collapsing a
% factor to zero. min/max ensure lb < ub regardless of sign.
lb_A = [-5*ones(nxM^2,1); min(beta_m_sign'.*beta_m_scale/10,
    beta_m_sign'.*beta_m_scale*5)];
ub_A = [ 5*ones(nxM^2,1); max(beta_m_sign'.*beta_m_scale/10,
    beta_m_sign'.*beta_m_scale*5)];

x0_A    = [zeros(nxM^2,1); beta_m_sign'.*beta_m_scale];
sig0_A  = [0.5*ones(nxM^2,1); beta_m_scale];

cmaOpts.LBounds = lb_A;
cmaOpts.UBounds = ub_A;
cmaOpts.PopSize = 4 + floor(3*log(length(x0_A)));

objA = @(p) sum(local_macroBlock_residuals(p, phiP_mm,
    sigma_macro, ...
                                B1m_OLS, B2m_OLS,
                                mature, nxM, ann)
                                .^2);
[xBestA, fBestA] = cmaes_dsgeDisplay(objA, x0_A, 1, sig0_A,
    cmaOpts);

Lambda_mm = reshape(xBestA(1:nxM^2), nxM, nxM);

```

```

beta_m      = xBestA(nxM^2+1 : end);
phiQ_mm     = phiP_mm - sigma_macro * Lambda_mm;
fprintf('  Macro CMA-ES SSR: %.2e\n', fBestA);
fprintf('  eig(phiQ_mm): %s\n', num2str(abs(eig(phiQ_mm))'));
fprintf('  beta_m: %s\n', num2str(beta_m'));

% Subproblem B: latent block (Lambda_ll_Jordan, beta_l)
% Parameter vector: p = [lev; lb_off; lc_off; beta_l]    (3 +
    nxL elements)
% Lambda_ll = [[lev, lb_off], [lc_off, lev]]    equal-diagonal
    Jordan form
% phiQ_ll    = phiP_ll - Lambda_ll
%
% IDENTIFICATION (H-W Propositions 1-2):
%   (a) H-W Proposition 2: for Nl=2, the Jordan
    parameterisation has
%       two observationally equivalent solutions related by
    transposing
%       phiQ_ll. Resolved by the canonical condition lb <= lc (
    i.e. the
%       (1,2) off-diagonal of Lambda_ll <= its (2,1)
    counterpart).
%   (b) Sign normalisation (H-W Proposition 1): beta_l[k] > 0
    for all
%       latent factors. Flipping sign of beta_l[k] and of row/
    col k of
%       phiQ_ll leaves B loadings unchanged    this is the
    remaining
%       rotation ambiguity for each factor independently.
    Resolved by
%       enforcing strict positivity via CMA-ES lower bounds.
%
% SCALE: |beta_l[k]| ~ diag(B11_chol)[k] / (ann * 0.86)

```

```

beta_l_scale = diag(B1l_chol) / (ann * 0.86);    % nxL x 1

% Strictly positive lower bound for beta_l (H-W sign
  normalisation).
% BETA_L FLOOR: 0.0001 on BOTH entries.
% This blocks the degenerate near-zero basin (beta_l~0) that
  CMA-ES
% The Jordan swap can permute the two entries, so both must be
  floored to be safe.
lb_B = [-5; -5; -5; 0.0001; 0.0001];
ub_B = [ 5;  5;  5; beta_l_scale(1)*5;  beta_l_scale(2)*5];

% THREE STARTS for robustness, where we keep the lowest SSR
  solution.
% Subproblem B inputs (B1B1_OLS, B2B1_OLS, phiP_ll) are
  identical to NS
% regardless of w_svy, so the V23 solution is always a valid
  starting point.
%
%   Start 1: canonical form      (b<c satisfied)
%   Start 2: pre-swap form       (b>c; same solution, different
    orientation)
%   Start 3: scale-based default (broad search fallback)
x0_B_s1  = [-0.0689; 0.1351; 0.1520; 0.00226; 0.00062];
sig0_B_s1 = [0.03;    0.03;    0.03;    0.00050; 0.00015];

x0_B_s2  = [-0.0689; 0.1714; 0.1157; 0.00232; 0.00038];
sig0_B_s2 = [0.03;    0.03;    0.03;    0.00050; 0.00010];

x0_B_s3  = [0; 0; 0; beta_l_scale];
sig0_B_s3 = [0.3; 0.1; 0.1; beta_l_scale];

cmaOpts.LBounds = lb_B;
cmaOpts.UBounds = ub_B;

```

```

cmaOpts.PopSize = 4 + floor(3*log(length(x0_B_s1)));

objB = @(p) sum(local_latentBlock_residuals(p, phiP_ll,
      B1B1_OLS, B2B1_OLS, ...
      mature, nxL, ann)
      .^2);

[xB_s1, fB_s1] = cmaes_dsgeDisplay(objB, x0_B_s1, 1, sig0_B_s1,
      cmaOpts);
[xB_s2, fB_s2] = cmaes_dsgeDisplay(objB, x0_B_s2, 1, sig0_B_s2,
      cmaOpts);
[xB_s3, fB_s3] = cmaes_dsgeDisplay(objB, x0_B_s3, 1, sig0_B_s3,
      cmaOpts);

fprintf('  Latent SSR: s1=%.2e  s2=%.2e  s3=%.2e\n', fB_s1,
      fB_s2, fB_s3);

[fBestB, iBest] = min([fB_s1, fB_s2, fB_s3]);
xBestBs = {xB_s1, xB_s2, xB_s3};
xBestB = xBestBs{iBest};
fprintf('  Latent best: start %d (SSR=%.2e)\n', iBest, fBestB);

lev      = xBestB(1);
lb       = xBestB(2);
lc       = xBestB(3);
beta_l   = xBestB(4:5);

Lambda_ll = [lev, lb; lc, lev];
phiQ_ll   = phiP_ll - Lambda_ll;

% Enforce H-W Proposition 2 canonical form: phiQ_ll[1,2] <=
      phiQ_ll[2,1]
%
% the canonical condition is on phiQ_ll itself, NOT on the

```

```

% Lambda_ll entries lb and lc. Since phiQ_ll = phiP_ll -
    Lambda_ll, and
% phiP_ll has non-zero off-diagonals, lb <= lc does NOT imply
% phiQ_ll[1,2] <= phiQ_ll[2,1] in general.
%
% The two equivalent solutions are phiQ_ll and phiQ_ll' (
    transpose),
% with beta_l permuted accordingly. We keep the representative
% with phiQ_ll[1,2] <= phiQ_ll[2,1].
if phiQ_ll(1,2) > phiQ_ll(2,1)
    phiQ_ll = phiQ_ll';          % transpose gives the other
    equivalent form
    beta_l = beta_l([2,1]);      % permute beta_l to match
    % Recompute Lambda_ll from the canonical phiQ_ll
    Lambda_ll = phiP_ll - phiQ_ll;
end
fprintf(' Latent CMA-ES SSR: %.2e\n', fBestB);
fprintf(' eig(phiQ_ll): %s\n', num2str(abs(eig(phiQ_ll))));
fprintf(' beta_l: %s (both > 0: %d)\n', num2str(beta_l'), all
    (beta_l > 0));
fprintf(' Jordan b<=c: b=%.4f, c=%.4f (satisfied: %d)\n', ...
    phiQ_ll(1,2), phiQ_ll(2,1), phiQ_ll(1,2) <= phiQ_ll
    (2,1) + 1e-10);

% Assemble full model matrices (block-diagonal: Q-independence)
phiQ = blkdiag(phiQ_mm, phiQ_ll);
beta = [beta_m; beta_l];

eigsQ = abs(eig(phiQ));
fprintf(' eig(phiQ) (moduli): %s\n', num2str(sort(eigsQ, '
    descend')));
fprintf(' beta = %s\n', num2str(beta));

%% Compute B loadings for downstream steps

```

```

[B1m_ann, ~, B1l_ann, B2l_ann] = local_computeB_ann(phiQ, beta,
    mature, nxM, nxL, ann);

%% Solve for (muQ, alpha) numerically

% Solve:  A2_star - A2_ann(alpha,muQ) + B2l_ann / B1l_ann *
    A1_ann = 0
% for x = [alpha; muQ_macro]    (muQ_latent = 0 by normalisation
    )

muQ = zeros(nx, 1);    % muQ_latent = 0 by normalisation;
    muQ_macro solved below

% Parameter vector: x = [alpha; muQ_macro]    (1 + nxM elements)
%
% ALPHA BOUNDS: narrow to economically plausible range.
%   H-W delta0 (alpha) = -0.0082 quarterly which is around r0 =
%   -3.3% ann?
%   The unconditional short rate is  $E[r] = \text{ann} * (\alpha + \beta' * (I - \phi P)^{-1} * \mu P)$ .
%   With  $E[y_{1yr}]$  4.86%, alpha should be in roughly -0.02
%   to +0.025 quarterly.
%   We use [-0.03, +0.03] to give CMA-ES adequate room without
%   allowing
%   alpha to go to -0.05 which results in degenerate solution
%   that occurs with wider bounds.
%
% UNCONDITIONAL MEAN SOFT CONSTRAINT:
%   The A-intercept moment conditions alone do not uniquely pin
%   down the
%   (alpha, muQ) decomposition: many combinations give the same
%   A2_star fit
%   but produce wildly different unconditional yield levels (we

```

```

    observed
%   alpha = -0.05,  $E[r] = -7.3\%$  when the data mean is  $+4.9\%$ ).
%   We add a soft penalty that anchors the unconditional short
    rate:
%   penalty =  $w * (ann * (alpha + beta' * (I - \phi P)^{-1} * \mu P) -$ 
    mean_y1yr)2
%   where w is chosen so the penalty is comparable to the
    A2_star residuals.
%   This does NOT change what is being estimated (it is not a
    hard constraint)
%   but prevents CMA-ES from settling on degenerate (alpha, muQ
    ) pairs.

mean_y1yr = mean(data(1,:));           % mean 1yr yield (
    annual decimal)

x0_7      = [mean_y1yr/ann; zeros(nxM,1)]; % alpha start: mean
    short rate / ann
sig0_7    = [0.005; 0.3*ones(nxM,1)];

% TIGHTENED muQ BOUNDS (from Survey experience):
%   muQ =  $c^Q$ : the constant market price of risk intercept.

%   Allowing 50 lets Step 7 find
%   large-muQ solutions that imply huge muP_1.
%   Bounding at 3 prevents this while giving 30 the HW
    magnitude.
lb_7      = [-0.03; -3*ones(nxM,1)];
ub_7      = [ 0.03;  3*ones(nxM,1)];

cmaOpts7      = cmaOpts;
cmaOpts7.LBounds = lb_7;
cmaOpts7.UBounds = ub_7;
cmaOpts7.PopSize = 4 + floor(3*log(length(x0_7)));

```

```

cmaOpts7.TolFun      = 1e-10;
cmaOpts7.TolX        = 1e-10;

% we have three soft constraints
%
% Constraint 1      alpha anchor (existing):
%   Penalise alpha deviating from mean_y1yr/ann.
%   Prevents alpha drifting to boundaries while muQ compensates
%   .
%
% Constraint 2      muQ shrinkage (from Survey V7):
%   Penalise large |muQ_macro|.  muQ enters A_ann through the
%   recursion;
%   large muQ -> large A_ann -> large muP_1 -> large E[x_latent
%   ] -> level wrong.
%   Even with tight bounds, shrinkage prevents muQ from sitting
%   at 3 .
%
% Constraint 3      FULL UNCONDITIONAL YIELD LEVEL:
%   Embed the Step 8 muP_1 recovery inside the objective so
%   that
%    $E[r] = ann * (\alpha + \beta * (1 - \phi_P)^{-1} * \mu_P)$  can be
%   directly penalised.
%   This breaks the circular dependency between Step 7 and Step
%   8 by
%   computing the implied muP_1 forward from any candidate (
%   alpha, muQ).
%   We anchor at the 10yr yield mean (data mean = 5.97%)
%   because the 1yr
%   is already anchored by Constraint 1, and the 10yr level
%   captures the
%   long-end slope problem most directly.
%
%   Implementation: for each candidate x=[alpha;muQ], compute

```

```

    A1_ann(x),
%   recover muP_l analytically, then compute E[r] and E[y_10yr]
    and penalise.
%
%   A2 residuals scaled by 1e6  -> w_mean and w_muQ in same
    units
%   Yield level in decimal      -> w_level scales to match 1e6
    -scaled residuals

w_mean  = 1e6;      % alpha anchor weight
w_muQ   = 1e4;      % muQ shrinkage weight
w_level = 5e5;      % level anchor weight (0.01% deviation = 1e-4
    decimal * 5e5 = 50 cost)

IminusPhiP_ll = eye(nxL) - phiP_ll;    % precompute for speed

% muP_m = Am_star (AP2003 independence: macro intercept = Block
    m OLS intercept).
% Defined here so it is available inside the step7_objective
    closure;
% Step 8 will also assign it explicitly for clarity.
muP_m = Am_star;

obj7 = @(x) step7_objective(x, mature, phiQ, beta, sigma, ...
    A2_star, A1_star, B1l_ann,
    B2l_ann, B1l_chol, ...
    phiP_ll, IminusPhiP_ll, muP_m, ...
    nxM, nxL, ann, mean_y1yr, ...
    w_mean, w_muQ, w_level);

[xBest7, fBest7] = cmaes_dsgeDisplay(obj7, x0_7, 0.5, sig0_7,
    cma0pts7);

alpha      = xBest7(1);

```

```

muQ(1:nxM) = xBest7(2:end);

fprintf('  CMA-ES SSR: %.2e\n', fBest7);
fprintf('  alpha = %.6f  (quarterly; r0 = %.4f%%)\n', alpha,
        alpha*ann*100);
fprintf('  muQ_macro = %s\n', num2str(muQ(1:nxM)'));

%% Recover muP analytically

% With the AP2003 independence assumption (phiP_m1 = phiP_lm =
%   0):
%
%   muP_m = Am_star    (macro intercept from block m OLS; note
%   xMacro has
%
%               zero mean by construction so Am_star
%   0)
%   muP_l = B1l_ann \ (A1_star - A1_ann + B1l_ann * phiP_ll /
%   B1l_ann * A1_ann)
%   muP    = [muP_m; muP_l]

[A1_ann, ~] = local_computeA_ann(alpha, muQ, phiQ, beta, sigma,
    mature, nxM, nxL, ann);

% muP_m = Am_star (independence: zero cross-block correction)
% muP_l uses B1l_chol (same rotation as phiP_ll, so frames are
%   consistent)
muP_m = Am_star;
muP_l = B1l_chol \ (A1_star - A1_ann + B1l_chol * phiP_ll /
    B1l_chol * A1_ann);
muP    = [muP_m; muP_l];

fprintf('  muP = %s\n', num2str(muP));

%% Compute bond pricing loadings (g0, gx, A, B)

```

```

% Bond pricing recursion (quarterly periods)
%
%   k=1:  A(1,1) = -alpha,      B(:,1) = -beta
%   k>1:  A(1,k) = -alpha + A(1,k-1) + B(:,k-1)'*muQ
%          + 0.5*B(:,k-1)'*sigma2*B(:,k-1)
%          B(:,k) = -beta + phiQ'*B(:,k-1)
%
% Here A and B store raw (per-quarter, unnormalised) recursion
% coefficients.
% Annualised yield loadings at maturity n quarters:
%   g0(n) = ann * (-A(1,n) / n)
%   gx(n) = ann * (-B(:,n)' / n)

maxMat = max(matSelect);
A       = zeros(1, maxMat);      % 1 x maxMat (matches 2F
convention)
B       = zeros(nx, maxMat);     % nx x maxMat

sigma2 = sigma * sigma';        % nx x nx innovation covariance

for k = 1:maxMat
    if k == 1
        A(1,k) = -alpha;
        B(:,k) = -beta;
    else
        A(1,k) = -alpha + A(1,k-1) + B(:,k-1)'*muQ + ...
            0.5*B(:,k-1)'*sigma2*B(:,k-1);
        B(:,k) = -beta + phiQ'*B(:,k-1);
    end
end

% Annualised yield loadings at the selected maturities
% Matches 2F convention:  g0 (numObs x 1),  gx (numObs x nx)

```

```

g0 = ann * (-A(1, matSelect)' ./ matSelect');
           % numObs x 1
gx = ann * (-B(:, matSelect)' ./ repmat(matSelect', 1, nx));
           % numObs x nx

% Short-rate loadings (annualised)      used in yield
decomposition
r0 = ann * alpha;
rx = ann * beta';    % 1 x nx

% Pack model struct
model.g0      = g0;
model.gx      = gx;
model.muP     = muP;
model.phiP    = phiP;
model.alpha   = alpha;
model.beta    = beta;
model.phiQ    = phiQ;
model.muQ     = muQ;
model.sigma   = sigma;
model.A       = A;
model.B       = B;
model.r0      = r0;
model.rx      = rx;
model.matSelect = matSelect;
model.nxM     = nxM;
model.nxL     = nxL;

%% Kalman filter

% Measurement equation:  data_t = g0 + gx * x_t + stdY * v_t
%   Y1 rows: measurement error = 0 (priced exactly)
%   Y2 rows: common scalar measurement error stdY

```

```

%
% Transition:   $x_{t+1} = \mu_P + \phi_P * x_t + \sigma * u_{t+1}$ 

stdY = mean(std(u2t_star));    % scalar common measurement error
                                std

% Measurement noise covariance Rv (numObs x numObs):
%   zero for Y1 rows, stdY^2 * I for Y2 rows
Rv = zeros(numObs, numObs);
Rv(idxError, idxError) = eye(Ne) * stdY^2;

outKF = local_KalmanFilter(data, g0, gx, Rv, muP, phiP, sigma2)
;
outKF.model = model;

fprintf('  Average log-likelihood: %.6f\n', outKF.sumLogL / T);

%% Local Functions

function F = local_macroBlock_residuals(para, phiP_mm,
                                sigma_macro, ...
                                B1m_OLS, B2m_OLS,
                                mature, nxM, ann)

% Residuals for the macro sub-problem in MCSE Step 2.
% para = [Lambda_mm(:); beta_m]  (nxM^2 + nxM elements)
% phiQ_mm = phiP_mm - sigma_macro * Lambda_mm
Lambda_mm = reshape(para(1:nxM^2), nxM, nxM);
beta_m     = para(nxM^2+1 : end);

phiQ_mm = phiP_mm - sigma_macro * Lambda_mm;
% No stationarity constraint on phiQ_mm. The Q-measure dynamics
% can legitimately have |eig(phiQ_mm)| > 1
% The B-loading recursion stays finite for

```

```

% finite maturities regardless of Q-measure stationarity.

% B macro columns only depend on phiQ_mm and beta_m (Q-
independence)
mats = [mature.exact, mature.error];
B_mac = zeros(length(mats), nxM);
phiQt = phiQ_mm';
for im = 1:length(mats)
    n = mats(im); S = zeros(nxM,1); P = eye(nxM);
    for j = 0:n-1; S = S + P*beta_m; P = P*phiQt; end
    B_mac(im,:) = (ann * S / n)';
end

% if B loadings diverged (phiQ_mm very explosive), return large
penalty
if ~all(isfinite(B_mac(:)))
    nEq = numel(B1m_OLS) + numel(B2m_OLS);
    F = 1e6 * ones(nEq, 1);
    return;
end

nE = length(mature.exact);
B1m_s = B_mac(1:nE, :);
B2m_s = B_mac(nE+1:end, :);

F = [(B1m_OLS - B1m_s) * 1e5; (B2m_OLS - B2m_s) * 1e5];
F = F(:);
end

function F = local_latentBlock_residuals(para, phiP_ll,
    B1B1_OLS, B2B1_OLS, ...
    mature, nxL, ann)
% Residuals for the latent sub-problem in MCSE Step 2.
% para = [lev; lb; lc; beta_1] (3 + nxL elements)
% Lambda_ll = [[lev, lb], [lc, lev]] (equal-diagonal Jordan

```

```

    form)
% phiQ_ll    = phiP_ll - Lambda_ll
lev = para(1); lb = para(2); lc = para(3);
beta_l = para(4:end);

Lambda_ll = [lev, lb; lc, lev];
phiQ_ll    = phiP_ll - Lambda_ll;

% HYBRID CONSTRAINT
%   Latent Q-block: enforce stationarity |eig(phiQ_ll)| <
    0.9999.

%   Without this constraint the landscape has degenerate local
    minima where
%   phiQ_ll is explosive and beta_l[2] collapses to zero,
    making the second
%   latent factor unidentified. The stationarity penalty
    prevents this.
if any(abs(eig(phiQ_ll))) >= 0.9999)
    nEq = nxL*(nxL+1)/2 + numel(B2B1_OLS);
    F = 1e8 * ones(nEq, 1);
    return;
end

mats = [mature.exact, mature.error];
B_lat = zeros(length(mats), nxL);
phiQt = phiQ_ll';
for im = 1:length(mats)
    n = mats(im); S = zeros(nxL,1); P = eye(nxL);
    for j = 0:n-1; S = S + P*beta_l; P = P*phiQt; end
    B_lat(im,:) = (ann * S / n)';
end

% if B loadings diverged (phiQ_ll very explosive) return large

```

```

    penalty
if ~all(isfinite(B_lat(:)))
    nEq = nxL*(nxL+1)/2 + numel(B2B1_OLS);
    F = 1e8 * ones(nEq, 1);
    return;
end
nE      = length(mature.exact);
B1l_s   = B_lat(1:nE, :);
B2l_s   = B_lat(nE+1:end, :);

% Symmetric part of B1l*B1l'
M1 = B1B1_OLS - B1l_s * B1l_s';
r1 = M1([1,2,4]) * 1e5;      % [M(1,1) M(2,1) M(2,2)] = upper-
    left, lower-left, lower-right

M2 = B2B1_OLS - B2l_s * B1l_s';
r2 = M2(:) * 1e5;

F = [r1(:); r2(:)];
end

function [B1m, B2m, B1l, B2l] = local_computeB_ann(phiQ, beta,
    mature, nxM, nxL, ann)
% Compute ANNUALISED B yield loadings for the MCSE moment
    conditions.
%
% Annual yield loading:  gx(n) = ann * (1/n)*sum_{j=0}^{n-1}(
    phiQ')^j * beta
%
% This returns B in annual decimal units, matching the OLS
    targets
% psiP_star1m and Omega_star1 which come from regressions on
    annual yields.
%
```

```

% Returns B partitioned into macro (m) and latent (l) column
    blocks,
% for maturities in mature.exact (Y1) and mature.error (Y2).

nx    = nxM + nxL;
mats = [mature.exact, mature.error];

B_all = zeros(length(mats), nx);
phiQt = phiQ';
for im = 1:length(mats)
    n    = mats(im);
    S    = zeros(nx, 1);
    pow = eye(nx);
    for j = 0:n-1
        S    = S + pow * beta;
        pow = pow * phiQt;
    end
    % Multiply by ann to convert per-quarter to annual units
    B_all(im,:) = (ann * S / n)';
end

nE = length(mature.exact);
B1 = B_all(1:nE, :);
B2 = B_all(nE+1:end, :);

B1m = B1(:, 1:nxM);
B1l = B1(:, nxM+1:end);
B2m = B2(:, 1:nxM);
B2l = B2(:, nxM+1:end);
end

function [A1, A2] = local_computeA_ann(alpha, muQ, phiQ, beta,
    sigma, ...
                                     mature, nxM, nxL, ann)

```

```

% Compute ANNUALISED A yield intercepts.
%
% Annual intercept:  $g_0(n) = \text{ann} * (-A\_bar(n)/n)$ 

sigma2 = sigma * sigma';
nMax    = max([mature.exact, mature.error]);

% Per-quarter recursion (same sign convention as main script
    Step 9)
A_raw    = zeros(1, nMax);
A_bar    = -alpha;
B_bar    = -beta;
A_raw(1) = -A_bar;      % = alpha

for k = 2:nMax
    A_bar    = -alpha + A_bar + B_bar'*muQ + 0.5*(B_bar'*sigma2
        *B_bar);
    B_bar    = phiQ' * B_bar - beta;
    A_raw(k) = -A_bar / k;
end

% Annualise
A_ann = ann * A_raw;

A1 = A_ann(mature.exact)';    % nxL x 1
A2 = A_ann(mature.error)';    % Ne x 1
end

function F = local_alphaMuQ_residuals(x, mature, phiQ, beta,
    sigma, ...
                                     A2_star, B1l_ann,
                                     B2l_ann, nxM, nxL, ann
                                     )
% Residuals of MCSE Step 4:  $A2\_star - A2\_ann + B2l\_ann/B1l\_ann$ 

```

```

    * A1_ann = 0
% All quantities in ANNUAL decimal units.
nx      = nxM + nxL;
alpha   = x(1);
muQ     = zeros(nx, 1);
muQ(1:nxM) = x(2:end);
try
    [A1_ann, A2_ann] = local_computeA_ann(alpha, muQ, phiQ,
        beta, sigma, ...
                                mature, nxM, nxL,
                                ann);
    F = (A2_star - A2_ann + B2l_ann / B1l_ann * A1_ann) * 1e6;
catch
    F = 1e10 * ones(length(mature.error), 1);
end
end

function out = local_KalmanFilter(y, g0, gx, Rv, h0, hx, Repts)
% Standard Kalman filter
% Measurement:  y_t      = g0 + gx*x_t      + Sv*v_t
% Transition:   x_{t+1} = h0 + hx*x_t + Seps*w_{t+1}

[ny, T] = size(y);
nx      = size(hx, 1);

x0      = (eye(nx) - hx) \ h0;
vecP    = (eye(nx^2) - kron(hx, hx)) \ reshape(Repts, nx^2, 1);
P0      = reshape(vecP, nx, nx);

xHat     = zeros(nx, T);
xBar     = zeros(nx, T);
PHat     = zeros(nx, nx, T);
SxHat    = zeros(nx, nx, T);
PBar     = zeros(nx, nx, T);

```

```

VaryBar    = zeros(ny, ny, T);
yBar       = zeros(ny, T);
logL       = zeros(T, 1);
K          = zeros(nx, ny, T);
predError  = zeros(ny, T);

for t = 1:T
    if t == 1
        xBar(:,t)    = h0 + hx*x0;
        PBar(:, :, t) = hx*P0*hx' + Repr;
    else
        xBar(:,t)    = h0 + hx*xHat(:,t-1);
        PBar(:, :, t) = hx*PHat(:, :, t-1)*hx' + Repr;
    end
    yBar(:,t)        = g0 + gx*xBar(:,t);
    VaryBar(:, :, t) = gx*PBar(:, :, t)*gx' + Rv;

    selectY    = ~isnan(y(:,t));
    ny_t       = sum(selectY);
    invVaryBar = VaryBar(selectY, selectY, t) \ eye(ny_t, ny_t)
        ;

    K(:, selectY, t) = PBar(:, :, t) * gx(selectY, :)' *
        invVaryBar;
    predError(:,t)   = y(:,t) - yBar(:,t);
    xHat(:,t)        = xBar(:,t) + K(:,selectY,t) * predError(
        selectY,t);
    PHat(:, :, t)    = PBar(:, :, t) - ...
        K(:,selectY,t)*VaryBar(selectY,selectY,
            t)*K(:,selectY,t)';

    % Ensure PHat stays symmetric and numerically positive
    % definite.
    % Symmetrise first to eliminate floating-point asymmetry.

```

```

PHat(:,:,t) = (PHat(:,:,t) + PHat(:,:,t)') / 2;
% chol with two outputs returns a *partial* factor (p x p)
% when the
% matrix is not PD, which would cause a size mismatch on
% assignment.
[Ltmp, info] = chol(PHat(:,:,t), 'lower');
if info == 0
    SxHat(:,:,t) = Ltmp;
else
    % PHat is not PD: add a ridge until it is
    ridge = 1e-12;
    for iTry = 1:20
        [Ltmp, info] = chol(PHat(:,:,t) + ridge*eye(nx), '
            lower');
        if info == 0
            SxHat(:,:,t) = Ltmp;
            break;
        end
        ridge = ridge * 100;
    end
end

logL(t,1) = -(ny_t/2)*log(2*pi) ...
    - 0.5*log(det(VaryBar(selectY,selectY,t))) ...
    - 0.5*predError(selectY,t)'*invVaryBar*
        predError(selectY,t);
end

sumLogL = sum(logL, 1);

out.data      = y;
out.xHat      = xHat;
out.PHat      = PHat;
out.Sx        = SxHat;
out.logL      = logL;

```

```
out.sumLogL    = sumLogL;
out.Kgain      = K;
out.predError  = predError;
out.VaryBar    = VaryBar;
out.xBar       = xBar;
out.PBar       = PBar;
out.g0         = g0;
out.gx         = gx;
out.h0         = h0;
out.hx         = hx;
out.Rv         = Rv;
out.Reps       = Reps;
out.yBar       = yBar;
end

function res = local_yieldCurveDecom(outKF, model)
% Decomposes model-implied yields into expected short rates and
% term premia.

xhat    = outKF.xHat;
[nx,T]  = size(xhat);
muP     = model.muP;
phiP    = model.phiP;
maxMat  = max(model.matSelect);

xExp = nan(nx, maxMat, T);
for t = 1:T
    for i = 1:maxMat
        if i == 1
            xExp(:,1,t) = xhat(:,t);
        else
            xExp(:,i,t) = muP + phiP * xExp(:,i-1,t);
        end
    end
end
```

```

end

rExp = nan(maxMat, T);
for t = 1:T
    for i = 1:maxMat
        rExp(i,t) = model.r0 + model.rx * xExp(:,i,t);
    end
end

yHat      = outKF.g0 + outKF.gx * outKF.xHat;
ny        = length(model.matSelect);
rExpAvg = nan(ny, T);
for t = 1:T
    for i = 1:ny
        rExpAvg(i,t) = mean(rExp(1:model.matSelect(i), t));
    end
end

res.yHat      = yHat';
res.rExp      = rExpAvg';
res.termPremia = res.yHat - res.rExp;
res.matSelect = model.matSelect;
end

function res = local_campbellSchiller(Data, m)

[T, n]      = size(Data);
matSelect_CS = m : m : n*m;
shortRate   = Data(1:T-m, 1);
numYields   = n;

CSBetta      = NaN(2, numYields);
CSBetta_tstat = NaN(2, numYields);
CSBetta_se    = NaN(2, numYields);

```

```

Ydata          = NaN(T-m, numYields);
Xdata          = NaN(T-m, numYields);
R2             = NaN(1, numYields);

for i = 1 : numYields - 1
    k = matSelect_CS(i+1);
    Y = Data(1+m:T, i) - Data(1:T-m, i+1);
    x = (Data(1:T-m, i+1) - shortRate);
    X = [ones(T-m, 1), x * m / (k - m)];
    resOLS = local_nwest(Y, X, m+1);
    CSBetta(:, i+1) = resOLS.beta;
    CSBetta_se(:, i+1) = resOLS.beta ./ resOLS.tstat;
    CSBetta_tstat(:, i+1) = resOLS.tstat;
    Ydata(:, i+1) = Y;
    Xdata(:, i+1) = x * m / (k - m);
    R2(:, i+1) = resOLS.rsqr;
end

res.CSBetta = CSBetta;
res.maturities = matSelect_CS;
res.CSBetta_se = CSBetta_se;
res.CSBetta_tstat = CSBetta_tstat;
res.CSBetta_CI95 = [CSBetta(2,:) - 1.96 * CSBetta_se(2,:);
    ...
                    CSBetta(2,:) + 1.96 * CSBetta_se(2,:)];
res.CSBetta_CI99 = [CSBetta(2,:) - 2.575 * CSBetta_se(2,:);
    ...
                    CSBetta(2,:) + 2.575 * CSBetta_se(2,:)];
res.Ydata = Ydata;
res.Xdata = Xdata;
res.R2 = R2;
end

function res = local_nwest(y, X, lag)

```

```

% Newey-West OLS

[T, k] = size(X);
beta    = (X'*X) \ (X'*y);
e       = y - X*beta;

S = (e .* X)' * (e .* X);
for l = 1 : lag
    w    = 1 - l / (lag + 1);
    Xl    = X(l+1:T, :);
    el    = e(l+1:T);
    X0    = X(1:T-l, :);
    e0    = e(1:T-l);
    Gl    = (e0 .* X0)' * (el .* Xl);
    S     = S + w * (Gl + Gl');
end

XpX_inv = (X'*X) \ eye(k);
V        = T * XpX_inv * S * XpX_inv;
se       = sqrt(diag(V) / T);

res.beta  = beta;
res.tstat = beta ./ se;
res.se    = se;
res.rsqr  = 1 - var(e) / var(y);
end

% Extended Step 7 objective with three soft constraints:
% 1: Standard A2_star moment conditions (A-intercept fit)
% 2: Alpha anchor: alpha      mean_y1yr/ann
% 3: muQ shrinkage: |muQ| small
% 4: FULL UNCONDITIONAL LEVEL: embed Step 8 muP_1 recovery to
    penalise
%      deviations of the model-implied 10yr yield from its

```

```

    sample mean.
%
% The level constraint breaks the circular dependency between
    Steps 7 and 8
% by computing the implied muP_l forward from any candidate (
    alpha, muQ),
% then evaluating  $E[y_{10yr}] = g0_{10yr} + gx_{10yr} * E[x]$ .

function val = step7_objective(x, mature, phiQ, beta, sigma, ...
    A2_star, A1_star, B1l_ann,
    B2l_ann, B1l_chol, ...
    phiP_ll, IminusPhiP_ll, muP_m,
    ...
    nxM, nxL, ann, mean_y1yr, ...
    w_mean, w_muQ, w_level)

alpha_c = x(1);
muQ_c = [x(2:end); zeros(nxL,1)]; % nxM macro + nxL=0
    latent muQ

% Standard A2_star residuals
r_A2 = local_alphaMuQ_residuals(x, mature, phiQ, beta, sigma,
    ...
    A2_star, B1l_ann, B2l_ann,
    nxM, nxL, ann);

val = sum(r_A2.^2);

% Constraint 1: alpha anchor
val = val + w_mean^2 * (alpha_c - mean_y1yr/ann)^2;

% onstraint 2: muQ shrinkage
val = val + w_muQ^2 * sum(x(2:end).^2);

% Constraint 3: full unconditional 10yr level

```

```

% Recover muP_1 implied by this (alpha, muQ) candidate (mirrors
  Step 8)
try
    [A1_ann_c, ~] = local_computeA_ann(alpha_c, muQ_c, phiQ,
        beta, sigma, ...
                                mature, nxM, nxL, ann);
    muP_1_c = B11_chol \ (A1_star - A1_ann_c + ...
        B11_chol * phiP_ll / B11_chol *
        A1_ann_c);
    muP_c    = [muP_m; muP_1_c];

% Unconditional factor means  $E[x] = (I - \phi P)^{-1} * \mu P$ 
% Use block structure:  $E[x_m] = (I - \phi P_{mm})^{-1} * \mu P_m$       0
%
%                                $E[x_l] = (I - \phi P_{ll})^{-1} * \mu P_l$ 
Ex_l = IminusPhiP_ll \ muP_1_c;    % nxL x 1

% Implied 10yr yield:  $g_0(10yr) + g_x(10yr, :) * E[x]$ 
% gx row index for 10yr: matSelect = [4,8,20,28,40,60]; 10
%   yr=40Q is index 5
gx_10yr = gx(5,:);    % 1 x nx  (annualised)
g0_10yr = gx_10yr * zeros(nxM+nxL,1); % placeholder;
%   actual g0 not yet computed

% Approximate  $E[y_{10yr}]$  using only the factor mean
%   contribution:
%  $g_x(10yr) = ann * (-B(:,40)') / 40$  from the phiQ/beta
%   recursion.
% We penalise  $g_x_{latent} * E[x_l]$  the latent drift
%   contribution to
% the 10yr yield. If this is large, the level will be off.
% Compute  $B(:,40)$  directly (40 quarters = 10yr):
n10 = 40;
B_k = -beta;
for kk = 2:n10

```

```
        B_k = -beta + phiQ' * B_k;
    end
    gx_10yr_lat = ann * (-B_k(nxM+1:end)' / n10);    % 1 x nxL

    latent_drift_10yr = gx_10yr_lat * Ex_1;    % scalar (
        annualised decimal)
    val = val + w_level^2 * latent_drift_10yr^2;
catch
    val = val + 1e20;    % penalise if recovery fails
end
end
```